Har Ghar Jal
Jal Jeevan Mission
Building Partnership
Changing Lives

सुजलाम् भारत

Operational Guidelines

Jal Jeevan Mission 2.0

Har Ghar Jal

सत्यमेव जयते

Government of India
Ministry of Jal Shakti
Department of Drinking Water & Sanitation
National Jal Jeevan Mission

2026

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आपो हिष्ठा मयोभुवस्ता न ऊर्जे दधातन।
महे रणाय चक्षसे।।1।।
यो वः शिवतमो रसस्तस्य भाजयतेह नः।
उशतीरिव मातरः।।2।।

(ऋग्वेद संहिता-10.9.1-2)

जल आनंद का स्रोत है, ऊर्जा का भंडार है।
कल्याणकारी है।।
पवित्र करने वाला है।
और माँ की तरह पोषक तथा जीवनदाता है।।

Water is the source of
happiness, energy, health and piety,
and is life giving as mother!

*...**Jal Jeevan Mission** marks a major step towards ensuring sustainable and reliable tap water supply for rural households across India. It shifts the focus from only infrastructure creation to citizen-centric service delivery supported by strong local governance and digital monitoring.*

The move will strengthen water security, improve health outcomes and empower rural communities, especially our Nari Shakti.

Shri Narendra Modi
Prime Minister of India

सत्यमेव जयते

प्रधान मंत्री

Prime Minister

MESSAGE

Water has been revered in our scriptures as the very source of life. As our scriptures say, शं नो देवीरभिष्टय आपो भवन्तु पीतये - May the divine waters bring us welfare and nourishment.

The Jal Jeevan Mission has already brought about a transformative change to crores of rural households by providing access to safe and adequate drinking water through individual household tap connections. Beyond improving the quality of life, it has also strengthened public health outcomes and empowered India's Nari Shakti by reducing the drudgery of fetching water from faraway sources.

Now, Jal Jeevan Mission 2.0 builds upon this strong foundation and takes this work to the next level. It establishes a robust institutional framework for drinking water governance, complemented by digital data governance for rural water supply systems. This phase introduces key reforms to enhance transparency, strengthen institutions and improve service delivery, supported by a comprehensive capacity-building framework.

In the true spirit of grassroots democracy, Gram Panchayats lie at the centre of this approach, envisioned as the owners and operators of village water supply systems. With the support of digital platforms such as 'Sujal Gaon', they will be better equipped to monitor, manage and maintain these systems. These tools will help connect village-level operations with the broader water supply ecosystem, ensuring accountability from source to household.

As India advances through the Amrit Kaal towards Viksit Bharat, Jal Jeevan Mission 2.0 represents a growth from enabling access through infrastructure creation to reliable and sustainable service delivery. This reflects our steadfast commitment to ensuring regular water supply, assured water quality, source sustainability and efficient system management for every household.

With continued community participation and the collective commitment of all stakeholders, Jal Jeevan Mission 2.0 will further strengthen India's journey towards a water-secure future.

(Narendra Modi)

New Delhi
चैत्र 19, शक संवत् 1948
09 April, 2026

सत्यमेव जयते

सी आर पाटिल
C R Paatil

जल शक्ति मंत्री
भारत सरकार
Minister of Jal Shakti
Government of India

22nd March, 2026

FOREWORD

Safe and reliable drinking water is the foundation of a healthy, dignified and prosperous society. Through the Jal Jeevan Mission, India has undertaken one of the largest transformations in rural water supply, bringing tap water to crores of households and significantly improving the lives of millions, especially women and girls.

As we move into Jal Jeevan Mission 2.0, we are entering a new era where access to potable drinking water for every rural household is becoming a reality. This is not merely about infrastructure; it is about ensuring a healthy and dignified life for every citizen.

In this phase, the focus must firmly shift to reliability of service delivery and Jan Bhagidari through GPs/ VWSCs participation. States/ UTs must accord the highest priority to adherence to technical standards and protocols for quality assurance. There must be regular and rigorous monitoring of works at all levels, ensuring that systems are built to last and perform efficiently over time.

The District Water and Sanitation Mission (DWSM) will play a central role in ensuring effective implementation and oversight. At the same time, State Governments must put in place strong mechanisms for supervision, verification and compliance. There must be zero tolerance for any irregularities or deviations from prescribed quality standards, as the safety and well-being of citizens depend on it.

The introduction of *Jal Utsav* and a mandatory 15-day trial period before handing over in-village infrastructure under *Jal Arpan*, reflects this commitment to accountability and performance. It ensures that systems are tested, meet service standards and deliver reliable supply before being entrusted to communities.

This has further strengthened the approach of Jal Sanchay, Jan Bhagidari & Catch the Rain Campaign in promoting water conservation efforts. States/ UTs have been advised to take ground water recharge measures including rainwater harvesting, borewell recharge and reuse of grey water, in convergence with schemes such as MGNREGS/ VB-G RAM G and SBM. These efforts will help ensure long-term sustainability of drinking water sources and reliable availability of water for tap connections under the Mission.

At the same time, citizen accountability is at the heart of JJM 2.0. Through *Jal Seva Aankalan*, communities will play an active role in assessing service delivery. These assessments will guide districts and States in preparing targeted service improvement plans, ensuring that feedback from the ground directly shapes policy and implementation. In this way, JJM 2.0 lays the foundation for strengthened rural drinking water governance, built on transparency, participation and continuous improvement.

(C.R. Paatil)

सत्यमेव जयते

केंद्रीय राज्य मंत्री (MoS)
जल शक्ति एवं रेल मंत्रालय
भारत सरकार
Union Minister for States (MoS)
Ministry of Jal Shakti and Railways
Government of India

वी. सोमन्ना
V. Somanna

FOREWORD

22nd March, 2026

A strong nation is built on the well-being of its people and access to safe and clean drinking water remains central to this goal. Water has always been an integral part of India's civilisation, shaping our lives, livelihoods and traditions, where it is respected, conserved and celebrated.

However, for many years, ensuring reliable access to safe drinking water remained a challenge in rural India, especially for women who bore the burden of fetching water.

The launch of the Jal Jeevan Mission marked an important step in addressing this challenge. In a short span, the Mission has made significant progress, reaching the majority of rural households and improving quality of life through the collective efforts of States, local institutions and communities. Its impact is visible across rural India, reducing drudgery, improving health and nutrition and strengthening community ownership.

As we move into the next phase, the focus is on sustaining these gains. Jal Jeevan Mission 2.0 marks a shift from infrastructure creation to reliable service delivery, with Gram Panchayats and communities at the centre.

Initiatives such as *Jal Arpan* and *Jal Utsav* are deepening Jan Bhagidari by fostering community ownership, while *Sujalam Bharat* reflects a vision of transparent, technology enabled and citizen-centric water governance.

The operational guidelines for Jal Jeevan Mission 2.0 provide a clear framework to strengthen institutions, enhance accountability and ensure long-term sustainability of rural water supply systems, supported by key structural reforms for effective operation and maintenance.

This Guidelines will play an important role in realising water security as a key pillar of inclusive development in India's journey towards Viksit Bharat @2047. By ensuring universal access to safe drinking water, India is contributing to Sustainable Development Goal 6 and demonstrating how people-led development, supported by a whole-of-government and whole-of-society approach, can deliver lasting outcomes.

With this, I urge all stakeholders, States, Districts, Gram Panchayats and Citizens, to work together with a shared sense of responsibility. Through Jan Bhagidari, we can ensure that every village not only has access to water, but also sustains it for the future.

Jai Hind!

(V. Somanna)

सत्यमेव जयते

अशोक के. के. मीना
Ashok K. K. Meena

सचिव
पेयजल एवं स्वच्छता विभाग
Secretary
Department of Drinking Water & Sanitation

22nd March, 2026

FOREWORD

The Jal Jeevan Mission has brought about a significant transformation in rural water supply by expanding access to tap water across the country. As we move into the next phase, Jal Jeevan Mission 2.0 seeks to consolidate these gains by focusing on sustainability, efficiency and accountable service delivery.

This phase places strong emphasis on institutional reforms to strengthen drinking water governance across all levels. A clear institutional architecture has been outlined, defining roles and responsibilities from the National to the Gram Panchayat level, with a particular focus on empowering local institutions for effective service delivery. The District Water and Sanitation Mission (DWSM) and District Technical Units (DTUs) will play a critical role in ensuring technical robustness and implementation oversight.

At the village level, the Gram Panchayat, through its standing committee for water and sanitation (VWSC), is positioned at the core of service delivery. The VWSC will function as the executive arm of the Gram Panchayat, responsible for day-to-day management, monitoring and upkeep of water supply systems. Strengthening these standing committees is central to ensuring ownership, accountability and sustainability of services at the grassroots level.

A key element of these reforms is the deepening of community participation. Community-based organisations such as Self-Help Groups (SHGs) and Primary Agricultural Credit Societies (PACS) will play an active role in supporting O&M, monitoring and local management overseen by VWSCs. This approach aims to build ownership, strengthen accountability and ensure long-term sustainability of water supply systems.

The guidelines also emphasise digital governance and data-driven monitoring, with the use of interactive dashboards at National, State, District and Panchayat levels. Integration with citizen interface platforms such as Meri Panchayat Mobile Application will enable real-time feedback, grievance redressal and participatory monitoring, thereby strengthening transparency and responsiveness in service delivery.

These reforms also focus on strengthening technical standards, improving operational efficiency and ensuring consistent service delivery outcomes. Together, all the measures aim to transition the sector from a scheme-based infrastructure creation approach to a service-oriented, lifecycle-based regular water supply systems in rural India.

Jal Jeevan Mission 2.0 represents a comprehensive effort to build resilient and sustainable rural water supply systems through strong institutions, empowered communities and effective governance mechanisms. I am confident that these guidelines will support States/ UTs in achieving the Mission's objectives and ensuring reliable drinking water services for every rural household.

(Ashok K. K. Meena)

सत्यमेव जयते

कमल किशोर सोन
Kamal Kishore Soan

अतिरिक्त सचिव एवं मिशन निदेशक (एनजेजेएम)
पेयजल एवं स्वच्छता विभाग
Additional Secretary & Mission Director (NJJM)
Department of Drinking Water & Sanitation

PREFACE

22nd March, 2026

Jal Jeevan Mission has significantly improved rural drinking water access by bringing tap water within household premises. In JJM 2.0, the focus now shifts from creating infrastructure to ensuring reliable and sustained water service delivery. The Cabinet has approved service-delivery orientation and the guidelines reflect that direction. The heart of JJM 2.0 is strong digital governance through the Sujalam Bharat digital framework. Every village is being assigned a Sujal Gaon ID, mapping the entire system from source to tap. This strengthens transparency, avoids duplication and supports better monitoring. Interactive dashboards and near real-time information flow are ensuring early flagging of problems and quick corrections, which is very important. Command and Support Centres to ensure that projects are built correctly and function as per plan. JJM 2.0 calls for stronger implementation systems and structural reforms. States/ UTs to prepare and follow Comprehensive Implementation and Reform Plans (CIRP) to document road maps for scheme completion, handing over to GPs and actions needed to keep those working.

Technical compliance remains a key requirement. Following CPHEEO norms and ensuring precise validation, States/ UTs will help in maintaining the construction quality, fair use of funds and sustainability. Decision Support Systems (DSS) will integrate groundwater status, scheme-level data and field observations to help DWSMs plan better and act faster, ultimately to support improved functionality. Districts are to provide technical support to GPs through District Technical Unit (DTU), monitor progress and guide O&M practices. The idea of district-level leadership was boosted during the 700-plus interactions with District Collectors held over last one year. Those interactions shaped the structural reforms, especially the shift towards accountability and monitoring. Initiatives like Peyjal Samvad further strengthened the message that DWSMs must develop practical solutions, improve preventive maintenance and adopt cost-effective practices suited to local needs. Similarly, village-level conversations like Sujal Gaon Samvad helped communities understand their roles in keeping systems functional and reporting service gaps. As JJM 2.0 places GPs at the centre of service delivery, schemes will be handed over through Jal Arpan, after 15 days trial run and verification to strengthen the O&M ecosystem through Jan Bhagidari. Villagers will come together in annual Jal Utsav, as a mark to restore commitment towards drinking water.

JJM 2.0 is a step toward a stronger, transparent, community owned and managed rural water service framework. As we begin this phase, our goal is clear: to build a service oriented, accountable and community led water supply system. The guidelines that follow reflect the structural reforms approved by the Cabinet and outline how States/ UTs, districts and GPs can work together to achieve this. I am confident that with collective effort, close attention to technical standards and steady implementation of CIRP, we will ensure that every rural household receives reliable drinking water, every day of the year.

(Kamal Kishore Soan)

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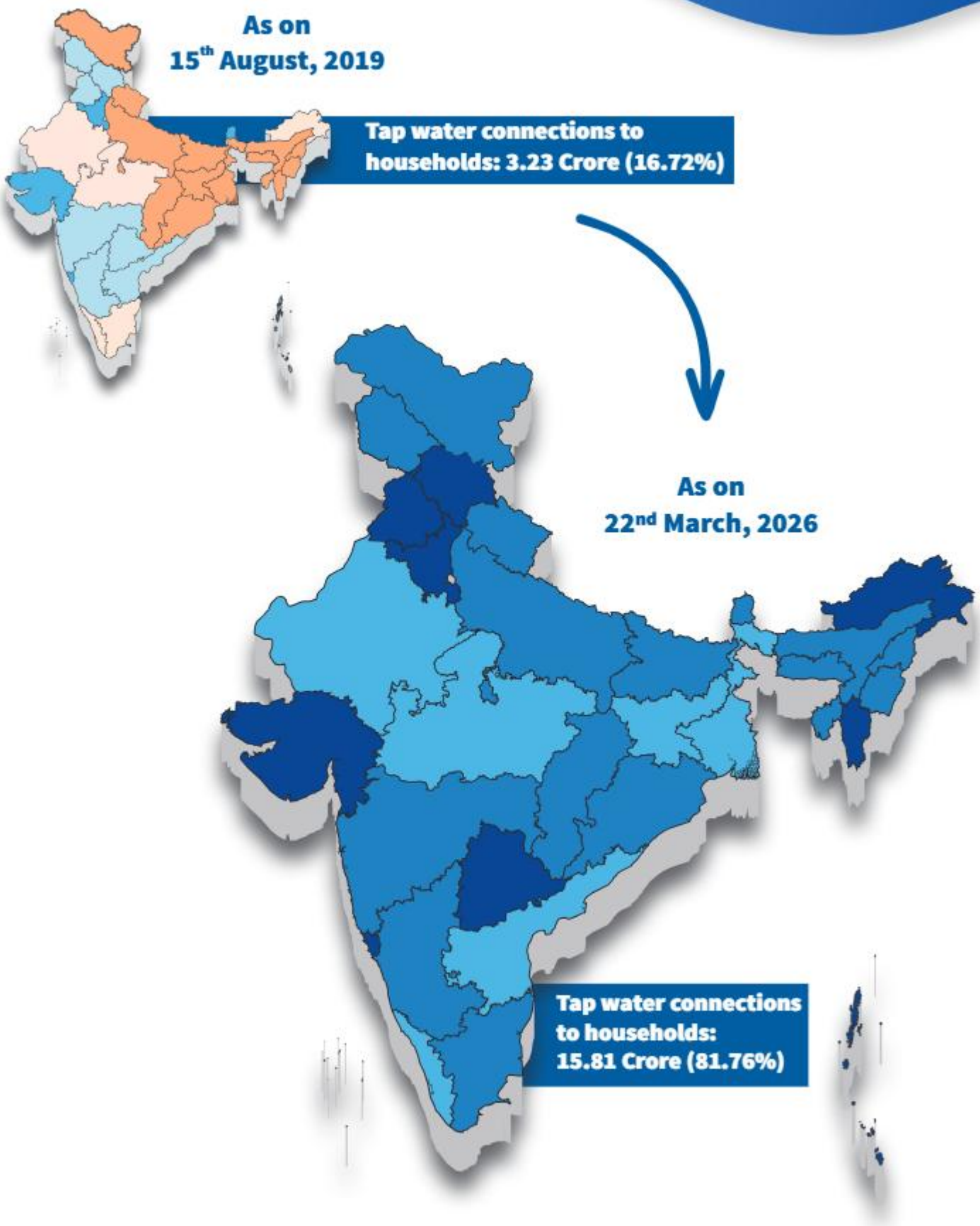
As on
15th August, 2019
Tap water connections to households: 3.23 Crore (16.72%)
As on
22nd March, 2026
Tap water connections to households: 15.81 Crore (81.76%)
0%-10%
11%-25%
26%-50%
51%-75%
76%-100%
100%

Abbreviations

AAP	Annual Action Plan
ACS	Additional Chief Secretary
AI	Artificial Intelligence
AICTE	All India Council for Technical Education
AMRUT	Atal Mission for Rejuvenation and Urban Transformation
ANM	Auxiliary Nurse Midwife
ANRF	Anusandhan National Research Foundation
API	Application Programming Interface
ASHA	Accredited Social Health Activist
ATR	Action Taken Report
AWS	Automatic Weather Station
AWW	Anganwadi Worker
BCC	Behaviour Change Communication
BIS	Bureau of Indian Standards
BWS	Bulk Water Supply
BWT	Bulk Water Transfer
C&AG	Comptroller and Auditor General of India
CA	Chartered Accountant
CBO	Community Based Organisation
CAPEX	Capital Expenditure
CEO-ZP	Chief Executive Officer-Zila Parishad
CGWB	Central Ground Water Board
CII	Confederation of IndianIndustry
CIRP	Comprehensive Implementation and Reform Plan
CLF	Cluster Level Federation
CPHEEO	Central Public Health and Environmental Engineering Organisation
CPO	Central Prabhari Officer
CSIR	Council of Scientific and Industrial Research
CSR	Corporate Social Responsibility
CWPP	Community Water Purification Plant
DA-JGUA	Dharti Abha Janjatiya Gram Utkarsh Abhiyan
DC/DM	District Collector/ Deputy Commissioner/ District Magistrate
DCSC	District Command and Support Centre
DDWS	Department of Drinking Water & Sanitation
DIP	District Improvement Plan
DLI	Disbursement Linked Indicator
DMF	District Mineral Development Fund
DPI	Digital Public Infrastructure
DPIIT	Department for Promotion of Industry and Internal Trade
DPMU	District Project Management Unit
DPR	Detailed Project Report
DSS	Decision Support System
DSSAP	District Source Sustainability Action Plan
DST	Department of Science and Technology
DTU	District Technical Unit
DWSM	District Water and Sanitation Mission
ESR	Elevated Storage Reservoir
FHTC	Functional Household Tap Connection
FICCI	Federation of Indian Chambers of Commerce and Industry
FRC	Free Residual Chlorine
FTK	Field Test Kit
GIS	Geographic Information System
GP	Gram Panchayat
GPDP	Gram Panchayat Development Plan
HGJ	Har Ghar Jal
IA	Implementing Agency
IAW	Internal Audit Wing
IDSP	Integrated Disease Surveillance Programme
IEC	Information, Education and Communication
IFMIS	Integrated Financial Management Information System
IIT	Indian Institute of Technology
IMD	India Meteorological Department
IMIS	Integrated Management Information System

INGRES	India-Ground Water Resource Estimation System
IoT	Internet of Things
IS	Indian Standard
ISO	International Organisation for Standardisation
ISRO	Indian Space Research Organization
ITDA	Integrated Tribal Development Agency
ITDP	Integrated Tribal Development Programme
ITI	Industrial Training Institute
IWMP	Integrated Watershed Management Programme
JJM	Jal Jeevan Mission
JJM-IMIS	Jal Jeevan Mission – Integrated Management Information System
JSA	Jal Seva Aankalan
JSIC	Jal Shakti Innovation Centre
LMS	Learning Management System
LPCD	Liters Per Capita Per Day
MAHA	Mission for Advancement in High-impact Areas
MBR	Master Balancing Reservoir
MeitY	Ministry of Electronics & Information Technology
MGNREGA	Mahatma Gandhi National Rural Employment Guarantee Act
MIS	Management Information System
ML	Machine Learning
MoEFCC	Ministry of Environment, Forest and Climate Change
MoHFW	Ministry of Health and Family Welfare
MoHUA	Ministry of Housing and Urban Affairs
MoJS	Ministry of Jal Shakti
MoU	Memorandum of Understanding
MoWCD	Ministry of Women and Child Development
MPLAD	Members of Parliament Local Area Development
MSDE	Ministry of Skill Development and Entrepreneurship
MVS	Multi Village Scheme

NCVET	National Council for Vocational Education and Training
NDWM	National Drinking Water Mission
NFHS	National Family Health Survey
NGO	Non-Governmental Organisation
NAQUIM	National Aquifer Mapping and Management
NHM	National Health Mission
NIRD&PR	National Institute of Rural Development and Panchayati Raj
NIT	National Institute of Technology
NITI Aayog	National Institution for Transforming India Aayog
NJJM	National Jal Jeevan Mission
NJM	Nal Jal Mitra
NJMP	Nal Jal Mitra Programme
NPMU	National Project Management Unit
NRDWP	National Rural Drinking Water Programme
NRLM	National Rural Livelihood Mission
NRW	Non-Revenue Water
NSQF	National Skills Qualifications Framework
NTC	National Technical Committee
NWE	National WASH Expert
O&M	Operation and Maintenance
OBF	Outcome Based Financing
OHSR	Overhead Service Reservoir
OPEX	Operating Expenditure
OTP	One Time Password
PACS	Primary Agricultural Credit Societies
PFMS	Public Financial Management System
PHE	Public Health Engineering
PHED	Public Health Engineering Department
PIA	Project Implementing Agency
PM-JANMAN	Pradhan Mantri Janjati Adivasi Nyaya Maha Abhiyan
PMKSY	Pradhan Mantri Krishi Sinchayee Yojana
PMKVY	Pradhan Mantri Kaushal Vikas Yojana
PPP	Public Private Partnership

PRA	Participatory Rural Appraisal
PRI	Panchayati Raj Institution
PS	Principal Secretary
PSA	Principal Scientific Adviser
PSU	Public Sector Undertaking
PWS	Piped Water Supply
QA	Quality Assurance
QC	Quality Control
QCI	Quality Council of India
R&D	Research & Development
RBI	Reserve Bank of India
RD	Rural Development
RDWSD	Rural Drinking Water Supply Department
RJJK	Rashtriya Jal Jeevan Kosh
RKVY	Rashtriya Krishi Vikas Yojana
RLB	Rural Local Body
RPL	Recognition of Prior Learning
RPWSS	Rural Piped Water Supply Scheme
RSETI	Rural Self Employment Training Institute
RWH	Rainwater Harvesting
RWPF	Rural WASH Partners' Forum
RWS	Rural Water Supply
SBM-G	Swachh Bharat Mission (Grameen)
SCADA	Supervisory Control and Data Acquisition
SCERT	State Council of Educational Research and Training
SCSC	State Command and Support Centre
SDG	Sustainable Development Goal
SHG	Self Help Group
SIDH	Skill India Digital Hub
SIP	Service Improvement Plan
SIRD	State Institute of Rural Development
SLA	Service Level Agreement
SLB	Service Level Benchmark
SLS	State Linked Scheme
SNA	Single Nodal Account
SOP	Standard Operating Procedure
SPM-NIWAS	Dr. Syama Prasad Mookerjee National Institute of Water and Sanitation
SPMU	State Project Management Unit
SPV	Special Purpose Vehicle
SRLM	State Rural Livelihood Mission
SSC	Source Sustenance Committee
SSDM	State Skill Development Mission
STC	State Technical Committee
STT	Short Term Training
SVS	Single Village Scheme
SWIC	State Water Informatics Centre
SWSM	State Water and Sanitation Mission
TAT	Turnaround Time
TOR	Terms of Reference
ToT	Training of Trainers
TPIA	Third Party Inspection Agency
UDISE	Unified District Information System for Education
UGC	University Grants Commission
UT	Union Territory
VAP	Village Action Plan
VB-G RAM G	Viksit Bharat-Guarantee for Rozgar and Ajeevika Mission (Gramin)
VEP	Village Entry Point
VGF	Viability Gap Funding
VHSNC	Village Health, Sanitation and Nutrition Committee
VSSC	Village Source Sustenance Committee
VWSC	Village Water and Sanitation Committee
WRIS	Water Resources Information System
WASH	Water, Sanitation and Hygiene
WCD	Women and Child Development
WDC	Watershed Development Component
WMPSC	Water Management & Plumbing Skill Council
WQM&S	Water Quality Monitoring and Surveillance
WQMIS	Water Quality Management Information System
WSO	Water Supply Operator
WTP	Water Treatment Plant
ZBR	Zonal Balancing Reservoir
ZP	Zila Parishad/ Zila Panchayat

Definitions

Annual Action Plan (AAP): A one-year operational programme prepared by States/ UTs within the CIRP framework.

Aquifer-Based Planning: A hydrogeological approach to design and manage groundwater-based water supply schemes using aquifer maps and tools (such as NAQUIM, India-WRIS), groundwater assessment data and aquifer characteristics (type, depth, seasonal behaviour, recharge potential, extraction stage) to inform siting of borewells, planning of recharge structures (check dams, percolation tanks, recharge shafts) and preparation of Sustainability Action Plans.

BIS IS:10500 (Drinking Water Quality Standard): The Bureau of Indian Standards specification IS:10500 (latest edition, read with latest amendments) that prescribes permissible limits for physical, chemical, bacteriological and radiological parameters of drinking water in India. All rural piped water supply schemes under JJM shall ensure that water supplied to households conforms to BIS publication IS:10500 standards at the delivery point.

Bulk Water Supply (BWS): It refers to the transfer of raw water from the point of intake through transmission lines upto the WTP.

Community Water Purification Plant (CWPP): A point-of-use or community-scale water treatment unit designed to remove specific chemical contaminants (arsenic, fluoride, iron, nitrates etc. as required) or microbial contamination from raw water to produce potable water. CWPPs are deployed as interim measures in quality-affected habitations providing 8–10 lpcd until a piped water supply system is established.

Comprehensive Implementation and Reform Plan (CIRP): A strategic, time-bound operational plan prepared by each State/ UT outlining scheme-wise completion timelines, financial allocations, reform commitments and governance milestones for the JJM 2.0 mission period.

Decision Support System (DSS) - Source Sustainability: A digital planning and analytical platform integrated with the JJM-IMIS framework that combines geo-spatial layers, hydrogeological parameters, CGWB groundwater status data, rainfall data, aquifer maps and drinking water source datasets to support evidence-based planning, monitoring and prioritisation of source sustainability measures from district to village level.

Design Capacity: Maximum hydraulic or treatment capacity for which a water treatment plant, pipeline, or reservoir is designed, usually expressed in million litres per day (MLD) or similar units to meet expected demand.

Design Life of Structures: The period for which water supply scheme structure or asset is designed to perform its intended function without repair/ reconstruction.

Designated Implementation Agency (DIA): The State authorized agency registered on the Sujalam Bharat PM GatiShakti App entrusted for GIS asset mapping.

Digital Twin: A dynamic virtual replica of a physical water supply system including sources, treatment plants, storage structures, transmission, distribution networks and service areas created using real-time and historical data to simulate, monitor and optimize system performance. It integrates data from sensors, SCADA systems, GIS platforms and operational records.

District Command and Support Centre (DCSC): The central operational hub established under the DWSM that provides near real-time monitoring of scheme functionality, water quality alerts, grievance tracking and coordinated corrective response for rural water supply across the district. The DCSC will be integrated with State and National centres to enable consolidated monitoring, advanced data analytics and two-way sharing of insights for informed decision making.

District Technical Unit (DTU): A technical committee constituted under the DWSM to provide continuous technical support to Gram Panchayat and VWSC-level micro utilities. The DTU coordinates between micro utilities (in-village infrastructure) and macro utilities (bulk/out-village infrastructure), oversees scheme commissioning and handover, manages asset registries, conducts source sustainability planning and operates District Command and Support Centres.

Domestic Demand: Water required for drinking, cooking, personal bathing, domestic sanitation, etc. Domestic demand does not include requirements of water for cattle, semi-commercial, commercial, institutional or industrial purposes.

Field Test Kit (FTK): A portable multi-parameter kit used for examination of physico-chemical contamination as initial screening.

Free Residual Chlorine (FRC): Free Residual Chlorine, is the low-level amount (0.2 mg/l as per BIS 10500) of chlorine remaining in the water after a certain period or contact time after its initial application. It constitutes an important safeguard against the risk of subsequent microbial contamination after treatment - a unique and significant benefit for public health.

Functional Household Tap Connection (FHTC): A household tap connection that provides potable water in adequate quantity (minimum 55 litres per capita per day), of prescribed quality (conforming to BIS IS:10500), on a regular basis at the delivery point.

Functionality Assessment: A periodic structured assessment to determine whether a village water supply scheme is delivering services meeting prescribed service level parameters: adequacy of supply (≥55 lpcd), quality compliance (BIS IS:10500), regularity of supply.

GIS-Based Asset Mapping: The geo-tagging or geo-referencing of all rural piped water supply infrastructure assets (sources, WTPs, pumping stations, MBRs, ZBRs, OHTs, distribution pipelines and household connections) using GPS-enabled mobile applications and satellite imagery, linked to their Asset IDs, Scheme IDs and Sujal Gaon IDs on the Sujalam Bharat platform.

Greywater Management: The collection, treatment and safe reuse of household wastewater generated from domestic activities (kitchen, bathing, laundry) excluding toilet waste (blackwater).

Har Ghar Jal (HGJ) Certification: A formal certification accorded to a Gram Panchayat or village where every rural household has been provided with a Functional Household Tap Connection (FHTC) supplying potable water at a minimum of 55 lpcd, of quality conforming to BIS IS:10500, on a regular basis.

In-Village Infrastructure: In-Village Infrastructure refers to the distribution of treated water from the Overhead Storage Reservoir (OHSR) to individual households within the village. This includes the internal distribution network, service connections and all components facilitating last-mile delivery of potable water to end users.

Inadmissible Expenditure (under JJM Central Share): Categories of expenditure that cannot be charged to the Central share under JJM.

Jal Arpan: The formal commissioning and community handover ceremony/protocol under which a completed rural piped water supply scheme (after a successful 15-day minimum trial run) is officially handed over by the implementing agency to the Gram Panchayat for long-term ownership, operation and maintenance.

Jal Seva Aankalan (JSA): A GP/ VWSC-led digital self-assessment tool and mechanism for evaluating the functionality, adequacy, quality compliance, institutional readiness and financial sustainability of rural water supply systems at the village level.

Jal Seva Adhyayan: Jal Seva Adhyayan is a programme under JJM 2.0 where engineering students, under faculty mentorship, visit villages to learn about the development of rural water supply infrastructure, undertake field-based assessment of rural water supply schemes, assess service delivery at the ground level and share their findings digitally to help improve transparency and system performance.

JJM-IMIS (Integrated Management Information System): The national web-based information management platform of the Jal Jeevan Mission that integrates data on scheme sanction, financial releases, physical progress, households covered, water quality monitoring (WQMIS), functionality assessments, grievances and training activities across all States/ UTs.

Key Performance Indicator (KPI): A measurable, time-bound metric used to evaluate the performance of a utility, scheme, or institutional body against defined service standards.

Life Cycle Cost (LCC) Assessment: An economic evaluation methodology that considers all costs associated with a water supply component over its

entire design life, including capital expenditure (CAPEX), operating expenditure (OPEX), rehabilitation and replacement costs and residual value.

Macro Utility: A State, Union Territory, or Regional-level professional body (parastatal agency, PHED, Jal Nigam, or specialised utility) responsible for planning, development, management and supervision of out-village water supply assets, including surface water intake structures, water treatment plants, bulk transmission systems, master balancing reservoirs and Multi-Village / Bulk Water Supply scheme networks.

Micro Utility: The village-level decentralised water service delivery unit constituted by designating VWSC (standing committee of the Gram Panchayat) as the service provider responsible for distribution of drinking water within the village, collection of service delivery charges, management of in-village infrastructure, minor repairs and maintenance, water quality monitoring and financial management.

Mother Sanction: The mother sanction in SNA SPARSH is the initial cumulative financial approval by DDWS through the PFMS to a State/ UT establishing the authorised expenditure limit for a given financial year.

Multi-Village Scheme (MVS): Comprehensive water supply scheme providing infrastructure and services to multiple villages or Gram Panchayats collectively, commonly used for regional water resource optimisation and economies of scale.

Nal Jal Mitra (NJM)/ Jal Vitaran Sanchaalak: A locally trained and NCVET-certified multi-skilled O&M personnel (NSQF Level-4, Job role: Jal Vitaran Sanchaalak) responsible for in-village water supply scheme operation and maintenance, including pump operation, basic plumbing repairs, routine maintenance, network upkeep, water quality reporting and grievance management.

Non-Piped Water Supply (Non-PWS) Scheme: A water supply arrangement that provides safe drinking water through community access points (stand posts, hand pumps, community water purification plants) in situations where household piped connections are technically unfeasible due to extreme terrain, climatic conditions, or dispersed habitations.

Non-Revenue Water (NRW): The difference between the volume of treated water put into the distribution system and the volume of water billed or accounted to consumers. NRW includes real losses (physical leakage from pipes, joints and storage structures), apparent losses (meter inaccuracies, unauthorized connections) and unbilled authorised consumption.

NSQF (National Skills Qualifications Framework): A nationally integrated competency-based framework that organizes qualifications according to a series of levels of knowledge, skills and aptitude, enabling standardization, mobility and progression across vocational education, training and general education systems.

Out-Village Infrastructure: Out-Village Infrastructure includes the transfer of treated water from the Water Treatment Plant (WTP) to the Master Balancing Reservoir (MBR) and from the MBR to the Overhead Tank (OHT) including all associated transmission lines.

Preventive Maintenance: The planned, periodic inspection, servicing, testing and timely repair or replacement of components of rural piped water supply systems, undertaken to prevent system failures, ensure water quality and functionality, reduce service disruptions and sustain reliable delivery of drinking water services over time.

Project Management Unit (PMU): A technical and managerial support unit established at the National, State and District levels, comprising technical, management and IT experts to support scheme implementation, structural reforms and data management, including functionality monitoring and technical support through command and support centres. It may also include experts for IEC, Capacity building and impact assessment.

Recognition of Prior Learning (RPL): A certification pathway through which individuals with prior informal work experience in O&M of water supply or related domains are assessed, upskilled through a structured 120-hour programme and awarded the nationally recognised NSQF Level-4 Jal Vitaran Sanchaalak certification without undergoing the full Short-Term Training (STT) of 510 hours.

Sanitary Inspection/ Sanitary Survey: A sanitary inspection is an on-site inspection of a water supply facility to identify actual and potential sources of microbiological contamination for evaluation of the physical structure and operation of the system and external environmental factors (such as toilet location).

SCADA (Supervisory Control and Data Acquisition): An automated control and monitoring system used to supervise, control and acquire operational data from distributed water supply infrastructure components such as pumps, valves, flow meter, pressure transmitters and treatment units.

Service Area: The defined geographical area comprising all villages, habitations, households and institutions planned to be served by the scheme, as per the approved technical design, source, treatment, storage and distribution infrastructure.

Service Improvement Plan (SIP): A structured remedial action plan prepared by DWSMs and SWSMs based on the findings of Jal Seva Aankalan, identifying service gaps (infrastructure, operational, financial, institutional) and defining time-bound corrective measures, responsible entities and resource requirements to restore or improve water supply service delivery.

Service Level Agreement (SLA): An agreement between service provider and contracting authority, clearly defining service delivery standards, responsibilities and measurable performance indicators.

Service Level Benchmarks (SLBs): Quantitative performance indicators defined by regulatory or programme authorities to measure the quality and efficiency of water supply services.

Service Level Norm (55 LPCD): The prescribed minimum quantity of water, i.e. at least 55 litres per capita per day (lpcd), of prescribed quality, i.e. conforming to BIS:10500 standards, on regular basis. The prescribed quantity includes water for drinking, cooking, personal hygiene and domestic sanitation needs etc.

Single Village Scheme (SVS): Water supply scheme which includes the local source, storage with or without treatment under the administrative jurisdiction of a Gram Panchayat.

SNA-SPARSH: An integrated financial management and payment mechanism that connects JJM-IMIS with the Public Financial Management System (PFMS), State Integrated Financial Management Information System (IFMIS) and the Reserve Bank of India's e-Kuber platform to enable scheme-wise, just-in-time payments, near-zero unspent balances and automated reconciliation of Central and State fund releases.

Source Protection Zone: A defined geospatial area surrounding a drinking water source (borewell, tube well, spring, or surface water intake) within which land use is regulated and certain activities are prohibited to prevent contamination of the drinking water source.

Source Sustainability: The ability to ensure the long-term reliability, adequacy and safety of drinking water sources throughout the full lifecycle of rural water supply schemes.

Source Sustenance Committee (SSC): A multi-tier committee constituted at the State, District and Gram Panchayat levels within the JJM institutional structure, responsible for planning, implementing, monitoring and coordinating source sustainability measures.

Sujal Gaon ID: A unique system-generated alphanumeric identifier assigned to each service area (habitation) within the Sujalam Bharat DPI framework, enabling complete source-to-tap digital mapping of rural piped water supply systems.

Sujal Gaon Utility App: The mobile application serving as the standardised transactional data-capture platform for village-level O&M under JJM, enabling digital recording of daily pumping operations, water quality, FTK/ lab testing results, chlorination logs, preventive maintenance activities, service delivery charge collection and grievances.

Sujalam Bharat (Digital Public Infrastructure-DPI): The national-level federated Digital Public Infrastructure for India's rural drinking water sector, developed as the digital backbone for GIS-based asset mapping, monitoring and lifecycle management of Rural Piped Water Supply Schemes (RPWSS). It assigns unique Sujal Gaon IDs to each service area, integrates interoperable digital registries, transaction platforms and decision-

support systems and provides democratised data access to multiple cross-sectoral stakeholders through open APIs.

Sujalam Shakti: Community-based water quality monitoring and surveillance constituted from women's Self-Help Groups (SHGs) under State Rural Livelihood Missions (SRLM), trained and certified at NSQF Level-1 in basic water quality testing and surveillance.

Third-Party Inspection Agency (TPIA): An independent technical agency empanelled to conduct quality assurance inspections of rural water supply scheme construction activities.

Treated Drinking Water: Water that has successfully undergone all required treatment processes in a Water Treatment Plant/ filtration units/ CWPP and meets accepted potable water quality standards as per BIS: 10500 suitable for human consumption.

Village Water & Sanitation Committee (VWSC)/ Pani Samiti: The village-level standing committee of the Gram Panchayat constituted to plan, implement, manage and oversee in-village rural piped water supply systems.

Water Budgeting: A structured, data-driven assessment & planning exercise to estimate total water availability (from rainfall, groundwater recharge, runoff, surface storage and greywater reuse) and total demand (domestic, livestock, irrigation and other uses) within a defined service area.

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SECTION 1
Background

CHAPTER 1.1

Introduction

Jal Jeevan Mission has brought a transformative change in rural India by ensuring access to safe and reliable tap water in more than 81% of the households. It has significantly reduced the drudgery of women and children who earlier spent hours fetching water, improved public health by providing clean drinking water and enhanced dignity and quality of life in villages. Beyond infrastructure, JJM has fostered community participation, strengthened local governance through Gram Panchayats and promoted sustainable water management practices. This holistic approach has not only improved service delivery but has also contributed to socio-economic development across rural India.

1.1.1. Historical Perspective of Rural Water Supply in India

Access to safe and adequate drinking water has always been an important development need in rural India because it affects health, daily life and livelihoods. For many years, people in villages depended on sources that were not reliable or safe, which created a need for planned and policy-based action in the water supply sector.

The below timeline presents the historical evolution of rural drinking water initiatives in India, reflecting a gradual transition from supply-driven public health interventions to community-based management and service-level delivery frameworks.

The first major step was the Accelerated Rural Water Supply Programme (ARWSP) launched in 1972. It aimed to supplement the efforts of State Governments, especially in areas of scarcity and those endemic to water borne diseases. Although it improved access, the approach was mainly supply driven and did not focus much on long term source sustainability or community ownership. Later programmes, such as the Technology Mission (1986)/ Rajiv Gandhi National Drinking Water Mission, added better technology, water quality testing and institutional systems, but many challenges remained.

A major change came with the Swajaldhara Programme (2002), which promoted a community led approach. It encouraged Gram Panchayats to take full responsibility for managing their schemes including operation and maintenance. While this improved ownership and reduced costs, many villages faced difficulties due to limited technical skills, financial issues and lack of long term support systems.

Figure 1: Historical Evolution of Rural Drinking Water Initiatives in India

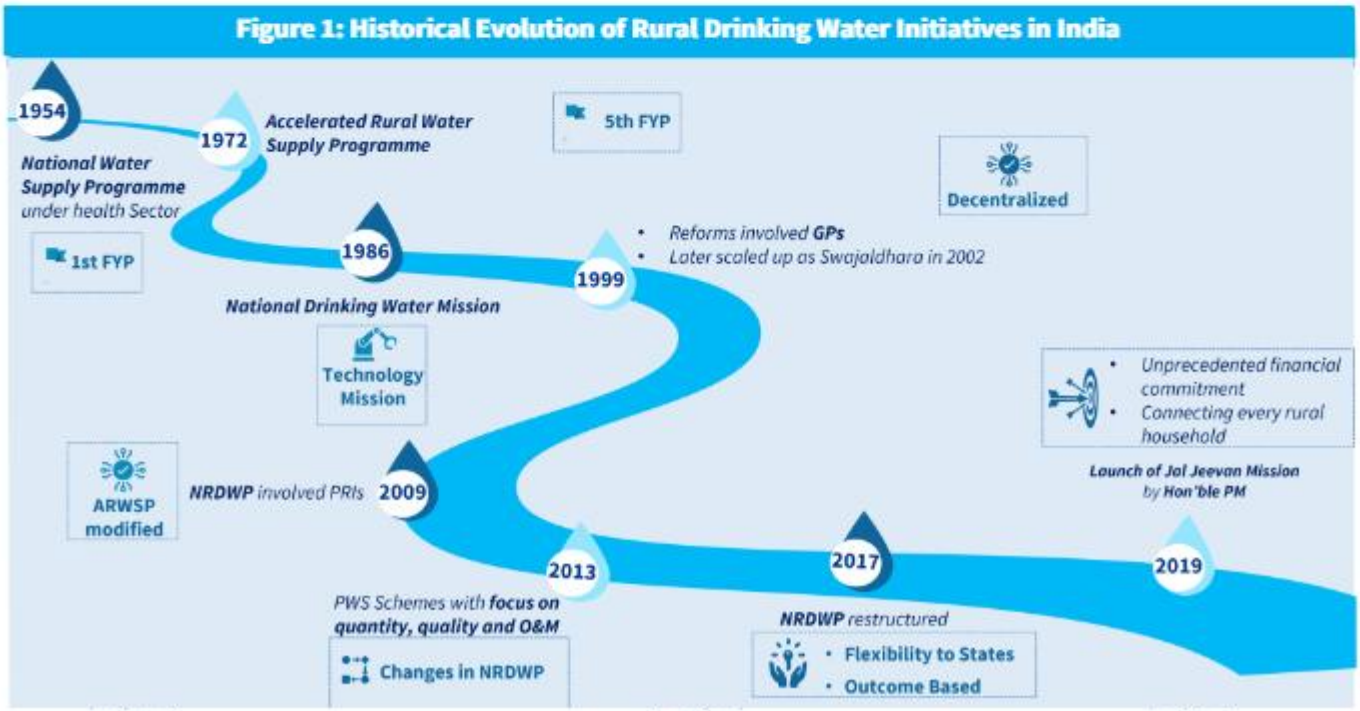
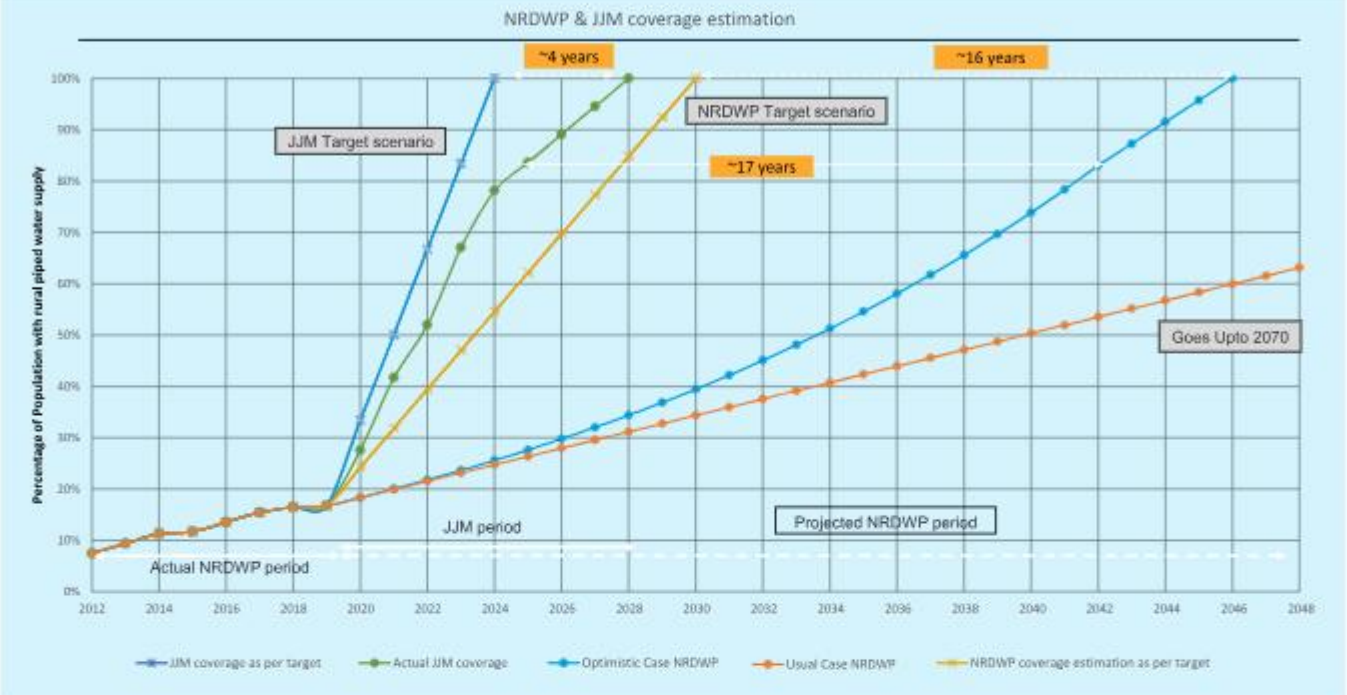


Figure 2: NRDWP and JJM PWS Coverage Estimation for Household Connections



The scenario assessment indicates that, had rural piped water supply expansion continued at the historical pace and investment levels of NRDWP, achieving universal coverage would have taken approximately 17 additional years compared to the progress under JJM. NRDWP followed a gradual, resource-constrained trajectory, whereas JJM represents a mission-mode national initiative characterized by a significantly higher scale of public investment.

The National Rural Drinking Water Programme (NRDWP, 2009) tried to bring together sustainability, decentralised planning and service delivery. However, problems related to functionality, source security and service levels continued.

Jal Jeevan Mission (2019) is the most ambitious reform in rural water supply sector. It focuses on providing every rural household with a Functional Household Tap Connection along with clear service standards, digital monitoring and stronger institutions. JJM builds on past lessons to ensure safe, reliable and sustainable drinking water for all rural households.

1.1.2. Jal Jeevan Mission - A Paradigm Shift

Jal Jeevan Mission aims to provide Functional Household Tap Connections (FHTCs) to every rural household, ensuring safe and adequate drinking water. The Mission marked a paradigm shift from infrastructure centric approach to a service delivery framework with defined standards for quality, quantity and reliability. Implemented in partnership with States/ UTs, JJM has significantly expanded rural water supply coverage through enhanced investments, strengthened institutional frameworks, community participation and sustainability measures. This transformation has led to notable improvements in public health, dignity and ease of living, laying a strong foundation for the next phase of the Mission.

A. Key Outcomes under JJM

Following are key outcomes achieved under JJM, reflecting a significant transformation in rural drinking water service delivery across the country:

i. **Universalisation of Household Tap Connection:** Shift from habitation level access to household level service delivery has enabled provision of tap water to rural households, ensuring potable water of prescribed quality, in adequate quantity, on a regular and long-term basis, thereby enhancing convenience and dignity;

Figure 3: JJM Solutions to NRDWP Challenges

ii. **Assurance of Minimum Service Level Standards:** Institutionalisation of a minimum service norm of 55 LPCD has ensured uniformity, adequacy and improved reliability in water supply, marking a transition towards assured service delivery;

iii. **Rapid Scale up of Coverage:** Tap water coverage has expanded from about 17% in 2019 to around 75% as on March 2024, with more than 11.39 crore additional households connected, reflecting strong Centre-State partnership and effective implementation;

iv. **Strengthening of Water Quality Systems:** Reduction in water quality affected habitations and universal coverage of Arsenic and Fluoride affected areas, supported by NABL accredited laboratories and community based surveillance, has enhanced public health safeguards and trust in tap water;

v. **Women Led Water Governance:** Ensuring at least 50% representation of women in VWSCs and their active role in water management has strengthened inclusivity, accountability and local ownership;

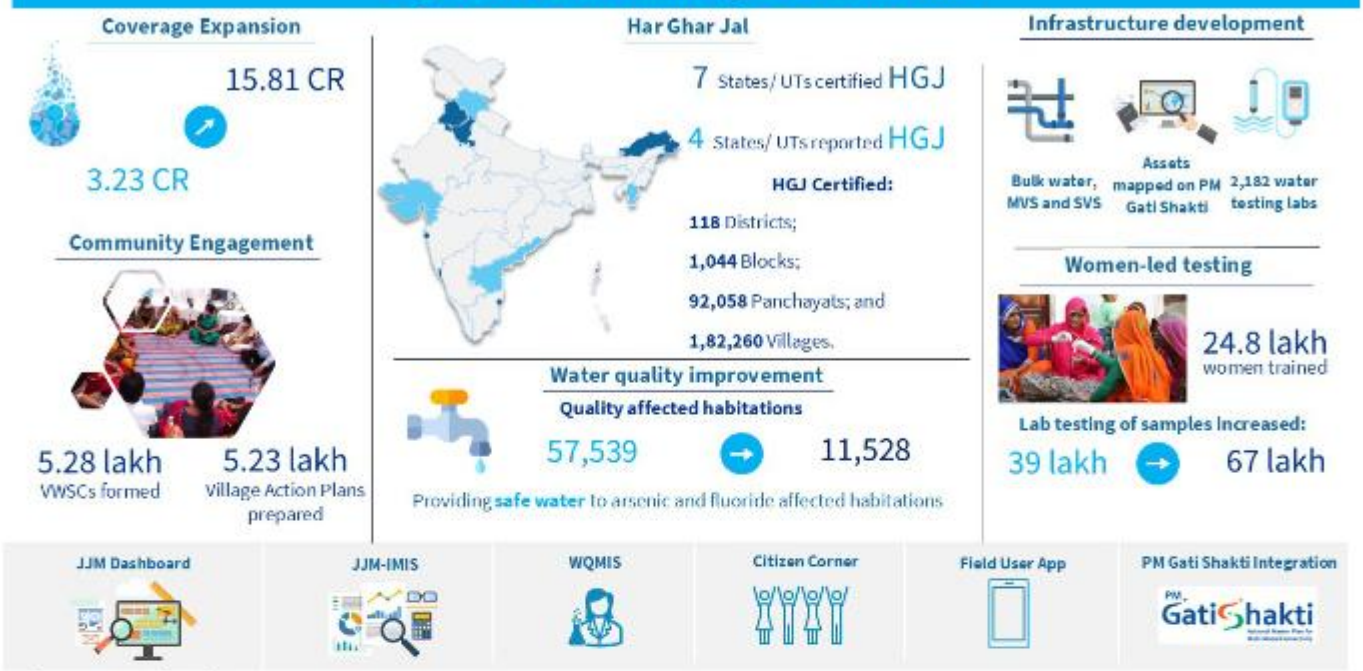
vi. **Integration of Source Sustainability:** Mainstreaming of source sustainability, including groundwater recharge, has improved resilience of water sources and ensured long-term water security;

vii. **Skilled Ecosystem for O&M:** Capacity building and initiatives such as Nal Jal Mitra Programme have created a local skilled workforce for operation, maintenance and minor repairs, ensuring continuity of services;

viii. **Expansion of WASH in Institutions:** Increased tap water access in schools, anganwadi centres and public institutions has improved hygiene, health outcomes and dignity, particularly for children and women;

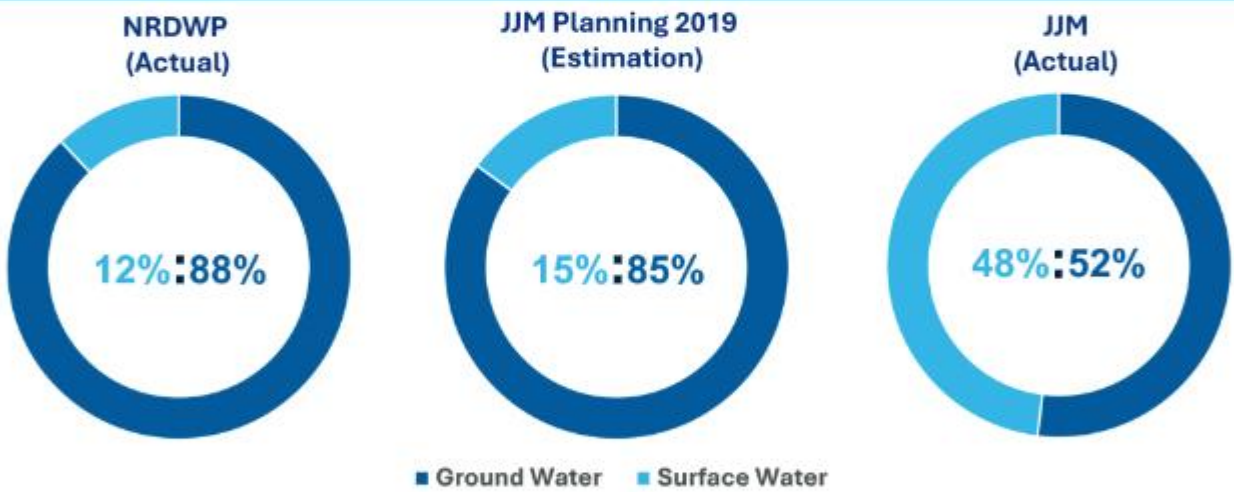
ix. **Shift to surface water-based schemes:** Under JJM, rural water supply schemes are designed to remain functional throughout the year, drawing lessons from the shortcomings of the erstwhile NRDWP, where the Performance Audit at the end of the 12th Plan period highlighted that 88% of piped water schemes were dependent on groundwater sources, affecting long term sustainability and safety and resulting in 4.76 lakh habitations slipping back during 2012–2017. Slip back occurred due to excessive groundwater extraction, inadequate attention to water quality aspects, poor sustainability of water sources and insufficient maintenance of existing systems, particularly in areas where agricultural withdrawal led to rapid depletion of groundwater, causing schemes to fail before their design period. Accordingly, States designed schemes under JJM, selecting sustainable sources leading to a paradigm shift from ground water to surface water-based schemes. Thus, surface water to groundwater ratio of 15:85

Figure 4: JJM – Transforming Rural Water Access



Data as on 20th March 2026

Figure 5: Scheme-wise Surface Water to Ground Water Ratio



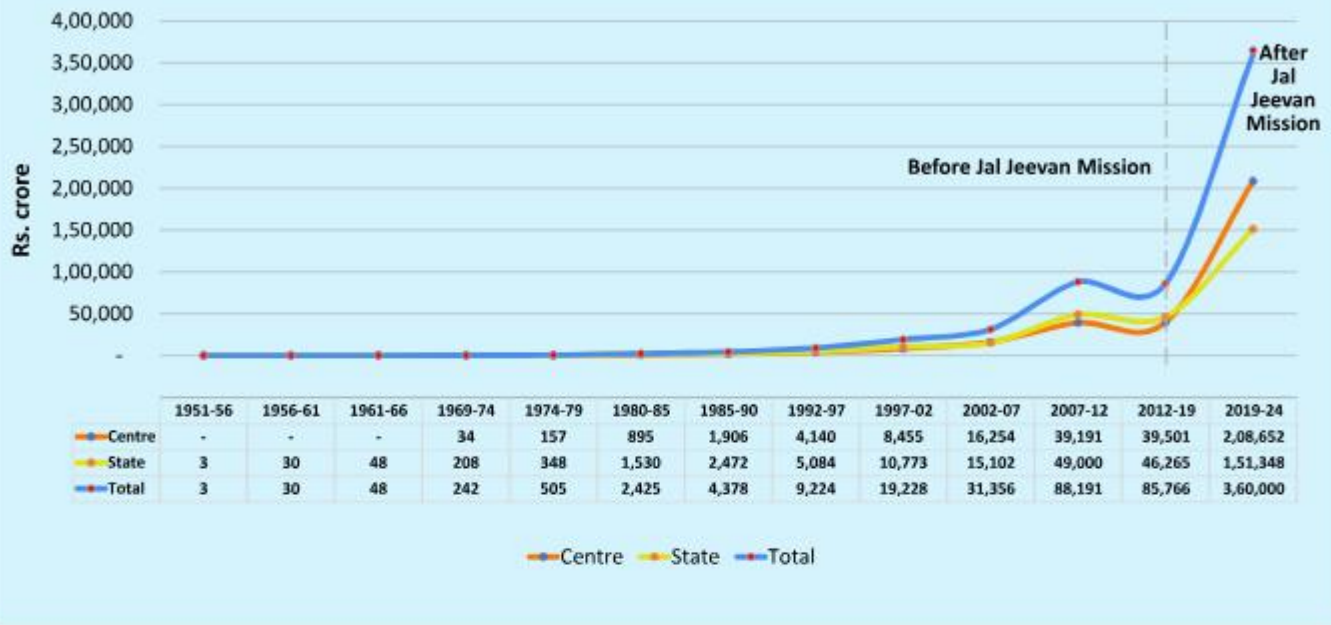
(as planned in initial stage of JJM) was changed to 48:52 (as approved during the mission period).

1.1.3. Scale of Infrastructure Planned

Jal Jeevan Mission (JJM) has driven a transformational expansion of rural drinking water infrastructure across India. From a baseline of 3.23 crore rural households (about 17%) with tap water connections in 2019, the Mission has enabled more than 12.58 crore additional households to receive tap water connections. As a result, over 15.81 crore rural households (81.73%), out of the 19.36 crore rural households identified by States and Union Territories are currently covered with household tap water connections.

Beyond the provision of new connections, significant efforts were undertaken to upgrade and augment existing water supply schemes. Several households that were earlier covered under centrally sponsored or State funded schemes were found to be non compliant with JJM service standards. These were subsequently integrated into JJM through strengthening of water sources, treatment facilities, storage and distribution infrastructure to meet prescribed service benchmarks.

Figure 6: Investments made for Rural Water Supply



	1951-56	1956-61	1961-66	1969-74	1974-79	1980-85	1985-90	1992-97	1997-02	2002-07	2007-12	2012-19	2019-24
Centre	-	-	-	34	157	895	1,906	4,140	8,455	16,254	39,191	39,501	2,08,652
State	3	30	48	208	348	1,530	2,472	5,084	10,773	15,102	49,000	46,265	1,51,348
Total	3	30	48	242	505	2,425	4,378	9,224	19,228	31,356	88,191	85,766	3,60,000

Under JJM, a total of 6,83,144 schemes have been planned, comprising 5,78,463 Single Village Schemes (SVS), 80,417 Multi Village Schemes (MVS), 2,590 Bulk Water Supply (BWS) schemes and 21,674 Non PWS schemes, representing one of the largest rural water infrastructure expansions globally. While scheme designs vary across regions to account for differences in hydrogeological conditions, terrain and cost structures, they adhere to uniform household level service delivery norms across the country.

A Piped Water Supply (PWS) scheme under JJM is conceived as a comprehensive and integrated system to deliver safe, treated and reliable drinking water from source to household. Such schemes comprise interconnected infrastructure including source abstraction structures, pumping systems, transmission networks, water treatment facilities, storage reservoirs and last mile distribution networks. These components are designed to address diverse local conditions, often involving long distance conveyance, multi stage pumping and advanced treatment processes wherever required. Infrastructure is typically organised as out village systems for bulk abstraction, treatment and transmission and in village systems for storage, distribution and household level service delivery, together forming a complex engineering and service delivery framework.

Implementation of the Mission has involved large scale infrastructure creation across rural India, including extensive village distribution networks, a more than fifteen-fold increase in water treatment capacity and a more than nine fold increase in planned and implemented water supply schemes. Parallel investments have been made to strengthen water quality monitoring and surveillance, with 2,874 water testing laboratories established including 685 laboratories located at water treatment plants, of which 1,709 are accredited or recognised. At the community level, approximately 5.28 lakh Village Water and Sanitation Committees (VWSCs)/ Pani Samitis have been constituted to institutionalise community participation in planning, implementation and operation and maintenance of water supply systems.

To realise the Mission's vision, States and Union Territories undertook infrastructure planning at an unprecedented scale, developing a wide range of solutions from single village schemes to multi village and large regional bulk water transfer systems suited to challenges such as water scarcity, quality constraints and difficult terrain. These schemes incorporate extensive water transfer networks, treatment plants and zonal storage infrastructure to ensure long term sustainability and functionality, particularly in areas with inadequate or unreliable local water sources. Bulk water supply systems were planned to support and sustain village level schemes, ensuring continuity and reliability of service delivery.

Figure 7: Quantum Jump in Water Supply Assets			
Assets	Unit of Measurement	Pre-JJM	Post-JJM*
Water Quality Laboratories	Nos.	2,071 *(619 accredited)*	2,874 *(1,709 accredited)*
Storage tanks & Reservoirs	MLD	1,200	34,000
Pipelines for Water Distribution Network (transmission and distribution)	KM	~ 5 lakh	~ 70 lakh
Water treatment	MLD	2,000	36,000
PWS schemes	Nos.	72,837	6,83,144
Population covered by PWS	%	38%	100%

**Above are the estimated figures based on approved schemes subject to final assessment of the mission period of JJM 2.0*

For the first time, the rural water supply sector has witnessed an unprecedented scale of investment under Jal Jeevan Mission, with total allocations rising sharply to nearly ₹3.60 lakh crore in the period 2019–24. This steep escalation marks a transformative shift from historically modest funding levels, enabling large-scale infrastructure creation and sustained service delivery.

1.1.4. Functionality Assessment

To assess compliance with the parameters, National Jal Jeevan Mission (NJJM) has undertaken functionality assessments through sample surveys by a Third Party Agency in 2021, 2022 & 2024. These assessments examined key service level indicators, including adequacy of supply, quality compliance, regularity of service and sustainability of the source and system. The scope of the surveys has progressively expanded over the time to improve representativeness and geographic spread.

NITI Aayog also conducted an independent survey (2025) across 34 districts of 12 States/ UTs. The findings of the assessments revealed that 87% of households have adequate access to drinking water with 82% of households satisfied with tap water quality, where connections have been provided.

The outcomes of the assessments have been utilized for identification of service gaps, prioritization of remedial actions and strengthening of operation and maintenance frameworks under the Mission.

1.1.5. Potential Socio-economic Impacts of JJM

Beyond the expansion of household tap water coverage, the mission is having a profound impact on the socio-economic development of rural communities, significantly enhancing their overall living standards. Access to safe and reliable drinking water has contributed to measurable improvements in public health and hygiene, reduction in disease burden, enhancement of

Table 1: Coverage of Functionality Assessment Surveys			
Parameters	**2020-21**	**2022**	**2024**
States Covered	31	33	34
Districts Covered	704	712	761
No. of villages Covered	6,992	13,299	22,812*
No. of HHs Covered	87,123	3,01,389	2,73,295

**of these, 19,812 are HGJ villages*

Table 2: Results of Functionality Assessments			
Parameters	**2021**	**2022***	**2024***
Working tap connections (Found working on the day of the survey)	94%	91%	87%
Adequate Quantity (i.e. 55 lpcd)	84%	84%	84%
No. of HHs Covered	61%	88%	80%
Full regular supply (% HH) (as per declared supply schedule)	87%	90%	76%
Overall Functionality	48%	69%	76%

**Survey outcome for Har Ghar Jal Villages*

Figure 8: Potential impacts of JJM

productivity and strengthening of local livelihoods. Evidence from research studies and sectoral assessments substantiates these outcomes and highlights the Mission's transformative impact on rural living standards.

Table 3: National Initiatives and Envisaged Impacts		
S.No.	**National Initiatives**	**Impact**
1.	SBM G	Piped water availability enhances sanitation and wastewater management.
2.	Mission Saksham Anganwadi & Poshan 2.0	Safe drinking water improves child nutrition and reduces malnutrition. Access to potable water at homes reduces the burden of pregnant and lactating mothers and in Anganwadi centers impact child nutrition and hygiene practices.
3.	National Health Mission (NHM)	Reduction in waterborne diseases lowers healthcare costs.
4.	National Rural Livelihood Mission (NRLM)	Engagement of SHGs in water testing, O&M etc. shall create entrepreneurship opportunities.
5.	VB-G RAM G	JJM promotes water conservation, groundwater recharge and source sustainability in convergence with VB-G RAM G.
6.	National Water Mission & Atal Bhujal Yojana	Promote source sustainability and groundwater recharge.
7.	Digital India	Digital monitoring of the operation and efficiency of infrastructure will enable rural India to connect with digital India.
8.	Skill India	Capacity building & training under Nal Jal Mitra Programme (NJMP)
9.	PM GatiShakti	GIS mapping of all the pipe water supply lines and assets of piped water supply schemes.
10.	Pradhan Mantri Awas Yojna (Grameen) – PMAY (G)	Providing water supply to all the households provided under PMAY (G).
11.	PM-JANMAN and DA-JGUA	Making water accessible to most marginalized communities of the society such as PVTGs.

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CHAPTER 1.2

Jal Jeevan Mission 2.0

1.2.1. Introduction

Since 2019, 15.81 crore households i.e. more than 81 per cent of India's rural population have been provided access to potable tap water connections. To achieve 100 per cent coverage extension of Jal Jeevan Mission has been approved for a period upto December 2028 with restructured implementation focusing on structural reforms in rural drinking water supply sector.

Enhancement of total outlay to ₹ 8.69 lakh crore has also been approved by the Government with total central assistance of ₹ 3.59 lakh crore enhanced from ₹ 2.08 lakh crore approved in 2019-20 i.e. additional central share of ₹ 1.51 lakh crore.

JJM 2.0 will focus on the quality of infrastructure and O&M of rural piped water supply schemes through Jan Bhagidhari by restructuring and reorienting the implementation of the JJM from infrastructure creation to a service delivery, supported by drinking water governance and institutional ecosystem for sustainable rural piped potable water supply. Separate MoUs shall be signed with States/ UTs, to ensure adherence to timelines, sustainability and citizen-centric water service delivery.

1.2.2. Vision

In continuation of the vision of the previous phase and to reorient the programme, JJM 2.0 aims that every rural household has access to drinking water supply in adequate quantity of prescribed quality on regular basis through community participation (Jan Bhagidari) and bringing in structural reforms for sustainable O&M of rural water supply systems, thereby enhancing the living standards of rural communities.

1.2.3. Objective

Aiming at universal and inclusive coverage, JJM 2.0 would facilitate the certification of all Gram Panchayats as Har Ghar Jal, by provisioning tap water connection to all the 19.36 crore rural households across the country. JJM 2.0 also aims at strengthening of institutional ecosystem by adoption of envisaged structural reforms through drinking water governance, to ensure functionality and sustainability of Rural Water Supply Schemes across the country.

1.2.4. Assessment of Implementation of JJM (2019-24)

Post the first phase of JJM (2019-2024) implementation programme evaluation was undertaken to address implementation gaps and shift the focus from infrastructure creation to building sustainable systems for operations. With the creation of such an extensive rural water supply infrastructure, the need for a robust and sustainable operation and maintenance (O&M) ecosystem becomes critical to ensure long term functionality and service delivery. Recognizing this, focused emphasis has been placed on building the institutional governance and community ecosystem necessary for sustainable O&M through structural reforms.

1.2.5. Structural Reform Linked Implementation Strategy

Jal Jeevan Mission 2.0 is a reform-linked programme, wherein financial assistance to States/ UTs will be aligned with the structural reforms undertaken to strengthen the rural piped water supply ecosystem. The Mission aims to transition towards a citizen-centric, service delivery-based system, anchored in professionally managed water utilities with active participation of local communities through village-level institutions. The reforms seek to establish a robust legal and regulatory framework, supported by accountable systems that build citizen trust and confidence in drinking water services.

At the same time, the Mission emphasises fostering a strong sense of ownership among communities and future generations through their involvement in key processes such as; commissioning and management of water supply systems. Participatory governance is further strengthened through initiatives such as Jal Arpan, Jal Utsav and Jal Seva Aankalan which institutionalise community engagement, transparency and continuous improvement in service delivery.

Further, digital data governance is also integrated for real-time monitoring & accountability along with establishment of research, innovation and knowledge ecosystem for sustainable, climate resilient and cost-

Figure 9: Pillars Of Structural Reforms

effective rural water service delivery systems in alignment with the vision of Atmanirbhar Bharat. The key pillars of structural reforms are indicated in the above Figure.

Key initiatives initiated by the Department during 2024-26 aiming towards envisaged Structural Reforms by strengthening service delivery systems, enhancing accountability, improving local institutional capacities and ensuring long-term financial and operational sustainability of rural water supply arrangements through Jan Bhagidari are as under:

i. **Development of Sujalam Bharat Module as Digital Public Infrastructure (DPI):** A National digital framework has been created wherein every service area of rural piped water supply scheme is assigned a unique Sujal Gaon ID, enabling complete end-to-end digital mapping of the water supply system, from source to household tap connection;

ii. **Launch of State, District and GP level Interactive Dashboards:** Interactive dashboards have been enabled at the State, District and Gram Panchayat levels to support real time monitoring, promote active local involvement and establish Command and Support Centres for data driven oversight;

iii. **Decision Support Systems for Source Sustainability Launched:** A Decision Support System (DSS) has been launched integrated with groundwater status data from the Central Ground Water Board (CGWB) to facilitate scientific planning and strengthening of source sustainability measures, especially for ground- water based schemes at the district level;

iv. **Digital Jal Seva Aankalan was launched for Functionality Assessment:** Jal Seva Aankalan has been introduced as a digital tool to enable Gram Sabhas to record service delivery responses, supporting objective assessment of functionality in Har Ghar Jal villages;

v. **Standardized Commissioning and Handover Protocols issued:** Protocols for commissioning and formal handover have been instituted through the Jal Arpan process, including a mandatory 15-day trial run with active participation of Gram Panchayats and Village Water & Sanitation Committees (VWSCs);

vi. **Independent Assessment through Jal Seva Adhyayan introduced:** Assessment of JJM infrastructure and its implementation is being undertaken through Jal Seva Adhyayan. involving engineering students to provide independent

insights and strengthen transparency and accountability;

vii. **Monthly Joint Inspections:** Monthly joint inspections by administrative and technical officials have been institutionalised;

viii. **Revised Terms of Reference for TPIAs issued:** Revised Terms of Reference for Third Party Inspection Agencies (TPIAs) were issued, introducing mandatory technical tests and enhanced administrative oversight by District Collectors/ District Magistrates to ensure higher quality assurance in scheme execution;

ix. **Regular Programme Evaluation through National WASH Experts and Central Prabhari Officers (CPOs):** A structured mechanism for systematic programme evaluation has been instituted through National WASH Experts and the Central Prabhari Officers of Aspirational District Programme/ Aspirational Block Programme;

x. **Programme Evaluation through Central Nodal Officers (CNOs):** A structured programme evaluation mechanism has been instituted through reviews conducted by Central Nodal Officers (CNOs). The department analysed CNO reports and has issued directives to States to address critical observations and implementation gaps identified during these assessments, thereby strengthening accountability and improving field-level performance;

xi. **Monthly Review Meetings with States:** The department holds monthly meetings with State Mission Directors, to review progress and resolve implementation issues flagged by the CNOs;

xii. **Advisories to District Collectors/ Magistrates:** Detailed advisories have been issued to all DCs/ DMs in States/ UTs to prepare source sustainability plans, protect water supply infrastructure, monitor quality of works and ensure PM GatiShakti-based asset mapping for all rural water supply schemes;

xiii. **Integration with e-Gram Swaraj Portal:** JJM portal has been integrated with the e-Gram Swaraj platform and States/ UTs have been directed to upload as-built drawings and adhere to standardized commissioning and handover protocols, mandating the presence of Gram Panchayats during the process;

xiv. **Strengthening Local Institutions through VWSCs and District Technical Units (DTUs):** VWSCs are to be integrated as part of the Standing Committees of Gram Panchayats to ensure their statutory role in planning, implementation and oversight of rural water supply systems. Further, District Technical Units (DTUs) are being established to provide technical support to Gram Panchayats through DWSM, enhance engineering capacities and strengthen field-level implementation and monitoring;

xv. **Community-Led Celebrations under Jan Bhagidari se Har Ghar Jal:** Standard Operating Procedures (SoPs) have been issued for community-led events to reinforce owner- ship and long-term service delivery outcomes. These include Jal Arpan (formal handing over of schemes), Jal Utsav (local celebrations to promote community ownership), Jal Bandhan (community commitment for asset maintenance) and Jal Sankalp (collective pledge for sustaining service levels);

xvi. **Community Radio for Behavioural Change and Awareness:** A dedicated community radio initiative has been launched in 13 languages and 34 dialects across 25 States/ UTs, 89 Districts and 100 Radio Stations (Pilot Phase) to enhance awareness, improve community participation and promote contextual communication on water service delivery and sustainability;

xvii. **Direct Outreach with DCs/DMs:** The Department has conducted 29 one-on-one interactions with more than 700 DCs/DMs to strengthen last-mile implementation, resolve bottlenecks and ensure effective field-level monitoring of rural water supply systems;

xviii. **Peyjal Samvad-Monthly Engagement with States/ Districts:** Regular Peyjal Samvad sessions are held every month with DCs/DMs and State Mission Directors aimed at strengthening district-led governance, promoting innovation, cross-sharing and peer learning among districts;

xix. **Sujal Gram Samvad-Direct Interaction with Beneficiaries:** The Department facilitates Sujal Gram Samvad, direct interactions with Gram Panchayat beneficiaries in their regional languages and local dialects, to encourage community ownership, gather feedback and strengthen transparency in scheme implementation;

xx. **Sarpanch Samvad through QCI Platform:** A nationwide engagement platform hosted by the Quality Council of India has been launched under Sarpanch Samvad, enabling more than 1 lakh

Sarpanches to participate in discussions on improving water service delivery governance and community oversight;

xxi. **Inclusion of Water, Sanitation and Hygiene (WASH) in School Curriculum:** Water, Sanitation and Hygiene components have been incorporated into the NCERT curriculum for Classes 1 to 8, fostering early awareness and behavioural change among school children as part of long-term sustainability efforts;

xxii. **High Level Policy Dialogue on Sustainable O&M:** Policy dialogue was conducted with State Ministers and senior officials of PHE/ RWS and Panchayati Raj Departments to strengthen understanding and commitment towards sustainable operation and maintenance of rural drinking water services. These engagements have reinforced the importance of institutional preparedness, long term financing and governance reforms required for reliable service delivery;

xxiii. **O&M Policy Framework Based on 19 Essential Components:** A model framework with 19 essential components of a comprehensive O&M Policy has been shared with all States/ UTs to assess their readiness for sustaining rural piped water supply systems. The draft policy framework and diagnostic matrix have been provided to guide States in aligning their systems with long-term sustainability requirements.

1.2.6. Sustainability of Schemes

JJM 2.0 also envisages establishment of Drinking Water Sustainability Framework for long term sustainability of schemes.

A Sustainability Framework for the Rural Drinking Water Sector defines sustainability as the ability of a rural water system to consistently deliver safe, adequate and reliable drinking water services over time, supported by arrangements that ensure:

i. Effective operation and maintenance of assets;

ii. Sufficient financing for routine and major costs;

iii. Long-term protection and management of water sources;

iv. Accountable institutions with strong citizen-facing service assurance.

It is structured around four core sustainability pillars namely Institutional & Service Delivery Sustainability, Financial Sustainability, Infrastructure Sustainability and Source Sustainability, enabled through four supporting pillars viz. Governance, Digital Systems, Community Participation and Citizen Connect to ensure service performance, transparency, responsiveness and continuous improvement.

Figure 10: O&M Policy- Essential Components

Figure: 11 Matrix for Pillars of Sustainability				
Support Initiatives → / Sustainability Pillars ↓	Governance	Digital Systems	Community Participation	Citizen Connect
Institutional & Service delivery Sustainability	• Role clarity • SLBs • Service Utilities • Capacity building	• Dashboards • Unified Asset Registry • Real-time alerts	• Gram Standing Committee • SHGs/ PACS • Nal Jal Mitras	• GRM • Satisfaction surveys • Citizen Charter/ Act
Financial Sustainability	• Service Delivery charge levy by Panchayat • Incentives/ VGF	• Digital billing • Volumetric metering • Energy tracking	• Service Delivery Charge awareness • Budget meetings • CBOs- revenue collection	• Feedback surveys • Functionality Ass. • Public disclosure
Infrastructure Sustainability	• Asset Mgmt. Plans • Preventive Maintenance • SOPs	• SCADA • Sensors • O&M digital logs	• Leak reporting • Community Surveillance	• Service Ratings • Downtime dashboards
Source Sustainability	• Source Protection • Water security • Water audits	• DSS • WQMS • GIS aquifer mapping	• Water budgeting • Jal Utsav • School eco-clubs	• Reports on contamination • Source updates

The detailed sustainability matrix is annexed as ***Annexure 1***

1.2.7. Outcomes

With the extension of the Jal Jeevan Mission and a restructured implementation framework supported by enhanced outlay and structural reforms in the rural drinking water sector, the envisaged outcomes are as under:

i. All Gram Panchayats are certified as Har Ghar Jal, with provision of Functional Household Tap Connections to all rural households across the country by December 2028;

ii. Scheme sustainability frameworks are established through adoption of envisaged Structural Reforms therein ensuring consistent delivery of safe, adequate and reliable drinking water through institutional, financial, infrastructure and source sustainability, supported by governance, digital systems, community participation and citizen-centric service assurance for rural water supply schemes.

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SECTION 2

Roadmap to Structural Reforms

CHAPTER 2.1

Establishing Institutional Architecture for Drinking Water Governance

Reform aims at:

- Establishing clear roles and responsibilities across governance, regulation and service delivery, replacing overlapping mandates;
- Transitioning from fragmented responsibilities to integrated, governance driven drinking water service delivery;
- Shifting implementation focus from an infrastructure creation approach to overall outcome-based service delivery;
- Establishing village level institutional architecture and enabling citizen participation;
- Strengthening regulatory oversight through a comprehensive legal framework.

2.1.1 Drinking Water Governance

Water, including drinking water supply, is a State subject under Entry 17 of the List II - State List of the Seventh Schedule to the Constitution of India, placing primary responsibility for governance and management of rural drinking water with State Governments. Further, the 73rd Constitutional Amendment Act, 1992, through Article 243G and the Eleventh Schedule, enabled States to devolve drinking water functions to Panchayati Raj Institutions, empowering Gram Panchayats to plan, implement, operate and manage local water supply systems in accordance with State laws. This constitutional framework provides the basis for decentralized, community-centric management of rural drinking water services.

The implementation of Jal Jeevan Mission has resulted in the creation of an extensive and complex rural drinking water infrastructure network, comprising household level tap connections through piped water supply systems and regional bulk supply arrangements with shared assets. Although geographically dispersed, these systems are functionally interconnected, often serving multiple habitations and administrative units and therefore require coordinated planning, operation and management.

To ensure sustainable drinking water service delivery, there is a need to move from the departmental and asset creation based approach to a citizen responsive governance led framework. In this context, drinking water governance means the way institutions, rules and systems are organised to plan, operate, monitor and manage water supply services, with clear responsibilities, coordination among stakeholders, accountability, proper financial management and regular monitoring to ensure reliable service delivery.

2.1.2. Key Dimensions of Drinking Water Governance

i. **Institutional Architecture and Functional Clarity:** Establishing a coherent, multi-tier institutional framework with clearly defined mandates, ownership and accountability across State, District and GP level ensuring alignment between policy, regulation and service delivery functions;

ii. **Assignment of Service Provider and Service Delivery Responsibility:** Clearly identifying the designated service provider(s) responsible for bulk water supply and in-village distribution and delineating their operational responsibility without any overlap, establishing clear accountability towards citizens and benchmarking the service delivery. It should also provide for technical support and day to day monitoring of water supply services through technical units at the District level integrated with Command and Support Centres;

iii. **Defined Policy, Legal and Regulatory Framework:** Creating a comprehensive O&M policy defining roles and responsibilities of service providers and ensure administrative oversight. It should also include service delivery and financial model, asset management protocols including preventive maintenance, energy efficiency and

lifecycle costing to ensure sustained performance of systems. It is imperative that States formulate legal and regulatory framework that sets service standards, regulates water utility operations, ensures quality compliance and establishes coordination mechanism among utilities, grievance redressal mechanism and provisions for protection of water sources and infrastructure to support reliable drinking water services;

iv. **Accountability and Transparency with Citizen Interface:** Institutionalising accountability mechanisms through service level disclosures, digital grievance redressal systems, citizen feedback loops and functionality assessment along with service improvement plans published on dashboards which ensures that the service providers remain responsive to end users;

v. **Participatory Governance (Jan Bhagidari):** Embedding community participation through Gram Panchayats and Village Water Sanitation Committees in planning, monitoring and local management, thereby strengthening ownership, behavioural change and long-term sustainability;

vi. **Inter-Departmental and Inter-Sectoral Convergence:** Establishing structured convergence across sectors such as water resources, rural development, sanitation, health, energy and environment to enable integrated water resource management, greywater reuse and optimal utilisation of public investments. It should also make provision for cross sectoral assessment of the impact of potable drinking water;

vii. **Data-Driven and Digital Governance Systems:** Leveraging digital public infrastructure i.e, Sujalam Bharat and decision support systems, to enable real-time monitoring, predictive analytics, asset tracking and evidence-based planning and policy decisions;

viii. **Financial Sustainability and Resource Management:** Ensuring efficient allocation and utilisation of financial, human and technical resources, supported by transparent accounting systems, sustainable O&M financing models and performance linked resource allocation;

ix. **Governance Responsiveness, Risk Management and Resilience:** Building institutional capacity to anticipate and respond to service disruptions, water quality risks, climate variability and system failures through early warning systems, contingency planning and adaptive governance mechanisms;

x. **Sustainability and Lifecycle Asset Management:** Ensuring long-term functionality through integrated source sustainability, lifecycle asset management, periodic maintenance and resilience planning to address climate and resource challenges.

2.1.3. Institutional Mechanism

With the objective of institutionalising drinking water governance, JJM 2.0 lays down an institutional framework across National, State, District and Gram Panchayat levels, with clearly defined roles and responsibilities. The framework is intended to guide programme implementation and systematically orient governance arrangements towards the operation and maintenance phase, ensuring coordinated management, accountability and sustainability of rural drinking water services.

A. National Level (NJJM)

i. The National Jal Jeevan Mission (NJJM), headed by Mission Director, will be responsible for steering and guiding the Mission at the National level. It will provide overall policy direction, strategic oversight and technical support to ensure effective implementation across States/ UTs. NJJM will coordinate closely with States/ UTs for planning, implementation and monitoring, while promoting standardisation, capacity building and knowledge sharing;

ii. NJJM shall provide financial assistance to States/ UTs through tranche-based disbursement of central assistance linked to CIRP, reform milestones and performance outcomes, supported by periodic review of progress to ensure accountability and efficiency;

iii. NJJM will support the development of regulatory frameworks, digital monitoring systems and performance assessment mechanisms to strengthen governance and ensure improved service delivery outcomes;

iv. A National Command and Support Centre will be setup for monitoring progress, tracking reform milestones and processing State-level information for policy decisions. It will be integrated with Command and Support Centres at State and District levels;

v. National Project Management Unit (NPMU) with engineering, management and IT experts, will assist NJJM and the Mission Director. The NPMU will support programme management, technical support, programme evaluation and development of IT monitoring architecture;

vi. Dr. Syama Prasad Mookerjee - National Institute of Water and Sanitation (SPM-NIWAS), an autonomous institution located in Joka, Kolkata, under the Ministry of Jal Shakti is established with an aim to address complex water and sanitation management challenges. It will play a crucial role in bridging the knowledge gaps in public health engineering, focusing on holistic management that includes health, law and public policy.

SPM - NIWAS will be responsible for building capacity, facilitating research and development related to low cost and efficient WASH technologies, enabling different Institutions to enhance their technical, financial and managerial skill in partnership with States/ UTs, academia, NGOs and other partner agencies.

B. State Level (SWSM, Apex and Executive Committees)

The State Water and Sanitation Mission (SWSM) comprise of Apex Committee and Executive Committee:

i. The Apex Committee, headed by the Chief Secretary and will comprise of Secretaries of PHE/ RWS Department, Rural Development, Panchayati Raj, Health, Finance, Planning, Education and other related departments, Representative of the Government of India and State Mission Director as Member Secretary:

a. It will guide policy, approve key decisions and support the functioning of water supply utilities under relevant State Acts and policies. It will also ensure coordination between departments. The Committee will issue directions on water security, regulatory measures, utility reforms and alignment with State priorities;

b. The State Level Scheme Sanctioning Committee (SLSSC) shall be merged with the Apex committee;

c. The Committee shall regularly meet on quarterly basis;

d. It will also review the progress of JJM implementation with DCs/ DMs periodically.

ii. The Executive Committee will be headed by the State Mission Director who will be responsible for overall planning, implementation and management of the Mission at the State level. The State Mission Director shall be equivalent to Joint Secretary to the Government of India or above with at least 2 years of experience as DC/ DM, ensuring adequate authority for inter-departmental convergence and effective decision making. The composition of the SWSM executive committee may be determined by the States/ UTs in line with the convergence requirements outlined in the guidelines, to enable effective coordination across relevant departments with mandatory participation from Rural Development and Panchayati Raj departments:

a. The Executive Committee will ensure alignment with JJM 2.0 guidelines, oversee the functioning of DWSMs and support planning, implementation and monitoring of schemes through effective operationalisation of District Technical Units. It will also facilitate engagement with district and Gram Panchayat institutions for timely execution of schemes and adherence to service delivery standards;

b. The SWSM shall conduct monthly review meetings and upload the Minutes of Meeting (MoM) on the SWSM Dashboard through the JJM-IMIS portal;

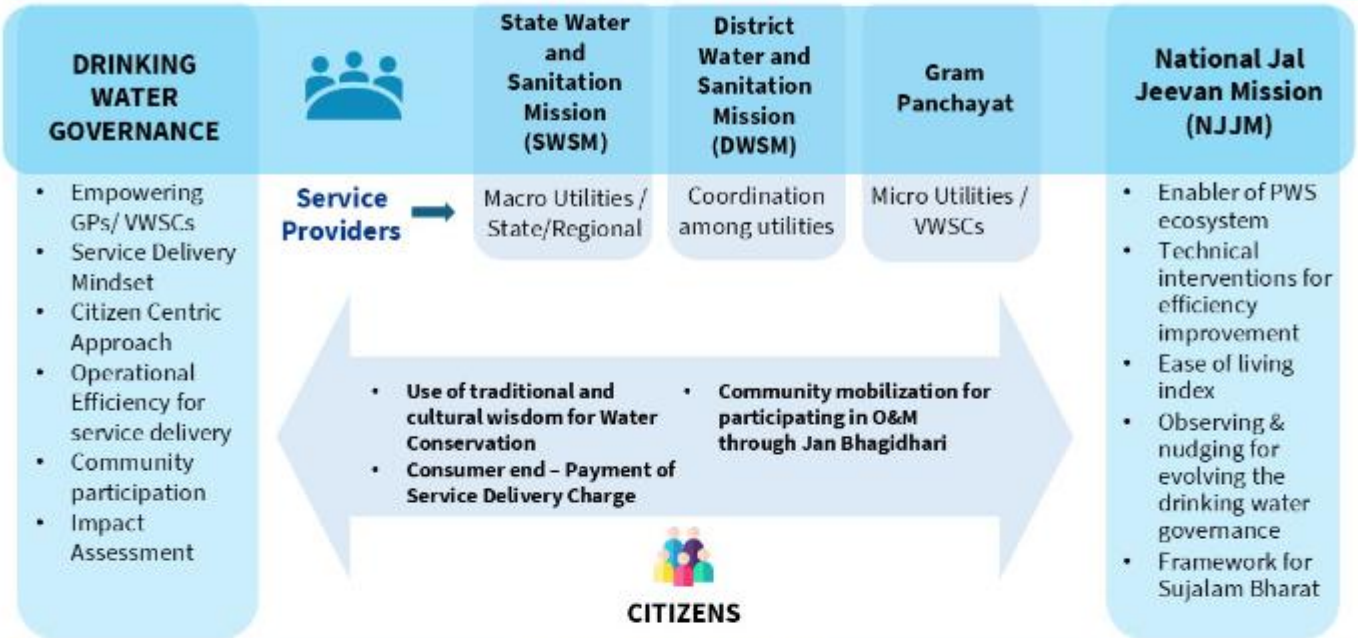
c. Coordination with NJJM will be institutionalised through regular interactions between the Mission Director, NJJM and State Mission Directors. Monthly review meetings, joint monitoring and digital platforms for real time data sharing will support aligned planning, reform implementation and data driven governance;

d. The Executive Committee will be supported by State Project Management Unit (SPMU) to provide technical, managerial and administrative support for planning, implementation, monitoring and evaluation of the Mission. The SPMU will also support data-driven decision-making, inter-departmental coordination and adherence to service delivery standards.

C. District Level (DWSM with DTU & DCSC)

i. District Water and Sanitation Mission (DWSM), chaired by the District Collector/ Deputy Commissioner, will be responsible for overall implementation, supervision and monitoring of the Mission at the district level. It will ensure convergence across departments and effective coordination with Gram

Figure 12: Governance Model under JJM



Panchayat institutions for timely and quality execution of schemes;

ii. DWSM shall comprise key district level officials, including the Chief Executive Officer (Zila Parishad)/ District Development Officer/ Chief Development Officer, Divisional Forest Officer, Project Director (in ITDA/ ITDP districts), District Medical Officer, District Education Officer, Executive Engineers from Water Resources/ Ground Water/ Irrigation, District Agriculture Officer, District Information and Public Relations Officer, representatives of relevant line departments, nominated technical experts and the Executive Engineer of PHE/ RWS Department as Member Secretary;

iii. DWSM shall conduct monthly review meetings with all stakeholders and upload the Minutes of Meeting (MoM) on the DWSM Dashboard through the JJM-IMIS portal;

iv. DWSM will operationalise the District Command and Support Centre (DCSC) as the central hub for planning, monitoring and oversight of Mission activities, supporting Jal Seva Aankalan, third-party inspections, grievance escalation and real-time data-driven decision-making;

v. Through the District Technical Unit (DTU) and District Project Management Unit (DPMU), the DWSM will provide technical and managerial support for O&M, asset management, sustainability, disaster preparedness and capacity building.

D. Gram Panchayat (GP) and Village Water and Sanitation Committee (VWSC)

i. **Institutional and Legal Framework:** While drinking water is delivered at the village level, its governance and service delivery are anchored in the constitutional framework of local self government. The Gram Panchayat, constituted under the 73rd Constitutional Amendment Act and respective State Panchayati Raj Act, is the primary authority responsible for ensuring the provision of safe and reliable drinking water services within its jurisdiction. The Gram Panchayat retains overall ownership of in village drinking water infrastructure and is accountable for service outcomes to rural households.

In accordance with earlier Jal Jeevan Mission guidelines, States/ UTs have been encouraged to constitute Village Water and Sanitation Committees (VWSCs) as legal entities empowered under the relevant State Panchayati Raj Acts or other applicable local government laws. Under this framework, VWSC functions as the standing committee of the Gram Panchayat for water and

sanitation, with composition, powers and functions as assigned under the State Act or rules framed thereunder. This arrangement enables decentralised service delivery while retaining democratic accountability at the Gram Panchayat level.

ii. **Legal Status and Role of Village Water and Sanitation Committee (VWSC):** VWSC shall function as the standing committee of the Gram Panchayat for water and sanitation, with statutory backing under the relevant State Panchayati Raj Act or rules framed thereunder:

a. The operational area of a VWSC may comprise a single village or a group of villages, as determined by the Gram Panchayat and the State Government, based on settlement patterns, population distribution and service delivery requirements;

b. Under JJM 2.0, VWSCs responsible for in-village water supply distribution are recognised as village-level service providers (micro-utilities). As micro-utilities, VWSCs shall act as the community interface for service delivery, ensuring service levels, citizen satisfaction and accountability to local users. This recognition enables clear separation of roles between governance and operations, supports professionalisation of service delivery at the village level and strengthens sustainability of rural drinking water services;

c. Where the Gram Panchayat enters into a tripartite agreement with the State Government and an external agency for operation and maintenance of drinking water supply systems, the Gram Panchayat shall assign the responsibility of monitoring compliance with Service Level Agreements (SLAs) and carrying out monthly functionality assessments at the village or habitation level to VWSC. VWSC shall act as the village level mechanism for tracking service performance and reporting deviations, if any;

d. The engagement of external agencies for O&M shall not result in discontinuation, dilution, or bypassing of VWSC. The VWSC shall continue to function irrespective of the O&M model adopted;

e. In line with the foundational principle of Jan Bhagidari under Jal Jeevan Mission, VWSC shall remain the primary community interface for drinking water service delivery. It shall ensure that prescribed service levels are met, citizen feedback is captured, local complaints are addressed and overall user satisfaction is monitored. The continued role of VWSCs as community institutions is essential for ensuring transparency, accountability and sustained service delivery, even when O&M is undertaken by external contractors.

iii. **Distinction of Roles:** Gram Panchayat and VWSC

a. **Governance Role of the Gram Panchayat:** Gram Panchayat shall perform managerial, supervisory, financial oversight and accountability functions for drinking water services, including:

1) Providing overall governance and leadership for village level drinking water service delivery;

2) Undertaking water budgeting to assess demand - supply gaps and prioritise drinking water requirements;

3) Approving village level plans for operation and maintenance, source sustainability and service expansion;

4) Monitoring overall service delivery performance and ensuring institutional accountability;

5) Approving provision of new household water connections in accordance with applicable norms;

6) Establishing grievance redressal and dispute resolution mechanisms;

7) Ensuring convergence of funds and prudent financial management, including oversight of service delivery charge and audits.

b. **Operations and Service Delivery Functions of VWSC:** VWSC shall be responsible for the following operational and service-level functions within its notified service area:

1) Supporting day-to-day management and monitoring of drinking water supply systems;

2) Ensuring equitable distribution of drinking water to all households, including those in scattered, remote, or difficult habitations;

3) Facilitating community access to safe drinking water in habitations where piped water supply is technically, commercially or geographically not feasible, through approved interim arrangements;

4) Conducting monthly functionality assessments at the village or habitation level, covering availability, quantity, quality and reliability of water supply;

5) Recording, tracking and facilitating resolution of consumer complaints at the local level;

6) Supporting routine water quality monitoring and disinfection as prescribed;

7) Collecting service delivery charges and maintaining basic operational records;

8) Undertaking IEC activities to promote water conservation, responsible use and protection of water sources.

c. **Monitoring, Jal Seva Aankalan and Community Accountability:** To ensure transparency and continuous improvement:

1) Gram Panchayat shall facilitate annual functionality assessment through Jal Seva Aankalan, covering all villages and habitations within its jurisdiction;

2) The findings of the Jal Seva Aankalan shall be presented village-wise in the Gram Sabha for public disclosure, citizen feedback and identification of corrective measures;

3) VWSC shall support the Jal Seva Aankalan process by providing monthly assessment data and community-level inputs.

d. **Community Engagement and Mission Events:** In furtherance of community participation and ownership:

1) Gram Panchayat shall facilitate Jal Arpan for formal commissioning and community handover of water supply schemes upon achieving service readiness;

2) Gram Panchayat and VWSC shall jointly promote Jal Utsav as an annual event to reinforce water conservation, service awareness and responsible use of drinking water services.

iv. **VWSC Engagement of Village level Support:** For effective discharge of its operational responsibilities as a village level service provider (micro utility), the Village Water and Sanitation Committee (VWSC) may engage and utilise locally available workforce and community-based organization. Such engagement shall support operation and maintenance, community interface and service assurance functions, while overall responsibility and accountability for drinking water services shall continue to rest with VWSC. VWSC may entrust specific tasks to the following village and cluster level human resources and community institutions, as appropriate and in accordance with State/UT approved norms.

a. **Nal Jal Mitra (NJM):** VWSC may engage Nal Jal Mitras, preferably local youth or ex-servicemen trained under the Nal Jal Mitra Programme (NJMP), to support day-to-day operation and maintenance of in-village drinking water supply infrastructure. Roles and responsibilities of Nal Jal Mitra may include:

1) Carrying out routine operation of piped water supply systems and associated infrastructure, including storage structures, pumps, pipelines and disinfection units;

2) Undertaking preventive maintenance and addressing minor technical issues in line with maintenance schedules and technical guidance from DWSM/DTU;

3) Supporting village level water quality monitoring, including coordination with trained community members using Field Testing Kits (FTKs);

4) Assisting VWSC in addressing user complaints and facilitating local grievance resolution;

5) Communicating field level updates, system status and issues to the District Technical Unit, Command and Support Centres and designated digital platforms.

b. **Community Based Organisations:** VWSC may outsource work of water testing, service delivery charge collection, functionality assessment, impact assessment to Self Help Groups (SHGs) (preferably part of SRLM/ Primary Agricultural Credit Society (PACS)/ Large Area Multi-Purpose Society (LAMPS) and other functions such as:

1) Mobilising community participation and raising awareness on safe drinking water practices, water conservation and optimal water use;

2) Supporting collection of service delivery charges and assisting VWSC in basic financial record keeping and communication with users;

3) Conducting water quality testing using the FTKs;

4) Providing facilitation and handholding support to VWSC in routine activities such as organising meetings, maintaining records and strengthening institutional functioning;

5) Supporting training needs assessment and facilitating capacity building activities related to operation and maintenance;

6) Assisting in coordination with Gram Panchayats, DWSM, DTU and other institutions for service

delivery support;

7) Facilitating participatory exercises such as water budgeting and village water security planning using participatory tools;

8) Supporting Government-led campaigns, IEC activities, documentation of good practices, organisation of exposure visits and community level impact assessment.

Community based organisations engaged by VWSC at the village level for supporting drinking water services shall be referred to as Sujalam Shakti Groups. This will help align these groups with the objectives of Jal Jeevan Mission and recognise them as field functionaries for the Mission. States may issue operational guidelines for these groups in coordination with SRLM.

This governance arrangement ensures that drinking water services are delivered through empowered local institutions, with the Gram Panchayat providing democratic oversight and accountability and VWSC acting as community-based organisation and engaging locals in routine operations and maintenance of water supply system instilling sense of ownership, public trust and confidence for long term sustainability.

2.1.4. Regulatory Framework of Rural Drinking Water

JJM 2.0 recognises household piped drinking water as the core element of water for life. States/ UTs shall ensure service delivery standards covering adequacy, regularity and BIS-10500 prescribed water quality. To support this, the following key regulatory measures shall be prioritised:

A. Domestic Water Allocation Priority

I. States/ UTs shall adopt a domestic water allocation priority principle, ensuring that drinking water supply to households is accorded the highest priority in allocation of both surface water and groundwater resources;

ii. In surface water-based schemes, especially those dependent on reservoirs and bulk transfer systems, dedicated allocations for rural drinking water supply shall be ensured and protected during periods of scarcity, in line with public health priorities and principles of equity.

B. Source Protection Framework

States/ UTs shall establish a statutory framework for drinking water source protection. This shall include:

i. Notification of Source Protection Zones around key drinking water sources such as borewells, springs and surface water intakes;

ii. Regulation of land use, extraction and potentially contaminating activities within these zones;

iii. Mandatory safeguards to prevent pollution and over-exploitation of sources;

iv. Empowerment of district authorities for enforcement and compliance.

C. Protection of PWS Infrastructure

States/ UTs shall establish regulatory provisions for protection of Piped Water Supply (PWS) infrastructure from damage, encroachment and unauthorised use. This shall include:

i. Notification of protective zones around critical infrastructure such as intake structures, treatment plants, transmission lines and storage reservoirs;

ii. Provisions to prevent physical damage, tampering, or illegal connections;

iii. Enforcement mechanisms at District and local levels to safeguard assets;

iv. Clear accountability for maintenance and protection of infrastructure.

D. Sustainability and Source Security

Rural water supply planning shall be based on aquifer and watershed principles, ensuring long-term sustainability of sources. All piped water supply schemes shall incorporate alternate sources to maintain continuity of service during source failure or seasonal variability.

E. Climate Resilience and Continuity Planning

States/ UTs shall integrate climate-resilient measures, including source diversification, aquifer recharge and local storage systems. Each scheme shall have an emergency continuity plan to ensure uninterrupted water supply during droughts, floods and other disruptions.

F. Operation and Maintenance (O&M) Policy

States/ UTs shall notify a comprehensive Operation and Maintenance (O&M) policy for rural drinking water supply systems, clearly defining roles and responsibilities across all tiers-State, District and Gram Panchayat/ standing committees. The policy shall establish accountability for service delivery, outline standard operating procedures for system upkeep and ensure sustainable financing mechanisms, including service

delivery charge and targeted support, to maintain functionality, reliability and long-term sustainability of water supply systems.

G. Public Service Guarantee and Grievance Redressal

i. States/ UTs shall bring drinking water services under Public Service Guarantee framework, prescribing defined timelines for provision of new household tap connections, extension of infrastructure and restoration of services. These service standards shall be notified and enforceable to ensure time bound and accountable delivery;

ii. A robust digital grievance redressal mechanism shall be established with defined timelines for resolution, supported by digital platforms for complaint registration, tracking and escalation. Provision for escalation at District and State levels shall be ensured to address unresolved grievances, thereby strengthening accountability, transparency and citizen trust.

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CHAPTER 2.2

Utility Based Approach for Drinking Water Service Delivery: Dual Utility Framework

Reform aims at:

- Transition towards a utility-based approach for sustainable and service-oriented drinking water delivery;
- Institutionalise and operationalise Micro Utilities for in-village drinking water supply operations and maintenance;
- Formation and operationalisation of State/ Regional-level macro utility constituted for out-village drinking water supply operations and maintenance;
- Performance Benchmarking and periodic assessment of utilities for service level improvement.

2.2.1. Background

Jal Jeevan Mission (JJM) aims to ensure provision of drinking water to all rural households in accordance with prescribed service delivery norms of adequacy, regularity and quality. This objective requires a transition from a predominantly scheme centric and infrastructure-focused approach to a service delivery-oriented framework that ensures sustained performance over the full lifecycle of water supply assets. Historically, rural water supply systems have been managed through department led, scheme-based arrangements which while successful in creating infrastructure, have faced structural and operational limitations such as weak coordination with Gram Panchayats, unclear assignment of roles and responsibilities, limited performance monitoring, operational inflexibility and uncertainty in operation and maintenance financing. With the increasing scale and complexity of systems, particularly in multi-village and bulk water supply schemes, these challenges have direct implications for the reliability, efficiency and sustainability of drinking water services.

In this context, a utility based approach to rural drinking water service delivery is essential to treat water supply as a continuous public service supported by well defined institutional arrangements, professional management practices and robust performance monitoring systems. The utility framework enables clear accountability, improved operational efficiency, predictable O&M mechanisms and stronger alignment between infrastructure investments and service delivery outcomes.

Integration of technology enabled platforms for monitoring, financial management, asset tracking and decision support further strengthen transparency and data driven planning. Aligned with the vision of Viksit Bharat 2047, the dual utility framework envisages community based institutions such as Village Water and Sanitation Committees functioning as Micro Utilities for last mile service delivery complemented by Regional or State level Macro Utilities responsible for bulk water supply and large scale system management. Together, this approach under JJM 2.0 seeks to ensure sustainable, efficient and accountable rural drinking water services while responding to evolving development priorities.

2.2.2. Dual Utility Framework

The rural water supply architecture under JJM 2.0 adopts a dual utility framework, comprising Micro and Macro Utilities, to address the distinct requirements of localized service delivery and system level management. This framework distinguishes between Micro Utilities and Macro Utilities based on the scale of operations, nature of functions and level of service delivery. The objective of this classification is to enable:

i. Clear delineation of roles and responsibilities;

ii. Improved operational efficiency and account-ability;

iii. Effective management of water supply systems across different levels of service delivery.

2.2.3. Village level Micro Utilities

A Micro Utility refers to a village level, consumer-centric service delivery entity responsible for ensuring provision of drinking water services within a defined service area typically corresponding to a Gram Panchayat/ Village/ a group of villages.

Under JJM 2.0, Village Water and Sanitation Committee (VWSC) constituted as a standing committee of the Gram Panchayat under the relevant State Panchayati Raj Act or rules shall function as the Micro Utility (as defined in Chapter 2.1). The Micro Utility shall be responsible for ensuring that drinking water services are delivered to all households in accordance with prescribed service norms relating to adequacy, regularity and quality.

In this regard, the Micro Utility shall undertake the following functions:

i. Operation and routine maintenance of in-village water supply infrastructure, including distribution networks, storage structures and associated components;

ii. Ensuring equitable and reliable distribution of water to all households, including those in remote or difficult habitations;

iii. Monitoring service delivery parameters at the habitation level and undertaking corrective measures, as required;

iv. Acting as the primary interface for consumers, including recording and facilitating resolution of grievances;

v. Supporting water quality surveillance, disinfection practices and adherence to safety standards;

vi. Promoting community participation, awareness and responsible use of water resources.

Micro Utility shall function as the last mile service provider and shall be directly accountable to the Gram Panchayat and rural households for delivery of services and maintenance of service standards.

2.2.4. State/ Regional level Macro Utilities

A Macro Utility refers to a system-level institutional entity responsible for planning, development and management of bulk water supply systems and providing technical, managerial and operational support across multiple service areas.

Macro Utilities may be established at the Block, District, Regional or State level, depending on the scale and complexity of water supply systems and may be housed within Public Health Engineering Departments (PHEDs)/ Rural Development/ Panchayati Raj Department, State Water Utilities, Special Purpose Vehicles (SPVs) or other designated agencies. Macro Utility shall be responsible for ensuring efficient and reliable functioning of large scale and technically complex infrastructure, including multi village schemes and bulk water transfer systems.

In this regard, the Macro Utility shall undertake the following functions:

i. Planning, design and execution of bulk water supply systems, including source development, treatment and transmission infrastructure;

ii. Operation and maintenance of out-village and large-scale infrastructure requiring specialised technical capacity;

iii. Ensuring availability and reliability of bulk water supply to service areas;

iv. Providing technical support, capacity building and guidance to Micro Utilities;

v. Monitoring system performance and identifying areas for improvement;

vi. Supporting financial planning, cost optimization, efficient resource utilization at an aggregated level and passing the benefits of economies of scale to the end-user through micro utilities.

The Macro Utility shall function as a system integrator and enabler, ensuring that infrastructure systems operate efficiently and support sustained service delivery at the local level.

2.2.5. Coordination between Utilities

Effective implementation of the dual utility framework under Jal Jeevan Mission (JJM) 2.0 requires establishment of structured coordination mechanisms between Micro Utilities and Macro Utilities to ensure seamless planning, operation and management of rural water supply systems.

The coordination framework shall aim to:

i. Ensure reliable and uninterrupted delivery of drinking water services;

ii. Facilitate integration between bulk water supply systems and in-village distribution networks;

iii. Enable timely resolution of operational and service delivery issues;

iv. Strengthen accountability and performance across utility levels.

A. Coordination between Micro and Macro Utilities

Micro Utilities (Gram Panchayat/ VWSC level entities) and Macro Utilities (regional or bulk water supply entities) shall function in a coordinated and complementary manner to ensure efficient and reliable service delivery.

Such coordination shall include:

i. Alignment of bulk water supply schedules with in-village distribution requirements;

ii. Sharing of operational and performance data across utility levels;

iii. Coordinated response to system failures and service disruptions;

iv. Joint planning for maintenance, augmentation and service improvements.

B. District Level Coordination Mechanism

District Water and Sanitation Mission (DWSM) shall serve as the primary coordination platform at the District level for integration of functions across Micro and Macro Utilities.

District Technical Unit (DTU) shall function as the technical coordination and support entity and shall:

i. Provide technical assistance to Micro Utilities;

ii. Facilitate coordination between Micro and Macro Utilities;

iii. Monitor service delivery performance at the District level;

iv. Support capacity building and troubleshooting.

In the event of disputes or operational issues between Micro and Macro Utilities such matters shall be referred to the DWSM for resolution with the DTU providing technical input and recommendations.

C. Role of State Water and Sanitation Mission (SWSM)

State Water and Sanitation Mission (SWSM) shall provide overall policy direction and oversight for implementation of the Utility Framework.

SWSM shall:

i. Approve plans and frameworks relating to utility operations;

ii. Monitor performance of utilities against defined benchmarks and service standards;

iii. Ensure adherence to O&M norms and financial sustainability frameworks;

iv. Facilitate inter-departmental coordination and policy alignment;

v. Establish a transparent and rational framework for service delivery charges between Macro Utilities and Micro Utilities. This framework shall:

a. Ensure financial viability of both Micro and Macro Utilities;

b. Reflect the cost of bulk water supply and system operations;

c. Maintain affordability for end users.

D. Integrated Governance Framework

The coordination mechanism shall operate within a multi-tier governance framework comprising:

i. Gram Panchayat/ VWSC (Micro Utility level) for service delivery;

ii. DWSM and DTU (District level) for coordination, technical support and monitoring;

iii. SWSM (State level) for policy direction, oversight and performance review.

This integrated framework shall ensure alignment of planning, implementation and service delivery functions across all levels, enabling efficient, accountable and sustainable rural drinking water services.

2.2.6. Benchmarking of Water Utilities

A. Purpose of Benchmarking

Benchmarking is an essential tool for assessing the performance of rural water utilities and enabling continuous improvement in service delivery. It provides a structured framework for evaluating utility performance against defined standards relating to service levels, operational efficiency and financial sustainability.

Benchmarking shall facilitate:

i. Objective assessment of utility performance across key parameters;

ii. Identification of performance gaps and operational inefficiencies;

iii. Setting of measurable improvement targets;

iv. Promotion of accountability, transparency and informed decision making.

B. Key Performance Areas

Benchmarking of Micro and Macro Utilities shall be undertaken across the following key dimensions:

Table 4: SMART Benchmarking Indicators of Utilities					
S. No.	Indicative Benchmarking	Basic Level	Improvement Level	Advanced Level	International Reference (Denmark) Indicators
1.	Percentage of NRW or unaccounted for water (UFW)	>25%	15 to 25%	<15%	Around 7% water loss
2.	Metering Coverage (%)	<20%	20 to 50%	>50%	66% remote read meters
3.	Service disruption (days of interruption/ month)	>3 days per month	1-3 days	<1 day	Around 46 min/ year interruption
4.	Water Quality Compliance (%)*	<97%	97 to 99%	>99%	99.4% samples compliant
5.	Energy Use (kWh/ m^3)	>0.75	0.55–0.75	<0.55	Around 0.44 kWh/m^3 average
6.	Number of leakages in service	>1 per month	Once in 3 months	Once in 6 months	-
7.	Financial Efficiency (Cost recovery)	<30%	30% to 50%	>50%	Managed by Utilities
8.	Grievance Redressal (Turn-around time)	>3 days	<3 days	<1 day	-

**Free Residual Chlorine 100%*

i. **Service Delivery:** Regularity, adequacy, quality and reliability of water supply;

ii. **Operational Efficiency:** Minimising non-revenue water or unaccounted for water, energy efficiency, system uptime and maintenance practices;

iii. **Financial Performance:** Cost recovery, O&M expenditure and revenue collection efficiency;

iv. **Customer Service:** Grievance redressal, response time and user satisfaction;

v. **Sustainability:** Source sustainability, water conservation and long-term system viability.

C. Benchmarking Approach

States/ UTs shall develop and adopt SMART (Specific, Measurable, Achievable, Relevant and Time bound) indicators for benchmarking of utilities.

The benchmarking framework shall be applied to both:

i. Micro Utilities (VWSCs) for village-level service delivery;

ii. Macro Utilities for system-level and bulk water management functions.

An indicative benchmarking matrix, including key indicators and target ranges may be developed by States/ UTs and periodically updated based on sectoral learning and performance trends.

An Indicative SMART benchmarking indicators for Utilities are provided in Table 4.

D. Use of Benchmarking

SWSM and DWSM shall utilize benchmarking as a core management tool for utility performance, including:

i. Conducting annual utility performance reviews based on defined indicators;

ii. Identifying underperforming utilities and providing targeted technical and managerial support;

iii. Setting of progressive improvement targets for utilities over defined time frames;

iv. Linking performance improvements with incentives, recognition and capacity building interventions;

v. Facilitating peer learning and dissemination of best practices across utilities.

E. Transparency and Public Disclosure

States/ UTs shall promote transparency in utility performance by:

i. Publishing periodic performance reports of utilities;

ii. Making key indicators available on digital platforms and dashboards;

iii. Enabling public access to performance data, wherever feasible.

Disclosure of utility performance shall encourage accountability, enhance citizen trust and promote competition among utilities for improved service delivery.

F. Learning from Best Practices

Best International and National practices may be considered for strengthening benchmarking systems. For instance, long term benchmarking practices in countries such as Denmark demonstrate that continuous performance measurement, universal metering, leak detection systems and public disclosure can significantly enhance efficiency and service reliability.

Such practices have enabled utilities to achieve:

i. Low levels of water loss;

ii. High levels of service availability;

iii. Efficient and responsive service delivery systems.

States/ UTs may draw upon such experiences, suitably adapted to local conditions to strengthen benchmarking frameworks for rural water utilities in India.

2.2.7. Adaptive and Phased Implementation by States/ UTs

States/ UTs may adopt a phased and context-specific approach for implementation of the dual utility framework, taking into account:

i. Existing institutional capacities;

ii. Typology of schemes (SVS/ MVS/ BWS);

iii. Geographical and demographic conditions;

iv. Availability of technical and financial resources.

The utility framework may be progressively strengthened over time with focus on improving service delivery outcomes, operational efficiency and sustainability.

A. Key Action Points for States/ UTs

i. Notify and operationalise VWSCs as Micro Utilities in accordance with State Panchayati Raj Acts;

ii. Define institutional arrangements for Macro Utilities at appropriate levels, where required;

iii. Establish clear reporting and coordination mechanisms between Micro and Macro Utilities;

iv. Issue detailed guidelines defining roles and responsibilities of Gram Panchayats, VWSCs, Macro Utilities and other stakeholders;

v. Ensure separation between governance (Gram Panchayat) and service delivery (Micro Utility) functions;

vi. Establish mechanisms for administrative oversight and performance monitoring;

vii. Develop standard operating procedures (SOPs) for operation and maintenance of water supply systems at both village and system levels;

viii. Define service level benchmarks and performance indicators for Micro and Macro Utilities;

ix. Establish protocols for coordination between bulk water supply systems and in-village distribution;

x. Undertake capacity building of VWSCs to function effectively as Micro Utilities;

xi. Create technical support structures at District/ Regional levels for supporting Macro Utility functions;

xii. Promote engagement of trained personnel such as Nal Jal Mitras and community-based institutions;

xiii. Ensure adequate provisioning for operation and maintenance (O&M) of water supply systems;

xiv. Establish mechanisms for collection and management of service delivery charge at the village level;

xv. Enable financial sustainability of utilities through convergence of resources and efficient fund utilisation;

xvi. Ensure integration of utility operations with digital platforms for financial management, physical progress monitoring, functionality and asset tracking;

xvii. Enable real-time data capture and reporting across Micro and Macro Utility levels;

xviii. Utilise digital dashboards for performance monitoring and decision-making;

xix. Establish performance monitoring frameworks for both Micro and Macro Utilities;

xx. Conduct periodic assessments of service delivery outcomes, including functionality and reliability;

xxi. Institute corrective action mechanisms for addressing performance gaps.

CHAPTER 2.3

Establishing Technical Support Framework for Gram Panchayats

Reform aims at:

- Improved quality and standardization of water supply schemes through strengthened technical adherence to CPHEEO norms;
- Implementation of rigorous and structured commissioning protocols ensuring system readiness;
- Enhanced sustainability of assets driven by systematic handing-over processes and strengthened community ownership;
- Strengthened technical capacity at District-level through institutionalisation of District Technical Units as continuous support systems;
- Improved operational efficiency and service reliability through better design.

JJM 2.0 is focusing towards building comprehensive framework for strengthening water supply systems by enhancing technical quality, operational reliability and long-term sustainability. It requires standardising planning and execution through strict adherence to CPHEEO norms, supported by structured commissioning protocols to ensure full system readiness. Emphasis also shifts for sustaining created assets through systematic handover processes and reinforced community ownership, while improving service efficiency through robust design validation. Central to this approach is the strengthening of District level technical capacity through the institutionalisation of District Technical Units, enabling continuous technical support and effective implementation at the grassroots level.

2.3.1. District Technical Unit (DTU) - Institutional Framework

States/ UTs shall undertake constitution of DTU[1] - Institutional Framework.

A. Background and Rationale

To ensure long-term functionality, sustainability and reliable service delivery of the created rural water supply infrastructure, JJM's extended phase necessitates a shift from a construction led, department driven model to a utility based, service-oriented governance framework. Accordingly, a DTU shall be established in every District as the technical backbone of the DWSM.

B. Composition and Role

The DTU shall comprise PHED officials at the District level, water quality officials, RD&PR representatives, Central/ State GWB officials, Irrigation/ Water resources department officials and technical consultants, as required. The DTU shall function as the extended technical arm of the PHED for rural water operations and shall report to DWSM on all technical and operational matters. The DTU shall coordinate between Micro Utilities (GP/ VWSC managed in-village infrastructure) and Macro Utilities (PHE/ RWS department managed out-village and bulk water infrastructure), in accordance with the dual utility system adopted under JJM.

C. Key Functions

To ensure long-term functionality, sustainability and reliable service delivery of the created rural water supply infrastructure, JJM's extended phase necessitates a shift from a construction led, department driven model to a utility-based, service-oriented governance framework. Accordingly, a DTU shall be established in every District as the technical backbone of the DWSM.

The DTU shall be responsible for inter-alia the following functions:

i. Scheme gap analysis, monitoring, commissioning and handover support;

ii. Technical review, performance assessment and functionality reporting support for functionality

[1] *'District Technical Unit (DTU) – Institutional Framework' as detailed in the letter Ref. DDWS letter No. WQ-11031/14/2025-WQ-DDWS-08 dated 12.01.2026'*

assessments (Jal Seva Aankalan) and SIPs preparation to DWSM;

iii. Water Quality Monitoring and Surveillance: The DTU should ensure the water quality monitoring and surveillance as per "Concise Handbook for Monitoring Water Quality of Piped Drinking Water Supply to Rural Household";

iv. Digital-first governance: All records on Sujalam Bharat Database, Sujal Gaon ID, GIS-linked digital asset registries, Meri Panchayat, JJM-IMIS, District/ GP dashboard etc. The DTU will consolidate all rural water supply schemes-including JJM, State, NRDWP and other schemes-into the Sujalam Bharat Database, creating a single unified digital registry for the District anchored to unique Sujal Gaon IDs;

v. Preventive & Predictive Maintenance: DTUs will shift from breakdown-based repair to planned maintenance using Sujalam Bharat asset data. AI-based analytics will help predict failures, plan timely replacements and manage spare parts efficiently, reducing downtime;

vi. Digital Ticketing for Technical Support: A digital helpdesk system will enable GPs/ VWSCs to raise and track technical issues quickly. Predictive tools will support faster diagnosis, timely resolution and better planning of repairs and asset replacement;

vii. Source sustainability planning using Decision Support System (DSS);

viii. Skilling, training and capacity building through NSQF Level-4 training for Nal Jal Mitra programme and NSQF Level 1 training for Sujalam Shakti group through Govt ITIs/ Polytechnics/ Govt, institutions;

ix. The DTU will play a key facilitative role in convergence of available funding streams- including JJM allocations, Finance Commission grants to GPs and other State or scheme specific funds;

x. Contract management for O&M: Service-level based, performance-linked: The DTU will guide districts and GPs in preparing O&M contracts whether executed directly by GPs/ VWSCs or through engaged technical agencies-covering scope of work, staffing norms, spares and inventory requirements and safety provisions;

xi. Standardisation of O&M practices, SOPs and District-specific O&M manuals: To ensure consistency and quality in operations across the District, the DTU will identify common and recurring O&M issues and issue SOPs for troubleshooting, preventive maintenance and emergency response. A District-specific O&M Manual, tailored to local technical and financial conditions, will also be prepared to serve as an operational reference for field functionaries and GP-level operators;

xii. Operation of District Command & Support Centres (DCSC) for performance monitoring: DTU will establish and operate DCSC that serve as the nerve centre for real-time monitoring of rural water supply scheme performance across the District.

2.3.2. Commissioning and Handing over Protocol

In order to ensure uniformity, technical compliance and sustainability in the commissioning of Rural Piped Water Supply Schemes (RPWSS), the commissioning process shall be completed in a structured manner over a period of approximately 13 weeks (3 months), divided into four sequential phases, each with defined activities, responsibilities and deliverables by implementing agencies, contractors and associated stakeholders. The detailed Commissioning Protocol may be seen at as prescribed in Guidelines for Commissioning and Handing Over Protocol & Community Ownership[2].The detailed checklist for commissioning of rural water supply scheme is annexed as **Annexure II**.

A. Commissioning Framework

i. Phase I: Pre-Commissioning Documentation (Weeks 1–3)

All implementing agencies shall ensure preparation and verification of comprehensive documentation prior to system testing. The implementing agencies shall finalize the as-built drawings incorporating all deviations from the approved design specifications, including pipe alignments, valve locations, pump configurations, treatment plant layouts and household connections. The implementing agencies shall complete geo tagged asset mapping of all infrastructure under RPWSS for integration with National platforms. All installed components shall be verified against approved designs and relevant BIS standards. A detailed visual inspection shall be conducted by independent technical personnel to identify defects, leakages, incomplete installations, or damages.

ii. Phase II: System Testing (Weeks 3–8)

All systems shall undergo rigorous testing to validate technical performance and service delivery standards.

[2] *'Guidelines for Commissioning and Handing Over Protocol & Community Ownership' circulated vide DDWS letter No. WQ-11031/14/2025-WQ-DDWS dated 19.11.2025.*

Hydrostatic pressure testing of pipelines and joints shall be conducted in accordance with applicable standards and norms. All household, institutional and community tap connections shall be inspected to ensure leak-free functionality. Flushing and disinfection of all reservoirs and pipelines shall be carried out in accordance with the CPHEEO Manual and WHO guidelines. The system shall be tested to ensure delivery of a minimum of 55 litres per capita per day (LPCD) at all consumer points, including tail-end households, with a minimum head of 7 metres. Water quality testing covering physical, chemical and biological/ bacteriological parameters shall be carried out through accredited laboratories. Residual chlorine levels shall be maintained at a minimum of 0.2 ppm at tail end consumer points. All electrical and mechanical systems, including pumps, motors, valves, sensors and control systems, shall be tested and validated for proper functioning. SCADA and automation systems shall be tested to ensure proper communication, cyber security compliance and functionality of human machine interfaces. All meters, gauges and sensors shall be calibrated against certified reference standards. System control parameters shall be adjusted and optimized to ensure efficient operation.

iii. Phase III: Trial Run (Weeks 9–11)

A trial run of not less than 15 days shall be conducted under actual operating conditions. During the trial run, continuous monitoring of flow rates, pressure levels, water quality, energy consumption and service reliability shall be carried out. The stability of the system shall be assessed with respect to pressure maintenance, flow consistency and treatment efficiency. All deficiencies, including pressure drops, leakages, pump failures and water quality deviations, shall be identified and rectified. Joint inspections shall be conducted involving engineers, contractors, Gram Panchayat representatives, VWSC members and community stakeholders. Information, Education and Communication (IEC) activities shall be undertaken to promote community awareness regarding safe water usage and acceptance of disinfection/ chlorination practices.

iv. Phase IV: Documentation and Compliance (Weeks 11–13)

Upon successful completion of the trial run, all required documentation and compliance measures shall be completed prior to final commissioning. A comprehensive commissioning report shall be prepared, including details of testing, results, compliance status and recommendations. Operation and Maintenance (O&M) manuals shall be developed in the local language, incorporating visual aids, troubleshooting procedures and maintenance schedules. All mandatory compliance certificates, including water quality certificates, pressure test reports and electrical safety certifications, shall be obtained. Training and capacity building of VWSC members and community stakeholders shall be conducted to ensure effective operation and maintenance of the system. The system shall be formally handed over to the community along with all relevant documentation, clearly defining roles and responsibilities for ongoing management.

B. Stakeholder Responsibilities

The commissioning process shall involve coordinated action among multiple stakeholders. Government and regulatory bodies shall be responsible for policy oversight, compliance verification and technical guidance. State and District implementation agencies shall be responsible for execution and supervision of commissioning activities. Technical experts and engineers shall be responsible for design validation, testing and system optimization. Quality assurance agencies shall undertake independent verification and certification. Community institutions, including Gram Panchayats and VWSCs, shall ensure local participation, oversight and long-term operation and maintenance. Digital and technology teams shall ensure asset mapping, automation and data management, while community mobilization agencies shall undertake IEC and behavioural change activities.

C. Standards and Compliance Requirements

All commissioning activities shall conform to BIS 10500:2012 for drinking water quality standards, ensuring zero tolerance for E. coli and total coliform. All technical and operational procedures shall comply with the latest CPHEEO Manual. Residual chlorine levels shall be maintained within the range prescribed under WHO guidelines, i.e., 0.2 to 0.5 mg/L.

D. Key Principles for Successful Commissioning

Technical integrity shall be ensured by strictly adhering to all prescribed standards and no system shall be commissioned without full compliance. Active community participation shall be ensured at all stages to facilitate ownership and sustainability. Transparency and accountability shall be maintained through mandatory joint inspections and proper documentation. All assets and operational parameters shall be integrated with designated digital platforms to enable monitoring and analytics.

E. Deliverables

The final commissioning package shall include verified checklists, detailed test reports, commissioning and compliance certificates, stakeholder mapping and responsibility matrix, operation and maintenance manuals, training records and a summary of timelines and performance benchmarks.

F. Applicability

These guidelines shall apply to all Rural Piped Water Supply Schemes (RPWSS) and shall be followed in entirety by all implementing agencies.

2.3.3. Handing over Protocol for Rural Piped Water Supply Schemes (RPWSS)

There shall be structured transfer of Rural Piped Water Supply Schemes from implementing agencies to Gram Panchayats and Village Water and Sanitation Committees under the Jal Jeevan Mission following 'Guidelines for Commissioning and Handing Over Protocol & Community Ownership'.[3] The protocol is intended to operationalize the principles of community ownership, sustainability and Jan Bhagidari by ensuring that water supply infrastructure is effectively managed by local institutions in a responsible and accountable manner. The framework seeks to enable long-term sustainability of assets through participatory governance and decentralized service delivery.

A. Expected Outcomes

The implementation of this protocol shall result in the successful and duly documented transfer of water supply infrastructure to legally constituted community institutions within the prescribed timelines. The Gram Panchayats and Village Water and Sanitation Committees shall develop adequate technical and managerial capacity to independently undertake operation and maintenance of water supply systems in accordance with prescribed standards. Community-based monitoring mechanisms shall be established to ensure regular assessment of service delivery, water quality and infrastructure performance. The protocol shall also result in improved financial sustainability of water supply systems through effective service delivery charge collection and prudent financial management. Enhanced community participation shall contribute to improved ownership and long-term sustainability of infrastructure, while also leading to reduction in water losses and improved system efficiency.

B. Pre-conditions for Handing Over

The handover of rural piped water supply schemes shall be undertaken only after fulfilment of all prescribed prerequisites. It shall be ensured that all infrastructure components, including source, treatment systems, storage facilities and distribution networks, are fully completed, tested and operational in accordance with approved designs and technical standards. It shall also be ensured that water supplied through the system meets prescribed quality standards and that baseline water quality data and monitoring protocols have been established. The Gram Panchayat or Village Water and Sanitation Committee shall be duly constituted, trained and equipped with clearly defined governance structures, byelaws and operational procedures. Trained manpower, including water operators, Sujalam Shakti group and Nal Jal Mitras, shall be in place and operation and maintenance manuals, standard operating procedures and maintenance schedules shall be available in documented form. A sustainable financial framework, including approved service delivery charge, dedicated bank accounts and accounting procedures, shall be established prior to handover. All required documentation, including asset registers, as-built drawings, technical assessment reports and compliance certificates shall be completed and shared with the receiving institution.

C. Trial Run and System Validation

A trial run of the water supply system shall be conducted for a minimum period of fifteen days under actual operating conditions prior to handover. During the trial run, system performance shall be continuously monitored with respect to flow rates, pressure levels, water quality, energy consumption and service reliability. The stability, efficiency and consistency of the system shall be assessed and any deficiencies such as leakages, pump failures or water quality deviations shall be identified and rectified. Community feedback shall be obtained during this phase to assess user satisfaction and acceptability of the service. All operational data generated during the trial run shall be documented to serve as a baseline for future monitoring and performance evaluation.

D. Community Integration and Verification

A joint inspection of the entire water supply system shall be conducted through a transect walk involving engineers, implementing agencies, Gram Panchayat representatives, Village Water and Sanitation Committee members and community stakeholders. This

[3] *'Guidelines for Commissioning and Handing Over Protocol & Community Ownership' circulated vide DDWS letter No. WQ-11031/14/2025-WQ-DDWS dated 19.11.2025.*

process shall ensure transparency, build confidence among stakeholders and facilitate consensus regarding system readiness for handover. Information, Education and Communication activities shall be undertaken to promote awareness regarding safe water practices, system usage and the importance of water quality, including acceptance of disinfection/ chlorination. Capacity building and on site training shall be provided to community representatives and operators to ensure their preparedness for managing the system independently.

E. Handover Process (Jal Arpan Framework)

The handover process shall be implemented in a structured and time bound manner through five sequential stages. The first stage shall involve pre-handover assessment and verification, during which all technical, institutional and financial readiness parameters shall be validated. The second stage shall involve community sensitization and participation, wherein awareness campaigns, consultations and training programmes shall be conducted. The third stage shall involve documentation and legal formalization, including execution of the handover deed or memorandum of understanding, Gram Sabha resolution and preparation of all technical and financial records. The fourth stage shall involve the formal handover ceremony, designated as Jal Arpan Diwas, during which assets, documents and responsibilities shall be formally transferred to the Gram Panchayat or Village Water and Sanitation Committee in the presence of the community. The fifth stage shall involve post-handover transition support, during which implementing agencies shall provide technical assistance, monitoring and troubleshooting support for a defined period.

F. Roles and Responsibilities

The Gram Panchayat and Village Water and Sanitation Committee shall be responsible for the day-to-day operation and maintenance of the water supply system, including infrastructure upkeep, financial management, water quality monitoring, record maintenance and grievance redressal. The Department/ DWSM/ DTU and associated technical agencies shall provide supervisory oversight, technical guidance, capacity building and support for major repairs and system improvements. The appointed operation and maintenance personnel shall be responsible for routine operation of the system, including valve operation, chlorination, monitoring of infrastructure and reporting of faults or irregularities.

G. Post-Handover Governance and Monitoring

A structured review mechanism shall be established for a period of three to six months following handover to assess system performance and sustainability. Periodic performance reviews shall be conducted to evaluate service continuity, water quality compliance, financial performance, infrastructure condition, community participation and institutional capacity. The findings of such reviews shall be documented and corrective action plans shall be prepared and implemented to address identified gaps. Continuous engagement between community institutions and government agencies shall be maintained to ensure sustained performance and improvement.

H. Key Principles

The handover process shall be guided by the principles of readiness prior to transfer, ensuring that no system is handed over without full compliance with technical and institutional requirements. The process shall ensure active community participation and acceptance through Gram Sabha involvement and transparent communication. Complete documentation and joint verification shall be ensured to maintain accountability and transparency. The long-term sustainability of the system shall be ensured through financial viability, technical soundness and institutional capacity. Post-handover support and monitoring shall be provided to ensure a smooth transition and sustained service delivery.

Figure 13: Commissioning, Handing over and Source Sustainability

Commissioning Readiness
(Design, Documentation, Compliance)

- Finalize **as-built** drawings; **geo-map** every asset (source→tap)
- Verify **design conformance** component-wise to standards.
- Close **defects** with evidence and sign-offs.
- Build **digital asset registry** with full metadata.
- Compile **commissioning pack** (tests, calibration, O&M SOPs, risk log).

System Validation & Trial Run
(Performance, Safety, Reliability)

- Prove pressure integrity and leak-free network.
- **Flush & disinfect**; confirm safe water and **RCL** at consumer points.
- Demonstrate **55 LPCD** at **tail ends** under real demand.
- Test **pumps/ motors/ valves/ sensors** and **automation**; tune controls.
- Run **trial** end-to-end – minimum 15 days; monitor KPIs; fix issues; issue **commissioning report**.

Handover
(Jal Arpan)
& Community Ownership
(Institutionalization)

- Lock prerequisites: functional infra, safe water, trained operators, preventive schedules, O&M financial plan with dedicated account.
- Complete **governance**: Special **Gram Sabha**, **MoU**, network diagram display, **user-charge** framework.
- Orchestrate **Jal Arpan**: technical brief, symbolic transfer, public pledge, documentation.
- Define interfaces: **Micro-utility (GP/VWSC)** for in-village O&M; **Macro-utility (Department/DTU)** for bulk, supervision, training, emergencies.
- Start **post-handover reviews** on continuity, quality, finances, assets, grievances.

Digital Governance & Performance Assurance
(Data Led Operations)

- Maintain **single source** of truth with geo-tagged assets and live updates.
- Use app-based **water-quality** stack with lab oversight and awareness.
- Provide **digital ticketing** with response/closure **KPIs** and dashboards.
- Shift to **predictive O&M**: preventive schedules, lifecycle replacements, spares forecasting.
- Stand up **command-and-control**: district dashboards, **SCADA** integration, exception management.

Source Sustainability
(DSS Led Resilience)

- Prepare DSS plan for source sustainability with hydro-geology and convergence.
- Set **annual calendar**: pre-monsoon inspection/cleaning; monsoon **recharge** works; post-monsoon recovery checks.
- Execute **catchment protection**, intake **desilting**, recharge structure upkeep, contamination pathway control.
- Close the **surveillance loop**: read trends; trigger timely remedies.
- Build **capacity & funding**: train operators, refresh skills, align multi-source O&M financing.

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CHAPTER 2.4

Establishing Citizen-centric Water Quality Governance

Reform aims at:

- Shift from reactive to preventive water quality management, with early detection and timely response to contamination risks;
- Empowered community-led monitoring, strengthening local ownership and ensuring last-mile vigilance through trained Sujalam Shakti Groups;
- Continuous monitoring, data-driven decision-making, enabled by digital platforms, geo-tagged reporting and integrated dashboards;
- Strengthened laboratory ecosystem and prompt corrective action, ensuring accuracy, standardisation and credibility of Water quality data;
- Enhanced transparency and citizen trust, through public dashboards, grievance responsiveness and accessible water quality information.

Strengthening Water Quality Monitoring & Surveillance (WQM&S) demands targeted investments to upgrade water supply infrastructure to serve water comply with BIS 10500 standards. While substantial progress has been made in expanding rural piped water supply systems, the long term sustainability and public confidence in these systems depend heavily on the reliability, transparency and responsiveness of water quality monitoring mechanisms. By integrating laboratory-based testing with community led surveillance through Field Test Kits (FTKs), the program reinforces shared responsibility and Jan Bhagidari.

Under JJM 2.0, States/ UTs shall progressively shift from periodic testing practices to a continuous, technology-enabled monitoring and response-based water quality governance system. The mission emphasises that each State/ UT must establish year-wise budgeted financial arrangements that enable a sustainable and accountable framework for Water Quality Monitoring & Surveillance (WQM&S) and digital reporting system are adequately resourced. This includes planning and securing long-term funds for human resources, laboratory operations, consumables, Field Test Kits (FTKs), digital portable field test kits and community-led monitoring & surveillance mechanisms. The strengthening of institutional capacities, supported by annually earmarked financial budget, shall ensure the sustained operation of continuous monitoring & surveillance systems and uphold the delivery of safe drinking water to all beneficiaries.

2.4.1. Key Features of the WQM&S Framework

A. Community Led Water Quality Surveillance

The village-level surveillance shall be undertaken through Community participation for routine water quality surveillance through trained women groups designated as Sujalam Shakti. The Sujalam Shakti groups shall undertake field level monitoring & surveillance and other activities as described in section 2.4.2 (part A). This mechanism is intended to strengthen local monitoring, enable early identification of water quality issues and promote community ownership. The details on Sujalam Shakti Group is annexed as **Annexure III**.

B. Digital Monitoring and Transparency

All water quality monitoring & surveillance activities shall be digitised to ensure uniform reporting and effective oversight. Monitoring data shall be captured through prescribed digital platforms, including the Sujal Gaon Utility App, by the Sujalam Shakti Groups. The captured data shall be integrated with centralised dashboards at the SWSM, DWSM and DTU levels to enable real time visibility of water quality status, alerts and remedial actions taken. This digital system shall be utilised to enhance transparency, accountability and evidence-based decision making at all levels. States may additionally customise Block/ Sub-divisional level

applications or digital modules to strengthen surveillance, reporting and response mechanisms based on local requirements.

C. Water Quality Laboratory Ecosystem

States/ UTs shall ensure strict compliance with the Water Quality Monitoring & Surveillance (WQM&S) requirements as prescribed in the 'Concise Handbook for Monitoring Water Quality of Piped Drinking Water Supply to Rural Households' released in December 2024 by DDWS. For this purpose, States/ UTs shall establish, strengthen and operationalise a multi-tier water quality laboratory network comprising:

i. State/ Regional-level laboratories;

ii. District-level laboratories;

iii. Block/ Sub-divisional level laboratories;

iv. In-house laboratories at water treatment plants;

v. Mobile water quality laboratories for remote areas, wherever required.

Such laboratories shall be State-owned or established through collaboration with accredited institutions, universities, or private laboratories. The laboratory network shall provide technical support for routine monitoring of water sources, conduct confirmatory testing in cases of suspected contamination and issue analytical reports to inform regulatory action and public health response.

D. Institutional Response Mechanism

The WQM&S Framework provides for a structured institutional mechanism to ensure timely and appropriate response to water quality monitoring results. Roles and responsibilities at the Gram Panchayat/ Village, District and State levels shall be clearly delineated and operationalised for source protection measures (refer Section 2.5.2 of Chapter 2.5) and for treatment/ disinfection optimisation, infrastructure repair and other corrective actions (refer Section 2.4.2). The integration of water quality monitoring outcomes with the institutional response mechanism shall enable timely corrective measures and ensure sustained compliance with prescribed drinking water quality standards.

2.4.2. Monitoring and Evaluation Framework

Monitoring and evaluation of the WQM&S framework under JJM 2.0 shall be undertaken through a combination of institutional monitoring mechanisms, community-based oversight and digital reporting systems across the four governance tiers. The monitoring framework shall utilise data captured through the Sujal Gaon Utility App, WQMIS portal and JJM-IMIS platform/ Sujalam Bharat data base, enabling continuous monitoring and timely technical response.

A. Village Level Monitoring & Surveillance

i. Sujalam Shakti groups shall conduct routine water quality testing using Field Test Kits (FTKs), including monitoring of residual chlorine levels, turbidity and other basic water quality parameters. Test results shall be recorded and uploaded through the Sujal Gaon Utility App, linked to the respective Sujal Gaon ID and supporting local response actions where water quality deviations are observed;

ii. GPs/ VWSCs shall review the water quality testing results reported by Sujalam Shakti groups and ensure appropriate actions such as disinfection, pipeline inspection, minor maintenance or source protection/ sanitation and cleaning measures taking assistance of DTU wherever required;

iii. Water quality monitoring outcomes and related actions shall be periodically reviewed in the Gram Sabha, ensuring community awareness, transparency and collective oversight of drinking water safety. Village level monitoring indicators shall include:

a. Number of FTK tests conducted per month as per prescribed testing frequency;

b. Residual chlorine and turbidity monitoring compliance;

c. Number of water quality contamination alerts reported for remediation action by DTU/ DWSM;

d. Timely reporting through Sujal Gaon Utility App.

B. District Level Monitoring

At the District level, monitoring shall be undertaken by the DTU in coordination with the DWSM to ensure timely technical response and system reliability.

The DTU and DWSM shall monitor and handhold/ assist for the following:

i. Monitoring of the completeness, timeliness and accuracy of water quality data reported through the Sujal Gaon Utility App and laboratory reporting systems;

ii. Verification of compliance with prescribed testing frequency for various water sources and supply systems, as specified in Concise Handbook;

iii. Monitoring the time taken for technical interven-

tion and corrective action following detection of contamination or water quality deviations;

iv. Ensuring timely submission of water samples which are in non-conformity at FTK testing stage for further confirmatory testing at laboratory and monitoring laboratory turnaround time (TAT) for reporting results;

v. Providing technical guidance to Gram Panchayats and VWSCs for addressing water quality issues through disinfection, infrastructure repair, treatment solutions or source protection measures;

vi. For Jal Arpan and HGJ declaration, DTU to analyse water quality monitoring data and coordinate technical interventions by mandatory issuance of disinfection protocols to HGJ villages and ensuring the training and orientation;

vii. DTU to analyse the feedback from GPs through Jal Seva Aankalan regarding Infrastructure repair or treatment system upgrades.

District level monitoring dashboards shall enable DTUs and DWSMs to identify high-risk locations, recurring contamination issues and operational gaps in water quality management.

C. State Level Monitoring

The SWSM shall undertake periodic review of water quality monitoring performance across Districts to ensure effectiveness and institutional accountability.

The State level monitoring framework shall include the following indicators:

i. Laboratory Performance Indicators:

a. Adequacy of water quality laboratory network to achieve the prescribed testing as per Concise Handbook;

b. Accreditation status of laboratories;

c. Turnaround time for laboratory testing;

d. Availability of trained laboratory personnel;

e. Compliance with testing protocols and reporting standards;

f. Number of water samples submitted by general public.

ii. Community Monitoring Coverage:

a. Number of villages with operational Sujalam Shakti groups and number of persons trained under Sujalam Shakti groups;

b. Percentage of villages undertaking routine water quality testing.

iii. State Level Response Mechanisms:

a. Effectiveness of DTU/ DWSM response to water quality alerts;

b. Timely communication of advisories to citizens in case of outbreaks of water borne diseases and/ or calamities/ disasters etc.;

c. Integration of water quality monitoring data with State dashboards.

SWSM shall also review trends in contamination events and develop corrective policy and operational measures where systemic issues are identified.

D. National Level Monitoring

The National Jal Jeevan Mission (NJJM) shall periodically review the overall performance of the Water Quality Monitoring & Surveillance ecosystem across States/ UTs.

The National-level review shall focus on the following parameters:

i. State Performance Indicators:

a. Progress of water quality monitoring & surveillance across rural habitations/ village;

b. Digital reporting compliance through Sujal Gaon Utility App;

c. Laboratory network functionality by indicators such as accreditation/ recognition status and TAT.

ii. Effectiveness of WQM&S Ecosystem:

a. Responsiveness of District-level technical/ DTU support systems;

b. Integration of community surveillance with laboratory confirmation;

c. Performance of digital monitoring platforms including adoption of Sujal Gaon Utility App.

iii. Public Health Outcomes:

a. Trends analysis of water quality contamination reported at delivery points in the village;

b. Trends analysis of water-borne disease risks associated with improved monitoring and response mechanisms.

The insights generated through National monitoring shall form policy guidance, technical advisories and capacity-building initiatives for strengthening water quality governance under JJM 2.0.

2.4.3. Utilisation of Water Quality Monitoring and Surveillance (WQM&S) Funds

The WQM&S component under Jal Jeevan Mission 2.0 is intended to support the establishment of a robust, citizen-centric water quality governance framework across rural drinking water supply systems. States/ UTs shall utilise these funds to build an integrated ecosystem comprising field-level surveillance, institutional monitoring, laboratory support systems and responsive remedial mechanisms, ensuring long-term safety and reliability of drinking water services.

A. Establishment of Community-Based Water Quality Surveillance

States/ UTs shall prioritise the establishment of field level surveillance mechanisms through community institutions. For this purpose, preferably women-led Self-Help Groups (SHGs), including women who have received prior training, will be reorganised as Sujalam Shakti Groups. These groups will support Gram Panchayats and Village Water and Sanitation Committees (GP/ VWSCs) in undertaking routine water quality monitoring, promoting community awareness and facilitating coordination for necessary remedial actions.

B. Capacity Building and Skill Development

To strengthen technical capacity at the community level, one to two members of each Sujalam Shakti group shall undergo structured training leading to certification under NSQF Level 1 in basic water quality monitoring and surveillance.

The training programme shall include:

i. Apply basic concepts of rural water supply and conduct drinking water quality testing using field test kit (FTK);

ii. Perform basic water disinfection using chlorine, bleaching powder etc;

iii. Understanding of basic water quality parameters and contamination risks;

iv. Water sampling protocol, turbidity and free residual chlorine monitoring and disinfection awareness;

v. Use digital, financial, communication skills, community engagement and customer service at the Workplace through the Sujal Gaon Utility App;

vi. Implement Source Contamination Prevention and Follow Regional Water Safety Guidelines.

The training shall be supported under the WQM&S component of JJM 2.0, with convergence and other Govt, training institutions to ensure scale and sustainability.

C. Incentive-Based Fund Release Linked to Governance Milestones

After supporting the establishment of community surveillance and training of field functionaries, the remaining funds under the WQM&S component shall be released to States in tranches linked to achievement of defined governance milestones. The milestone-based incentive structure is intended to support States in systematically establishing the institutional components required for effective water quality governance.

Indicative milestones may include:

i. Operationalisation of Sujalam Shakti groups and digital reporting through the Sujal Gaon Utility App;

ii. Strengthening State (building cost permitted only for State Level Laboratory), District, Regional, Subdivisional/ Block and mobile level laboratory monitoring systems;

iii. Adoption of standard water quality testing protocols and remedial response mechanisms;

iv. Strengthening of laboratory networks with adequate technical manpower;

v. Accreditation of laboratories and compliance with national testing standards;

vi. Development of State-level protocols for water quality surveillance, preventive maintenance and response mechanisms;

vii. Establishment of sustainable State budgetary provisions for long-term water quality monitoring and surveillance.

D. Strengthening Institutional Water Quality Governance

The remaining funds under the WQM&S component shall be released to States in tranches linked to achievement of defined governance milestones. The milestone-based structure is designed to ensure that States progressively establish field-level surveillance, laboratory monitoring systems, technical response mechanisms and sustainable institutional arrangements for water quality governance.

States shall utilise WQM&S funds to ensure that water quality monitoring systems are supported by clear institutional arrangements, including:

i. Technical oversight by District Technical Units

(DTUs);

ii. Laboratory support for confirmatory testing;

iii. Monitoring through State Water and Sanitation Mission (SWSM);

iv. Digital reporting integrated with the Sujalam Bharat platform and JJM-IMIS systems.

This approach will strengthen public confidence in drinking water safety and ensure long-term sustainability of rural piped water supply systems under Jal Jeevan Mission 2.0.

2.4.4. Transparency and Citizen Information

The Citizen Corner- JJM WQMIS will provide a platform to States/ UTs to enable access to citizens regarding information on:

i. Water quality testing results: Water quality data generated through community testing by Sujalam Shakti groups, routine laboratory analysis and surveillance activities shall be digitally linked with the Sujal Gaon service area architecture under the Sujalam Bharat platform. Relevant water quality information pertaining to supplied drinking water shall be regularly displayed on the Citizen Corner to enable citizens to access the latest available testing results for their respective service areas;

ii. Status of remedial actions: District Technical Units, DWSM and SWSM shall monitor the responsiveness of this system to ensure timely updating of test results, effective communication with citizens and prompt remedial action in cases where water quality deviations are reported;

iii. Functionality of local water supply systems: Functionality assessment of water supply Systems through Jal Seva Aankalan shall strengthen public confidence in drinking water safety and promote community participation in water governance.

2.4.5. Roadmap for Strengthening WQM&S

The following is the yearwise roadmap for strengthening WQM&S, considering

i. Community monitoring and digital enablement;

ii. Laboratory strengthening and technical response;

iii. Institutionalisation and sustainability.

Figure 14: Roadmap for Strengthening WQM&S

Year 1 *(by December 2026)*

Community Monitoring and Digital Enablement

1. Operationalisation of Sujalam Shakti Groups through SHGs, with training and certification (at least 1–2 women per village at NSQF Level 1), including Field Test Kit (FTK) usage.
2. Roll-out of digital reporting systems through the Sujal Gaon Utility App, enabling geo-tagged sampling and geo-referenced water quality reporting for both laboratory and FTK testing, including parameters such as turbidity and residual chlorine.
3. Establishment of coordination mechanisms among Sujalam Shakti Groups, VWSCs, and Gram Panchayats for effective WQM&S implementation.
4. Adoption of State-level WQM&S protocols and SOPs, aligned with the Concise Handbook for Monitoring Water Quality of Piped Drinking Water Supply to Rural Households.
5. Operationalisation of District Technical Units (DTUs) for monitoring, technical support, and contamination response.

Year 2 *(by December 2027)*

Laboratory Strengthening and Technical Response

1. Establishment and upgradation of district and block/sub-divisional/cluster laboratories, along with deployment of trained laboratory personnel.
2. Achievement of ISO/IEC 17025 accreditation for water quality laboratories.
3. Integration of laboratory data with JJM IMIS and Sujalam Bharat platforms for seamless data flow and analysis.
4. Implementation of remedial action protocols and monitoring of laboratory performance indicators.
5. Deployment of automated dashboards providing near real-time alerts and enabling data-driven technical decision support systems.

Figure 14: Roadmap for Strengthening WQM&S

Year 3
(by December 2028)

Institutionalisation and Sustainability

1. Implementation of preventive maintenance protocols to ensure sustained water quality assurance in water supply systems.
2. Activation of citizen-facing services through JJM IMIS Citizen Corner and utility applications.
3. Ensuring digital transparency and advanced monitoring, including public dashboards, self-diagnostic tools, predictive analytics, groundwater integration, digital twins, and AI-enabled support systems.

2.4.6. WQM&S during Emergency and Disasters

During emergencies and disasters, the Water Quality Monitoring and Surveillance framework under Jal Jeevan Mission shall be activated on a priority basis to ensure the continued safety of drinking water supplies. States and implementing agencies shall undertake immediate risk assessment of water sources and infrastructure, with enhanced frequency of water quality testing, particularly for bacteriological contamination. Disinfection measures, including chlorination and point of use water treatment measures shall be ensured at all supply points and alternative safe water arrangements shall be made where contamination is detected or anticipated including areas where the sources and/ or supply systems are defunct due to disaster. Village Water and Sanitation Committees (VWSCs) shall be mobilized for community-level surveillance and reporting, while District and State laboratories shall operate in a mission mode for rapid testing and response.

Timely dissemination of advisories to the public regarding safe water handling and usage shall be ensured to prevent water-borne diseases.

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CHAPTER 2.5

Source Sustainability & Water Security Framework

Reform aims at:

- Decision Support System (DSS) introduced as a digital platform for source monitoring and planning, integrating multi-source datasets under the Water as One Data approach;
- Aquifer-based planning institutionalised to guide scientifically suitable recharge and conservation measures;
- GIS-based identification of water-stressed villages and vulnerable sources for prioritised sustainability measures;
- District Source Sustainability Action Plans (DSSAP) introduced for systematic identification of critical sources and phased sustainability measures;
- Strengthened institutional and convergence framework, including Source Sustenance Committees and alignment with National and State programmes for recharge and conservation works.

One of the critical pillars of the sustainability of the schemes is ensuring reliability and quality of all drinking water sources throughout the scheme life cycle. The focus shall progressively shift from source finding to source sustenance through scientific assessment, aquifer-based planning, climate risk screening and proactive management of withdrawals and recharge.

Source sustainability refers to ensuring the long-term reliability, adequacy and safety of drinking water sources of rural water supply schemes. It requires maintaining both the quantity and quality of the source at levels sufficient to meet present and future demand, while preventing ecological stress and avoiding disruptions in scheme functionality.

2.5.1. Key Features of Source Sustainability

i. Ensures adequate and reliable quantity and quality of drinking water through sustained source yield, seasonal stability, regular testing and timely corrective measures;

ii. Supports scheme functionality by reducing breakdowns, supply interruptions and dependence on temporary water arrangements;

iii. Requires annual and seasonal assessments of groundwater levels, yield, quality and recharge behaviour for all drinking water sources;

iv. Enables predictive identification of scheme criticality through continuous and real time monitoring, helping detect declining yields, seasonal stress, or emerging quality risks;

v. Strengthen implementation through convergence with different ministries and departments, such as VB-G RAM G, Watershed Development Component-Pradhan Mantri Krishi Sinchayee Yojana (WDC-PMKSY) 2.0, etc., for recharge, watershed development and catchment improvement;

vi. Promotes multi-department coordination for recharge planning, source protection, water budgeting, demand management and climate resilient interventions;

vii. Integrates water budgeting and community participation, enabling Gram Panchayats to monitor water availability, demand and sustainability measures;

viii. Utilises GIS based mapping and annual risk categorisation to identify water stressed villages and prioritise interventions;

ix. Relies on continuous monitoring of groundwater and surface water sources throughout the scheme life cycle;

x. Achieved through a combination of technical measures (recharge structures, aquifer-based

planning, source protection) and institutional mechanisms (SSC, DWSM reviews and governance systems).

2.5.2. Institutional Framework

Source sustainability shall be supported through clearly defined roles at the State, District and Gram Panchayat levels, ensuring that technical review, coordination and monitoring are part of regular planning and implementation processes.

Source Sustenance Committee: As envisaged in the Operational Guidelines for the implementation of Jal Jeevan Mission, para 5.2, Source Finding Committees were constituted at the planning stage of each scheme to identify and recommend the availability of potable water in adequate quantity in prescribed quality for the scheme design period. In Operation & Maintenance phase of Jal Jeevan Mission, the Source Sustenance Committee (SSC) shall be constituted at Gram Panchayat, District and State level in alignment with the JJM institutional structure to ensure periodic assessment, protection and long-term sustainability of drinking water sources. At the Gram Panchayat level, VWSC shall act as the Source Sustenance Committee and shall be responsible for source-level monitoring, protection and reporting activities. At the District level, the SSC shall function under the District Water and Sanitation Mission (DWSM) supported by the District Technical Unit (DTU) to plan, implement and monitoring of source sustainability measures. At the State level, the SSC shall function under the Executive Committee of the State Water and Sanitation Mission (SWSM) to provide policy guidance, technical oversight and coordination with relevant State departments.

A. Gram Panchayat Source Sustenance Committee

The Gram Panchayat Source Sustenance Committee shall periodically assess and support the following:

i. Maintenance of basic source records and seasonal registers for drinking water sources within the Gram Panchayat, including providing inputs for digital monitoring;

ii. Ensure protection and upkeep of drinking water sources and supports water quality testing (labs/ FTK) with prompt reporting of any quality issues or supply distress;

iii. Contribute to GPs/ Village water budgeting and assist in preparation of GPDP;

iv. Assist in the upkeep of minor recharge or conservation works approved by the District and escalate emerging risks (drying, turbidity, leakage);

v. Plan and oversee recharge and source protection measures at GP level, including structures implemented through convergence with VB- G RAM G, WDC-PMKSY 2.0 and other schemes.

B. District Source Sustenance Committee

The District Source Sustenance Committee shall periodically assess and support:

i. Review the seasonal performance of all drinking water sources and assess sustainability, especially during lean periods;

ii. Plan and oversee recharge and source protection measures, including structures implemented through convergence with VB-G RAM G, WDC-PMKSY 2.0 and other schemes;

iii. Identify at risk sources and GPs/ Villages facing depletion or quality deterioration and recommend appropriate corrective actions;

iv. Facilitate water budgeting, demand management measures and inter-departmental coordination for recharge, conservation and catchment protection;

v. Submit recommendations to the DWSM for approvals, convergence planning and required follow-up actions.

C. State Source Sustenance Committee

The State SSC shall:

I. Review consolidated District reports on source sustainability and track overall progress;

ii. Issue policy guidance and technical advisories to districts based on emerging challenges and priorities;

iii. Provide support to districts facing recurring or widespread sustainability issues, including technical inputs and coordination assistance;

iv. Facilitate convergence and coordination with State level departments responsible for water resources, groundwater, watershed management and environmental protection.

D. Digital Monitoring and Support System

Digital monitoring and support for source sustainability under Jal Jeevan Mission shall be enabled through a Decision Support System (DSS) developed on the JJM-IMIS platform. The DSS shall act as an enabling digital platform under the "Water as One Data" approach, integrating geo-spatial layers, hydrogeological parameters, rainfall information,

aquifer data and drinking water source datasets from various National and Departmental platforms. This includes integration of CGWB groundwater status data, Sujalam Bharat source mapping and digitised inspection inputs from districts. Accessible through dashboards for District Water and Sanitation Missions (DWSMs) and Gram Panchayats, the system shall support evidence-based planning, monitoring and prioritisation of source sustainability measures from District to Village level.

The DSS shall generate composite risk indicators, early warning alerts and possible intervention options to help identify vulnerable sources and prioritise recharge and conservation measures. By bringing together data from multiple departments in a single platform, the system shall support coordinated planning, improve transparency and strengthen proactive management of drinking water sources to ensure sustained functionality of rural water supply systems under Jal Jeevan Mission.

2.5.3. Aquifer Based Planning

States/ UTs shall adopt aquifer-based planning to ensure that drinking water sources are developed and managed in accordance with local hydrogeological conditions. Districts shall use NAQUIM aquifer maps and groundwater assessment data from CGWB/ INGRES to understand aquifer types, seasonal groundwater behaviour and long-term trends, particularly in semi-critical, critical and over-exploited areas. Recharge and conservation measures such as check dams, percolation tanks, recharge shafts and spring-shed interventions shall be selected based on aquifer characteristics, depth to water table and recharge potential. The Decision Support System (DSS) shall integrate these datasets to support informed siting of drinking water sources, planning of recharge structures and preparation of Sustainability Action Plans under JJM.

2.5.4. Sustainability Action Plans

District Water and Sanitation Missions (DWSMs) shall prepare a District Source Sustainability Action Plan (DSSAP) to ensure the long-term reliability and functionality of drinking water sources under Jal Jeevan Mission. The Action Plan shall identify vulnerable or critical sources and assess hydrogeological conditions, groundwater trends, seasonal source performance and water quality risks and outline appropriate corrective and preventive measures along with responsible departments and implementation timelines. Preparation of the Action Plan shall also incorporate local knowledge and traditional water management practices to support locally appropriate and community responsive sustainability measures. Districts shall utilise datasets and analytical outputs available through the Decision Support System (DSS) to support evidence-based prioritisation of sustainability interventions. Information on drinking water sources mapped under the Sujalam Bharat platform shall also be reviewed to assess source performance and criticality, in order to guide timely sustainability planning.

The Action Plan shall adopt a phased approach covering near-term stabilisation of vulnerable sources, short-term implementation of recharge and conservation measures and medium-term aquifer-scale and watershed interventions, while promoting convergence with relevant programmes and integration with GPDPs and District Action Plans to strengthen long-term water security. States and Districts shall also promote Jal Utsav/ Jal Arpan Diwas as a periodic village-level initiative for inspection, cleaning and preventive maintenance of drinking water sources, recharge structures and related infrastructure to support sustained source functionality.

2.5.5. Commissioning, Handover and Annual Source Maintenance

Source sustainability planning shall be aligned with the commissioning and handover processes under Jal Arpan. Prior to commissioning of a drinking water scheme, a successful trial run shall be undertaken to verify the performance of the source, treatment system and distribution network. Upon commissioning, water supply assets shall be formally handed over to the Gram Panchayat/ VWSC for operation and maintenance, including responsibilities for source protection, monitoring and upkeep of associated recharge and conservation structures.

To support sustained functionality of drinking water sources, Jal Utsav shall be organised periodically as a village level initiative for inspection, cleaning and maintenance of water sources, recharge structures and related infrastructure. Such activities shall reinforce community participation, preventive maintenance and long-term sustainability of rural drinking water sources under Jal Jeevan Mission.

2.5.6. Source Protection and Contamination Measures

In accordance with the Source Protection Framework outlined in Chapter 2.1, Para 2.1.4 (B), States/ UTs shall implement statutory and technical measures for protection of drinking water sources to ensure the safety and long-term sustainability of groundwater used for drinking water supply. States/ UTs shall undertake source protection and contamination prevention measures in compliance with the relevant guidelines and standards issued by the Central Public Health and Environmental Engineering Organisation (CPHEEO), Central Ground Water Board (CGWB) and Bureau of Indian Standards (BIS).

Appropriate minimum sanitary protection distances shall be maintained around drinking water sources such as tube wells and borewells to minimize the risk of contamination from anthropogenic activities.

District Administrations and Gram Panchayats shall utilize geospatial information and datasets available through the Decision Support System (DSS) and other digital platforms to:

i. Map drinking water source locations;
ii. Monitor surrounding land use patterns;
iii. Identify potential contamination risks;
iv. Support planning and implementation of appropriate source protection measures.

The surroundings of drinking water sources shall be maintained in a clean and well-kept condition and may be suitably developed or beautified to promote community ownership, responsible behaviour and protection of the source.

To prevent contamination of drinking water sources, the following safeguards shall be ensured within the notified protection zone:

i. Prohibition of landfill sites or solid waste disposal facilities in proximity to drinking water sources;
ii. Prevention of disposal or storage of hazardous or polluting substances near tubewells and borewells;
iii. Avoid construction of soak pits, leach pits, septic tanks, magic pits or other waste disposal structures within the sanitary protection radius of drinking water sources. Where such structures cannot be avoided due to site/ area constraints, appropriate protective and mitigation measures shall be adopted to prevent contamination of the drinking water source.

Further, to prevent over extraction of groundwater and hydraulic interference between wells, installation of additional tubewells or borewells within the prescribed safe distance from an existing drinking water source shall be reviewed in accordance with CPHEEO, CGWB and BIS technical standards.

These measures are essential to maintain source integrity, prevent contamination and ensure long-term sustainability of groundwater based rural drinking water supply systems.

2.5.7. Water Budgeting

Water budgeting at the Gram Panchayat (GP) level shall be undertaken using a structured, data driven approach to assess water availability and demand across domestic, livestock, irrigation and other uses. Priority shall be given to allocation of water for drinking water supply ensuring that domestic water requirements under Jal Jeevan Mission are safeguarded before planning for other uses. GPs shall use relevant datasets available through the Decision Support System (DSS) such as rainfall, groundwater trends, land use and aquifer information along with local data on population, livestock, crop patterns and water storage structures to estimate water availability from rainfall, groundwater recharge, runoff, surface storage and greywater reuse. The resulting water balance can support evidence-based decision making, prioritisation of interventions and preparation of Village Water Security Plans under JJM.

2.5.8. Greywater Management

Effective management of household greywater shall be an important component of source sustainability under Jal Jeevan Mission. Gram Panchayats shall promote safe collection, treatment and reuse of greywater through appropriate systems such as soak pits, leach pits, drainage channels and nature-based treatment options, in convergence with programmes such as SBM(G) Phase II. Proper greywater management helps prevent contamination of groundwater and drinking water sources, reduces stagnation and local pollution around water supply infrastructure and enables beneficial reuse for activities such as irrigation of plantations and community green spaces. These measures contribute to improved village sanitation, protection of drinking water sources and longterm water security.

2.5.9. Convergence with Government Schemes/ Grants

States/ Districts/ GPs shall plan and implement source sustainability measures through convergence with Central and State programmes supporting groundwater recharge, catchment protection, greywater management, water conservation and demand regulation ensuring that investments are directed to priority locations. Programmes such as VB-G RAM G, WDC-PMKSY 2.0 and Finance Commission Grants may be leveraged for this purpose. As per the Ministry of Rural Development Notification dated 23.09.2025 amending Schedule I, Para 4(2) of the MGNREGA Act, 2005 (Gazette of India – Extraordinary), a minimum share of expenditure on water related works shall be prioritised in groundwater assessment units (blocks) based on their groundwater status 65% in over exploited and critical blocks, 40% in semi-critical blocks and 30% in safe blocks thereby directing recharge and conservation investments towards water stressed areas and strengthening long term sustainability of drinking water sources under JJM.

An indicative list of schemes under which activity/ fund convergence can be prioritized to improve the sustainability of villages is attached as **Annexure IV.**

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Har Ghar Jal
Jal Jeevan Mission

CHAPTER 2.6

Establishing Digital Data Governance in Rural Drinking Water Supply Systems

Reform aims at:

- Shift from fragmented data systems towards objectives of Sujalam Bharat which include enabling end to end visibility of rural water supply systems on an integrated digital ecosystem;
- Enhanced real time monitoring and decision making, through multi-level dashboards tracking service delivery and scheme performance;
- Complete asset and network visibility with GIS-based registries and digital mapping of hydraulic connectivity across assets;
- Improved monitoring, operational efficiency and accountability, driven by digital capture of operational, financial and water quality transactions at the service provider level;
- Strengthened citizen engagement and transparency, through integrated platforms for feedback, grievance redressal and participatory monitoring.

Across sectors globally, governments and utilities are increasingly adopting digital frameworks to manage large scale service delivery networks through integrated data systems, operational workflows and citizen facing interfaces. Drinking water sector with its geographically dispersed and hydraulically interlinked assets, requires similar digital governance mechanisms to ensure reliability, accountability and long-term sustainability. Unlike traditional monitoring systems, drinking water sectoral Digital Public Infrastructure (DPI) enables unified management by linking physical infrastructure, operational data, financial information and citizen feedback within a single digital ecosystem. Sujalam Bharat framework builds a population scale digital public infrastructure for water governance, establishing a national IT architecture for end-to-end digital management of Rural Piped Water Supply Systems under JJM 2.0. The architecture standardises digital mapping of schemes, sources, assets and service areas through interoperable registries and GIS validated hydraulic connectivity, creating a common foundational layer on which operational monitoring, utility management, water quality systems, financial tools and decision support platforms can be built.

Designed to be modular and interoperable the Sujalam Bharat DPI enables States to develop additional applications using common data standards. By integrating GIS-based asset layers with stakeholder transactions supported through the Sujal Gaon Utility Application, the framework institutionalises a unified, data driven governance mechanism aligned with the vision of Viksit Bharat @2047. Collectively, this digital ecosystem strengthens accountability, transparency and lifecycle-based service management, enabling structured, real time and sustainable governance of rural drinking water supply systems.

Following digital reforms have been introduced under JJM 2.0:

2.6.1. Sujalam Bharat Digital Public Infrastructure (DPI) for Rural Drinking Water Sector

Sujalam Bharat is being developed as a federated Digital Public Infrastructure (DPI) to enable lifecycle based, service oriented and source-secure management of rural piped water supply systems across the country. It integrates interoperable digital asset registries, geo spatial asset mapping, transaction platforms and decision support systems to create a unified yet flexible national digital backbone for water service delivery. Such a digital architecture enables seamless digital visibility of Rural Piped Water Supply Schemes (RPWSS) at all levels of governance and to the local communities. Earlier, it was a challenge as the data systems were fragmented with non-standardized

digital representation of water supply schemes which were a result of multiple schemes executed at different time periods, thereby limiting source to tap visibility and lifecycle management. At the same time institutional silos, asset agnostic grievance systems and limited regulatory oversight further impede timely decision making and preventive maintenance.

These gaps underscored the need for an integrated, scalable DPI to shift from asset centric monitoring to service oriented water governance. Addressing this need, Sujalam Bharat DPI integrated with PM GatiShakti platform provides a GIS validated national architecture that assigns unique digital identities to rural water supply schemes (Sujalam Bharat ID) and service area (Sujal Gaon ID), enabling source to tap mapping through structured hydraulic linkages which supports integrated planning, accountable O&M and data driven governance nationwide.

Key features and functional enablers of Sujalam Bharat DPI:

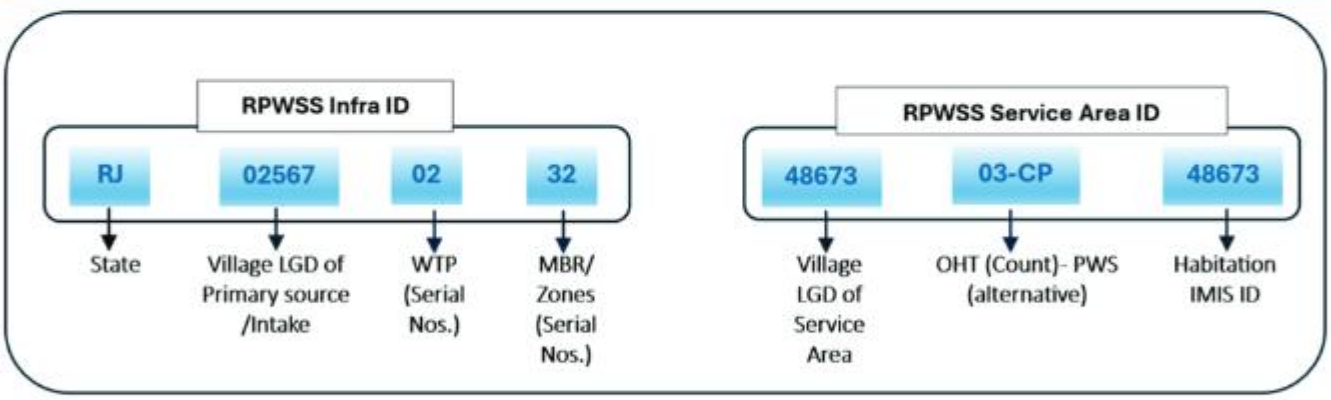
i. **Hydraulic Connectivity:** A structured parent node and child node model was introduced to represent linear hydraulic connectivity across assets, enabling complete source to tap mapping with service area linked representation of each RPWSS;

ii. **Integrated GIS and Geo-Referencing Layer:** A GIS validated interface with geo coordinate capture functionality was integrated through the Sujalam Bharat-PM GatiShakti Application to enable field-based geo referencing of assets with interoperability and spatial data integration with PM GatiShakti;

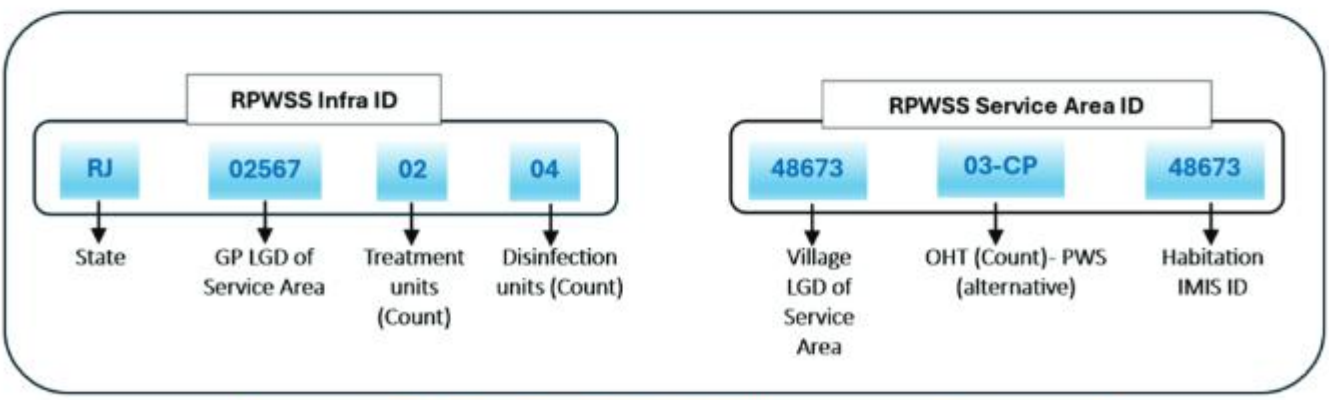
iii. **Role-Based Access Control (RBAC):** Clearly defined access and operational roles for State Water & Sanitation Missions (SWSM) and District Water & Sanitation Missions (DWSM) were incorporated to strengthen accountability asset data verification and workflow management;

iv. **Service Area - Asset Linkage:** Habitation level mapping was redesigned with transparent parent

Figure 15: System Generated Sujal Gaon IDs Syntax Logic



Sujal Gaon ID: with Out-Village and In-Village components



Sujal Gaon ID: with only In-Village components

child logic to enable precise tracking of water supply pathways from source to each habitation and institutional endpoint;

v. **Village Entry Point (VEP):** The digital mapping of Village entry point clearly demarcates the linkage between in-village water supply distribution network with its out-village components. It also clears the boundaries for infrastructure management between Macro and Micro Utility;

vi. **Enhanced System Visualisation:** The module enables generation of comprehensive digital architecture diagrams for each RPWSS allowing planners and administrators to visualize functional, hydraulic and spatial relationships of assets and service areas through a unified interface;

vii. **Modular and Scalable Design:** It follows a modular architecture enabling independent yet interconnected data layers for schemes, sources, assets and service areas. This modularity allows flexible data structuring and analysis at National, State, District and scheme levels and supports scalability for future expansion. The design enables schemes and asset registries to be visualised or shared in modular components based on administrative, planning or analytical requirements without altering the integrity of the core architecture;

viii. **Citizen Empowerment:** Enables citizens with real-time access to service level information on water supply, quality and functionality by integrating Sujalam Bharat DPI with GP dashboard and Sujalam Bharat-PM GatiShakti Application. It strengthens social accountability by enabling community monitoring, grievance redressal and ownership of rural water assets;

ix. **Empowerment of Non-Governmental Stakeholders:** Provides standardized, interoperable data access to private sector, civil society and community institutions, enabling innovation for local solutions and participatory governance. It creates an enabling ecosystem for startups, Community Based Organizations and VWSCs to contribute to planning, monitoring and operations of piped water supply systems;

x. **Strengthening Community Institutions:** Village Water & Sanitation Committees (VWSCs), SHGs and Panchayats are empowered with actionable data for decentralized planning, operation and maintenance of water supply systems. This fosters community led sustainable service delivery;

xi. **Enabling Market and Innovation Ecosystem:** Provides a trusted digital backbone for private players to develop technology solutions such as IoT-based monitoring, technical support for O&M, predictive analytics and asset management systems thereby catalyzing a water tech innovation and R&D ecosystem;

xii. **Facilitating Data Driven Financing and Partnerships:** Reliable and standardized data generated through DPI supports improved risk assessment, outcome based financing and public-private partnerships, enabling greater participation of financial institutions and development.

Roles and responsibilities of States/ UTs:

i. **Data Entry and Standardisation:** States/ UTs shall ensure that all data entered in the Sujalam Bharat (RPWSS) Module is accurate, complete and updated as per approved SOPs. This will help represent all layers in rural piped water schemes viz., Bulk Water, Out-Village and In-Village components connecting the infrastructure uniquely and distinctively with the service area IDs;

ii. **Mapping as per Ground Reality:** States/ UTs shall follow the Sujalam Bharat DPI framework to map water supply systems from source to habitation. Asset mapping should match the actual ground situation and work of the scheme to ensure correct linking of Scheme and Assets with Sujalam Bharat ID. Thereafter, the system will assign Sujal Gaon IDs to service area which are mapped to the respective scheme;

iii. **Managing Designated Implementation Agencies (DIA):** States/ UTs shall onboard and guide Designated Implementation Agencies (DIA) for asset GIS mapping on PM GatiShakti layer through Sujalam Bharat-PM GatiShakti Application. DIAs shall prepare and submit data (including KML files where required) while approval and final validation will be done by State/ District authorities;

iv. **Classification of Infrastructure:** States/ UTs shall classify infrastructure as Out-Village or In-Village based on its function and O&M responsibility. Assets serving multiple Gram Panchayats (such as treatment plants and bulk supply systems) should be marked as Out-Village while assets serving a single GP and managed locally may be marked as In-Village;

v. **Review and Validation of Data:** States/ UTs shall review and verify entries made at District and field

levels to ensure they correctly represent the scheme design, water flow and source to tap connectivity. This will help maintain overall data quality and consistency;

vi. **Correct Sequencing of Infrastructure:** States/ UTs shall ensure that all assets are mapped in the correct order as per actual scheme design and water flow. This will enable accurate representation of how the system functions;

vii. **Geotagging and Data Accuracy:** States/ UTs shall ensure that all assets are properly geotagged/ georeferenced through designated implementing agencies and linked with the correct Scheme and Asset IDs mapped to the Sujalam Bharat IDs. All submissions should be reviewed in a timely manner and corrected where required;

viii. **Following Geospatial Standards:** States/ UTs shall ensure that mapping follows required standards such as GPS accuracy, correct location within administrative boundaries, proper pipeline mapping and validation of spatial data. A maker-checker system should be followed to maintain data quality;

ix. **Capacity Building and Digital Awareness:** States/ UTs shall support training and digital capacity building of Gram Panchayats and VWSCs in coordination with NJJM. This will help improve their ability to use Sujalam Bharat DPI effectively;

x. **Use of DPI for Planning and Improvement:** States/ UTs shall use Sujalam Bharat DPI data to develop local solutions and improve planning, monitoring and service delivery of water supply systems;

xi. **Use of Sujal Gaon ID for User Database:** States/ UTs shall enable Gram Panchayats to use Sujal Gaon IDs and Sujal Gaon Utility App to create and maintain a database of households/ users within the service area. This will help in better planning, tracking of services and management of water supply systems in future.

A. Project Monitoring Module as a Use Case for Sujalam Bharat DPI

The Project Monitoring Module in the Sujalam Bharat-PM GatiShakti Application is an extended feature designed to track the progress of assets under Rural Piped Water Supply Schemes as mapped in the Sujalam Bharat DPI framework. The module builds on the existing system where geocoordinates of assets, including transmission and distribution pipelines are mapped against Sujalam Bharat IDs. The module shall enable regular updating of physical progress of assets under construction. This will support monitoring of ongoing schemes and asset/ infrastructure components completion status under JJM 2.0 and provide a clear view of implementation status at different levels.

Key Features:

i. **Single Digital Backbone:** All schemes and assets shall be linked to Sujalam Bharat IDs, ensuring continuity from source to distribution and enabling clear mapping of scheme wise asset data;

ii. **Lifecycle Tracking of Assets:** The module shall capture the status of assets viz. Not-Started, Ongoing and Completed during the CAPEX phase while in the OPEX phase this Application can used for tracking repairs and maintenance;

iii. **Physical Progress Monitoring:** States/ UTs shall update the physical progress of ongoing assets through the module enabling real time tracking of scheme implementation;

iv. **Geo-Referenced Asset Tracking:** The module shall build on existing geo tagged/ georeferenced data of assets and pipelines which is linked to the Sujalam Bharat IDs to provide accurate location-based monitoring of infrastructure;

v. **Transparency and Public Information:** Progress of schemes shall be reflected in Gram Panchayat dashboards and the Meri Panchayat Application enabling citizens to view status of works and improving transparency;

vi. **Audit and Record Management:** The system shall maintain a digital record of project updates and changes, supporting transparency, accountability and ease of review.

2.6.2. Interactive Dashboards and Decision Support Systems

Building on the Sujalam Bharat DPI as the core data system, interactive dashboards shall be operationalised at Gram Panchayat, District, State and National levels for real time monitoring and decision making in rural water supply systems. These dashboards will convert system data into simple, actionable information to support better planning, monitoring and service delivery.

The interactive dashboards shall be integrated with service provider applications and field level systems such as asset management tools, SCADA/ IoT systems,

water quality monitoring platforms and grievance redressal systems. This will ensure seamless data flow from field level operations to decision makers across all levels. They will function as the digital backbone for Command and Support Centres, enabling continuous monitoring and coordinated response. The dashboards will support a shift from reactive problem solving to proactive, predictive and data driven management, improving efficiency, reliability and accountability of water supply services.

The dashboards will enable:

i. **Reducing Water Loss:** Using data on water flow and consumption to identify leakages and reduce non-revenue water(NRW);

ii. **Preventive and Predictive Maintenance:** Monitoring asset performance and alerts to carry out timely repairs and prevent system failures;

iii. **Monitoring Service Delivery:** Tracking key parameters such as quantity, quality, continuity and pressure of water supply;

iv. **Performance Monitoring and Accountability:** Assessing performance of implementing agencies and service providers against defined standards;

v. **Water Quality Monitoring:** Tracking water quality data to ensure safe drinking water and timely corrective action;

vi. **Grievance Redressal:** Monitoring complaints and their timely resolution to improve service delivery and citizen satisfaction;

vii. **Better Planning and Resource Allocation:** Using data to identify gaps, prioritise investments and allocate resources efficiently;

viii. **Early Warning and Risk Management:** Identifying potential risks such as source issues or system stress in advance to enable timely action;

ix. **Support for Utility-Based Service Delivery:** Enabling tracking of service areas, user databases and consumption patterns to support improved service management and future readiness for utility-based systems;

x. **Convergence and Integrated Governance:** Supporting integration with other sectoral systems to enable coordinated planning and holistic water resource management.

Types of Dashboards

i. **NJJM Dashboard:** Provides a national level view of performance enabling monitoring, comparison of States/ UTs and informed policy decisions. It shall aggregate data from State dashboards to enable benchmarking, trend analysis and early identification of performance gaps;

ii. **SWSM Dashboard:** Supports State level monitoring, data validation, performance review and coordination with Districts and Gram Panchayats. It shall integrate with regional SCADA and other monitoring systems to capture key parameters of system functionality and performance of bulk and large-scale utilities;

iii. **DWSM Dashboard:** Enables District level monitoring of scheme functionality, grievance management and technical support. It should also provide a system for District Technical Units (DTUs) to record and track their responses to performance issues and technical support needs of service utilities, especially micro utilities;

iv. **Panchayat Dashboard:** Provides Gram Panchayats with simple, real time information on water supply status, quality and system performance, enabling local decision making and community monitoring.

The dashboard ecosystem shall enable seamless integration across Gram Panchayat, District, State and National levels, creating a unified flow of information and decision-making that strengthens coordination, improves responsiveness and supports effective governance of rural water supply systems. This shall institutionalize real-time governance and coordinated decision making mechanisms.

Roles and responsibilities of States/ UTs:

i. **Dashboard Based Monitoring and Governance:** States/ UTs shall establish dashboard-based monitoring mechanisms at State and District levels to track grievance pendency, service level compliance, repeated issues and resolution performance. These dashboards shall be used to identify persistent service failures and initiate timely corrective actions;

ii. **Issuance and Adherence to Standard Operating Procedures (SOPs):** States/ UTs shall issue Standard Operating Procedures defining grievance categories, routing mechanisms, escalation timelines and closure verification processes across Gram Panchayat, District Water and Sanitation Mission (DWSM), District Technical Units (DTUs) and State Water and Sanitation Mission (SWSM);

iii. **Promotion of Citizen Feedback Platforms:** States/ UTs shall promote the use of citizen feedback mechanisms such as Jal Seva Aankalan, Meri Panchayat Application and e-GramSwaraj, integrated with Sujal Gaon IDs. This shall enable systematic capture of community feedback related to drinking water services;

iv. **Traceability of Grievances through Sujal Gaon IDs:** States/ UTs shall ensure that all grievances related to rural water supply are mandatorily linked to Sujal Gaon IDs. This shall enable precise localisation of service issues, identification of affected habitations and targeted resolution of service breakdowns;

v. **Strengthening Institutional Coordination for Time Bound Resolution:** States/ UTs shall strengthen coordination among SWSM, DWSM, DTUs and Gram Panchayats to ensure timely resolution of community reported issues. DTUs shall provide necessary technical support to local institutions to address operational and system level challenges effectively;

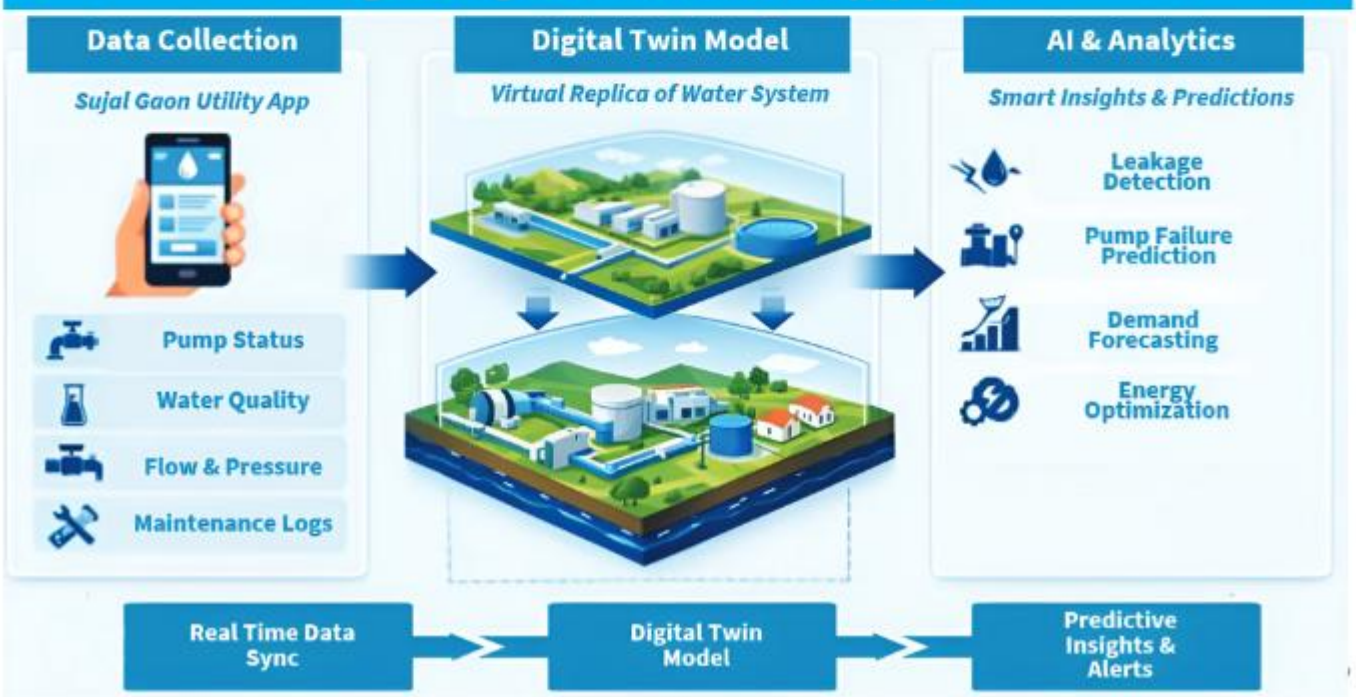
vi. **Establish District Technical Units (DTUs):** States/ UTs shall establish DTUs to monitor and manage data flow from field-level data entry through the Sujal Gaon Utility Application. The DTU shall act as the convergence point between macro and micro utilities. DTUs shall analyse dashboard data to identify system inefficiencies and risks, coordinate with utilities and Gram Panchayats ensure data validation, support operational decision making and troubleshooting and facilitate digital capacity building of field level stakeholders. DTUs shall also ensure that data insights are translated into timely and actionable decisions at the operational level.

vii. **Establish Command and Support Centres:** States/ UTs shall establish State-level and District level Command and Support Centres and organize regular guidance of DTUs for their effective operation, maintenance and management. Command and Support Centres shall support routine system monitoring and timely responses by DTUs and enable continuous monitoring of water supply system performance, data driven planning and prioritisation of interventions, coordination across departments and administrative levels and prompt response to system failures, emergencies and service disruptions.

2.6.3. Digital Transaction Data Capture by Service Utilities

To enable real time monitoring and effective governance, Sujalam Bharat DPI shall support structured capture of operational transactions by service utilities across all levels of rural water supply systems. These transactions include daily activities such as water

Figure 16: Digital Twin for Rural Water Supply Systems



supply operations, disinfection, distribution, water quality testing, maintenance and grievance handling. This system shall ensure that day to day functioning of water supply systems is digitally recorded, creating a reliable and continuous flow of data for monitoring, planning and decision-making.

A. Micro Utilities at Village Level-Sujal Gaon Utility Application

At the village level, micro-utilities and community-based service providers shall use the Sujal Gaon Utility Application to record daily operations and service delivery activities in a simple and structured manner.

The application shall capture:

i. **Water Supply Operations:** Daily valve and pump operations, water supply hours and distribution schedules, along with reporting of breakdowns and restoration timelines;

ii. **Maintenance Activities:** Routine and preventive maintenance activities to ensure system functionality and reduce service disruptions;

iii. **Water Quality Management:** Chlorination, disinfection and field level water quality testing, along with recording of contamination and corrective actions;

iv. **Revenue and Local Financial Records:** Collection of service delivery charge and basic financial records to support transparency and sustainability of services;

v. **Community Monitoring and Issue Reporting:** Reporting of leakages, infrastructure issues and service gaps by local stakeholders for timely action;

vi. **Grievance Redressal and Citizen Engagement:** Registration and tracking of complaints, along with actions taken by VWSCs to resolve them;

vii. **Scheme Monitoring and Local Planning:** Periodic assessment of scheme functionality, public sharing of water supply information and village-level water budgeting and planning.

This will ensure that last mile service delivery is digitally captured, regularly monitored and made transparent and accountable.

B. Macro Utilities and Bulk Water Supply Transfer – Integration of data from Monitoring Mechanism Systems (SCADA/ IoT)

At the bulk water transfers level, States/ UTs shall integrate macro utilities with digital interactive dashboards to capture operational data from large scale water supply systems.

Data shall be captured through SCADA, IoT and telemetry-based systems for the following aspects:

i. **Water Production, Treatment and Storage:** Real time monitoring of bulk water systems and treatment processes;

ii. **Transmission and Bulk Distribution:** Tracking water movement across major pipelines and storage systems;

iii. **Energy and Operational Efficiency:** Monitoring of energy use, pump performance and system efficiency;

iv. **System Performance and Supply Reliability:** Ensuring consistent bulk supply to downstream distribution systems;

v. **Water Quality Monitoring:** Water quality data status from laboratories, monitoring of emerging contaminants and geogenic contaminants enabling water quality data surveillance as per standards.

This will enable continuous monitoring of bulk water systems and improve coordination between bulk supply and local distribution.

C. Integrated Data Flow and Decision Support for Service Utilities:

Data captured from village level applications (micro utilities) and bulk systems (macro utilities) through integration of data from automated systems like SCADA/ IoT, shall be integrated within the Sujalam Bharat DPI to provide a complete, real-time view of system performance across the entire source to tap chain. This data shall flow into interactive dashboards, where it will be analysed and used for decision-making. Command and Support Centres shall use these insights to monitor performance, identify issues and take timely action.

The continuous capture of transaction data shall create a strong data foundation for advanced analytics. Over time, this will enable development of machine learning and AI-based systems for generating alerts, predicting failures, identifying risks and supporting better decision-making.

i. **Roles and Responsibilities of States/ UTs:**

a. Adoption of Standard Digital Platform: States/ UTs shall adopt and notify the Sujal Gaon Utility Application as the standard platform for capturing village level O&M and service delivery transactions;

b. Standardisation of Data and SOPs: States/ UTs shall issue SOPs and data standards for daily and periodic entries, including reporting frequency, alert thresholds and escalation protocols;

c. User Onboarding and Role-Based Access: States/ UTs shall ensure onboarding and authorization of all users with role-based access, OTP authentication and audit trails;

d. Integration of Macro Utility Systems: States/ UTs shall integrate bulk water systems with SCADA/ IoT platforms to capture real time data on production, treatment, storage and transmission;

e. Data Quality and System Integration: States/ UTs shall ensure validation and integration of all data with Sujalam Bharat DPI and dashboards for reliable monitoring;

f. Dashboard Governance and Accountability: States/ UTs shall define KPIs and assign responsibility for monitoring alerts, service delivery, water quality, grievances and financial performance;

g. Closed-Loop Monitoring and Response: States/ UTs shall ensure time bound response to alerts with defined escalation mechanisms and tracking of outcomes;

h. Technical Support through DTUs: States/ UTs shall direct DTUs to provide technical support based on system alerts and performance trends;

i. Financial Monitoring and Sustainability: States/ UTs shall ensure regular monitoring of demand, collection and expenditure data and take corrective actions where needed;

j. Capacity Building and Digital Enablement: States/ UTs shall train and support all stakeholders for effective use of digital systems;

k. Use of Data for Planning and Improvement: States/ UTs shall use transaction data and analytics for planning, performance improvement and resource allocation.

ii. **Roles and Responsibilities of Districts (DWSM / DTUs):**

a. Supervision of Data Capture: Districts shall ensure that all service providers regularly record transactions through the Sujal Gaon Utility Application and related systems;

b. Data Review and Validation: Districts shall review data entries for accuracy and completeness and coordinate with field functionaries for corrections;

c. Technical Support and Response (DTUs): DTUs shall analyse system alerts and trends, provide technical guidance, support fault diagnosis and assist in preventive maintenance and major repairs, especially for micro utilities;

d. Monitoring of Scheme Functionality: Districts shall use dashboard data to monitor performance of schemes, identify issues and take timely action;

e. Grievance Monitoring and Resolution: Districts shall track complaints and ensure timely resolution through coordination with GPs and service providers;

f. Coordination and Escalation: Districts shall coordinate between State and GP levels and escalate unresolved issues as per defined protocols;

g. Support for Capacity Building: Districts shall support training and handholding of field functionaries and local institutions.

iii. **Roles and Responsibilities of VWSCs and Service Providers (Micro Utilities):**

a. Daily Transaction Data Entry: VWSCs, Water Supply Operators, SHGs and other service providers shall record daily activities such as water supply operations, maintenance, disinfection and testing through the Sujal Gaon Utility Application;

b. Water Quality and Maintenance Reporting: They shall regularly record water quality testing, chlorination, breakdowns and corrective actions;

c. Grievance Registration and Resolution: VWSCs shall register complaints, track their status and ensure timely resolution at the local level;

d. Revenue and Financial Records: service delivery charge collection and basic financial records shall be maintained digitally;

e. Community Monitoring and Reporting: Local stakeholders shall report leakages, infrastructure issues and service gaps for timely action;

f. Use of Data for Local Planning: VWSCs shall use available data for village level planning, water budgeting and improving service delivery.

2.6.4. Citizen Participation and Feedback Systems

Citizens shall be empowered as active participants in planning, monitoring and decision making. The Meri Panchayat Application shall serve as the primary

citizen interface providing a single digital window for accessing information on water supply services and for submitting feedback and grievances.

Sujalam Bharat DPI shall enable transparent access to service delivery information and establish structured mechanisms for citizen feedback and grievance redressal. All citizen inputs captured through digital platforms shall be linked to defined service areas through Sujal Gaon IDs, enabling precise identification of issues and targeted response. Multi-level dashboards shall provide visibility of feedback, grievance status and service outcomes, supporting data driven governance.

Key Features:

i. **Citizen Interface through Meri Panchayat Application:** It shall provide citizens with access to village level information on water supply schemes, water quality status, service delivery details and information on VWSCs/ Pani Samitis. It shall also enable citizens to submit feedback, report issues and track responses making it the primary platform for citizen engagement;

ii. **Service Area-Based Feedback and Grievance Linkage:** Sujal Gaon IDs serve as the common reference for linking all citizen feedback and grievances to specific habitations and schemes. This will enable precise issue identification and targeted action by Gram Panchayat, District and State authorities;

iii. **Jal Seva Aankalan as Local Feedback Tool:** Jal Seva Aankalan functions as a digital tool for Gram Panchayats and VWSCs to record functionality status and service issues. It shall also enable notification to District authorities and DTUs for timely technical support and corrective action;

iv. **Integration with Project Monitoring Module:** The Meri Panchayat Application shall be integrated with the Sujalam Bharat Project Monitoring Module to enable citizens to track physical progress of ongoing schemes. Linkage with Sujal Gaon IDs shall provide visibility of source to tap connectivity, while integration with the Sujal Gaon Utility Application shall provide information on day-to-day service delivery;

v. **Closed Feedback and Accountability Loop:** Citizen feedback and grievances shall be captured, tracked and linked to action at appropriate levels. This will ensure a closed loop system where citizen inputs lead to measurable actions and improved service delivery;

vi. **Real-Time Monitoring through Dashboards:** Multi level dashboards (NJJM, SWSM, DWSM and GP) shall display real time status of grievances, pendency, escalations and resolution timelines, strengthening transparency, accountability and evidence-based governance;

vii. **Citizen Feedback for AI-Based System Learning:** Citizen feedback and grievance data shall serve as an important input for advanced analytics and AI systems. Over time, this data shall help in identifying recurring issues, improving service delivery patterns, generating predictive alerts and strengthening decision support systems, enabling continuous improvement of water supply services.

Roles and Responsibilities of States/ UTs:

i. **Enable Citizen Interface:** States/ UTs shall ensure adoption and effective use of the Meri Panchayat Application as the primary citizen interface for accessing information, submitting feedback and tracking grievances;

ii. **Ensure Data Integration and Availability:** States/ UTs shall ensure integration of Sujalam Bharat DPI, Project Monitoring Module and Sujal Gaon Utility Application with citizen platforms so that accurate and up-to-date information on schemes, service delivery and progress is available;

iii. **Link Feedback with Service Areas:** States/ UTs shall ensure that all citizen feedback and grievances are linked to Sujal Gaon IDs to enable precise identification of issues and targeted resolution;

iv. **Establish Grievance Redressal Mechanism:** States/ UTs shall define clear workflows, timelines and escalation mechanisms for addressing citizen feedback across Gram Panchayat, District and State levels;

v. **Monitor Feedback and Response:** States/ UTs shall use dashboards to monitor grievance status, resolution timelines and service delivery outcomes and ensure timely action by responsible authorities;

vi. **Strengthen Accountability:** States/ UTs shall assign clear responsibilities to Districts, VWSCs and service providers for responding to citizen feedback and regularly review performance;

vii. **Capacity Building and Awareness:** States/ UTs shall undertake awareness campaigns and training to encourage citizens and local institutions to use digital platforms for feedback and participation;

viii. **Use Feedback for System Improvement:** States/ UTs shall analyse citizen feedback and grievance data to identify recurring issues and improve service delivery, planning and system performance;

ix. **Development of Integrated Citizen Service Interfaces:** States/ UTs may progressively develop and promote integrated digital interfaces for citizens to access a full range of water supply services, including applications for new connections, connection restoration, service requests, grievance registration and tracking of resolution. Leveraging interoperability of Sujalam Bharat DPI and Sujal Gaon IDs, States may build context specific applications and use cases that enable end-to-end service delivery at the household level.

a. These interfaces aim to provide a single, user-friendly platform for all citizen interactions with water supply systems, enabling transparency, ease of access and timely service delivery. Over time, such systems may support features such as service status tracking, user account management, payment and billing integration and personalised service information. States/ UTs shall also ensure inclusion through local language support, low-bandwidth compatibility and assisted access through Gram Panchayats or community institutions;

b. This will support the gradual transition towards citizen-centric rural water utilities where service delivery is structured, accountable and accessible through integrated digital platforms.

x. **Citizen Data Ownership and Consent Based Data Sharing:** States/ UTs may progressively adopt frameworks that recognise citizens as owners of their service-related data and enable secure, consent based data sharing across platforms. Systems may be designed to allow citizens to view, control and selectively share their water service data through trusted digital interfaces. This approach will promote transparency, build trust and enable responsible use of data for improving service delivery. Over time, it may support personalised services, targeted interventions and improved planning, while ensuring privacy and data protection.

2.6.5. Evolution Towards AI-Enabled Water Management Systems

The Sujalam Bharat DPI, along with transaction-based data capture and interactive dashboards, establishes a strong digital foundation for advanced analytics and intelligent water management. As data from village level operations and bulk supply systems grows in volume and quality, it shall enable the development of machine learning and artificial intelligence (AI) based systems.

These systems will utilise historical and real-time data to identify patterns, generate insights and support predictive and data driven decision making across rural water supply systems.

i. **From Data Capture to Intelligent Systems:** Continuous capture of transaction data on water supply operations, water quality, maintenance and system performance shall create a structured and reliable dataset. This shall enable identification of recurring issues, understanding of demand patterns and detection of anomalies, forming the basis for machine learning models;

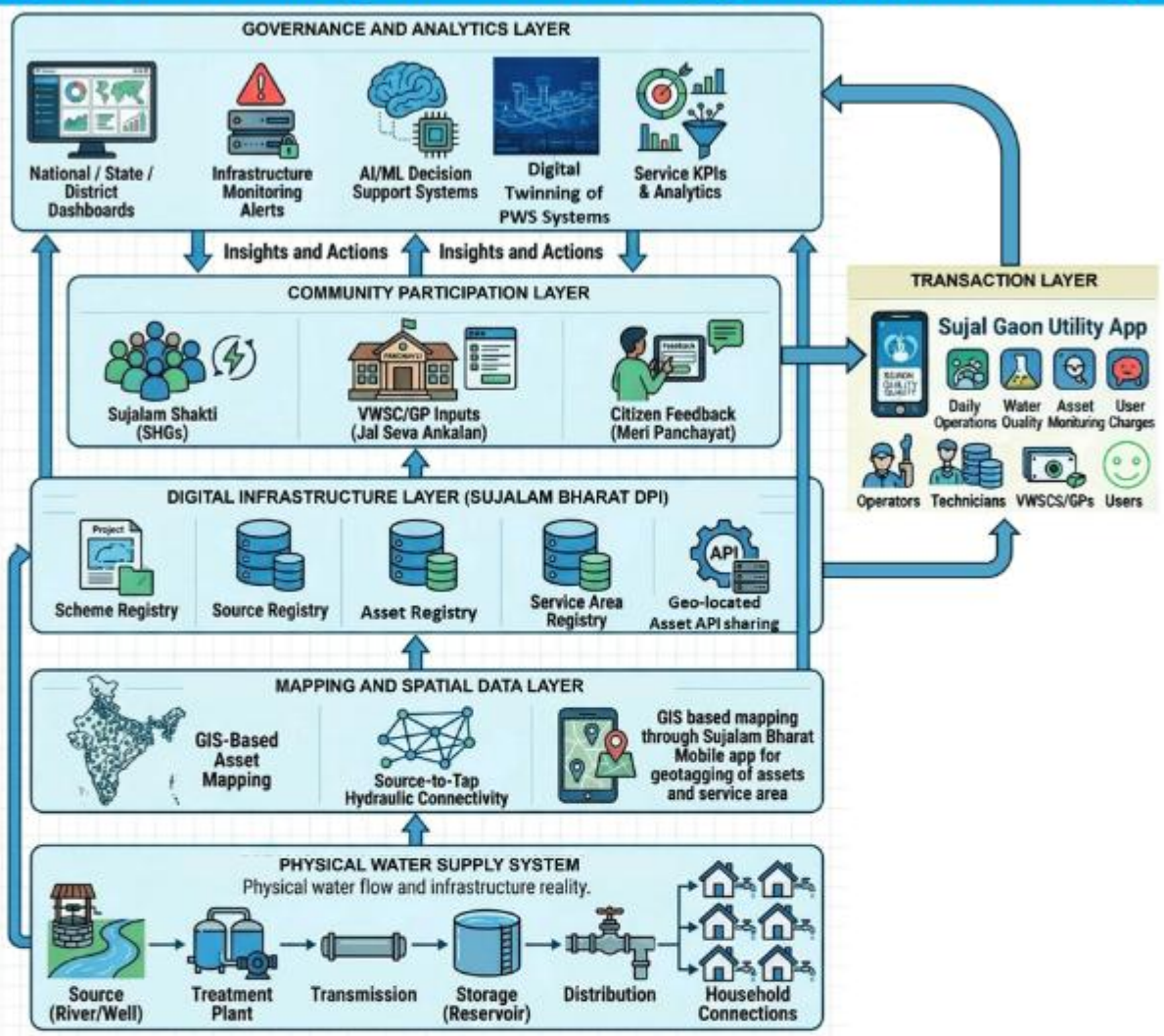
ii. **Predictive and Preventive System Management:** AI based systems shall support predictive maintenance of assets, early detection of leakages, identification of water quality risks and forecasting of demand. This shall enable a shift from reactive response to proactive and preventive system management;

iii. **Intelligent Alerts and Decision Support:** Advanced analytics shall enable automated alerts and actionable insights for decision makers. Dashboards and Command and Support Centres (CSCs) shall evolve into decision support systems to prioritise issues, recommend actions and support timely and informed decision-making;

iv. **Digital Twin of Water Supply Systems:** States/ UTs may progressively develop digital twins of water supply networks using integrated data from DPI, service provider applications and SCADA/ IoT systems. These digital twins will support simulation of system behaviour, scenario analysis and optimisation of operations and infrastructure planning;

v. **Integrated and Advanced Water Management:** These systems shall enable end to end visibility from source to tap, integration of bulk and distribution systems, data driven planning and improved operational efficiency. They shall also support convergence with related sectors such as groundwater, climate resilience and energy management;

Figure 17: Overview of the Sujalam Bharat DPI Illustrating System Layers & Functional Linkages



vi. **Phased and Scalable Approach:** States/ UTs shall adopt a phased approach for implementation, starting with strengthening of data capture and integration followed by use of dashboards for monitoring, deployment of AI/ ML models for predictive insights and gradual development of advanced systems such as digital twins;

vii. **Data Governance and Quality Assurance:** States/ UTs shall establish strong data governance frameworks to ensure accuracy, completeness, consistency and standardisation of data. Regular validation, audit trails and feedback mechanisms shall be implemented to maintain data quality;

viii. **Interoperability and Open Standards:** All digital systems shall follow interoperable frameworks and open standards to enable seamless data exchange and integration across platforms. This shall support scalability, flexibility and future integration of emerging technologies;

ix. **Human-in-the-Loop Decision Systems:** While AI systems shall generate alerts and recommendations, decision-making shall remain with designated officials and service providers. Systems shall support informed decision making by combining human judgment with machine generated insights;

x. **Cybersecurity and Data Privacy:** States/ UTs shall ensure appropriate safeguards for data security, privacy and system resilience including secure access controls, data protection measures and risk mitigation mechanisms;

xi. **Continuous Learning and System Improvement:** AI and analytics systems shall be designed to continuously learn from new data and feedback. System performance, field inputs and outcomes shall be used to refine models and improve accuracy over time.

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CHAPTER 2.7

Establishing Participatory Governance through Jan Bhagidari

Reform aims at:

- Strengthening participatory governance through Jan Bhagidari by making communities active partners in managing and sustaining water supply systems;
- Positioning IEC as a key enabler for making informed citizens leading to community ownership;
- Establishing a structured communication framework for transparency, accountability and continuous engagement;
- Adopting Jal Utsav as a national framework for sustained community participation, cultural connect and periodic system review;
- Leveraging Jal Arpan and Jal Seva Aankalan to reinforce ownership, accountability and performance monitoring;
- Institutionalising Samvad platforms for dialogue, peer learning and continuous programme feedback;
- Promoting digital platforms for wider outreach, localisation and informed decision making.

2.7.1. Participatory Governance in context of JJM

Participatory governance refers to a system where government, citizens and community institutions, particularly Gram Panchayats and VWSCs, actively participate in planning, decision making, implementation and oversight of public services such as water, promoting transparency, accountability and sustained community ownership. Participatory governance lies at the core of JJM 2.0, marking a paradigm shift from the government working for the people to working with the people, where communities are active partners in decision making, implementation and long-term management of water services. The Mission goes beyond infrastructure creation to empower communities to take ownership of their water supply systems and become responsible custodians for sustained drinking water service delivery.

Rural piped water supply systems involve multiple technical and operational components and their sustainability depends on how well communities understand their schemes, engage with them and take responsibility for their operation and maintenance. Accordingly, under JJM 2.0, participatory governance through Jan Bhagidari is adopted as a core approach to enable meaningful community involvement in managing and sustaining rural water supply systems. By ensuring access to relevant and reliable information, simplifying technical processes and enabling continuous interaction through structured platforms, the Mission aims to build informed communities who are equipped to participate effectively in the management and sustainability of their water supply systems. This is supported through a structured framework of Information, Education and Communication (IEC).

2.7.2. Information, Education and Communication (IEC) Framework for Participatory Governance

Information, Education and Communication (IEC) under JJM serves as a key enabler of participatory governance by connecting communities with their water supply systems and supporting their informed participation. It is not a one-time awareness activity but a continuous and structured process that enables communities to understand their water supply systems, build capacity to manage them and actively engage in their monitoring and improvement. Through a combination of information dissemination, experiential learning and two-way communication platforms, IEC facilitates sustained community involvement thus promoting Jan Bhagidari in managing and sustaining rural water supply systems.

While operationalising this, States/ UTs shall ensure that all materials and communication efforts are delivered in local languages and contextualised to local cultural settings so that messages remain accessible, relatable and rooted in community practices.

Details of each of these three components are outlined below:

A. Information: Enabling Awareness of Water Supply Systems and Processes

Information under JJM focuses on making communities aware of their own water supply systems, what infrastructure exists, how it functions and where water comes from. When people become aware of the source of their drinking water, the infrastructure created and the processes involved in supplying water to their households they are better able to connect with the system and recognise its value.

To support this, JJM has enabled access to information through platforms such as Sujalam Bharat database, Sujal Gaon Utility App, Dashboards at each governance level, Meri Panchayat App and other digital platforms along with public display of key scheme details at the Gram Panchayat level. These initiatives allow communities to access and understand essential information including the source of water, system components and service delivery arrangements. The objective is to ensure that every citizen is aware of their water system as this awareness forms the foundation for developing a sense of responsibility and ownership towards its operation and sustainability.

As part of strengthening informed participation, a Quarterly Public Disclosure Mechanism shall be instituted. Every Gram Panchayat and District shall ensure disclosure of key service delivery parameters once every quarter to promote transparency and accountability. The parameters shall include water quality testing status, functionality status (through Jal Seva Aankalan or as specified from time to time), grievance redressal data, collection and utilisation of service delivery charges and other O&M funds and actions taken for source sustainability. States/ UTs shall ensure that information related to village water supply systems is made easily accessible and understandable to communities through digital platforms, public display at the Gram Panchayat level, Gram Sabha or Jal Chaupal meetings and other local dissemination mechanisms. Efforts shall also be made to ensure that every household is aware of the source, system components and service delivery arrangements of their water supply scheme.

In addition, Community radio has been initiated by NJJM as a local and accessible medium for dissemination of information and awareness on rural water supply systems. States/ UTs may leverage community radio to strengthen last mile outreach, promote localised messaging and enhance community engagement, particularly in remote areas.

B. Education: Building Understanding and capacity for Informed Participation

Education under JJM 2.0 focuses on building a clear understanding of how water supply systems function and what is required for their effective operation and maintenance. It aims to simplify technical aspects and enable communities to learn through active participation.

This is facilitated through structured platforms such as Jal Utsav, comprising of Jal Mahotsav (National level), Rajya Jal/ Nadi Utsav (State level) and Lok Jal Utsav (Gram Panchayat level), where learning takes place through community engagement, interaction and activities that support long-term sustainability of water supply systems. These platforms also help strengthen the community's connection with water as a shared resource.

In addition, under JJM 2.0, processes such as Jal Arpan which mark the formal handover of schemes to the community are designed to integrate technical steps like trial runs, system checks and document verification with participatory activities such as transect walks and Gram Sabha/ Jal Chaupal discussions. These approaches aim to enable communities to observe, understand and engage with system functioning in a simple and practical manner.

Education under JJM also extends beyond community engagement to building long term awareness and responsibility across generations. The Mission promotes integration of water related learning into school education through SCERT or State School Education Board, so that both present and future generations develop awareness and understanding of water systems and their role in sustaining them. States/ UTs shall encourage schools to engage students through practical activities such as water audits, rainwater harvesting demonstrations, water quality testing using FTKs and participation in local initiatives like Jal Utsav and source protection drives. Such experiential learning shall help build early ownership and nurtures responsible water usages among young citizens. Such

continuous learning will enable communities to understand their roles in operation and maintenance, thereby strengthening their ability to participate effectively.

States/ UTs shall facilitate this continuous community learning by operationalising platforms such as Jal Utsav, Jal Arpan ensuring that participatory processes like trial runs, transect walks and Gram Sabha/ Jal Chaupal discussions are conducted effectively. They shall promote simplified and locally contextualised learning materials to help communities understand system functioning and their roles in operation and maintenance. States/ UTs shall also coordinate with school education departments to integrate water related learning into school curricula through SCERT frameworks and encourage engagement of youth and schools in water awareness activities.

C. Communication: Enabling Participation, Feedback and Collective Responsibility

Communication under JJM 2.0 is not limited to dissemination of information; it is a two-way process that enables interaction between the government and the community which means information is shared with communities and feedback is received from them.

Under JJM 2.0, structured platforms have been established to ensure that communities can express their experiences, raise concerns and contribute to improving service delivery. These include:

i. Jal Seva Aankalan (JSA), which allows communities to assess their own systems periodically and give feedback to District;

ii. Dialogue platforms such as DC/ DM Peyjal Samvad and Sujal Gram Samvad, which enable sharing of experiences and peer learning;

iii. Sarpanch Samvad App facilitated by QCI, may be leveraged to enable continuous peer learning and sharing of good practices among Panchayats.

Besides, Gram Sabha/ Jal Chaupal should be institutionalised as key platforms for community dialogue, decision making and validation enabling collective discussion on water supply systems and strengthening transparency and accountability. Through these mechanisms, communities not only be informed but also heard actively. Their feedback will contribute to planning, improves responsiveness and strengthening accountability. This continuous interaction builds trust and ensures that communities remain actively engaged in the governance of their water systems.

States/ UTs shall play a proactive role in strengthening communication systems by institutionalising regular platforms for dialogue and feedback at State, District and Gram Panchayat levels. They shall ensure that feedback from communities and stakeholders is systematically captured, analysed and used for timely decision making and course correction.

Further, States/ UTs shall document and disseminate good practices, innovations and field learnings to promote cross learning and continuous improvement across regions.

2.7.3. Integrating IEC Framework with Key Governance Tools

Under JJM 2.0, Information, Education and Communication (IEC) shall be operationalised through key governance tools such as Jal Arpan, Jal Utsav, Pey Jal Samvad, Sujal Gram Samvad and Jal Seva Aankalan. This integration will ensure that community engagement is embedded across the entire service delivery lifecycle from planning and implementation to operation, handing over schemes to monitoring and ensuring sustainability rather than being treated as a standalone activity. IEC shall also be aligned with structured processes so that community participation becomes an integral part of system functioning and continuous improvement.

The following governance tools operationalise this approach by integrating IEC with specific stages of the service delivery of piped water supply system:

A. Jal Arpan: Linking Commissioning and Handing Over Protocols with Community Participation

Commissioning and handing over of rural water supply systems under JJM 2.0 shall not be treated as purely technical milestones but as community centric processes that establish ownership, accountability and readiness for sustained service delivery. Jal Arpan marks the formal handing over of completed schemes to the Gram Panchayat/ community. Thus, Jal Arpan framework shall serve as the institutional mechanism to integrate community participation into commissioning and handing over protocols, ensuring that Gram Panchayats and communities are actively involved in the transition from construction to operation and maintenance.

Accordingly, States/ UTs shall ensure that community participation is embedded in commissioning and handing over processes through the following:

i. **Community Involvement during Trial Run:** Gram Panchayat/ VWSC shall participate in trial run (minimum of 15 days) to verify service levels (supply hours, pressure, quality) and identify gaps;

ii. **Validation through Gram Sabha/ Jal Chaupal:** Commissioning status, service parameters and readiness for O&M shall be presented and discussed in Gram Sabha or Jal Chaupal for community validation;

iii. **Transparency of System Details:** Key documents such as As-built drawings, asset registers, water quality reports and O&M plans shall be shared with the community and made accessible at the Gram Panchayat level;

iv. **Orientation of Local Stakeholders:** VWSC members, operators (Nal Jal Mitra) and community representatives shall be oriented on their RPWSS, operation, maintenance protocols, water quality monitoring and grievance redressal mechanisms etc. associated with their piped water scheme;

v. **Formal Handover through Jal Arpan:** Completed schemes shall be formally handed over to the Gram Panchayat/ community through Jal Arpan, reinforcing collective ownership and responsibility;

vi. O&M fund arrangements and roles and responsibilities shall be clearly established prior to handover;

vii. **Public Disclosure and Community Awareness:** Service levels, water quality status and grievance mechanisms shall be publicly disclosed as part of Jal Arpan activities.

Water quality compliance shall also be ensured during Jal Arpan, as detailed under Chapter 2.4. While Jal Arpan is institutionalised as a one-time event for formal handover, States/ UTs shall integrate it with Jal Utsav so that it can also serve as a platform for wider community engagement and periodic review of piped water supply scheme performance.

B. Jal Utsav: Framework for continuous Community engagement and Cultural Connect with Water

Jal Utsav provides a structured framework for continuous community engagement, linking culture, celebration and governance around drinking water. Recognising that community ownership is central to long term sustainability, the focus extends beyond infrastructure creation to effective operation and maintenance and protection of water sources with celebrations promoting a sense of pride and emotional connection that supports sustained community led management of Rural Piped Water Supply Schemes.

Jal Utsav is implemented across three levels, National, State and Panchayat creating a continuous cycle of engagement. At the national level, it builds momentum and convergence; at the State level, it connects water with regional culture and traditions; and at the Panchayat level, it becomes a people's celebration rooted in local practices, folklores and collective action.

Since, these are recurring events, this framework provides an opportunity for communities to review their drinking water systems, assess performance and take corrective actions while reinforcing their cultural connection with water. By blending festive traditions with governance processes, Jal Utsav promotes awareness, pride and shared responsibility for sustaining rural water supply systems.

i. **Jal Mahotsav (National level):** National focus and thematic direction; Jal Mahotsav shall be positioned not merely as an annual event but as a national platform to guide States, Districts and Gram Panchayats in institutionalising structured community engagement for effective operation and maintenance of rural water supply systems.

   National level dialogues shall be organised to set the overall vision and facilitate convergence, complemented by activities at the State and Panchayat levels. The Mahotsav shall be observed annually from 8th March to 22nd March with a specific theme identified each year to guide focused actions.

   The emphasis shall be on strengthening governance processes, converging with other aligned Missions and programmes, promoting regular system review and enabling community led monitoring and accountability while also promoting a sense of pride and connection with water through locally relevant cultural associations and reinforcing source sustainability for long term water security;

ii. **Rajya Jal Utsav/ Nadi Utsav (State/ UT level):** Regional context and traditions; At the State/ UT level, Rajya Jal Utsav/ Nadi Utsav shall be organised to reflect regional water traditions, local river systems and culturally rooted conservation practices including alignment with traditional festivals associated with water. State specific calendars shall be developed to ensure convergence across departments and active participation of districts and local institutions.

Figure 18: Jal Utsav- Framework for Continuous Community Engagement and Cultural Connect with Water

States/ UTs may integrate regionally significant events such as Kumbh, Pushkarams, Chhath Puja, Boita Bandana, Narmada Jayanti and other local water centric festivals to strengthen community connect. Such platforms shall also be used to promote river and source strengthening including water body rejuvenation, cleanliness drives and protection of local water sources. The emphasis shall be on linking cultural relevance with governance processes to improve operation and maintenance ensure source sustainability and strengthen community led management of rural water supply systems;

iii. **Lok Jal Utsav (GP/ Village level):** Community led engagement rooted in local culture; At the Gram Panchayat/ Village level, Lok Jal Utsav shall be organised as community led water festivals rooted in local traditions, folklore and customary practices associated with water, reflecting the community's cultural and historical relationship with its water sources. These platforms shall combine awareness, learning, celebration and review of piped water supply systems. Gram Panchayats shall prepare annual Lok Jal Utsav calendars to ensure regular engagement through activities such as Gram Sabha/ Jal Chaupal, water quality demonstrations, asset awareness and collective actions for water conservation and source protection. Districts and States will ensure this is included in the annual action plan and get implemented. Such engagements shall strengthen community ownership, promote effective operation and maintenance and reinforce the role of local traditions and folklore in sustaining water systems.

C. Jal Seva Aankalan (JSA): Community led Assessment and Feedback

Jal Seva Aankalan shall be institutionalised as a community led governance tool under JJM 2.0 to enable Gram Panchayats and VWSCs to undertake periodic self-assessment of rural drinking water service delivery. The process shall promote community participation, strengthen local accountability and support informed decision making for sustained operation and maintenance of piped water supply systems.

Jal Seva Aankalan shall cover key service delivery parameters including regularity, adequacy and quality of water supply, operation and maintenance practices, financial management, service delivery charge collection, grievance redressal and source sustainability measures.

The assessment shall be carried out in all Har Ghar Jal panchayats by the Gram Panchayat/ VWSC with participation of community members and the findings shall be presented in Gram Sabha/ Jal Chaupal and publicly disclosed at the Gram Panchayat level to ensure transparency, community validation and collective ownership.

The outcomes of Jal Seva Aankalan shall be used to identify gaps in service delivery and to prepare Service

Improvement Plans (SIPs) or other corrective action plans at the Gram Panchayat, District and State levels, as applicable.

Aggregated findings of Jal Seva Aankalan shall support monitoring, review and planning at higher administrative levels and shall contribute to strengthening the overall accountability framework under JJM 2.0.

Structured dialogue platforms such as Pey Jal Samvad and Sujal Gram Samvad shall utilise the findings of Jal Seva Aankalan to facilitate experience sharing, peer learning and continuous improvement across States, Districts and Gram Panchayats.

Detailed description on Jal Utsav, Jal Arpan, Jal Seva Aankalan and related activities are provided in the Handbook titled "Community Managed Piped Water Supply Systems in Rural India"[4] released by NJJM in November 2025 which may be referred to, along with other advisories issued by the NJJM.

D. Samvad Platforms: Learning, Sharing, Recognition and Scaling Good Practices

Samvad platforms have been initiated under JJM as structured forums for learning, experience sharing, recognition and scaling of good practices across States, Districts and Gram Panchayats and shall be further strengthened and institutionalised. These platforms function as two-way communication channels, enabling communities, elected representatives and field functionaries to directly interact with NJJM and share their field experiences, challenges and innovations. At the same time, they provide a mechanism for dissemination of guidance, learnings and good practices from the national level to the field. This two way exchange supports cross-learning, strengthens feedback loops and enables continuous improvement in operation and maintenance, service delivery and source sustainability.

Two key platforms shall be operationalised:

i. **DC/ DM Peyjal Samvad:** This platform facilitates sharing of District level experiences at the national level, highlighting administrative innovations, governance practices, O&M models and convergence approaches adopted by Districts. It shall enable DCs/ DMs, DWSMs and field teams to exchange learnings and replicate successful interventions across regions;

ii. **Sujal Gram Samvad:** This platform showcases community led initiatives and best practices from Gram Panchayats/ VWSCs at the National level, including examples of effective O&M, service delivery charge collection, water quality management, source sustainability and local innovations driven by community participation.

States/ UTs shall play a key role in identifying and curating good practices from districts and Gram Panchayats, facilitating participation in Samvad platforms and ensuring dissemination and replication of successful models. States shall also strengthen feedback loops so that learnings from these platforms inform policy, planning and implementation.

Similarly, the Meri Panchayat App, an integrated digital governance platform developed by Ministry of Panchayati Raj is being leveraged as a citizen interface and through initiative such as Jal Seva Aankalan, enables community led assessment of drinking water services with a built in citizen feedback mechanism. The platform is envisaged to further evolve for grievance redressal.

Apart from this, Sarpanch Samvad App which is a Digital platform facilitated by the Quality Council of India (QCI) shall be leveraged to enable continuous peer learning, documentation of best practices and wider dissemination of knowledge among Panchayats and stakeholders.

These platforms shall promote peer learning, recognise exemplary efforts and support scaling of proven approaches thereby strengthening participatory governance and improving service delivery outcomes.

2.7.4. Leveraging Digital Tools for Participatory Governance

States/ UTs shall promote the creation and dissemination of short videos, including reels, testimonials and case-based documentation to document and showcase community led initiatives, success stories and good practices under JJM. Such content shall be used to enhance outreach, encourage peer learning and motivate wider community participation.

2.7.5. Convergence for Strengthening Community led IEC to promote Participatory Governance

i. JJM 2.0 shall adopt a whole of government approach to communication by strengthening coordination across Ministries and Departments. A

[4] *Scan the QR code to access Jan Bhagidari Handbook (Vol-1)*

unified WASH communication framework shall be followed to ensure consistent messaging, coordinated outreach and effective dissemination;

ii. Sujalam Shakti refers to a women led group, constituted from existing community-based institutions such as Self Help Groups (SHGs) formed under State Rural Livelihood Mission (SRLM), Primary Agricultural Credit Societies (PACs) or other similar Community Based Organizations (CBOs) formed under different Missions/ Programme. These groups shall support the Village Water and Sanitation Committee and the Gram Panchayat in the operation and management of rural piped water supply schemes, including activities related to water service delivery, basic maintenance, community engagement and water quality monitoring;

iii. States/ UTs may develop appropriate incentive mechanisms, in convergence with SRLM or the case may be, to support engagement of Sujalam Shakti Groups in water quality monitoring, IEC activities and operation and maintenance of rural piped water supply systems. Such incentives may be supported through Support and Water Quality Monitoring and Surveillance (WQMS) funds, as per guidelines and this must be part of Annual Action Plan/ CIRP;

iv. Field level IEC under JJM 2.0 shall be rooted in community ownership and driven through convergence with trusted local institutions and frontline functionaries, bringing them to the center of engagement. Anganwadi Workers (AWWs), ASHAs, ANMs, Kisan Mitras, Rozgar Sahayaks, NJJM functionaries, Sujalam Shakti Groups, Village Health, Sanitation and Nutrition Committees (VHSNCs), Poshan Samiti, Self Help Groups (SHGs) and their federations such as Cluster Level Federations (CLFs) under SRLM along with School Management Committees (SMCs) shall act not merely as facilitators but as community champions of safe water and service accountability;

v. Drawing on their strong community connect, they shall enable sustained engagement, promote mind set change and promote collective responsibility for water service delivery and source sustainability, making IEC participatory and strengthening convergence, ownership, inclusion and accountability at the grassroots level;

vi. Impact assessment shall be undertaken in convergence with relevant programmes, schemes and Ministries/ Departments having close linkages with drinking water such as POSHAN 2.0 under WCD, National Health Mission (NHM) under MoHFW, Swachh Bharat Mission Grameen (SBM-G) under DDWS, SRLM under Rural Development & Panchayati Raj etc. as part of a whole-of-government approach. The assessment shall draw upon outcomes and indicators from these programmes, particularly in areas of nutrition, health, sanitation and hygiene practices, to evaluate the broader impact of safe and reliable drinking water services;

vii. Aligned communication calendars, common branding and messaging and coordinated media engagement shall be ensured across all convergence efforts to promote clarity, credibility and greater impact, while integrating water related messaging with nutrition, health and education and aligning with related Schemes, Departments and National Missions.

2.7.6. Collaboration with SRLM and other similar Programmes in Participatory Governance

Under JJM 2.0, States/ UTs shall strengthen implementation by collaborating with SRLM and engaging Cluster Level Federations (CLFs) of SHGs to support participatory governance and other IEC activities.

i. Such collaboration shall facilitate community mobilisation, support in undertaking activities under IEC framework and capacity building of GP/ VWSCs, Sujalam Shakti Groups and other community level institutions working under JJM to strengthen ownership and participation in operation and maintenance;

ii. Coverage and deployment of CBOs such as CLF or PACs shall be planned at cluster level as defined under SRLM or the relevant Programme considering the local situation, to ensure continuous engagement and effective implementation;

iii. States/ UTs may utilise available Support funds, as per guidelines for engagement of such CBOs to ensure sustained community mobilisation, IEC activities and capacity building at the grassroots level.

2.7.7. Partnership with Development Partners and Rural WASH Partners Forum (RWPF)

Collaboration with RWPF and other Development Partners shall be promoted to support implementation through pilot local innovations and solutions, capacity building and knowledge sharing. RWPF shall serve as a platform for enabling partners to contribute to IEC, community engagement, documentation and dissemination of good practices.

These partnerships shall support demonstration of locally relevant solutions, generate field level learnings and strengthen local institutional capacities. This will enable States/ UTs to adopt and scale effective practices for improved service delivery, operation and maintenance and source sustainability.

2.7.8. Outcome Measurement and Performance Indicators

The impact of participatory governance shall be assessed, based on outcomes in system performance and service delivery including functional VWSCs, strengthened O&M practices, improved regularity, quality and pressure of supply, sustained community participation and timely issue resolution. These outcomes shall be monitored through tools such as Jal Seva Aankalan, public disclosure mechanisms and periodic assessments enabling data driven decision making and continuous improvement.

2.7.9. Integration with Annual Action Plan/ CIRP for utilisation under Support Fund

i. All IEC and participatory governance activities and action plans at State, District and Gram Panchayat levels shall align with CIRP implementation and the reform matrix;

ii. States/ UTs shall ensure that all convergence-related activities are clearly reflected in the Annual Action Plan (AAP) with defined timelines, roles and expected outcomes. These plans shall be prepared in consultation with concerned Departments, frontline functionaries and local institutions to ensure practicality, ownership and effective implementation;

iii. While planning activities, priority shall be given to Har Ghar Jal Gram Panchayats.

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CHAPTER 2.8

Capacity Building Framework for Continuous Service Improvement

Reform aims at:

- Shift from generic training to function and competency based capacity building, ensuring role clarity and measurable outcomes;
- Professionalisation of service delivery systems with utility oriented, citizen centric skill development for both micro and macro utilities;
- Creation of a scalable and standardised training ecosystem, through a structured ToT based hub and spoke model;
- Strengthened district level technical capacity, enabled by data driven assessments;
- Improved accountability and tracking of capacity outcomes through time bound training targets supported by digital monitoring tools.

2.8.1. Introduction

With the substantial expansion of rural piped water supply infrastructure under the Jal Jeevan Mission, the focus is now shifting from infrastructure creation to sustained service delivery. As systems mature, ensuring their long-term functionality, efficiency and reliability becomes critical.

This requires systematic capacity building and continuous competency enhancement of all stakeholders across governance and service delivery levels. Strengthened institutions supported by trained human resources and clear accountability mechanisms will be essential to sustain services beyond the Mission period. States/ UTs shall adopt a structured competency-based approach to capacity building to ensure effective operation and maintenance of rural piped water supply systems.

2.8.2. Vision and Guiding Framework

Rural piped water supply systems range from simple village level operations to complex multi-stakeholder service delivery arrangements. Effective service delivery therefore requires both individual competency and strong institutional coordination.

The capacity building framework under JJM 2.0 envisions empowered, accountable and well coordinated institutions at all levels delivering reliable, sustainable and citizen centric rural water services. Stakeholders shall be oriented not only to their individual responsibilities but also to the coordination required across various levels for effective service delivery.

Governance institutions shall lead this transformation by driving change management, ensuring transparency and enabling service utilities to function efficiently. This shall be supported through continuous training, technical support and structured capacity-building systems.

2.8.3. Key Stakeholders

Capacity building under JJM 2.0 shall cover stakeholders across community, service utility and governance levels, each performing distinct but interlinked roles in rural water service delivery. The framework recognises both utility functions and governance responsibilities to ensure effective operation, oversight and sustainability of rural piped water supply systems.

At the community, service utility and governance levels, institutions ranging from Gram Panchayats and VWSCs to Micro and Macro Utilities and higher-level governance bodies shall collectively ensure sustained rural water service delivery outcomes. This integrated institutional approach shall enable effective coordination across levels and support the transition from infrastructure creation to sustained service delivery. The key stakeholders, along with their indicative roles and responsibilities are as follows:

Table 5: Stakeholders and Roles in Rural Water Service Delivery at Different Levels			
S.No.	**Level**	**Key Stakeholders**	**Roles & Responsibilities***
1.	Community Level	• GPs/ VWSCs • Sujalam Shakti Group[5] such as SHGs/ PACS etc. • Local Functionaries such as AWWs/ ASHAs/ Other Front-line workers engaged in other Programmes • NJM/ Local operators & technicians • Volunteers	• Community engagement/ mobilization & support • Service delivery charge collection • Conduct Water quality testing • Daily operation of village water systems • Undertake minor repairs & maintenance. • Contract/ vendor management
2.	Service Utility Level	Micro Utility • GPs/ VWSC/ any other Village-level micro-utilities • Sujalam Shakti Group (if engaged in service delivery) Macro Utility • PHED or State Department responsible for Rural Piped Water Supply • Professional water service agencies	• Daily operation of village water systems • Minor repairs & maintenance • Contract/ vendor management • Operation & maintenance of infrastructure • Asset management & preventive care • Contract/ vendor management • Ensuring reliable service delivery
3.	Governance Level	• Gram Panchayat/ VWSC • DWSM and DTU • SWSM	• Policy implementation & oversight • Monitoring and evaluation. • Capacity-building and change management support. • Financial, technical facilitation

**This is indicative, roles and responsibilities may vary depending on State/ UT level policy*

2.8.4. Capacity Building Approach

Capacity building under JJM 2.0 shall move beyond one time training programmes and adopt a continuous, structured and competency-based learning approach. This approach shall focus on strengthening functional capacities of stakeholders in alignment with their roles in rural water service delivery, ensuring sustained improvements in system performance and service outcomes. All capacity building activities shall be planned as part of the States/ UTs Action Plan under the Support Fund.

The capacity building strategy shall be structured around the following key components to ensure systematic planning, effective delivery and continuous improvement.

A. Competency Framework

Capacity building shall be anchored in clearly defined competencies across State, District and Gram Panchayat levels. At the State level, competencies shall focus on policy formulation, sector planning and enabling institutional systems. At the District level, emphasis shall be on technical support, coordination, performance monitoring and service improvement planning. At the Gram Panchayat and Village levels, competencies shall focus on operation and maintenance, financial management, water quality assurance and community engagement.

Across all levels, competency frameworks shall cover technical, institutional, financial, governance and

[5] *For detail information of Sujalam Shakti Group kindly refer to Chapter 2.4 of this guidline*

digital domains and shall be guided by periodic capacity gap assessments. Indicative competency areas of stakeholders involved at Community and District level are annexed as **Annexures V** and **VI** respectively.

B. Training Content Development

A structured and standardised training content framework shall be developed at the national level by NJJM with support from SPM-NIWAS, Development and International Partners and other technical/ academic institutions. States/ UTs shall customise this content through their Administrative Training Institutes, State Institutes of Rural Development and other relevant institutions to reflect local contexts, including language, geography and socio-cultural conditions.

Training content shall be aligned with identified competencies and shall address role specific requirements such as operation and maintenance, water quality management, financial sustainability, use of digital platforms, source sustainability or any other challenges come across during implementation. Digital learning modules may also be developed to support continuous and scalable learning and States/ UTs may explore establishing Learning Management Systems for structured delivery and tracking.

C. Training Delivery

Training delivery shall follow a structured cascade approach based on Training of Trainers, ensuring outreach to all levels of stakeholders. Master Trainers shall be developed at National and State levels and shall support training delivery at District and Gram Panchayat levels.

Training delivery shall be anchored in a hub and spoke institutional framework to ensure standardisation, scalability and sustained field-level support.

2.8.5. Operationalisation of Hub and Spoke Model

Capacity building shall be implemented through a hub and spoke model to ensure standardised content development and decentralised training delivery. Hubs shall provide technical expertise, content and training delivery, while spokes shall enable contextualised delivery and field level support, ensuring quality, scale and continuous handholding.

States/ UTs shall identify appropriate institutions as hubs and spokes across levels based on capacity and training needs and implement a cascading approach for effective outreach without compromising quality.

At the National level, the National Jal Jeevan Mission shall function as the hub for development of standardised training frameworks, resource materials and capacity building strategies for the senior management and policy makers.

At the State/ UT level, Administrative Training Institutes, State Institutes of Rural Development and other identified academic and technical premier institutions working in WASH sector shall function as hubs responsible for contextualisation of training content, development of master trainers and overall coordination.

At the District and Sub-district levels, institutions such as DWSMs, DTUs, training centres and other local institutions shall be responsible for training delivery, field level mentoring and continuous technical support to Gram Panchayats and frontline functionaries.

States/ UTs shall ensure that clear linkages are established between hubs and spokes for continuous knowledge flow and feedback. Training delivery shall be supported by field level handholding, on-site demonstrations and regular technical support. Existing institutional platforms such as SRLMs, RSETIs, ITIs and PRI training systems shall be leveraged to expand outreach and strengthen last mile capacity.

States/ UTs shall ensure convergence with capacity building programmes of related Ministries/ Departments having linkages with drinking water including the Ministry of Women and Child Development, Ministry of Health and Family Welfare and Ministry of Education to promote integrated and holistic service delivery.

In addition to structured training programmes, continuous orientation and awareness shall be reinforced through community engagement framework such as Jal Arpan and Jal Utsav as specified in Chapter 2.7 Participatory governance for Jan Bhagidari.

RWPF may also be utilized as a partnership platform to support capacity building through knowledge sharing, technical assistance and demonstration and scaling of innovative practices. The Forum shall facilitate collaboration among partners to strengthen competencies for sustainable rural water supply management.

2.8.6. Training Cycle and Assessment

Capacity building shall be implemented through structured training cycles consisting of orientation and refresher training. Each field level functionary shall undergo at least one orientation training followed by periodic refresher training based on identified capacity

gaps. Assessment of capacity building shall focus not merely on the number of trainings conducted, but on demonstrated enhancement of competencies in service delivery, system performance and institutional effectiveness.

Digital platforms such as e-GramSwaraj shall be leveraged for tracking training progress and integration with Jal Jeevan Mission digital systems shall be explored to enable comprehensive monitoring and reporting.

Further, for HGJ certification of any Gram Panchayat or Village, all designated field level functionaries shall complete the required trainings either prior to certification or within six months from the date of HGJ certification to ensure readiness for sustained operation and management of rural drinking water supply systems.

2.8.7. Key Monitoring Points

To strengthen the overall effectiveness of the Comprehensive Implementation Reform Plan (CIRP) and ensure sustainable service delivery under the mission, it is essential to prioritize capacity building across all levels of field functionaries. Hence, following milestones should be prioritized:

i. All field functionaries in Har Ghar Jal (HGJ) Gram Panchayats shall be trained and training completion records shall be maintained on e-GramSwaraj/ GP Dashboard;

ii. At least one round of training shall be completed in all HGJ GPs by March 2027. Refresher training shall be conducted after six months or as decided by the State/ District.

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CHAPTER 2.9

Establishing Human Resource Skilling Ecosystem

Reform aims at:

- Availability of certified Human Resources with all the RLBs/ GPs for operations of water supply schemes;
- NSQF aligned certification through NCVET-Jal Vitaran Sanchaalak as a recognised job role with a valid and portable qualification;
- Recognition of Prior Learning (RPL), a structured pathway to certify the existing informal O&M workforce already working in villages;
- Engineering Elective and Operator Diploma in formal Academic Institutions, 'water-supply O&M' embedded into AICTE affiliated engineering colleges and polytechnic diploma programmes creating a long term institutional pipeline;
- End-to-end digital tracking lifecycle monitoring from enrolment through training, certification and GP wise deployment through SIDH and the DWSM dashboards respectively.

2.9.1. Introduction and Objective

Under JJM, substantial progress has been achieved in creation of rural drinking water supply infrastructure. With the transition to JJM 2.0, the focus is on ensuring sustained functionality, service delivery and long-term operability of assets created. In this context, establishment of a robust human resource skilling ecosystem is critical to ensure that adequately trained personnel are available at all levels for O&M of rural water supply systems continuously.

The main strategy is to:

i. Ensure availability of trained and certified human resources at Gram Panchayat level;

ii. Promote community ownership and decentralized O&M through RLBs/ VWSCs/ Pani Samitis;

iii. Enable standardisation of training through NSQF aligned courses and certification;

iv. Facilitate local employment opportunities and O&M skill-based livelihoods.

2.9.2. Multi-tier Human Resource Framework

States/ UTs shall adopt a multi-tier workforce structure aligned with infrastructure typology. The Table-6 describes workforce categories may be filled through locally mobilised eligible candidates including rural youth, ex-servicemen, women, etc. to ensure sustainability and local ownership of O&M systems.

Table 6: Multi-tier Human Resource Framework for O&M

Level	Role	Qualification	Scope of Work
In Village	Nal Jal Mitra (Jal Vitaran Sanchaalak)	NSQF Level 4	In village SVS: pumps, pipelines, OHTs, water quality reporting
Out Village	Multi skill Technician/ Operator	ITI/ Diploma Polytechnic/ equivalent NSQF level course	WTPs, BWS, MVS, MBRs: routine maintenance, monitoring and minor repairs; day to day WTP, pump, valve operations and routine checks
Supervision	Engineer	Engineering Elective in Degree College	SCADA operations, quality control, scheme supervision at WTPs
Village (Cross cutting)	Sujalam Shakti Groups - FTK Testing & Monitoring	Micro skilling equivalent equivalent NSQF Level 1 course	WQ testing using FTK: identified members of Sujalam Shakti Group

2.9.3. Nal Jal Mitra Programme (NJMP)

Nal Jal Mitra Programme (NJMP) aims to provide skill-based trainings to local persons to equip them with a comprehensive set of skills, so that they can function as scheme operators and are able to carry out minor repairs and maintenance including preventive maintenance of the piped water supply scheme(s) in their village. States/ UTs may encourage mobilisation of candidates from rural youth, ex-servicemen, women, members of SHGs and other locally available skilled or semi-skilled persons. Ex-servicemen having experience in handling mechanical and electrical systems may be suitably encouraged to participate in the programme wherever available. The Nal Jal Mitra Programme is to implement an NSQF Level-4 multi-skilling programme under the job role Jal Vitaran Sanchaalak approved by NCVET to make available local multi-skilled personnel for in-village O&M. There are two pathways for imparting the training under NJMP:

i. Short Term Training (STT)[6] of 510 hours for fresh candidates;

ii. Recognition of Prior Learning (RPL) & Upskilling[7] of 120 hours for experienced candidates.

2.9.4. Multi skill Technician Training/ Operator Diploma Programme

Multi skill Technicians are for O&M of out-village components including Bulk Water Schemes (BWS), Multi Village Schemes (MVS), Water Treatment Plants (WTPs), Elevated Storage Reservoirs (ESR) and Master Balancing Reservoirs (MBRs). As per the guidelines, the out-village infrastructure is envisaged to be maintained by technical agency such as PHE/ RWS department/ Jal Nigam/ Boards etc. The State governments may develop specialised courses in ITIs/ polytechnic for providing above mentioned skills and utilise such trained personnel for O&M of out-village infrastructure. The existing engineering colleges may also be encouraged by States/ UTs to offer such courses.

2.9.5. Engineering Elective on Water Supply Infrastructure Management

Government Technical Universities and Engineering Colleges, UGC recognized Universities and AICTE affiliated Engineering Colleges may be engaged for adoption and integration of an elective course on Rural Water Supply Infrastructure Management within undergraduate engineering programmes. These electives may cover design principles, treatment technologies, distribution network management, automation and SCADA systems, water quality management and regulatory frameworks etc.

2.9.6. Micro-skilling of Sujalam Shakti Groups for WQM&S

The identified members of Sujalam Shakti groups in village will be trained for WQM&S using Field Test Kits, especially to monitor bacteriological contaminants and residual chlorine. The NJJM will endeavour to develop an NSQF Level-1 certified course in collaboration with relevant sector skill council. The States may utilise their Water Quality laboratories to provide training on testing especially residual chlorine along with other parameters.

2.9.7. Institutional Arrangement

A. National Level

NJJM, DDWS in convergence with the MSDE and other relevant ministries shall provide technical and financial support for effective implementation of skilling of human resources.

B. State Level

State Water and Sanitation Missions (SWSMs) shall identify the State Skill Development Missions (SSDMs) or State Rural Livelihood Mission (SRLM) as Programme Implementing Agencies (PIA) and coordinate with them for planning and delivery of all trainings. Annual district-wise training plans are to be prepared covering all workforce categories and submitted to NJJM as part of Annual Action Plans.

C. District Level

District Water and Sanitation Missions (DWSMs) are responsible for implementation, coordination and monitoring of skilling activities at the District level. Each DWSM is to designate DTU as a nodal for skilling. The DTU has to coordinate with SSDMs, ITIs, Polytechnics, RSETIs and other implementing agencies for preparation of proposal and training delivery. For this, at least one person from DTU may be identified for analysis, planning, coordination, implementation and monitoring of the skill requirement of the districts. All skilling activities are to be entered and reflected on the DWSM dashboard.

[6] Refer NJMP guidelines circulated through email on 16.08.2023
[7] Refer RPL guidelines circulated vide letter No. W-11011/6/2023-JJM-III-DDWS-Part(1) dated 28.02.2024

D. Gram Panchayat/ Village Water and Sanitation Committee

Gram Panchayat/ Village Water and Sanitation Committee shall mobilize and recommend candidates for the NJMP/ Women - FTK Testing & Monitoring from Sujalam Shakti Group. While mobilising candidates, Gram Panchayats may give preference to local youth, ex-servicemen, women and members of community-based institutions to strengthen community ownership and ensure availability of trained personnel within the village. GP may deploy the candidates post successful certification.

2.9.8 Training Infrastructure

All trainings to be delivered through government institutions - ITIs, Polytechnics, RSETIs and other government training centres. All training centres are to obtain necessary accreditation and meet infrastructure standards prescribed under guideline before commencement of training.

2.9.9 Financing

Common Cost Norms notified by MSDE from time to time, shall be applicable for training of human resources, wherever not explicitly indicated in this Guidelines, including for difficult geographies. The financial support for skilling programme under JJM 2.0 will be met from below-mentioned funding sources:

i. Finance Commission tied grants;

ii. JJM Support component[8];

iii. Convergence through PMKVY, RSETIs, other skilling programmes of government.

Further, detailed SOP along with schedule of implementation will be issued separately.

2.9.10. Course Adoption and Certification

NSQF aligned courses approved by NCVET shall be adopted for training progamme. The assessor of candidates should meet the requirements and standards prescribed by Water Management & Plumbing Skill Council or any other body approved by NCVET. The certificate to the candidates is to be issued upon successful completion of assessment. The Skill India Digital Hub (SIDH) platform shall be utilised for lifecycle tracking of training activities of candidates from registration training, assessment and certification. The deployment may be monitored through DWSM dashboard of JJM-IMIS. The State government may endeavor to develop Operator Diploma Programme and Engineering Elective course that may be adopted by universities, polytechnic, engineering colleges etc. for skills mentioned in para 2.9.4. The NJJM will coordinate with AICTE/ NCVET to develop such programme at national level.

2.9.11. Monitoring and Reporting

A. Monitoring

i. **National Level:** Monitor gaps in training and deployment of skilled human resources at various level;

ii. **State Level:** At the State level, SWSM under the leadership of ACS/ PS/ Secretary, PHE/ RWS department, shall periodically monitor (preferably on a quarterly basis) the State Action Plans. The State shall ensure timely data reporting, review district performance and facilitate coordination with training institutions and other skilling programmes;

iii. **District Level:** At the District level, DWSM under the leadership of the District Collector/ Deputy Commissioner/ CEO-ZP, shall be responsible for implementation, monitoring, validation of progress and ensuring convergence with relevant government skilling initiatives.

B. Reporting

All skilling related activities shall be reported through the SIDH portal and reflected on the DWSM dashboard to ensure real-time monitoring and transparency. At the District level, digital monitoring shall be undertaken through the DWSM dashboard. The following data points shall be maintained and updated regularly (preferably monthly):

i. Number of requirements of O&M personnel assessed and mapped (category-wise and scheme-wise, Village/ GP wise);

ii. Candidates enrolled, trained, assessed and certified (category-wise);

iii. Training institutions engaged, including accreditation status, number of batches formed, trained and certified;

iv. Deployment status of Nal Jal Mitras and Multi-skill Technicians (GP-wise and scheme-wise);

v. Women identified under Sujalam Shakti Groups trained for FTK testing and monitoring (village wise);

[8] *A detailed SoP on utilisation of funds under JJM Support component for skilling activities shall be issued by DDWS.*

vi. Convergence achieved with other government skilling programmes [Name of programme, number of human resources trained (category wise), fund utilized].

Districts shall ensure completeness, accuracy and timeliness of data entry for effective review at State and Central levels.

2.9.12. Key Action Points for the States/ UTs

i. Ensure NJMP training in all Har Ghar Jal villages with at least one trained and certified Nal Jal Mitra per Gram Panchayat;

ii. Undertake assessment of human resource requirement at all levels for O&M of schemes and compute District-wise training gaps;

iii. Adopt and rollout the training courses across identified training institutions in all States/ UTs;

iv. Formulate policy for engagement of human resources at rural water service utility levels;

v. Appoint Programme Implementing Agencies (PIA) for planning, coordination and delivery of all training;

vi. Review state-wise progress against NJMP enrolment, training and certification targets -in alignment with Annual Action Plans;

vii. Monitor deployment status of certified Nal Jal Mitras and multi-skill technicians mapped against scheme-wise O&M requirements;

viii. Ensure convergence with other skilling programmes of government;

ix. Monitor quality of training delivery through assessment pass rates, dropout rates and feedback from certified personnel and Gram Panchayats and community institutions in skilling programmes to ensure availability of locally deployable trained personnel;

x. Ensure adequacy of training infrastructure including accredited institutions, operational batches and geographic coverage across Blocks and Districts;

xi. Encourage participation of rural youth, ex-servicemen, women and community institutions in skilling programmes to ensure availability of locally deployable trained personnel.

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CHAPTER 2.10

Operational and Financial Sustainability of Water Supply Schemes

Reforms aims at:

- Shift from reactive repairs to preventive maintenance driven operations, reducing breakdowns and extending asset life;
- Reduction in O&M costs through efficient system management, including pressure control, leak detection and lowering Non-Revenue Water (NRW);
- Transition from ad-hoc financing to structured and sustainable O&M funding, ensuring uninterrupted and reliable service delivery;
- Improved scheme functionality and reduced slippage through regular monitoring, planned operations and lifecycle based asset management;
- Strengthened community led oversight and accountability with Gram Panchayats actively engaged in supervision, management and sustainability of system.

Earlier, many rural water supply schemes implemented under NRDWP became non-functional due to weak monitoring, inadequate preventive maintenance planning and limited community ownership. The Operation and Maintenance (O&M) financing was also insufficient as there was no structured mechanism to ensure dedicated funds resulting in inefficient operations and high costs. Limited focus on infrastructure sustainability led to rapid asset deterioration further affecting service reliability.

Under the reform-based approach of JJM, States now have to ensure structured O&M financing for uninterrupted services, mandatory O&M budgeting for every scheme and systematic planning to control O&M costs. Greater emphasis is to be placed on preventive maintenance, long term asset management and cost efficiency. The Gram Panchayats play more active role in supervision, surveillance, day to day operations and management.

To take it forward the following key principles are envisaged for both micro and macro utilities:

i. **Transparency in Accounts:** All financial records, receipts and payments must be clear, traceable and maintained in an open manner for easy verification;

ii. **Accurate and Timely Record Keeping:** Revenue, expenditure and balances must be recorded regularly without delays to support correct decision making;

iii. **Billing and Payments:** For water service bills must be generated for each customer and receipts should be given and recorded. The digital tools for billing, collection, receipts, expenditures for vendor payments should be encouraged to reduce errors, improve traceability and increase collection efficiency;

iv. **Cost Control and Efficiency:** The cost of operations per household or per cubic meter of water supply should be minimized. The measures should be taken for leakage, energy efficiency, preventive maintenance and optimized manpower deployment. Every year the targets should be provided to utilities to improve cost efficiency;

v. **Regular Monitoring and Audits:** Conduct routine financial reviews and conduct annual audits to identify gaps, ensure compliance and strengthen accountability;

vi. **Planning of Funds:** Assessment of actual requirements of funds for each utility should be carried out. The availability of funds from various sources should be assessed and any gap may be addressed through monitoring.

2.10.1. Efficient Financial Management

Financial management in the drinking water sector refers to the ability of a water supply system to generate, retain and efficiently utilize financial resources to

meet its full operation, maintenance, lifecycle replacement and service improvement costs on a long-term basis without depending on unpredictable external subsidies or ad-hoc fiscal support.

A. Revenue Adequacy

States/ UTs should aim to cover all Operation & Maintenance (O&M) expenses including electricity, Honorarium/ professional fees/ service charges to human resources, water quality testing, routine repairs and emergency needs through service delivery charges for both Micro and Macro Utility. If the gap persists the Finance Commission funds, State funds etc. should be utilized.

Collection of Water Distribution Charges/ service delivery charges for in-village schemes and for Bulk Water Transfer should be steady and timely with strong collection systems, digital payments and incentives for on-time payment. The institutions and other commercial connections should also be charged. The revenue generated as a percentage of overall O&M fund requirement/ expenditure.

B. Development of Financial Tools for effective O&M

The State should develop a financial planning tool that is directly linked to water supply scheme ID-Sujal Gaon ID used under the Jal Jeevan Mission for capturing component wise village level data. By connecting this data to a financial tool, the State can generate accurate estimates of the annual O&M needs for each village and for the State as a whole. This tool can automatically fetch technical and operational information from Sujalam Bharat platform and convert it into yearly budget requirements for power, manpower, water quality and repairs. It will help the State prepare consistent, data-based O&M plans, identify villages with higher needs and support transparent budgeting.

C. State Budget Linkage for Rural Water Supply

Assessment of actual requirements of funds for each utility should be carried out. The availability of funds from various sources such as finance commission, DMF, CSR etc. should be assessed for each utility and any gap should be met with the State financial resources. For this purpose, separate budget head should be created.

D. Proper Use of Finance Commission Funds

The State must ensure that tied grants from the Finance Commission are utilised strictly for water and sanitation purposes as per prescribed guidelines with clear financial records, regular audits and proper functioning of Panchayati Raj institutions. The State should help the Rural Local Bodies (RLBs) to meet these rules so that funds continue without stoppage.

E. Ring Fencing of Funds in Asset Management

Ring fencing O&M funds means a dedicated and transparent account that prevents diversion of resources and ensures that financing is always available for essential activities like preventive maintenance, electricity payments, water quality testing and emergency repairs. Therefore, a separate bank account should be maintained by the micro utilities.

Similarly, for macro utilities which carry out more complex operations at various locations should prepare annual financial statement for O&M separately.

2.10.2. Operation and Maintenance (O&M) Budgeting Process

O&M budgeting begins at the village level where each VWSC prepares a simple, need based annual estimate for every scheme using SWSM formats. The DWSM then compiles and verifies these budgets adds the O&M needs of bulk systems and submits a consolidated District requirement to the State. The SWSM aggregates District figures, assesses available resources from service delivery charges, Finance Commission grants, State allocations, CSR and convergence schemes and identifies the O&M funding gap. This bottom-up process ensures realistic field-based planning, clear gap identification, structured resource convergence and mandatory digital entry through the Sujal Gaon ID financial tool ensures transparency, uniformity and strong financial governance.

2.10.3. Cross-subsidized Financing for Financial Sustainability

A cross-subsidized model can significantly strengthen the financial sustainability of rural and peri-urban water supply schemes by using revenue from peri-urban, commercial and industrial users to support schemes that cannot fully recover their O&M costs. This approach works effectively only when planning is consolidated at the State, District and Scheme levels, allowing authorities to clearly identify which schemes generate surplus revenue and which operate at a deficit. Through systematic assessment of low revenue or high-cost schemes, States/ UTs can allocate additional income from higher tariff user categories to ensure stable funding for essential services in underserved areas. This model not only helps bridge O&M gaps but also promotes equity, service reliability

and long-term financial viability across the entire water supply system.

The following key financial monitoring indicators may be used for efficient financial management:

2.10.4. Efficient Operation Management

Operations Management focuses on ensuring that the water supply system functions smoothly every day without interruption. Skilled operators, properly trained technicians and functional VWSCs are essential for daily operations. Standard Operating Procedures (SOPs) must be followed for pump operation, chlorination, valve control, water quality testing and customer grievance handling.

Real time monitoring tools such as flow meters, pressure sensors, online water quality devices and SCADA, help detect issues early and improve decision making. Record keeping must be systematic so that energy use, water production, service hours and breakdowns can be tracked regularly. Daily data entry of operational activities on Sujal Gaon App is essential for operational sustainability and ensuring consistent, safe and reliable water supply to households as per JJM service standards.

If operational efficiency is not maintained, the overall expenditure of the water supply system can increase very quickly. Uncontrolled pressure, irregular operations, neglected maintenance and frequent breakdowns can sharply raise O&M costs and reduce the life of the system. Therefore, for long-term financial sustainability, it is essential to follow efficient financial management practices together with strong operational discipline. When both finances and operations are managed efficiently, schemes run smoothly, water losses are minimised and resources are used wisely ensuring reliable service delivery at a lower cost.

For effective operation management, following measure are to be taken:

A. Community Vigilance and Surveillance

Community vigilance is an important part of effective operation and management of water supply schemes. When community members regularly observe and monitor the functioning of their scheme, disruptions, leakages, pressure changes or breakdowns can be detected early and corrected quickly.

This active involvement reduces the chances of major failures, prevents water loss and lowers overall O&M costs. Communities must be encouraged and engaged in routine checks, reporting issues and supporting timely repairs. Through strong community surveillance, schemes become more reliable, responsive and financially efficient.

B. Asset Management & Lifecycle Planning

Asset management under JJM requires maintaining a complete, updated record of all scheme assets pumps, motors, pipelines, storage tanks, treatment units, meters and source structures through a detailed asset register capturing installation date, condition, repair history and expected lifespan. Lifecycle planning must guide timely rehabilitation and replacement with depreciation provisions to ensure future funding. Effective asset management prevents sudden failures, reduces long term costs and supports sustainable O&M.

The DTU must maintain District wise, updated asset and inventory records, verify assets on the ground, update condition data and ensure timely entry of new and replaced components. Efficient inventory management ensures availability of spares, faster

Table 7: Key Financial Monitoring Indicators for Micro and Macro Utility

Indicator	Description/ Formula
O&M Cost Recovery Ratio	Service delivery charges collected (÷) Total O&M expenditure
Electricity Cost Ratio	Electricity expenditure (÷) Total O&M cost
Collection Efficiency	Amount collected (÷) Amount billed
O&M Budget Utilization	Actual expenditure (÷) Approved O&M budget
Non-Revenue Water (NRW)	Difference between water produced and water billed
Timeline of Financial Reporting	Compliance with monthly financial reporting

repairs and better use of funds. DTU must also use the PM-Gati Shakti platform for GIS-based mapping of all pipelines, pump houses, tanks, sources and treatment units, ensuring every asset is geo-tagged and visible on the Sujalam Bharat Platform to support transparent planning and long-term sustainability.

C. Monitoring the Efficiency of Water Supply Schemes

Monitoring the efficiency of water supply schemes is essential to ensure that systems remain functional, financially sustainable and capable of delivering safe water every day. Regular monitoring helps detect issues early, plan timely maintenance, control O&M costs and keep service levels consistent with JJM standards. A structured approach supported by financial tools, preventive maintenance, budgeting discipline and community involvement strengthens long-term performance and reduces breakdowns.

2.10.5. Preventive Maintenance

Preventive maintenance is a key sustainability measure required under JJM. It involves planning and conducting regular servicing and inspection activities to avoid sudden breakdowns and costly emergency repairs. Activities include:

i. Scheduled service of pumps and motors;

ii. Leak detection;

iii. Pipeline flushing;

iv. Valve operation cycles;

v. Tank cleaning;

vi. Electrical inspection;

vii. Calibration of meters and sensors.

Predictive maintenance tools like vibration monitoring and pressure analysis help detect early warning signs of failure. A preventive maintenance schedule must be prepared by the DTU. Conducting preventive maintenance on time extends the life of assets, reduces O&M costs and ensures uninterrupted water supply.

2.10.6. Key Action Points for the States/ UTs

i. State may adopt financial monitoring tools;

ii. Complete assessment of all Har Ghar Jal schemes;

iii. Ensure annual O&M budgeting is provisioned in the State budgets;

iv. DTU to publish a preventive maintenance schedule for all schemes;

v. Engage the community in regular surveillance and vigilance to identify disruptions early;

vi. States to adopt structural reforms alongside providing funds, function and functionaries for sustainable O&M of rural piped water supply infrastructure.

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CHAPTER 2.11

Research, Innovation & Knowledge -Ecosystem

Reform aims at:

- **Institutionalized Technology Appraisal:** Establishment of a structured technology appraisal system through the National Technical Committee (NTC) and State Technical Committees (STCs) to guide systematic evaluation and adoption of technologies;
- **Structured Validation and Adoption Framework:** Introduction of processes that enable solutions to progress through structured development, assessment, piloting and field based learning prior to wider deployment;
- **Collaborative Innovation and Knowledge Sharing:** Strengthening collaboration with academia, research institutions, startups and industry, along with systematic documentation and National dissemination of innovations to facilitate wider sectoral learning;
- **Outcome-Based Financing for R&D Proposals:** Introduction of a results-linked financing approach that aligns R&D funding with clearly defined milestones and measurable outcomes, ensuring that technologies progress through evidence-based development and systematic field validation.

2.11.1. Objective

Jal Jeevan Mission (JJM) shall establish and promote a structured ecosystem dedicated to research, innovation and knowledge management in the rural water supply sector.

The ecosystem shall facilitate the following outcomes:

i. Development and deployment of low cost, scalable and locally appropriate technologies;

ii. Strengthen indigenous innovation, design and manufacturing capabilities;

iii. Institutionalize evidence-based validation field testing and piloting of emerging technologies;

iv. Foster partnerships among academia, industry, startups and implementation agencies;

v. Facilitate systematic documentation and dissemination of best practices, operational learnings and knowledge resources across all levels.

This structured approach shall guide States/ UTs, institutions and partners to establish sustainable, climate resilient and cost-effective rural water service delivery systems in alignment with the vision of Atmanirbhar Bharat.

2.11.2. Institutional Framework for Innovations

In 2014, the then Ministry of Drinking Water and Sanitation constituted a Standing Committee for Examination of best Technologies concerning Water and Sanitation. The committee was chaired by Dr. Mashelkar, former Director General, Council for Scientific and Industrial Research. Many technologies on water and sanitation were considered by the Committee and approved for use in the sectors.

Earlier, a National Technical Committee was constituted under the chairmanship of PSA to Govt. of India to examine and review the R&D proposals. Presently, the National Technical Committee has been reconstituted and shall function as the apex body for technology appraisal and innovation promotion in WASH under the chairmanship of Member, NITI Aayog.

A. National Technical Committee (NTC)

The National Technical Committee has been reconstituted and shall function as the Apex body for technology appraisal and innovation promotion in rural WASH sector.

The NTC shall undertake functions, including but not limited to:

i. Evaluate and recommend innovative technologies and R&D proposals;

ii. Facilitate technology validation and adoption across States/ UTs;

iii. Promote indigenous technology development;

iv. Enable collaboration with National and International research institutions;

v. Issue advisory notes on emerging technologies and sectoral innovations;

vi. Review and evaluate applied research studies and reports submitted by JJM Professor Chairs as mentioned in the verification protocol of their Disbursement Linked Indicators.

B. State Technical Committee (STC)

States/ UTs shall constitute a State Technical Committee (STC) aligned with the National framework. The STC shall report to the State Water and Sanitation Mission (SWSM) to ensure alignment with state priorities and to facilitate coordinated review, piloting and adoption of appropriate technologies.

The STC shall undertake functions, including but not limited to:

i. Evaluate region specific technologies and innovations;

ii. Review proposals for pilot implementation of new technologies;

iii. Assess suitability of solutions based on local hydrogeological conditions;

iv. Facilitate adoption of innovations recommended by the NTC;

v. Recommend scale up of validated technologies;

vi. STC may refer R&D proposals to the NTC for their consideration.

C. Role of SPM-NIWAS

SPM-NIWAS shall provide secretarial support for the compilation, shortlisting and facilitation of the evaluation of innovative technologies and R&D proposals. It shall also function as a hub and spoke platform to enable collaboration among research institutions, industry, startups and implementation agencies including facilitating international collaboration and knowledge exchange. SPM-NIWAS shall coordinate joint research initiatives, innovation challenges and support water technology innovators and startups.

D. Role of National Jal Jeevan Mission

National Jal Jeevan Mission (NJJM) shall function as a National platform to promote the development and adoption of innovative technologies in the rural water supply sector. NJJM shall provide strategic guidance to SPM-NIWAS for coordinating sectoral innovation through collaboration among research institutions, academia, startups and industry for the development, piloting and validation of solutions, integrating evidence-based knowledge and facilitating sectoral learning and the exchange of global and indigenous best practices. Where required, NJJM may also undertake or facilitate such initiatives to support convergence and technology adoption.

2.11.3. Research & Development in Water Sector Technologies

A. Priority Areas for Research

To support technological advancement and strengthen service delivery in rural water supply systems, research and innovation efforts shall focus on the following thematic areas:

i. **Water Source Security and Quality:**

a. Source Sustainability and Aquifer Management: Strengthening understanding of groundwater systems and supporting long term sustainability and security of drinking water sources;

b. Emerging Contaminant Management: Improving detection, monitoring and mitigation of emerging contaminants affecting drinking water quality;

c. Low-cost Advanced Treatment Technologies: Development of efficient, affordable and context-appropriate treatment solutions suitable for rural water supply systems.

ii. **Water supply system efficiency and climate resilience:**

a. Climate Resilient Water Supply Systems: Advancing approaches that enhance resilience of water supply systems to climatic variability;

b. Energy efficient Pumping and Treatment: Promoting solutions that reduce energy consumption and improve operational efficiency in water supply systems.

iii. **Digital and Operational Innovation:**

a. Predictive Maintenance and Digital Monitoring: Enabling digital tools for real time monitoring, predictive diagnostics and improved reliability of water supply infrastructure;

b. Cost-effective O&M Models: Supporting innovative approaches that strengthen operational sustainability and reduce lifecycle costs.

B. State Driven R&D Initiatives

States/ UTs shall undertake R&D initiatives aligned with local hydrogeological conditions, operational requirements and sectoral priorities identified under the Mission. These efforts shall include identifying areas requiring research attention, initiating need-based R&D projects.

States shall take up various research-oriented solution streams, including:

i. **Service centric solution:** Development and piloting of technologies that enhance the efficiency, reliability and operational management of rural water supply systems including tools for monitoring, asset management and operational decision support;

ii. **Citizen centric solutions:** Innovations that strengthen user participation, transparency and accountability in water service delivery, behavior change interventions, community based monitoring tools and strengthened grievance redress system;

iii. **Sustainability centric solutions:** Research and pilots that improve long term water security through improved source protection, aquifer understanding, water conservation approaches and resource efficient technologies;

iv. **Climate resilient solutions:** Development and assessment of approaches and technologies that enhance the resilience of rural water supply systems to climate variability and extreme events including tools for risk appraisal, drought responsive planning, diversified sourcing and resilient infrastructure design.

2.11.4. Support for Water Technology Startups

To promote indigenous innovation, the Mission shall facilitate joint research initiatives, innovation challenges and mentorship activities to encourage water tech startups in developing indigenous technologies.

A. Convergence and Collaboration with Government Innovation Platforms and Industry Bodies

i. **Department for Promotion of Industry and Internal Trade (DPIIT):** To leverage the National startup ecosystem, incubation networks and innovation initiatives for development, piloting and scaling of emerging technologies relevant to rural water supply;

ii. **Federation of Indian Chambers of Commerce and Industry (FICCI):** To facilitate industry engagement, technology partnerships and knowledge exchange for promoting innovative and scalable solutions in the water sector;

iii. **Confederation of Indian Industry (CII):** To enable private sector participation, technology demonstrations and industry led innovations that support improved efficiency, sustainability and operational performance of water supply systems;

iv. **Jal Shakti Innovation Centre (JSIC):** The JSIC shall be leveraged for technology incubation, field level demonstration, validation and the scaling of indigenous innovations;

v. **Department of Science & Technology (DST):** National innovation and research platforms including the Anusandhan National Research Foundation (ANRF)-Mission for Advancement in High impact Areas (MAHA) Water Program shall be leveraged to support the development and deployment of field validated solutions for water security. The MAHA Water Program facilitates multidisciplinary and multi-institutional consortia to advance priority technologies from early stage research to field ready applications, including assessment, validation, prototyping and scalable deployment.

2.11.5. Piloting and Scaling of Innovative Technologies

Emerging technologies shall progress through structured pilot testing prior to large scale adoption. The pilot framework shall include:

i. Technical appraisal by the National and State Technical Committees;

ii. Performance monitoring under field conditions;

iii. Assessment of performance & reliability, cost effectiveness, ease of O&M and scalability and replicability.

Following successful pilots, subsequent steps shall include:

i. Development of standard implementation protocols;

ii. Preparation of O&M guidelines;

iii. Phased scaling in similar hydrogeological regions;

iv. Dissemination through National and State knowledge platforms;

v. Support for scaling through industry partnerships and manufacturing ecosystems.

Outcomes from pilot projects shall be documented and shared with the National Jal Jeevan Mission to support wider adoption, strengthen evidence-based decision making and enhance National State convergence.

2.11.6. Collaborative Research and Knowledge Platforms

A. Collaboration with Academia, Research Institutions and Line Departments

i. To strengthen research, innovation and evidence-based decision making in the rural water supply sector, the Mission shall establish and institutionalize collaborations with IITs, NITs, CSIR laboratories leading Central and State universities and international academic institutions to support the development, field validation and standardization of affordable and low-cost technologies for the rural water sector;

ii. The NJJM shall undertake collaborative research programmes in convergence with the Department of Science and Technology (DST), Indian Space Research Organization (ISRO), Ministry of Earth Sciences and the Ministry of Housing and Urban Affairs (MoHUA) and other allied departments to leverage their specialized domain expertise;

iii. SPM-NIWAS shall collaborate with industry bodies such as FICCI and CII to leverage their water sector platforms and initiatives including the FICCI Water Mission, India Industry Water Conclave & awards and programs of the CII Triveni Water Institute to promote technology innovation, technology transfer and knowledge sharing in the water sector.

B. JJM – Professor Chair

JJM Professor Chair established by the DDWS shall be engaged to strengthen the research, innovation and knowledge ecosystem in the rural water sector.

Performance of DDWS-supported Chair Professors shall be linked to measurable deliverables including applied research studies, technology validation pilots, policy inputs, State technical engagements and knowledge repository contributions. Disbursement and continuation of support shall be contingent upon achievement of annually defined Disbursement Linked Indicators (DLIs) verified by SPM-NIWAS.

C. International Collaborations on Global exchange of Technologies

International collaboration under the Jal Jeevan Mission shall act as a strategic driver for improving service delivery, sustainability and innovation in the rural water supply sector. These partnerships shall strengthen knowledge exchange, promote global good practices and provide access to emerging technologies while aligning with India's decentralized and community led implementation model.

i. Priority Areas for Collaboration:

a. Access to good global practices, structured knowledge exchange and emerging technologies;

b. Technical and advisory support from multilateral institutions, partner countries, research bodies and development agencies;

c. Focus on water quality surveillance, digital monitoring systems, climate resilient infrastructure and utility-oriented service delivery models;

d. Use of platforms such as Indian Technical and Economic Cooperation (ITEC) and other cooperation frameworks for capacity building, knowledge management and international exchange;

e. Engagement with countries experienced in community-based management, greywater reuse, private sector participation and performance based O&M systems;

f. Sharing JJM's scalable practices: community-led planning, tech-enabled digital monitoring (Sujalam Bharat), water quality monitoring and surveillance through community testing and source sustainability convergence.

D. Knowledge Engagement Framework

SPM-NIWAS shall develop an annual calendar of conferences, workshops and knowledge exchange platforms focused on water technologies and R&D aligned with priority issues identified by States/ UTs. The activities shall include:

i. **National and State Knowledge Platforms:** Structured knowledge exchange and sector learning shall be facilitated by SPM-NIWAS through periodic knowledge engagement platforms. An annual framework shall include the following activities;

Table 8: Disbursement Linked indicators and Performance Deiverables for JJM Professor Chair				
DLI No.	**Activity / Deliverable**	**Annual Target per Chair**	**Verification Protocol**	**Disbursement Trigger**
DLI-1	Applied research studies addressing operational challenges in rural water supply	Minimum one applied to research study annually in the identified thematic area	Study report submitted and reviewed by National Technical committee	Acceptance of research report
DLI-2	Technology pilots / demonstrative models / field validation studies	At least one field pilot or prototype validation per year	Field validation report, supporting documentation, State certification	Completion of pilot activities
DLI-3	Technical notes, guidance documents or policy inputs	Minimum one technical guidance document annually	Document endorsed by DDWS/SPM-NIWAS	Publication or adoption in sector advisory
DLI-4	State/ UT technical engagement	Engagement with minimum two States/ UTs annually through workshops or advisory sessions	Submission of engagement summaries	Completion of engagements
DLI-5	Policy or programme integration	At least one research output is integrated into advisory notes or pilot programmes	Certification by DDWS/SPM-NIWAS	Adoption into programme implementation

ii. **State level thematic workshop: These workshops shall be convened periodically, focusing on:**

a. Low-cost treatment technologies including water quality monitoring and surveillance;

b. Source sustainability;

c. Digital monitoring systems;

d. Climate-resilient infrastructure;

e. Energy efficiency in water supply.

iii. **Stakeholder Consultation with Industry and Startups:** Structured engagement with water tech startups, relevant departments, industry bodies such as DPIIT, FICCI and CII and technology providers shall support innovation promotion, technology transfer, standardization and scaling opportunities. The objective of the consultation shall be:

a. Promote technology transfer;

b. Encourage indigenous manufacturing;

c. Identify scaling pathways for innovation.

iv. **Knowledge Exchange Forums:** Collaboration with IITs, NITs, CSIR laboratories, JJM Professor Chairs and other National and International research institutions shall enable the exchange of R&D outcomes, emerging technologies and applied research relevant for field adoption. Knowledge exchange forums shall further strengthen cross learning and dissemination of sector wide experiences. The objectives include:

a. Disseminating research outcomes;

b. Sharing implementation experiences;

c. Facilitating Cross State learning.

v. **Innovation Grand Challenge:** Launch of problem statements in coordination with NJJM, academia and States/ UTs to support development, prototyping and field validation of innovative and cost-effective solutions. The objective of the Grand challenge shall be:

a. Identify solutions to specific operational challenges;

b. Support prototype development;

c. Enable field testing of innovative technologies.

vi. **India International WASH Conference:** India International WASH Conference shall serve as a National platform for knowledge exchange and sectoral innovation. The conference shall facilitate:

a. Presentation of research papers;

b. Technology demonstrations;

c. Global knowledge exchange;

d. Dissemination of innovation compendiums.

E. Knowledge Management and Dissemination

A National knowledge repository shall be maintained to systematically document and disseminate sector knowledge, innovations and implementation experiences. The repository shall include:

i. Technologies and innovations relevant to rural water supply;

ii. Outcomes of pilot projects and demonstration initiatives;

iii. Case studies and best practices from States/ UTs;

iv. Technical standards, guidelines and reference materials.

2.11.7. Cross Learning and Field Assessment

The Mission shall promote continuous cross-learning and robust field assessments by leveraging the diverse expertise of academic programs, National WASH partner networks and specialized line departments.

A. Jal Seva Adhyayan Programme

To strengthen evidence-based improvements in service delivery, the Jal Seva Adhyayan initiative shall be undertaken through AICTE impaneled engineering institutions to support field assessment and experiential learning.

i. Key Features of Jal Seva Adhyayan:

a. This shall be conducted with AICTE empaneled engineering institutions to utilize academic and technical expertise for field-based learning and assessments;

b. At least two field assessment cycles shall be conducted each year to enable regular learning and feedback from implementation sites;

c. Faculty led student teams shall be undertaken structured technical field studies as part of experiential learning and sector knowledge generation.

ii. Areas of Assessment:

a. Scheme functionality: Assessments shall examine functionality, performance and technologies used in drinking water supply schemes;

b. Water quality management: Reviews shall cover monitoring, treatment processes and compliance with water quality standards;

c. Source sustainability: The studies shall examine measures taken to ensure long-term sustainability of water sources;

d. Operation and maintenance practices: The studies shall review existing O&M practices to identify gaps and improvement areas.

iii. **Findings of the Jal Seva Adhyayan shall be utilized to:**

a. Improve design protocols: Insights generated through the program shall be leveraged to inform improvement and refinement of technical designs and planning approaches;

b. Strengthen O&M frameworks: It shall be used to support strengthening of operation and maintenance frameworks and institutional arrangements;

c. Identify replicable best practices: The program shall facilitate identification and documentation of good and replicable practices for wider dissemination.

B. Rural WASH Partners Forum (RWPF)

RWPF shall be leveraged as a National knowledge and collaboration platform for:

i. **Promoting Innovation:** RWPF shall facilitate exchange of experiences and ideas to support development, piloting and adoption of innovative and context appropriate solutions in the rural WASH sector;

ii. **Development of knowledge products:** RWPF shall support preparation and dissemination of guidance documents, technical notes, case studies and other knowledge products;

iii. **Capacity building of sector institutions:** RWPF shall enable knowledge sharing, peer learning and exposure to good practices aimed at strengthening institutional capacities at State and local levels;

iv. **Dissemination of evidence based practices:** RWPF shall support sharing of research findings, field evidence and documented good practices to inform policy, program design and implementation under the Mission.

C. Cross learning in convergence with line Department and other Ministries

i. **Ministry of Housing and Urban Affairs (MoHUA) (Atal Mission for Rejuvenation and Urban Transformation (AMRUT) and Smart City Mission):** Use of GIS mapping, IoT-based monitoring and data driven for smart water management;

ii. **Department of Water Resources, River Development & Ganga Rejuvenation**: Sustainable water use, conservation, surface and groundwater planning and management, including river basin organizations and Water Resources Information System (WRIS);

iii. **Ministry of Science and Technology**: Advanced water treatment technologies like membrane filtration and oxidation processes;

iv. **Ministry of Electronics & IT (MeitY)**: Digital tools, real time analytics and IoT-based water monitoring;

v. **Ministry of Environment, Forest and Climate Change (MoEFCC)**: Sustainable watershed management and climate resilient infrastructure;

vi. **State Water Departments and Panchayats:** Localized water conservation and decentralized management models.

2.11.8. Outcome-based Financing for R&D Proposals

Outcome-Based Financing (OBF) shall be adopted to link financial support for selected R&D projects from States/ UTs with the achievement of defined milestones and measurable results. This approach encourages structured progression from concept development to field validation and ensures that technologies demonstrate verifiable performance.

i. **Key Principles**: Funding for research and innovation projects shall be linked to the achievement of clearly defined outcomes and milestones. This principle emphasizes measurable progress, encourages efficient use of resources and aligns financial support with demonstrated results;

ii. **Indicative Financing Structure:** The State/ UT shall earmark support funds for research and piloting initiatives in the rural water supply sector with approval, oversight and tranche release undertaken at the State level, including clearance by the State Technical Committee. Such R&D proposals shall be integrated into the State/ UT Annual Action Plan (AAP) to enable structured planning, financial provisioning and coordinated implementation and shall align with the Comprehensive Implementation and Reforms Plan (CIRP) which serves as a strategic, time bound roadmap for the Jal Jeevan Mission by outlining utilization of earmarked Central and State funds and key governance, service delivery, sustainability and implementation milestones and deliverables with adherence linked to the release of payments.

a. Proposal approval: Initial seed grant (up to 30%);

b. Prototype development: Milestone based tranche;

c. Field validation: Disbursement linked to pilot performance;

d. Independent evaluation: Performance verification by premier technical institutions such as IITs, NITs, CSIR labs and/ or any other institutes as recommended by NTC;

e. Scaling recommendation: Final tranche upon demonstrated impact.

iii. **Outcome indicators:** R&D proposals shall be assessed based on clearly defined outcome indicators that demonstrate measurable improvements in cost efficiency, service quality and system performance. Outcome measurement shall be benchmarked against baseline values established at the start of each R&D project to enable consistent and transparent assessment of progress. These indicators shall include:

a. Reduction in capital or O&M costs: Extent to which the solution lowers infrastructure, operational or lifecycle expenditure;

b. Improvement in water quality compliance: Ability to enhance adherence to drinking water standards through better treatment, monitoring or contamination control;

c. Enhancement in energy efficiency: Reduction in energy consumption in pumping, treatment or distribution systems;

d. Improvement in system reliability: Contribution to consistent service delivery and reduced operational disruptions;

e. Reduction in non-functional schemes: Impact on restoring or sustaining functionality of rural water supply systems;

f. Replicability across States/ UTs: Suitability of the solution for adoption in diverse hydrogeological and operational contexts.

iv. **Monitoring of R&D Projects:** The R&D portal shall be utilized as a central platform for monitoring the progress of approved R&D proposals. It shall enable systematic tracking of milestones, documentation of field progress, submission of periodic updates and review of technical outputs. The portal shall maintain digital records of evidence, pilot data, verification reports and audit trails to ensure transparency;

v. **Verification Mechanism:** Outcomes shall be validated through evidence-based processes to ensure transparency, technical robustness and credibility of results. It shall include:

a. Independent technical evaluation: External assessment of the technology's performance, design adequacy and suitability for rural water supply conditions;

b. Pilot performance data: Analysis of quantitative and qualitative data generated during pilot implementation to assess operational performance under field conditions;

c. Field validation reports: Structured on-site assessments documenting functionality, user experience, efficiency and overall system performance;

d. Certification by the State Technical Committee: Formal technical review and endorsement confirming alignment with sector standards and readiness for wider adoption.

2.11.9. Key Action Points for the States/ UTs

i. Notification of the STC with clear TORs for appraisal, pilot approvals and scale up recommendations aligned to NTC;

ii. STC shall notify with approved Terms of Reference aligned to NTC;

iii. Launch of structured program for State led R&D, product development and pilot implementation using an Outcome Based Financing (OBF) approach;

iv. Call for Research Proposal Outcome Based Financing (OBF) approach, covering concept development, prototyping and field validation;

v. Approval of Research Proposal;

vi. Verification, standardization and phased scaling of the developed product across suitable hydro geological regions, based on independent evaluations and approvals from the State Technical Committee;

vii. Prototype development and piloting;

viii. Independent evaluation report completed;

ix. STC approval/ certification;

x. Documentation including Technology Protocols and SoPs;

xi. Scale up plan implemented in at least one cluster.

The implementation of research, innovation and knowledge ecosystems shall be done as per SOPs issued by the Department.

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SECTION 3

Programme Implementation

CHAPTER 3.1

Memorandum of Understanding (MoU) between DDWS and States/ UTs

For effective implementation of JJM 2.0, it has been envisaged that separate MoUs will be signed with States/ UTs, to ensure sustainability and citizen-centric water service delivery. The MoU formalizes the partnership between the Government of India and the State/ UT Governments.

3.1.1. Objective

The MoU aims to ensure effective implementation and robust monitoring of works undertaken under JJM, including qualitative and timely completion of schemes; proper commissioning and handover of in-village piped water supply infrastructure to GPs or VWSCs; sustained functionality and efficient operation of schemes; and adoption of the Drinking Water Sustainability Framework to drive structural reforms at the GP, District and State levels. The MoU envisages achievement of Har Ghar Jal certification of every Gram Panchayat by December 2028. The following are the components of MoU:

3.1.2. Roles and Responsibilities of DDWS

i. Issue operational guidelines for JJM 2.0 and such other directions/ instructions as may be required from time to time for achieving the objectives of the Mission;

ii. Facilitate scheme wise release of Central Grants-in-aid to States/ UTs subject to timely fulfilment of eligibility criteria;

iii. Monitor and review the progress of works under the Mission and extend requisite support to States/ UTs, wherever required.

3.1.3. Roles and Responsibilities of States/ UTs

A. Policy, Institutional and Governance Reforms

i. Notify and operationalise a structured Operations & Maintenance (O&M) policy covering the nineteen (19) essential components as outlined by DDWS to strengthen long-term drinking water sustainability;

ii. Bring in structural reforms in the rural water supply sector by making suitable statutory and regulatory provisions for development of a dual utility framework comprising:

a. Micro utilities at the village level;

b. Macro utilities at the State/ Regional level. The in-village water supply infrastructure shall be owned and managed by the GP, which shall designate its standing committee, i.e., VWSC, to function as the micro service utility responsible for distribution, O&M and service delivery within the village. State Governments shall make enabling provisions, as required, under the respective Panchayat Acts or relevant regulations to operationalise such village-level utilities.

iii. Enact or suitably amend applicable legal, regulatory and administrative provisions to:

a. protect drinking water sources from contamination;

b. safeguard rural piped water supply infrastructure from damage or misuse; and

c. ensure provision of assured drinking water services through designated service delivery utilities, while prescribing norms to ensure cost-efficient and sustainable operations.

B. Asset Ownership, Management and Technical Support

i. Adhere to the standard protocols for commissioning and handing over as issued by DDWS for rural water supply schemes. As per these protocols, infrastructure for out-village and in-village infrastructure created by the executing agency shall be formally handed over to the State and Gram Panchayat respectively, after completion of Sujal Gaon ID creation, successful minimum 15 days trial run and orientation of designated village-level utilities for operations and maintenance;

ii. Establish District Technical Unit (DTU) as technical committee of DWSM to provide continuous

technical support to village-level utilities, including engineering guidance, water quality oversight, digital monitoring support and coordination with macro utilities responsible for bulk water transfer systems.

C. Planning, Financing and Programme Implementation

i. Prepare a Comprehensive Implementation and Reform Plan (CIRP), including scheme-wise allocation of financial resources, to ensure timely completion in the extended mission period, functionality and sustainability of rural water supply systems, with the objective of achieving Har Ghar Jal certification across all Gram Panchayats;

ii. Ensure that duplication of infrastructure is avoided and that conjunctive or augmentation schemes are undertaken, only wherever required, on a reimbursement basis, adhering to principles of financial prudence, optimal resource utilization and cost-efficient design of water supply schemes;

iii. Certification by State/ UT of compliance to Central Public Health & Environmental Engineering Organisation (CPHEEO) manuals, guidelines, design norms and standards in the planning and design of water supply systems to ensure cost-efficient infrastructure development, along with any technical advisory issued by NJJM thereby, optimising capital expenditure (CAPEX) and taking up apportionment of cost wherever required with the approval of Apex Committee as per recommendations of NJJM and thus reducing long-term O&M (OPEX) costs. This is to be completed by August 2026;

iv. Undertake a comprehensive assessment of financial requirements for the O&M phase of rural water supply systems and ensure long-term financial sustainability through appropriate budgeting of revenue and expenditure at State, District and GP levels. States shall promote efficient financial management practices for village-level utilities. Any financial support if extended by the State to utilities shall be linked to measurable operational efficiency parameters, including functionality, service continuity, energy efficiency and improved cost recovery, to incentivise sustainable and accountable service delivery;

v. Undertake an activity-wise reconciliation of funds utilised during the first phase of the Mission under the Support and Water Quality Monitoring & Surveillance (WQM&S) sub-components to ensure transparency, accuracy of accounts and alignment with intended objectives. Based on such reconciliation, States shall prepare a utilisation plan for the pending funds available under these components. Under the CIRP, future disbursements under the Support and WQM&S components will be directly linked to the reforms planned and undertaken by the States. The CIRP will therefore serve as the basis for assessing the implementation of reform commitments and for determining eligibility for subsequent allocations under these components;

vi. States to prepare Drinking Water Sustainability Framework for long-term scheme sustainability with four pillars as indicated below:

a. Institutional and Service Delivery Sustainability: Enable water supply systems to operate professionally, accountable governance and assured standards;

b. Source Sustainability: Securing and protecting water resources year-round;

c. Financial Sustainability: Transparent revenue recovery models with community ownership and participation; benchmarking operational efficiencies of service utilities to reduce O&M cost;

d. Infrastructure Sustainability: Systematic asset registries, preventive maintenance and lifecycle efficiency.

D. Digital Systems, Data Governance and Monitoring

i. Map all the schemes and associated assets on the Sujalam Bharat - Digital Public Infrastructure (DPI) platform with assignment of Sujal Gaon IDs to ensure transparency and accountability. Further, all scheme assets shall be geo-referenced on Sujalam Bharat PM Gati-Shakti App, integrated with the Sujalam Bharat Module, to enable unified asset visibility and planning;

ii. Establish and operationalise State and District Command & Support Centres, supported by interactive dashboards at State, District and Gram Panchayat levels. Progress and functionality information shall be made available on GP dashboards through the JJM IMIS, with a citizen-facing interface provided through the Meri Panchayat Application;

iii. Institutionalise an integrated functionality assessment mechanism through Jal Seva Aankalan, to be undertaken at the Gram Sabha level, followed by preparation of Service Improvement Plans (SIPs) at District and State levels. Assessment outcomes and corresponding SIPs shall be periodically reviewed and placed in the public domain to strengthen transparency and continuous service improvement.

E. Skilling of Manpower

i. Formulate and notify appropriate policy and guidelines for engagement of human resources at rural water service utility levels, particularly for GPs and their designated micro utilities (VWSCs) under State Panchayati Raj Act. Such policies shall clearly define modalities of engagement, roles and responsibilities, tenure, remuneration norms and safeguards to ensure operational efficiency without creating unintended legal or employment liabilities;

ii. Adopt and implement the Nal Jal Mitra/ Jal Vitaran Sanchaalak training programme aligned with National Skills Qualifications Framework (NSQF) Level–4 for village-level water supply operators. Such training and certification shall be undertaken through Government ITIs, Polytechnics and other approved government technical institutions to ensure standardized skilling, certification and quality assurance;

iii. Undertake a comprehensive assessment of human resource requirements across operational, supervisory and managerial levels for sustainable water supply management and develop a structured State-wide skill development framework. This shall include integration of regular training modules within technical institutions and continuous capacity enhancement of personnel engaged in planning, operations, monitoring and management of rural water supply systems.

F. Water Quality Surveillance

i. Establish a water quality testing and surveillance system, with tagging of water samples collected for testing to household locations, to enable effective monitoring of remedial actions and to ensure long-term sustainability of schemes, supported by adequate financial provisions and strengthened institutional and human resource arrangements;

ii. Build public confidence by strengthening responsive mechanism on citizen corner (WQMIS) and integrating it with Meri Panchayat app/ any citizen interface app for displaying quality and testing reports;

iii. Ensure convergence with the State Rural Livelihood Mission (SRLM) to form and strengthen village-level Self Help Groups for periodic water quality testing of drinking water sources and household supply using FTKs and for reporting results on designated digital platforms for timely action; and coordinate with VWSCs, Gram Panchayats and technical functionaries for prompt remedial measures in cases of contamination or service disruptions.

G. Source Sustainability and Water Security

i. Ensure that water conservation works planned and implemented are reflected in the District Action Plan and further consolidated into the Annual Action Plan for source sustainability;

ii. Undertake village-level water budgeting and source sustainability planning, converge water conservation initiatives and make provisions to prioritise and allocate adequate water for drinking water requirements;

iii. Operationalise the Decision Support System (DSS) platform developed by DDWS as a digital planning platform at District and GP levels for source sustainability, integrated with Sujalam Bharat and National water datasets.

H. Communication and Public Confidence

i. Endeavour to build upon traditional wisdom and cultural ethos by engaging communities and individuals through campaigns such as Jal Utsav by assimilating local cultural significance of water;

ii. Ensure conduct of Jal Arpan for completed piped water supply schemes to formally hand over piped water supply infrastructure to Gram Panchayats/ VWSCs and the community;

iii. Ensure community ownership of the system from the date of Jal Arpan, with GPs/ VWSCs responsible for day-to-day operation and maintenance, supported by the State/ UT through continued technical guidance and capacity building;

iv. To bring transparency and accountability during the implementation of schemes, the scheme-wise progress to be reflected on GP dashboards (JJM-IMIS) and information to be provided to citizens through Meri panchayat application/ any citizen interface application;

vi. Undertake Jal Seva Aankalan at GP level to record the responses in service delivery and reflecting the outcomes to citizens through Meri panchayat application/ any citizen interface application;

vii. Establish citizen-centric and feedback-enabled grievance redressal mechanisms at State, District and GP level.

I. Adherence to Guidelines

The State/ UT Government shall adhere to and comply with the operational guidelines of JJM 2.0.

3.1.4. Finance and Costs

i. The financial provisions and conditions stipulated in the operational guidelines of JJM 2.0 shall be adhered to;

ii. No cost and time overrun shall be supported under JJM.

3.1.5. Review

The mid-term evaluation of the commitments under the MoU along with physical and financial progress of the schemes shall be undertaken in mid-2027 and the outcome shall be placed before the competent authorities for information and further directions.

3.1.6. Miscellaneous

i. In case of any dispute or claim relating to any matter connected with the MoU or any activity envisaged therein, including its execution, interpretation, performance or non-performance, the decision of DDWS, Government of India shall be final and binding;

ii. This MoU shall be construed and governed in accordance with the laws of India.

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CHAPTER 3.2

Comprehensive Implementation and Reform Plan (CIRP)

3.2.1. Comprehensive Implementation and Reform Plan (CIRP) Framework

Comprehensive Implementation and Reform Plan (CIRP) is a structured and integrated framework to be prepared by each State/ UT for effective planning and execution under JJM 2.0.

The CIRP shall comprise of the following two components:

i. **Scheme Implementation Matrix:** Detailing timelines and milestones for all approved and ongoing schemes to ensure their timely completion;

ii. **Reform Implementation Matrix:** Outlining reform actions, milestones and corresponding timelines linked to sectoral improvements and institutional strengthening.

CIRP shall serve as the principal guiding framework for mission planning, prioritisation, implementation, fund allocation and monitoring. It shall facilitate the timely completion of all approved and ongoing schemes, including retrofitted schemes, while ensuring alignment with the broader structural reform objectives of JJM 2.0.

Under this framework, States/ UTs shall be accountable for:

i. Achieving Har Ghar Jal certification of all Gram Panchayats through completion of all approved and ongoing schemes, including retrofitted schemes;

ii. Ensuring functionality and long-term sustainability of the created water supply infrastructure;

iii. Establishing and operationalising robust Operation & Maintenance (O&M) systems.

Accordingly, the CIRP shall function as a comprehensive roadmap for both infrastructure delivery and sectoral reforms, ensuring sustainable and outcome-oriented implementation of JJM.

3.2.2. Scheme Implementation Matrix

The Scheme Implementation Matrix shall categorize schemes into following categories:

A. Scheme Implementation Matrix Framework

i. This framework shall outline the utilization plan for the allocated Central share, along with the corresponding State-level financial planning to meet the balance funding requirements, ensuring alignment between financial provisioning and implementation priorities;

Table 9: SVS, Non-PWS Schemes (community access points/stand posts) and MVS(Q&Q) - Upfront payment mode

Scheme Type	Financial Progress	Target Completion Date	Commissioning of scheme	Handing Over of scheme	Physical Completion Deadline (As per NJJM)	Remarks
SVS	0–25%	April 2026 +11 months	[auto] + 1 month	[auto] +3 month	Mar-27	• If final deadline needs to be exceeded, it will require the approval of NJJM. • After final deadline the payment will not be released
	25–50%	April 2026 +11 months	[auto] + 1 month	[auto] +3 month	Mar-27	
	50–75%	April 2026 +9 months	[auto] + 1 month	[auto] +3 month	Jan-27	

Table 9: SVS, Non-PWS Schemes (community access points/stand posts) and MVS(Q&Q) - Upfront payment mode						
Scheme Type	**Financial Progress**	**Target Completion Date**	**Commissioning of scheme**	**Handing Over of scheme**	**Physical Completion Deadline (As per NJJM)**	**Remarks**
	75–100%	April 2026 +6 months	[auto] + 1 month	[auto] +3 month	Oct-26	
	Work order not awarded	April 2026 +18 months	[auto] + 1 month	[auto] +3 month	Oct-27	
Non PWS	0-100%	April 2026 +6 months	[auto] + 1 month	[auto] +3 month	Oct-26	• After final deadline the payment will not be released
MVS In Q&Q Affected areas	0–25%	April 2026 +20 months	[auto] + 1 month	[auto] +3 month	Dec-27	• Water Q&Q Affected areas as per JJM IMIS; if different from JJM records then, NJJM approval to be obtained. • If final deadline needs to be exceeded, it will require the approval of NJJM.
	25–50%	April 2026 +20 months	[auto] + 1 month	[auto] +3 month	Dec-27	
	50–75%	April 2026 +18 months	[auto] + 1 month	[auto] +3 month	Oct-27	
	75–100%	April 2026 +11 months	*[auto]* + 1 month	*[auto]* +3 month	Mar-27	

Table 10: MVS & BWS– Reimbursement Payment Mode						
Scheme Type	**Financial Progress**	**Target Completion Date**	**Commissioning of scheme**	**Handing Over of scheme**	**Physical Completion Deadline (As per NJJM)**	**Remarks**
MVS	0–25%	April 2026 +26 months	*[auto]* + 1 month	*[auto]* +3 month	Jun-28	• If final deadline needs to be exceeded, it will require the approval of NJJM. • After final deadline the payment will not be released. • In case PPP mode is adopted; VGF payment will be released and same shall be flagged to NJJM.
	25–50%	April 2026 +23 months	*[auto]* + 1 month	*[auto]* +3 month	Mar-28	
	50–75%	April 2026 +20 months	*[auto]* + 1 month	*[auto]* +3 month	Dec-27	
	75–100%	April 2026 +11 months	*[auto]* + 1 month	*[auto]* +3 month	Mar-27	
	Work not Awarded	April 2026 +29 months	*[auto]* + 1 month	*[auto]* +3 month	Sep-28	
BWS	0–25%	April 2026 +26 months	*[auto]* + 1 month	*[auto]* +3 month	Jun-28	
	25–50%	April 2026 +23 months	*[auto]* + 1 month	*[auto]* +3 month	Mar-28	
	50–75%	April 2026 +20 months	*[auto]* + 1 month	*[auto]* +3 month	Dec-27	
	75–100%	April 2026 +11 months	*[auto]* + 1 month	*[auto]* +3 month	Mar-27	
	Work not Awarded	April 2026 +29 months	*[auto]* + 1 month	*[auto]* +3 month	Sep-28	

Table 11- All the Retrofitting Schemes/ PM-JANMAN and DA-JGUA						
Scheme Type	**Financial Progress**	**Target Completion Date**	**Commissioning of scheme**	**Handing Over of scheme**	**Physical Completion Deadline (As per NJJM)**	**Remarks**
Retrofitted Schemes	0-100%	April 2026 +11 months	*[auto]* + 1 month	*[auto]* +3 month	Mar-27	• The State shall complete these schemes from its state budget.
New Schemes for PM-JANMAN/ DA-JGUA Villages	New scheme	-	-	-	Mar-28	• Schemes shall be taken up based on the recommendations of the Committee under the co-chairpersonship of Secretary, DDWS and Secretary, MoTA.

Note:

- *Total completion is estimated based on the percentage of financial progress entered in JJM-IMIS.*
- *A period of one (1) month for commissioning, followed by two (2) months for completion of the handing-over process after the physical completion deadline will be entered automatically.*

ii. A dedicated CIRP module shall be operationalised on the JJM Portal for the preparation, submission and monitoring of CIRP. States/ UTs shall project completion timelines for all approved and ongoing schemes, ensuring that timelines are defined at the individual scheme level. Further, States/ UTs shall finalize scheme-specific timelines aligned with the overall Mission period;

iii. States/ UTs shall ensure timely and accurate updation of physical and financial progress on Project Monitoring and SNA SPARSH payment module. The data captured shall form the basis for periodic review, performance assessment and decision-making at the National level;

iv. The implementation matrix developed on a scheme-wise, basis shall reflect timelines for achievement of Har Ghar Jal status in accordance with the completion schedules finalised by the States/ UTs;

v. To enhance transparency and citizen engagement, scheme-wise physical and financial progress captured through the SNA financial platform and the project monitoring module shall be made available at the GP dashboard. The dashboard shall leverage the linkage between Sujal Gaon ID and Sujalam Bharat ID to present service delivery status alongside infrastructure progress. The GP dashboard shall be integrated with the Meri Panchayat App to enable Gram Panchayats and citizens to view and track scheme progress;

vi. The CIRP shall also include details of state budgetary provisions for retrofitting and other state-funded schemes, ensuring full visibility of the financial commitments made by States/ UTs for sustaining and augmenting rural drinking water supply infrastructure. This information shall form an integral part of financial planning under the CIRP to ensure coherence between centrally supported activities and State-financed interventions;

vii. A mid-term evaluation of the programme shall be undertaken in 2027 to assess physical and financial progress of schemes against the CIRP and to facilitate course correction, wherever required.

3.2.3. Reform Implementation Matrix

Each State/ UT shall prepare a detailed reform implementation matrix as an integral part of the CIRP, outlining actionable components under each of the eleven reform areas, along with clearly defined

milestones, measurable indicators, time-bound targets and identification of responsible institutions at the State, District and Local levels. The reform implementation matrix shall be closely mapped with the scheme-wise implementation matrix to ensure that reform actions progress in parallel with infrastructure development, avoiding any gap between asset creation and institutional readiness.

Table 12: Reform Matrix

S.No.	Reform	Measuring Evidence & Indicator	Broad Timeline by NJJM
Chapter 2.1: Drinking Water Governance			
2.1.1	State/ UT O&M Policy with incorporation of all the 19 points	• Gazette notification of O&M Policy	Jun-26
2.1.2	Drinking water allocation for assured water supply in all the surface water- based schemes	• State government order for drinking water allocation under relevant State policy	Jun-26
2.1.3	Inclusion of Drinking water services in Public Service Guarantee Act/ Citizen Charter	• State Government Gazette notification and amended provisions in Public Service Guarantee Act/ Citizen Charter	Jun-26
2.1.4	District Technical Units (DTUs) Established	• Order issued by government with details of its staff deployed with roles and responsibilities	Jun-26
2.1.5	VWSC recognized as statutory standing committee of GP	• State Government Gazette notification along with relevant provisions incorporated under State Panchayati Raj Act	Aug-26
Chapter 2.2: Service Utility Framework			
2.2.1	Formation of State/ Regional-level Parastatal body (macro utility) constituted for out-village drinking water supply operation and maintenance Note: Applicable to States where BWS and high number of MVS are planned	• Gazette notification of formation of Parastatal body	Feb-27
2.2.2	Performance benchmarking of Macro Utility/ Utilities for out-village drinking water supply operation and maintenance	• Notification of performance benchmark guidelines by State • Guidelines uploaded on respective State/ SWSM dashboard	Aug-26
2.2.3	Institutionalise Micro Utilities (VWSC) for in-village drinking water supply operation and maintenance	• In case of contractual arrangements for O&M, Tripartite agreement between Agency, VWSC/GP and Department • Designate VWSC as micro-utility	Sep-26

Table 12: Reform Matrix			
S.No.	**Reform**	**Measuring Evidence & Indicator**	**Broad Timeline by NJJM**
2.2.4	Performance benchmarking of Micro- Utilities (VWSC) for in-village drinking water supply operation and maintenance	• Notification of Performance benchmark guidelines by DTU • In case of O&M contract managed by an agency or Parastatal body, tripartite agreement between District, Agency/ Parastatal body and GP for in-village O&M	Jul-26
Chapter 2.3: Technical Support Framework for Gram Panchayats			
2.3.1	Adopt Commissioning & Handing Over Protocol	• State/ UT Gazette notification issued based on guidelines issued by NJJM	Aug-26
2.3.2	Quarterly TPIA Effectiveness Reviews by DWSMs	• Quarterly TPIA review reports on DWSM dashboard	Quarterly
2.3.3	Reform/ corrective action taken on advisory issued by NJJM based on field visit observations	• Adoption/ advisory issued by State/District based on the NJJM advisory • Advisory and ATRs uploaded on DWSM dashboard	Quarterly
2.3.4	Reform/ corrective action taken on advisory issued by NJJM based on the observations under Jal Seva Adhyayan	• Adoption/ advisory issued by State/District based on the NJJM advisory • Advisory and ATRs uploaded on DWSM dashboard	Bi-annual
Chapter 2.4: Establishing Citizen-Centric Water Quality Governance			
2.4.1	Community based Water Quality Testing and Surveillance	• Notification by State for WQM&S protocols/SOPs as per Concise Handbook issued by NJJM • All HGJ GPs adopting water quality testing protocol	Dec-26
2.4.2	Training of Sujalam Shakti groups (via SHGs) for train/certify 1-2 women per GP (NSQF L-1) including FTK use	• Sujalam Shakti groups (via SHGs) trained in all HGJ GPs • Training completion records on e-Gram Swaraj/ GP dashboard	Dec-26
2.4.3	ISO Accreditation of laboratories	• All WQ labs accredited (Chemical and bacteriological) • Uploading of accreditation certificates; ISO records as per WQMIS	Dec-27
Chapter 2.5: Source Sustainability & Water Security			
2.5.1	Source Sustenance Committees-SSCs constituted at State (SSSC), District (DSSCs), and GP/ Village levels (GP/VWSCs)	• Notification of SSC constitution at all three levels; along with TOR documents.	Sep-26 Phased Manner

Table 12: Reform Matrix			
S.No.	**Reform**	**Measuring Evidence & Indicator**	**Broad Timeline by NJJM**
2.5.2	District Source Sustainability Action Plans (DSSAPs)-prepared by District committee (DSSC) and approved by DWSMs	• DSSAP document approved by DWSM and uploaded on DSS.	Dec-26
2.5.3	Decision Support System (DSS)	• DTU training for adoption of DSS. • All HGJ Panchayats trained to use DSS	Oct-26
2.5.4	Village Water Budgeting- GP- level water budgeting conducted using DSS data	• Village water budgeting documents uploaded on JJM IMIS GP dashboard • DSS adoption in all HGJ GPs	Jan-27
Chapter 2.6: Digital Governance - Sujalam Bharat			
2.6.1	Sujal Gaon IDs - Created for all service areas with complete source-to-tap digital Mapping	• Creation of all Sujal Gaon IDs for HGJ GPs	Mar-27
2.6.2	GIS-Based Asset Mapping - All RPWSS assets geo-tagged through Sujalam Bharat Mobile App	• GIS-tagged asset records on Sujalam Bharat Mobile App for HGJ GPs	Mar-27
2.6.3	Sujal Gaon Utility App adopted as standard transactional platform for O&M data capture	• Sujal Gaon App adopted in all HGJ GPs	Dec-26
2.6.4	Project Monitoring Module - Physical progress monitored through Sujalam Bharat module	• Adoption of app and regular updation of physical progress on Sujalam Bharat Project Monitoring Module in HGJ GPs	Aug-26
2.6.5	District Command and Support Centre (DCSC) - Established under DWSM for real- time operational monitoring	• DCSC establishment order; operational status report • Integration of response mechanism for grievances	Dec-26
2.6.6	State Command and Support Centre (SCSC) - Established under SWSM with API-based data transfer to NJJM	• SCSC establishment order; API integration with SWSM dashboard • Integration of response mechanism for grievances	Dec-26
2.6.7	Citizen Interface for grievance redressal - Meri Panchayat App and e-GramSwaraj linked to Sujal Gaon IDs for grievance and feedback	• Integration of response mechanism for grievances	Dec-26

Table 12: Reform Matrix			
S.No.	**Reform**	**Measuring Evidence & Indicator**	**Broad Timeline by NJJM**
Chapter 2.7: Participatory Governance			
2.7.1	Public disclosure of all the important parameters of drinking water supply - Meri Panchayat App	• Public disclosure of all important parameters of drinking water institutionalized. (Circular Issued by State) • All HGJ GPs and Districts publicly display 5 parameters as per JJM guidelines: i. Water Quality Testing Status ii. Functionality Status (HH tap supply days/hours, quantity/ quality, pressure) iii. Grievance Redressal Data iv. O&M Fund Collection and Utilisation v. Source Sustainability Actions	Quarterly
2.7.2	Jal Utsav Events	• Approved annual calendars (State/ District/ GP) in AAP	Jun-26
2.7.3	GP/ Community Ownership through Jal Arpan	• Adoption of SoP issued by NJJM by States/ UTs • Issue of template of Jal Arpan by State/ District • Jal Arpan Calendar prepared for eligible villages/ GPs • Jal Arpan (handing over of schemes) conducted in HGJ GPs as per protocol	Mar-27
2.7.4	Inclusion of WASH curriculum in schools	• SCERT circular issued for implementation	Mar-27
2.7.5	Jal Seva Aankalan - JSA conducted for all HGJ villages with Service Improvement Plans (SIPs)	• All HGJ GPs completed JSA • All districts submitted SIP • State submitted their SIP • Evidence based assessment of SIP	Annual
2.7.6	Impact Assessment – HGJ	• Impact Assessment tool developed and used in HGJ GPs • State level Events organised in convergence with relevant Depts. • Impact Assessment conducted in sample Districts/GPs. • District Impact Assessment report submitted to SWSM • Reports published in public domain	Mar-27 (To be conducted annually)

Table 12 : Reform Matrix			
S.No.	**Reform**	**Measuring Evidence & Indicator**	**Broad Timeline by NJJM**
2.7.7	State Impact Assessment (IA) Report	• State Impact Assessment Report prepared • Reports published in public domain	Jun-27 (To be conducted annually)
2.7.8	Two-way Communication through Samvad Platforms	• No. of Districts covered under DC/ DM Pey Jal Samvad • No. of GPs covered under Sujal Gram Peyjal Samvad	Monthly
Chapter 2.8: Capacity Building			
2.8.1	Training of HGJ Village Field Functionaries	• Training completion of all field functionary in HGJ GPs • Training completion records on e-Gram Swaraj/ GP dashboard. • One round of minimum training to be done in all HGJ GPs by March 2027. • Gap Assessment conducted for trained field functionaries and conduct refresher training after 6 months or as decided by States/ Dist.	Mar-27
Chapter 2.9: Skilling & Human Resource Development			
2.9.1	District-wise O&M Human Resource (HR) Gap Assessment & Training Plans	• Approved district gap matrix & AAP submitted • Personnel required vs mapped (tier-wise)	Aug-26
2.9.2	Adoption and roll-out of Training Courses	• State notification for adoption of NSQF Level-4 Jal Vitaran Sanchaalak course and other NSQF aligned courses across training institutions in all the districts. • Training infrastructure and training institution related notification • Appointment of 'Programme Implementing Agencies' (PIA) by States/ Districts for planning, coordination and delivery of all training.	Jun-26
2.9.3	HR Policy notification and deployment	• State to formulate and notify appropriate policy and guidelines for engagement of human resources at rural water service utility levels • At least 1 trained and certified NJM	Dec-27

Table 12: Reform Matrix			
S.No.	**Reform**	**Measuring Evidence & Indicator**	**Broad Timeline by NJJM**
		deployed in all HGJ GP.	
Chapter 2.10: Operational & Financial Sustainability			
2.10.1	Financial Monitoring Tool for O&M of drinking water supply	• State notification for use of financial monitoring tool in all HGJ GPs.	Oct-26
2.10.2	Annual O&M Financial Provisioning	• Ensure funding gaps for O&M are addressed through State financial resources. • Provision of annual O&M funds in separate State budget head.	Dec-26
2.10.3	Preventive Maintenance (PM) Schedule for all HGJ villages	• Publish scheme-wise preventive maintenance schedule with the help of DTU.	Dec-26
2.10.4	Operational efficiency	• Develop SOPs for energy use, water production, service hours, and breakdowns.	Dec-26
Chapter 2.11: Research, Innovation & Knowledge			
2.11.1	State Technical Committee (STC)- Notified with TOR aligned to National Technical Committee	• STC notification order; TOR document	Sept-26
2.11.2	Outcome Based Financing (OBF) R&D Programme	• OBF Framework by States/ UTs in line with NJJM • State/ UT Govt. notification & Call for Proposals issued • Proposals received/ shortlisted/ approved • Budget released	Mar-27

3.2.4. CIRP: Implementation, Funding and Monitoring Framework

i. **Planning Framework and Timeline:** CIRP shall cover the entire Mission period up to December 2028 and shall form the basis for preparation of Annual Action Plans (AAPs). AAPs shall be system-generated from the CIRP and shall be used for determining State-wise financial allocations, subject to approval by DDWS;

ii. **Alignment of Implementation with CIRP:** Adherence to the CIRP shall guide overall programme implementation. States/ UTs shall ensure that scheme execution, reform actions and milestone achievements remain aligned with the timelines and targets defined in the CIRP;

iii. **Funding Linkages and Performance-Based Releases:** Release of funds and subsequent financial allocations shall be aligned with achievement of agreed timelines/ milestones, timely utilisation of funds and regular updation of progress on the digital platform. Further, fund releases may be phased, with subsequent instalments contingent upon demonstrable reform progress, physical achievements and timely reporting across all modules;

iv. **Non-Compliance and Funding Consequences:** In case of non-compliance with CIRP milestones, delayed reporting, or inconsistencies between reported and validated progress, fund releases may be withheld, deferred, or reduced until corrective actions are completed and verified. Persistent non-compliance may lead to reprioritization reallocation of funds, along with enhanced monitoring and corrective action requirements as deemed necessary by DDWS;

v. **Digital Integration and Data Architecture:** CIRP module shall be integrated with relevant digital systems to enable real-time monitoring and data-driven programme management, forming part of an interoperable digital public infrastructure for rural water service delivery. All schemes and habitations shall be mapped within this integrated architecture, wherein each habitation is assigned a Sujal Gaon ID and all assets are tagged with a Sujalam Bharat ID, ensuring end-to-end traceability;

vi. **Monitoring Framework and Review Mechanisms:** States/ UTs shall also establish a robust monitoring framework to track reform progress, scheme implementation status and Har Ghar Jal (HGJ) certification. GP and District dashboards shall serve as primary monitoring tools, providing real-time updates on habitation saturation, scheme-wise progress and reform milestones;

vii. **Validation and Verification Mechanisms:** Reported progress shall be validated through digital verification on the Project Monitoring module, periodic field inspections, third-party assessments and structured review meetings at State and National levels. All monitoring and validation activities shall be aligned with the integrated digital architecture to ensure transparency, reliability and real-time oversight;

viii. **Corrective Actions and Course Correction:** In case of deviations from planned timelines or implementation challenges, States/ UTs shall undertake corrective measures, including revision of milestones and submission of time-bound action plans in consultation with NJJM. Any gaps, delays or discrepancies identified through monitoring or validation shall be addressed within the CIRP framework through timely corrective actions;

ix. **Internal Review and Accountability:** States/ UTs shall establish internal review mechanisms to periodically track progress against the reform implementation matrix, enabling timely interventions and ensuring alignment with overall CIRP commitments.

3.2.5. Har Ghar Jal Certification

A. Process for Har Ghar Jal Certification

i. **Completion of Scheme:** A scheme shall be considered complete only after ensuring provision of FHTCs to all households, as well as to schools, Anganwadi centres, Community Health Centres, GP building/ Panchayat Ghars and other public institutions;

ii. **Commissioning of Scheme and Capacity Building of GPs/ VWSCs:** Upon completion of construction and installation, the scheme shall be taken up for commissioning. A mandatory trial run of minimum 15 days shall be conducted to test all components of the system and ensure reliable and sustained service delivery. Simultaneously, the GP and/ or its VWSC shall undergo training and capacity building for management of water supply schemes. In cases where O&M is to be handed over

to other agencies, a formal agreement to this effect shall be executed and the GP shall be provided with the details of such agencies responsible for O&M prior to handover;

iii. **Handover of Scheme:** After successful completion of the trial run, the scheme shall be formally handed over to the Gram Panchayat through a Jal Arpan ceremony. Thereafter, the responsibility for O&M of in-village infrastructure shall vest with the GP and/ or its Standing Committee, i.e. VWSC;

iv. **Gram Sabha Resolution & Certification:** Once all schemes in the village have been completed, commissioned and handed over, the GP shall convene a Gram Sabha in the presence of a DWSM representative and pass a resolution confirming that:

a. All households, public institutions and PM-JANMAN/ DA-JGUA habitations have been covered with functional tap water connections;

b. All habitations, including those with dispersed populations or less than 20 rural households forming part of a revenue village, have been covered to ensure access to safe drinking water; wherever provision of piped water supply is not techno-commercially or geographically feasible, such access has been ensured for drinking and cooking purposes (minimum 8–10 LPCD) through appropriate community-based solutions, local innovations, or alternate technological interventions, along with suitable arrangements for other domestic uses;

c. Any exception to provision of piped water supply has been limited to cases, where techno-commercial or geographical constraints have been duly assessed and appropriate alternate service levels have been ensured as per norms;

d. The Village/ Gram Panchayat qualifies as Har Ghar Jal Certified.

The copy of resolution passed by Gram Sabha and a brief video capturing the Gram Sabha proceedings shall be uploaded on JJM Dashboard and village shall be marked as certified in JJM-IMIS.

v. **Gram Panchayat Level Certification:** Where all villages within a GP are certified as Har Ghar Jal, a resolution declaring the Gram Panchayat as Har Ghar Jal Certified shall be passed.

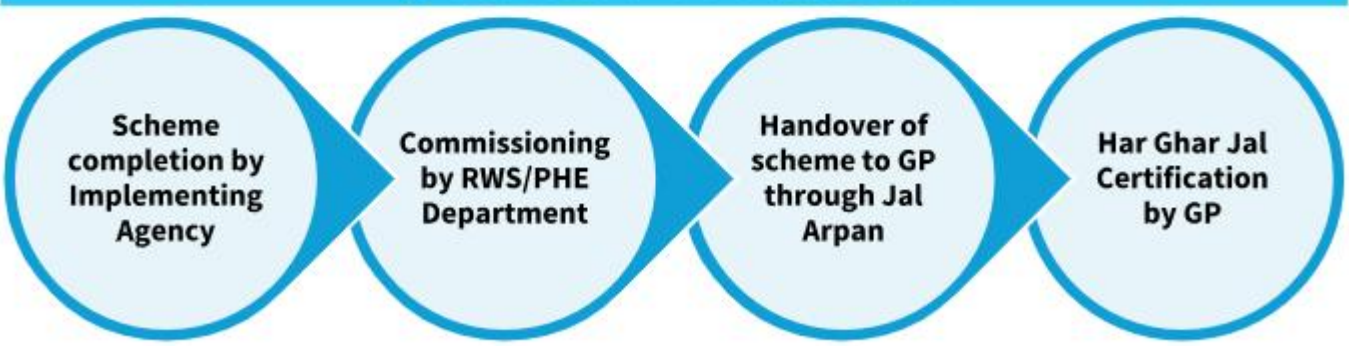
vi. **Measurement of Progress:** Progress and achievement under the Mission shall be assessed based on the number of GPs achieving Har Ghar Jal certification status.

3.2.6. Conclusion

The CIRP under JJM 2.0 shall function as the integrated platform linking reforms, infrastructure development and financing, ensuring a coherent and coordinated approach to programme implementation. Release of funds through SNA-SPARSH shall remain contingent upon measurable and verifiable progress in reform milestones and adherence to timelines defined in the CIRP.

This reform- linked financing framework will reinforce accountability and promote timely progress, ensuring that JJM 2.0 delivers sustainable, resilient and future-ready rural drinking water supply systems.

Figure 19: Process for Har Ghar Jal Certification



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CHAPTER 3.3

Financial Planning and Funding

For achieving the envisaged objectives of JJM 2.0 in a time-bound manner, it is imperative to have robust financial planning, assured funding at central and State level and adequate resource mobilization.

In this regard, the SWSM shall prepare a Comprehensive Implementation and Reform Plan (CIRP) to ensure timely completion, functionality and sustainability of rural water supply schemes, with the objective of achieving Har Ghar Jal certification across all Gram Panchayats. The CIRP may further include dedicated state funding, in addition to the prescribed matching State share, to ensure expeditious completion of priority schemes.

3.3.1. Outlay

During the previous phase of the Mission, i.e., up to March 2024, States/ UTs reported approval of more than 6.83 lakh rural water supply schemes through their respective State Level Scheme Sanctioning Committees (SLSSCs) for provision of tap water connections, with an estimated cost of ₹8.07 lakh crore (excluding inadmissible components).

In view of the financial requirements for coverage components, as well as for undertaking structural reforms through the Support Activities and Water Quality Monitoring & Surveillance (WQM&S) activities, the overall financial outlay of the Mission has been enhanced from ₹3.60 lakh crore to ₹8.69 lakh crore, with the Central share increasing from ₹2.08 lakh crore to ₹3.59 lakh crore. The details of total approved outlay are as under:

Amount in ₹ lakh crore

Component	Outlay
Coverage	8.07
PM-JANMAN, DA-JGUA	0.05
Support and WQM&S sub schemes	0.57
Total	**8.69**

3.3.2. Financial Release Strategy

The additional Central assistance of ₹1.51 lakh crore is intended to facilitate certification of all Gram Panchayats as 'Har Ghar Jal' by providing tap water connections to all 19.36 crore rural households across the country by December 2028. This shall be achieved through a calibrated financial release strategy, in accordance with the existing funding pattern, as outlined below:

i. **Priority for Completion of Ongoing and Critical Schemes:** For extending central assistance, priority shall be accorded to completion of ongoing Single Village Schemes (SVSs), schemes providing safe drinking community water access to remote and difficult areas and Multi Village Schemes (MVSs) addressing quantity and quality-affected areas. Scheme-wise payments shall be made through SNA-SPARSH;

ii. **Reimbursement-Based Support:** Ongoing MVSs, in conjunction with groundwater-based rural water supply schemes, shall be supported through central assistance on a reimbursement basis;

iii. **Promotion of PPP and Annuity-Based Models:** In continuation of the previous JJM guidelines on alternative financial models, States shall be encouraged to adopt annuity-based PPP models for the approved bulk water transfer schemes. Any Viability Gap Funding (VGF) provided by the State Government shall be eligible for central sharing in accordance with the approved Centre - State funding pattern and extant guidelines;

iv. **No Central Assistance for Retrofitting Schemes:** No Central assistance shall be provided for retrofitting schemes. Any pending liabilities under such schemes shall be met by the respective state governments from their own financial resources.

3.3.3. Financial Safeguards for Release of Central Assistance

The disbursement of Central assistance shall be contingent upon adherence to the following conditions:

i. **Scheme Validation:** All schemes proposed for central assistance shall be validated against the Sujalam Bharat GIS-linked Asset Registry to ensure uniqueness of service areas and prevent overlap or duplication;

ii. **Technical Compliance:** Certification of compliance with CPHEEO design norms by state government and any technical advisories issued by NJJM;

iii. **Phased Release Linked to Milestones:** Central assistance shall be linked to clearly defined structural reforms milestones, duly verified through JJM Portal;

iv. **Cost Rationalization/ Apportionment:** Projects costing ₹ 100 crore and above and schemes addressing urban/ sectoral water demand or higher service levels, shall be scrutinized by the CPHEEO, National Jal Jeevan Mission Joint Technical Committee. Based on its recommendations, NJJM will propose cost rationalization or apportionment to the State-level Apex Committees for concurrence prior to release of funds.

3.3.4. Allocation of Fund

A. Basis of Allocation

Allocation of central funds to States/ UTs shall be based on the approved Comprehensive Implementation & Reform Plan and Annual Action Plan, within the overall central outlay available for the respective financial year, with the approval of Hon'ble Union Minister Jal Shakti, Government of India .

B. Allocation Modalities

Based on the CIRP and AAP of the States/ UTs, the annual allocation shall be appropriately earmarked under the following distinct modalities:

i. **Upfront Releases:** For financing ongoing and approved activities in a need-based, demand-driven manner through issue of mother sanction;

ii. **Reimbursement Mode:** For reimbursement of eligible expenditure incurred by States/ UTs under Multi-Village Schemes (other than Q&Q blocks) and Bulk Water Supply Schemes, as per approved provisions;

iii. **Settlement of Central Share (Post-Mission Period):** For settlement of expenditure incurred by States/ UTs in lieu of the Central share beyond the conclusion period of JJM and up to 31st March, 2026, subject to eligibility and verification.

C. Interim Allocation Arrangement

Considering that the CIRP is an extensive and consultative exercise covering the entire Mission period under JJM 2.0 and may require adequate time for finalisation and approval, allocations for the initial period shall be made on an interim basis to ensure continuity of ongoing works and implementation momentum. For this purpose, allocation for the first quarter of the financial year 2026-27 shall be based on data available on the JJM IMIS, pending finalization of CIRP.

D. Review and Reallocation

Annual allocation shall be reviewed biannually based on the progress of States/ UTs against CIRP/ AAP targets. Based on such performance assessment, the department may propose reallocation of funds among States/ UTs to optimise utilisation and accelerate achievement of Mission objectives.

3.3.5. Modes of Payments under JJM 2.0

Central assistance under JJM 2.0 shall be extended scheme-wise through multiple funding modes, based on the nature and priority of schemes. The objective is to ensure efficient fund flow, promote accountability and align financial support with physical and reform milestones. The modes of assistance are as follows:

i. **Upfront Payment Mode (Proposed for SVSs, MVSs in quality/ quantity-affected areas and Non-PWS schemes including Support & WQM&S components):** Payments shall be processed as per the approved funding pattern, wherein Central and State shares are released simultaneously. Central assistance shall be routed through the SNA-SPARSH platform to States/ UTs. Prior to each Mother Sanction, the performance of the State/ UT against the respective CIRP shall be assessed. Mother Sanctions issued by the department shall serve as the authorized drawing limit for the States/ UTs under the Mission. For UTs such as Ladakh, Andaman & Nicobar Islands, Lakshadweep and Dadra & Nagar Haveli and Daman & Diu (DNH&DD), payments shall be made through SNA-SPARSH/ Letter of Authorization (LoA);

ii. **Reimbursement Mode (Proposed for other MVSs and BWS):** Expenditure incurred under approved components of MVS/ BWS schemes shall be released in phased instalments strictly on a reimbursement basis, with no upfront release of Central share. The eligible Central share shall be released based on actual financial expenditure and proportional physical progress, as reported and validated through the Programme Monitoring Module on the Sujalam Bharat PM Gatishakti App. The number and size of instalments shall be determined based on the scale, cost and progress of individual projects. This mode shall also be

adopted in the case of funding of Central share in PPP and Annuity-Based Models.

3.3.6. Programme Fund Utilisation

A. Coverage

i. Available central assistance may be utilized only for the schemes sanctioned by SLSSCs on or before 31st March 2024 and reported as such on JJM-IMIS. However, No Central assistance shall be available for retrofitting schemes or schemes sanctioned under NRDWP;

ii. Works approved by SLSSC during the Mission period but pending for award as on 31st March 2024 will be re-examined at the NJJM level before extending any Central assistance;

iii. Any tender premium approved for schemes during the first phase of the Mission Period (2019–2024) will be admissible only until the initial contract period. If the scheme faces a cost overrun due to extension of contract period, the central assistance will be limited to the originally estimated cost;

iv. If project costs decrease due to lower material costs, a lower discovered cost than estimated, or other unforeseen factors, the NJJM must be informed at the earliest;

v. States will utilize National and State Disaster Response Funds to restore water supply infrastructure damaged by natural calamities or internal disturbances;

vi. New schemes shall be considered only in PM-JANMAN and DA-JGUA areas that were left uncovered during the first phase, subject to submission of due justification by the States/ UTs explaining the reasons for their exclusion. For this purpose, a committee under the chairpersonship of Secretary, DDWS shall be constituted;

vii. For providing FHTCs to PMAY(G) houses the JJM funds shall be used in convergence mode. However, in such cases, tap water connections shall be provided from the existing village distribution network and no creation of new distribution network shall be allowed for this purpose.

B. Support and Water Quality Monitoring & Surveillance (WQM&S)

i. **Preconditions for Fund Release:** Release of funds under Support and WQM&S shall be subject to completion of activity-wise reconciliation of Phase I expenditure, submission of Utilisation Certificates (UCs) and audited financial statements and reporting of physical and financial progress on the JJM IMIS;

ii. **Reconciliation and Validation:** States shall undertake detailed reconciliation to verify expenditures against approved and reform-linked activities, identify unspent balances and liabilities and ensure alignment between financial reporting and physical progress reflected on the JJM IMIS. Only validated expenditure shall be considered for subsequent releases;

iii. **Utilisation of Funds and Unspent Balances:** States shall prepare a utilisation plan aligned with Section 2 reforms and CIRP-linked outcomes, prioritising critical gaps in monitoring, surveillance and system strengthening. All proposed activities shall be mapped and tracked on the JJM IMIS to enable transparency and effective oversight;

iv. **Milestone-Based and Reform-Linked Disbursement:** Expenditure under Support and WQM&S components shall be linked to achievement of defined milestones, implementation of structural reforms as outlined in Section 2 and verified progress on the JJM IMIS, ensuring outcome-oriented fund utilisation;

v. **Utilisation of Support Funds:** Support funds can be utilised for activities aligned with the reform framework under Section 2, which outlines the 11 key structural reforms with defined activities and outcome linkages under CIRP. States shall ensure that utilisation is mapped to reform-linked activities, is outcome-driven and is mandatorily reported on the JJM IMIS. Indicative areas include digital governance and monitoring systems, institutional strengthening, capacity building, IEC/ BCC and community mobilisation, research and innovation and water quality monitoring and surveillance;

vi. **Financial Management and Compliance:** States shall ensure that funds are utilised in a timely, efficient and outcome-oriented manner in alignment with approved plans and reform priorities. Financial management systems shall enable transparency, accountability and end-to-end traceability of expenditures through PFMS-compliant processes, duly aligned with data reported on the JJM IMIS. Accurate and up-to-date reporting shall support monitoring, performance assessment and informed decision-making, while ensuring optimal utilisation of available resources;

vii. **Performance-Linked Review:** Subsequent fund releases shall be based on utilisation of previous instalments within prescribed timelines, achievement of reform milestones and outcomes and verified performance data on the JJM IMIS;

viii. **Audit and Accountability:** Expenditure shall be subject to third-party verification, audit and social audit mechanisms, with cross-verification against data.

3.3.7. Reimbursement of Additional Expenditure Incurred by States

i. **Timeline for Approval of Works and Expenditure Reporting:** The approval of works under JJM was concluded up to March 2024. However, the remaining Central outlay was released up to March 2025 with the approval of the Department of Expenditure (DoE). In this context, States/ UTs shall update the expenditure incurred against all Central assistance released up to March 2025;

ii. **Recording of Additional State Expenditure:** States/ UTs shall ensure that any expenditure incurred upto March 2026, over and above the available Central share, on admissible components and along with the corresponding state share as per the approved funding pattern, is duly reflected as additional state expenditure in the Financial Reconciliation Module. This is essential to ensure accuracy, transparency and accountability prior to submission of reimbursement requests;

iii. **Submission and Prioritization of Reimbursement Claims:** After appropriate apportionment of the eligible central share against the reported expenditure, States/ UTs shall submit reimbursement requests for the admissible amount as per the prescribed procedure. Priority for reimbursement shall be accorded to single village schemes;

iv. **Digital Fund Request Mechanism for Reimbursement:** The release of funds for settlement of additional expenditure incurred by States/ UTs shall be based on a digitally generated Fund Request Form for reimbursement. This form shall serve as the primary instrument for seeking Central share and shall be integrated with the Sujalam Bharat database and the Financial Reconciliation Module to enable automated validations;

v. **Technical Compliance for Schemes Below ₹100 Crore:** For fund requests pertaining to schemes below ₹100 crore, a Technical Compliance Certificate shall be incorporated within the digital Fund Request Form, confirming adherence to prescribed technical standards and norms;

vi. **DPR Appraisal and Approval for Projects Above ₹100 Crore:** For projects above ₹100 crore, Detailed Project Reports (DPRs) shall be examined and such cases shall be recommended by NJJM to the Apex Committee of SWSM for ensuring technical compliance and apportionment, wherever applicable;

vii. **Ensuring Transparency and Accountability in Fund Release:** This integrated framework is intended to ensure end-to-end transparency, accuracy and accountability in the reimbursement and fund release process under JJM 2.0.

3.3.8. Fund Sharing Pattern

The existing Central-State fund-sharing pattern will continue under JJM 2.0 as follows:

Sub-components	Funding Pattern
Coverage	• 100:0 for UTs without legislature • 90:10 for NE & Himalayan States & UTs with legislature • 50:50 for other States
Support WQM&S	• 100:0 for UTs without legislature • 90:10 for NE & Himalayan States & UTs with legislature • 60:40 for other States

3.3.9. Other Release Conditionalities

i. **Governing Framework and Financial Compliance:** The sanction, release and utilization of funds shall be governed by the operational guidelines of JJM 2.0, Ministry of Finance (MoF) instructions and the provisions of the General Financial Rules (GFR), as amended from time to time. All claims and releases of Central grants-in-aid shall strictly adhere to these frameworks;

ii. **Phased Release under SNA-SPARSH:** Funds earmarked under SNA-SPARSH for upfront payment shall be sanctioned in two instalments of 50% each, with each instalment further released in two equal tranches. Approval of the Competent Authority and concurrence of IFD shall be obtained prior to release of each instalment;

iii. **Release of Funds under Reimbursement Mode:** Funds earmarked for reimbursement mode shall be released in accordance with a detailed Standard Operating Procedure (SoP) to be issued by the Department with the approval of the Competent Authority, in line with the prescribed procedure;

iv. **Documentation Requirements for Availing Central Assistance:** Proposals submitted by States/ UTs for availing Central assistance under JJM shall, inter alia include the following documents:

a. Provisional Utilization Certificate (UC) for the current financial year;

b. Audit Report from the Accountant General (AG) for the year preceding the previous year, or an Audit Statement of Accounts (ASA) from a CAG-empanelled Chartered Accountant (CA) for the previous year(s);

c. Final Central and State Utilization Certificates for the previous year(s), along with e-UCs submitted through PFMS.

3.3.10. SNA-SPARSH Framework and Payment Mechanism

The SNA-SPARSH system has been envisaged as an integrated financial framework connecting the Public Financial Management System (PFMS), State Integrated Financial Management Information System (IFMIS) and the Reserve Bank of India's (RBI) e-Kuber platform. The SNA-SPARSH payment mechanism enables NJJM to ensure just-in-time payments, thereby significantly reducing unspent balances to near zero and enhancing financial prudence, efficiency and transparency. Key Features of SNA-SPARSH Implementation are as under:

i. **Integration with JJM-IMIS for Scheme-wise Monitoring:** The SNA-SPARSH module of PFMS shall be integrated with JJM-IMIS to enable scheme-wise payments and facilitate real-time monitoring and reporting of financial progress;

ii. **Structured Banking Arrangements Based on Funding Pattern:** States/ UTs with different funding patterns for coverage and Support & WQM&S components shall maintain separate SLS and RBI accounts. States/ UTs with uniform funding patterns shall maintain a single SLS and RBI account for both components. The department shall operate a single RBI account for release of the central share;

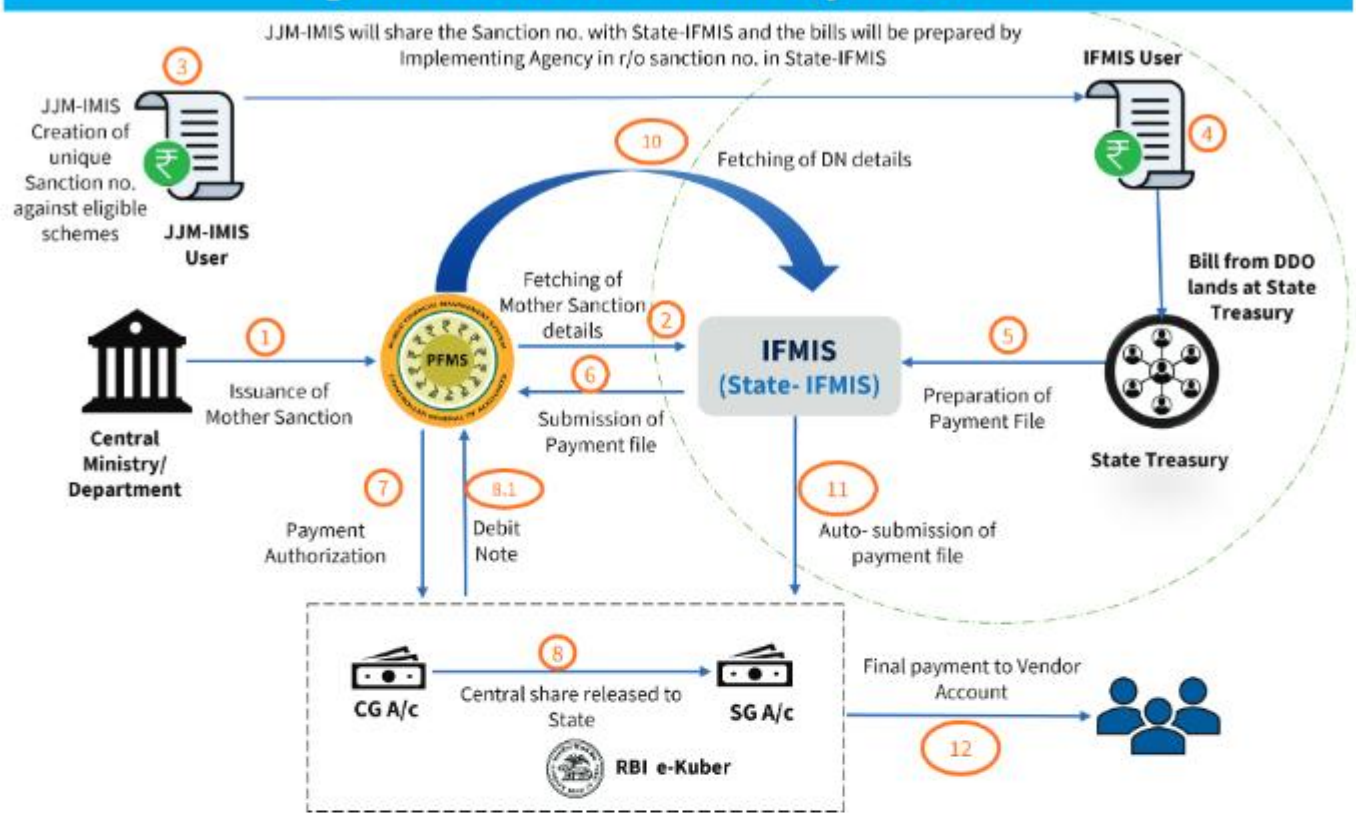
iii. **Mother Sanction and Tranche-wise Drawing Limits:** The Centre shall issue a Mother Sanction (tranche-wise) through PFMS, which shall function as a State-wise drawing limit for each tranche;

iv. **Claim Submission by Implementing Agencies:** Implementing Agencies (IAs) shall raise claims through JJM-IMIS and the respective state IFMIS platforms;

v. **Mandatory Onboarding:** States/ UTs shall ensure that all schemes are mandatorily onboarded onto the SNA-SPARSH module through the prescribed formats, by duly transitioning them from the CIRP module to the SNA-SPARSH module;

vi. **Verification Responsibility at Implementing Agency Level:** The responsibility for verification of bills, claims, invoices and supporting documents shall rest with the Implementing Agencies of the respective States/ UTs.

Figure 20: SNA-SPARSH Framework and Payment Mechanism



3.3.11. Community Contribution and Incentives under JJM

As per the provisions of the earlier JJM guidelines, community contribution towards in-village piped water supply infrastructure and associated source development, implemented by the Gram Panchayat and/ or VWSC was prescribed at 5% of the capital cost (in cash, kind, or labour) for hilly and forested areas, North Eastern and Himalayan states and villages with more than 50% SC/ ST population and 10% for all other villages.

Based on entries authenticated in the JJM-IMIS (Form 10-B), it has been observed that community contribution has not been collected in a significant number of schemes. In this regard, it has been conveyed that where a State/ UT decides to exempt the collection of community contribution, the corresponding amount shall be borne by the State/ UT from its own resources.

To encourage community ownership and ensure long-term sustainability of water supply systems, the provision for incentivizing the community shall continue to be emphasized. Post commissioning of the scheme, the community shall be rewarded in a phased manner up to 10% of the capital expenditure of the in-village water supply scheme. This amount shall be maintained as a revolving fund at the village level to meet emergency repair and maintenance needs and shall be replenished over time through service delivery charge and contributions from the local community. However, the incentives fund clause will not apply in cases where community contribution has not been received.

3.3.12. Rashtriya Jal Jeevan Kosh (RJJK)

As part of India's commitment to ensuring safe drinking water access, various individuals, corporate and industrial houses, as well as charitable institutions, contribute towards strengthening rural drinking water systems. To effectively facilitate and manage such contributions, the RJJK has been established under the NJJM.

3.3.13. Audit

i. In case of society, SWSM will ensure that the accounts are audited by a Chartered Accountant firm empaneled by the Comptroller and Auditor General within six months of the close of the financial year. The audit will be supported by a statement of reconciliation with the accounts of implementing agency and a certificate of the Chartered Accountant on its accuracy;

ii. Accounts of Gram Panchayat and/ or its standing committee, i.e. VWSC shall be operated and managed at GP level;

iii. DWSM to ensure that every account/ ledger being maintained by Gram Panchayat for receiving community contribution, incentive money etc. is audited every financial year;

iv. Concurrent auditing of GPs or it's standing committee's accounts be done regularly with the help of retired local audit officials;

v. A format for the Audit Report from States with respect to the JJM is annexed as **Annexure VII.** In addition to the Audit by the Chartered Accountant, the works under this mission would be subject to audit by the Comptroller and Auditor General of India (C&AG) and Internal Audit Wing (IAW) of the Ministry of Jal Shakti. The Audit by the C&AG may cover aspects of quality of the work done, in addition to financial audit;

vi. In case the report from Accountant General/ CA is not received because of any unforeseen reason, the release will not be withheld, if State Government/ UT Administration is able to provide specific reasons for delay and gives undertaking for furnishing the same after the receipt of the same from the office of the Accountant General/ CA. In case, in the report of AG/ CA, some discrepancies/ deficiencies are reported, the same will be adjusted in subsequent releases/ follow up action taken by State.

3.3.14. Departmental-level activities

Departmental-level activities at the National Jal Jeevan Mission (NJJM) include, but not limiting to, administrative and capital expenditure related to SPM-NIWAS; engagement of National Project Management Units and other professional services; monitoring and evaluation, including third-party functionality and impact assessments; Information, Education and Communication (IEC) and capacity-building initiatives; research and development (R&D); human resource development (HRD); computerization and management information systems (MIS); development of IT systems and Digital Public Infrastructure; establishment and support of Centres of Excellence and Professor Chairs; Jal Seva Adhyayan and skill development initiatives; communication and promotion of Jan Bhagidari under JJM; development of National sustainability and climate resilience frameworks; support for rollout of the District Technical Unit (DTU) framework; strengthening of asset management, lifecycle costing and preventive maintenance frameworks for cost-efficient operation and maintenance (O&M); organization of conferences, workshops, seminars, exhibitions and policy dialogues; and international collaboration and knowledge exchange, including sharing of India's JJM experience with multilateral and bilateral partners and other administrative expenditure.

3.3.15. Inadmissible Expenses

A suggestive list of inadmissible expenses includes:

i. Expenditure on purchase of land, vehicles, etc;

ii. Payment of centage charges;

iii. Construction, renovation, or repair of buildings, offices, conference halls, guest houses or residential buildings;

iv. Tender premiums;

v. Diversion of JJM funds to state schemes;

vi. Expenditure exceeding the estimated or approved cost of schemes;

vii. Salary payments for permanent employees;

viii. Operation and maintenance (O&M) of schemes;

xi. Provision of service levels beyond 55 lpcd;

x. Meeting urban/ other sectoral water demand.

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CHAPTER 3.4

Whole of Government Approach

Under JJM 2.0, strategic convergence has been envisaged through "Whole-of-Government" approach. This approach builds upon the essential work of Central or State departments by recognizing the fact that ensuring sustainable and reliable drinking water supply is a collective National priority that required coordinated action across sectors. It aims to align drinking water interventions with broader developmental goals such as public health, education, rural development, environmental sustainability, etc. thereby contributing to the overall well-being of citizens.

3.4.1. National Level

At the National level, effective drinking water management under JJM envisages continuous and seamless coordination among multiple Ministries and Departments. Key stakeholders include the Department of Water Resources, River Development & Ganga Rejuvenation, Ministry of Rural Development, Ministry of Panchayati Raj, Ministry of Education, Ministry of Environment, Forest and Climate Change, among others. Each of these institutions plays a complementary role ranging from source sustainability and watershed management to financing, decentralized governance and environmental regulation. Convergence across these entities will ensure that policies are harmonized, financial resources are optimally utilized and implementation mechanisms are efficient, transparent and outcome-oriented.

To operationalize this convergence, JJM 2.0 envisages the establishment of robust institutional mechanisms for coordination, monitoring and review at the National level. These mechanisms will facilitate regular information sharing, joint planning and real-time tracking of progress. Periodic inter-ministerial reviews will further ensure alignment of priorities and enable necessary course corrections in planning and execution of water supply schemes.

The structural reforms underpinning JJM 2.0 already elaborated in Section 2 and detailed in the reform implementation matrix provide the foundation for this convergence framework. These reforms emphasize service delivery, sustainability of water sources, strengthened operation and maintenance systems and community ownership. The key areas for convergence are annexed as **Annexure VIII**, which need to be reviewed and implemented at the State, District and Gram Panchayat levels as well.

3.4.2. State Level

At the State level, a similar convergence framework shall be institutionalized through the State Water and Sanitation Mission. States shall develop clear protocols for inter-departmental coordination, ensuring alignment of schemes related to water resources, rural development, agriculture, environment, etc. Dedicated convergence plans, with defined roles, responsibilities and timelines, shall be prepared and periodically reviewed. States are also encouraged to integrate JJM planning with State-specific priorities and flagship programmes to maximize impact and avoid duplication of efforts.

3.4.3. District Level

At the District level, the District Water and Sanitation Mission shall serve as the nodal platform for convergence. The DWSM will coordinate with line departments to ensure that water supply schemes are supported by complementary interventions such as watershed development, groundwater recharge and sanitation measures. District-level implementation plans shall incorporate convergence components with measurable outcomes and regular review meetings of DWSM shall be conducted to monitor progress and resolve implementation challenges.

3.4.4. Gram Panchayat Level

At the Gram Panchayat level, convergence will be operationalized through Gram Panchayat and its standing committees including Village Water and Sanitation Committees and other committees. GPs shall integrate drinking water supply planning with local development plans, including source sustainability measures, greywater management and community participation initiatives. A specific reform implementation schedule shall be developed at the GP level, outlining key activities, timelines and responsibilities. This will ensure that convergence is translated into tangible outcomes on the ground and that communities play an active role in planning, implementation and O&M of water supply systems.

In conclusion, the Whole-of-Government approach under JJM 2.0 represents a paradigm shift from isolated programme implementation to integrated and collaborative governance. By fostering convergence across sectors and levels of administration, it aims to ensure that every rural household receives reliable, safe and sustainable drinking water services, while simultaneously advancing National development priorities.

Under JJM 2.0, strategic convergence has been envisaged through a Whole-of-Government approach. This approach builds upon the essential work of Central or State departments by recognizing the fact that ensuring sustainable and reliable drinking water supply is a collective National priority that required coordinated action across sectors. It aims to align drinking water interventions with broader developmental goals such as public health, education, rural development, environmental sustainability, etc. thereby contributing to the overall well-being of citizens.

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Har Ghar Jal
Jal Jeevan Mission

CHAPTER 3.5

Monitoring of the Programme

Reform aims at:

- **Shift from FHTC tracking to monitoring key scheme implementation milestones,** including infrastructure creation, commissioning, handover and Har Ghar Jal certification;
- **End-to-end monitoring of rural water supply systems from source to tap** through the Sujalam Bharat Database and Sujal Gaon ID–based asset registry;
- **Quality Assurance Monitoring:** Mandatory monitoring by **DWSM** of the effectiveness of TPIA to ensure quality of implementation;
- **Progress-linked financial monitoring through SNA-SPARSH,** enabling scheme-wise expenditure tracking and automatic reconciliation of payments;
- **Transactional and near real time assessment of functionality and service improvement planning** to ensure sustained service delivery outcomes;
- **Institutionalising National State and District command and support centres,** focusing on monitoring of responsiveness, asset managementandoperational efficiency.

3.5.1. Introduction

The monitoring architecture is planned to enable all administrative levels to review progress and quality of programme implementation and functionality of rural water supply systems including scheme commissioning, handover to Gram Panchayats, service delivery monitoring and institutional monitoring through interactive dashboards, Meri-Panchyat Application, Jal Seva Aankalan and Sujalam Bharat PM Gati-Shakti Project Monitoring Module. Used together, these mechanisms enable timely identification of implementation gaps, support evidence-based corrective actions required to be taken by States/ UTs and strengthen accountability across all levels.

This approach strengthens outcome-oriented programme management and would ensure that service delivery benchmarks are delivered as assured.

3.5.2. Scheme Implementation Monitoring

Monitoring under JJM 2.0 is undertaken through an integrated set of complementary mechanisms spanning infrastructure progress, financial utilisation, service delivery outcomes and institutional performance.

A. Monitoring through Project Monitoring Module

Progress of scheme implementation is monitored through the Project Monitoring Module on the Sujalam Bharat, PM GatiShakti Application as mentioned in Section 2, captures component-wise progress of infrastructure creation and key milestones. It also records scheme completion, commissioning and handing over of schemes to Gram Panchayats or designated service utilities. The Project Monitoring Module will integrate asset registry, milestone-based progress tracking and recording of Third Party Inspection Agency (TPIA) observations. DWSM shall ensure validation, oversight and review of progress, including examination of TPIA findings recorded on the module and initiation of corrective action.

B. Monitoring through field visits

Field-level verification mechanisms complement digital monitoring and provide independent assessment of implementation progress. These include:

i. Inspection by National team, National WASH Expert (NWE) visits, Central Prabhari Officer (CPO) visits;

ii. Joint inspection of schemes by State and District authorities.

The teams assess infrastructure quality, implementation progress and compliance with technical standards. The dedicated modules for NWE visits, CPO visits and Joint Inspection of schemes have been developed on the JJM-IMIS platform. Observations recorded through

these field verification mechanisms are made visible to the District Water & Sanitation Mission and State Water & Sanitation Mission for administrative oversight and necessary corrective actions. Action Taken Reports (ATRs) against these observations can also be uploaded on the module, enabling systematic tracking of compliance. The module further serves as a digital repository of inspection reports, observations and supporting documents.

C. Quality of Implementation Monitoring through DWSM

Quality assurance during scheme implementation is supported through Third Party Inspection Agencies (TPIAs), which conduct independent technical inspections of works.

TPIAs undertake field verification of construction activities, review quality assurance records and identify deviations from approved technical standards. Their observations, along with supporting documentation and photographs, are uploaded on the Project monitoring module on Sujalam Bharat PM GatiShakti Application.

The DWSM/ DTU is responsible for reviewing these observations and monitoring the corrective actions taken at the scheme implementation level. Action Taken Reports are periodically reviewed to ensure that identified gaps in construction quality or technical compliance are addressed in a timely manner. This will also provide the basis to DWSM for quarterly assessing the effectiveness of TPIA deployment. This Quality Assurance Protocol ensures improved quality control and sustainability of infrastructure created under JJM.

D. Financial Monitoring through SNA-SPARSH

Financial monitoring under JJM 2.0 is undertaken through SNA-SPARSH, which integrates JJM-IMIS, State IFMIS, PFMS and the RBI e-Kuber platform to enable real-time tracking of programme expenditure.

The framework enables scheme-wise and activity-wise monitoring of payments and provides the following safeguards:

i. Real-time visibility of expenditure across programme components;

ii. Milestone based payments especially for MVS & BWT(PPP);

iii. Automatic reconciliation between sanctioned funds and payments made;

iv. Compliance monitoring with prescribed funding patterns

v. Prevention of misclassification of expenditure;

vi. Improved transparency and audit readiness.

This progress-linked financial monitoring strengthens financial discipline and supports evidence-based programme management.

3.5.3. Service Delivery Monitoring

Monitoring under JJM 2.0 also focuses on service delivery outcomes through Sujal Gaon Application together with independent audit by Community through Jal Seva Aankalan (JSA), students (Jal Seva Adhyayan) etc. Data for Jal Seva Aankalan is collected digitally through the Panchayat Secretaries on the Panchayat Dashboard of JJM IMIS or through the Sujal Gaon Application and transmitted in near real-time to the Sujalam Bharat Database, enabling District and State Authorities to review assessment findings promptly and initiate targeted corrective action. The integration of JSA with the broader digital monitoring architecture ensures that service delivery evaluation is not a one-off exercise but a continuous feedback loop that strengthens accountability at the Gram Panchayat and VWSC level.

3.5.4. Establishing Continuous Monitoring Structure for O&M phase

JJM 2.0 introduces a structured system of State and District Command and Support Centres to strengthen operational monitoring of rural water supply systems.

Operational data from digital systems such as the Sujalam Bharat database, utility applications and other monitoring systems is integrated at District and State Command and Support Centres. These centres support near real-time monitoring of operational issues and facilitate timely response to service delivery challenges.

A. District Command and Support Centre (DCSC)

The DCSC serves as the operational monitoring centre for rural water supply at the district level under DWSM. It acts as the first point of response for service delivery issues reported from Gram Panchayats and VWSCs.

Key functions:

i. Near real-time monitoring of scheme functionality using Sujalam Bharat Database, Sujal Gaon Application and IoT systems where available;

ii. Receiving and tracking grievances and ensuring timely resolution;

iii. Coordinating with field engineers and GP-level

functionaries to address breakdowns, leakages and other operational issues;

iv. Monitoring of water quality test results and initiating corrective action where required;

v. Maintaining inventory of spare parts and O&M resources for quick response;

vi. Submitting periodic operational reports to the State CSC.

B. State Command and Support Centre (SCSC)

The SCSC operates under the State Water & Sanitation Mission (SWSM) and provides State-level operational oversight by consolidating information from DCSCs. It ensures early identification and resolution of systemic issues and acts as a link between districts and the National monitoring system.

Key functions:

i. Aggregating District-level data to monitor scheme functionality, water supply continuity, water quality compliance and O&M performance across the State;

ii. Addressing escalated complaints and service failures beyond the capacity of District CSCs and facilitating technical or administrative support;

iii. Monitoring trends in water quality alerts, complaints and seasonal disruptions and issuing alerts to districts;

iv. Overseeing State-level O&M planning, including procurement, deployment of maintenance teams and monitoring service providers;

v. Reporting State-level performance to the National Ministry through the Sujalam Bharat platform;

vi. States/ UTs shall ensure that Operation and Maintenance contractors managing SCADA systems, particularly for bulk water transfers, share real-time functionality data with the District and State Support and Command Centre's to enable integrated monitoring. Further, data from allied devices such as flow meters, sensors and other field instruments shall also be integrated into the system to facilitate comprehensive oversight and ensure seamless coordination among service utilities.

C. National Command and Support Centre (NCSC)

At the National level, monitoring focuses on the effectiveness of both tiers of command centres, including, inter alia, indicators such as:

i. Response time for technical support requests;

ii. Water quality surveillance, response and remedial action at village level;

iii. Asset and inventory management practices.

These indicators enable assessment of institutional performance in managing rural water supply infrastructure.

The detailed indicative monitoring parameters at National, State, District and GP level are annexed as **Annexure IX**.

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CHAPTER 3.6

Evaluation and Impact Assessment

Reform aims at:

- **Shift to service outcome evaluation:** regularity, adequacy, potability and sustainability;
- **Community-led assessment through Jal Seva Aankalan:** validated in Gram Sabha and used for improvement planning;
- **Independent evaluation mechanisms :** NITI Aayog, engineering institutions (Jal Seva Adhyayan) and CPO assessments;
- **CIRP linked departmental evaluation:** of State reform commitments and scheme performance.

Under JJM 2.0, monitoring provides continuous oversight of infrastructure creation and service delivery performance, evaluation provides a structured, periodic assessment of whether the programme is delivering intended service outcomes i.e., whether rural households are receiving regular, adequate, potable drinking water on long term & sustainable basis. The evaluation framework under JJM 2.0 is organised across three complementary pillars.

Figure 21: Three Pillar Evaluation Framework JJM 2.0

3.6.1. Jal Seva Aankalan

Jal Seva Aankalan (JSA) shall serve as the primary community-led evaluation mechanism under JJM 2.0. It provides a structured process through which Gram Panchayats and Village Water & Sanitation Committees assess the functionality, adequacy, quality compliance, institutional readiness and financial sustainability of rural water supply systems. Assessment shall be undertaken at Gram Panchayat level, validated through Gram Sabha and submitted digitally through the JJM-IMIS panchayat dashboard by the Panchayat Secretary. The submissions shall thereafter be reviewed at District level through the District Technical Unit for technical diagnosis of service delivery gaps. Based on the DTU review, the District Water & Sanitation Mission shall undertake District-Level Analysis and prepare District Improvement Plan, while the State shall review district findings and prepare State Improvement Plans based on identified gaps and proposed corrective actions.

Citizen feedback is additionally enabled through the Meri Panchayat platform.

JSA shall initially be conducted periodically or as announced by NJJM. In its initial phase, JSA is being operationalised in Har Ghar Jal reported Gram Panchayats. States shall progressively expand coverage to all Gram Panchayats. States shall publish the annual Functionality Assessment Report based on

Figure 22: Jal Seva Aankalan: Four-tier Evaluation & Improvement Flow

Tier 1: GP/ VILLAGE	Tier 2: DTU	Tier 3: DWSM	Tier 4: SWSM
Gram Panchayat & VWSC	District Technical Unit	District Water & Sanitation Mission	State Water & Sanitation Mission
• Conduct periodic assessment of water supply, service covering adequacy, quality, O&M and sustainability • Validate in Gram Sabha • Submit digitally on JJM-IMIS	• Review village submissions • Diagnose technical reasons for service gaps • Record diagnostic observations for DWSM consideration	• Review DTU findings • Deliberate on systematic issues • Prepare District Improvement Plan (DIP) with defined responsibilities	• Review and aggregate district findings • Prepare State Improvement Plan (SIP) • Provide policy direction and resource coordination

aggregated JSA data, which shall be a Key Performance Indicator for fund release under JJM 2.0.

As digital service delivery systems mature, JSA may progressively use operational data captured through digital platforms such as Sujal Gaon App, including supply hours, water quality records, grievance data and service delivery charge collection, to strengthen evidence-based community validation.

3.6.2. Third-Party Evaluation Mechanisms

To ensure independent assessment of programme outcomes, the following third-party evaluation mechanisms have been institutionalized:

A. NITI Aayog - Outcome Evaluation

NITI Aayog, through its Development Monitoring & Evaluation Office (DMEO), shall undertake independent outcome evaluation of JJM aligned with the Output-Outcome Monitoring Framework.

A comprehensive evaluation framework shall be developed in collaboration with NITI Aayog to assess programme outcomes beyond infrastructure creation, with focus on service delivery, sustainability and socio-economic impact.

The framework may include, inter-alia:

i. **Health outcomes:** Reduction in water-borne diseases and improvement in public health indicators;

ii. **Education outcomes:** Improved school attendance and reduced burden of water collection;

iii. **Women's participation:** Engagement of women in VWSCs, O&M roles and water quality monitoring;

iv. **Service delivery innovation:** Adoption of efficient and low-cost O&M systems;

v. **Local economic impact:** Growth of local supply chains, manufacturing and livelihood opportunities.

B. Jal Seva Adhyayan-Scheme Assessment by Engineering Students

Under Jal Seva Adhyayan, engineering students shall undertake structured field assessments of rural water supply schemes under faculty mentorship, in collaboration with AICTE and participating technical institutions. The assessment shall cover infrastructure quality, operational performance, water quality compliance and household-level service delivery. Students shall conduct field visits using a standard digital assessment format and reports shall be submitted through JJM-IMIS for review by District and State authorities.

C. Central Prabhari Officers-Evaluation in Aspirational Districts/ Blocks

Senior Government of India officers designated as Central Prabhari Officers (CPOs) assigned under Aspirational Districts / Blocks Programme shall review JJM implementation during field visits. A digital reporting module shall be used to capture observations on service delivery, infrastructure quality and citizen satisfaction.

CPOs shall review JJM implementation in Aspirational Districts and Blocks, with observations submitted through a digitised module on JJM-IMIS. Going forward, CPO reporting formats shall progressively transition from scheme-level monitoring forms to structured evaluation forms designed to assess the overall impact of JJM on Aspirational Districts - covering household water access, health outcomes and community service satisfaction. Findings shall be consolidated into the departmental mid-term evaluation.

3.6.3. Impact Assessment of Jal Jeevan Mission

i. Impact assessment under JJM 2.0 shall move beyond scheme-level outputs to capture the

broader socio-economic and developmental outcomes arising from assured drinking water supply. It shall be undertaken using an Output-Outcome-Impact framework, linking infrastructure creation to improvements in service delivery, quality of life and human development indicators. While monitoring provides continuous oversight, impact assessment shall enable periodic evaluation of whether the Mission is translating into tangible benefits for rural households, particularly in terms of health, gender equity, education and livelihoods. District-Level Impact Evaluation Matrix (Indicative) has been attached annexed as **Annexure X;**

ii. The impact assessment shall be undertaken at the district level under the overall supervision of the District Water & Sanitation Mission. Each District shall designate a Nodal Officer responsible for coordinating data collection across departments, ensuring validation of findings and preparation of the District Impact Assessment Report. These reports shall be consolidated by the State Water & Sanitation Mission into State- level assessments, which shall be further compiled at the National level by DDWS/ NJJM. States/ UTs shall ensure submission of State Impact Assessment reports by September 2027 to enable timely consolidation for mid-term evaluation. The findings from these assessments shall form an important input into the mid-term evaluation of JJM 2.0 and shall support evidence-based policy refinement and programme strengthening;

iii. Impact assessment shall be undertaken through convergence with relevant Ministries and Departments so as to capture cross- sectoral outcomes of improved water access. In the health sector, for instance, analysis may draw upon data from systems such as the Integrated Disease Surveillance Programme (IDSP) and the Health Management Information System (HMIS) to examine trends in waterborne diseases, child morbidity and public health outcomes in areas with improved water supply. Similarly, in relation to women and child development, insights may be derived from platforms such as the Poshan Tracker and Anganwadi records to understand changes in improvements in nutrition indicators, child well-being and access to safe water for nutrition and hygiene, along with other related indicators;

iv. In the education sector, impact assessment may consider information available through UDISE+ and school-level reporting systems to assess improvements in school attendance, especially among girls and the availability of functional drinking water facilities in schools. In the rural development sector, data from platforms such as VB-G RAM G and NRLM/ Aajeevika may be used to examine linkages between water access, time savings, livelihood participation and convergence with water conservation and source sustainability works. Likewise, data available within the Jal Shakti ecosystem, including JJM-IMIS, the Sujalam Bharat Digital Public Infrastructure, Water Quality Management Information System and databases of agencies such as the Central Ground Water Board (CGWB) may be leveraged to assess functionality of systems, water quality compliance and sustainability of water sources;

v. In the socio-economic impact assessment, livelihood creation activities may be considered for assessment due to provision of assured drinking water supply by leveraging available datasets, primary questionnaire on citizen perception and ground-level information collection and reports under the National Rural Livelihoods Mission (NRLM) and the corresponding State Rural Livelihoods Missions (SRLMs). Using NRLM/ SRLM household profiling, SHG-level livelihood tracking and institutional data, the DWSM shall examine outcomes such as reduction in time spent particularly by women on water collection, increased participation of women in livelihoods and self-employment activities, improved attendance in SHG meetings and skill development programmes and diversification of household income sources attributable to improved water access and reduced drudgery.

The assessment may further evaluate the role of women's collectives and SHG institutions, as reflected in NRLM/ SRLM records, in strengthening community-level water governance and source sustainability, including participation of SHG members in Village Water & Sanitation Committees, involvement in operation and maintenance activities, service delivery charge collection, water quality surveillance and grievance redressal mechanisms. These indicators, analysed by the DWSM, shall inform assessments of institutional readiness, financial sustainability and community ownership of rural water supply systems;

vi. In addition to administrative datasets, impact assessment shall incorporate citizen-centric evidence, feedback received through digital platforms like the Meri Panchayat Application and other local surveys. Such inputs may be used to assess parameters such as regularity and adequacy of supply, user satisfaction, reduction in time spent on water collection, changes in household expenditure on water and effectiveness of grievance redressal systems, among others. Districts may also supplement these with sample surveys or focused studies, wherever required, to better understand local impacts;

vii. The overall approach shall be based on triangulation of multiple data sources, combining administrative records, community-based assessments and sectoral data- sets. Districts shall ensure that data from different departments is appropriately harmonised and interpreted, with due regard to local context. The findings of the impact assessment shall be placed in the public domain to enhance transparency and account- ability and shall be used to prepare Service Improvement Plans, prioritisation of interventions under the CIRP and broader policy decisions at State and National levels.

3.6.4. Conclusion

Significant progress has been achieved under the Mission in improving the functionality of rural piped water supply systems and in generating positive impacts across public health outcomes and quality of life. However, as the programme transitions from infrastructure creation to sustained service delivery, it is critical that evaluation assessments going forward are undertaken in scientific and evidence-based manner, while also ensuring meaningful community participation. This approach shall strengthen credibility of assessments, enhance accountability and ensure that programme outcomes remain aligned with the needs and expectations of rural households.

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Detailed Sustainability Matrix

Drinking Water Sustainability Matrix:

i. **Institutional and Service Delivery Sustainability:** The institutional framework and governance mechanisms that enable water supply systems to operate professionally, accountably and sustainably, while delivering assured service standards to all households:

a) Institutionalise accountability at all levels: State, District, PRI, VWSC/ GP and operators;

b) Assure service delivery and standards.

ii. **Source Sustainability:** The long-term security of water sources through groundwater recharge, source protection and integrated water resource management ensuring adequate water availability year-round:

a) Secure water sources through hydrogeology-based planning, recharge structures, water quality surveillance and infrastructure protection mechanisms.

iii. **Financial Sustainability:** The ability of water supply schemes to generate sufficient revenue through service delivery charges to cover operational costs and maintenance, without perpetual subsidy dependence:

a) Ensure financial sustainability by instituting transparent tariff structures and strengthening revenue collection mechanisms, while effectively communicating the intrinsic value of water services to end-users.

iv. **Infrastructure Sustainability:** Systematic management of water supply infrastructure assets through digital registries, preventive maintenance scheduling, lifecycle costing and performance tracking ensuring long asset life and operational efficiency:

a) Strengthen long-term asset management through asset and inventory management registry, O&M, preventive maintenance and lifecycle costing.

Support Ecosystem of Sustainability Matrix:

i. **Digital Systems:** States to progressively adopt integrated digital systems to strengthen planning, monitoring and accountability of rural water services. Development of GIS based asset registries, service area identifiers, dashboards and decision support tools can enhance asset visibility, support informed decision making, improve water quality surveillance and enable transparent service delivery across all levels;

ii. **Governance:** States to strengthen governance structures by clarifying institutional roles, enhancing coordination across departments and aligning water policies with sustainability principles. Empowerment of local institutions, promotion of regulatory mechanisms for source protection and standardization of asset management and O&M practices can support resilient service delivery and long term drinking water security;

iii. **Community Ownership:** States to encourage greater community ownership by enabling Gram Panchayats, village institutions and local youth to participate actively in the management of rural water supply systems. Structured capacity building programmes, clear operating procedures, local stewardship of water sources and community engagement initiatives can foster asset ownership and sustainable operations;

iv. **Citizen Feedback:** States to strengthen citizen centric service assurance by institutionalizing feedback and grievance redress mechanisms. Regular functionality assessments, transparent tariff communication, service performance ratings, supply information systems and structured commissioning and handover processes can improve responsiveness, build trust and support continuous service improvement.

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Annexure-II

Checklist for Commissioning of Rural Water Supply Scheme

S. No.	Activity / Item	Details / Verification Points	Status (✓/×/Remarks)
A. Documentation and Pre-Commissioning Preparations			
1.	As-built Drawings	Verified availability and accuracy of as-built drawings for all components (intake, pumping stations, pipelines, storage tanks, treatment units, distribution network, electrical layouts).	
2.	Design Compliance Inspection	All infrastructure components inspected for conformity with design specifications, technical standards and quality norms as approved by competent authority.	
3.	Component-Wise Inspection	Pipelines, pumps, valves, storage tanks, treatment units, meters, electrical panels and control systems (all civil, electrical & instrumentation works) verified for installation completeness and workmanship.	
4.	Visual Defect Check	Visual inspection for defects, leaks, incomplete works, rusting, damages or misalignments.	
B. Technical Validation Tests			
1.	Hydrostatic Pressure Test	Conducted as per standard protocols to confirm pipeline integrity under design pressure. Results recorded.	
2.	Tap Connection Leak Test	All household tap connections checked for leakage or improper fittings.	
3.	Flushing & Disinfection	All pipelines, storage tanks and treatment units flushed and disinfected using approved procedures (e.g., chlorination).	
4.	Flow and Capacity Test	Conducted to verify hydraulic performance, flow rates and pressures throughout network against design values.	
5.	Water Quality Testing	Chemical and bacteriological analysis conducted at source, treatment plant outlet, distribution points and household taps.	
6.	Residual Chlorine Test (RCL)	Residual chlorine levels measured at multiple points; levels within prescribed limits (0.2–0.5 mg/l).	

S. No.	Activity / Item	Details / Verification Points	Status (✓/×/Remarks)
C. Electrical, Mechanical & Automation Systems			
1.	Pump & Motor Testing	Functional testing of all pumps and motors; checking voltage, current, vibration, alignment and noise levels.	
2.	Valve & Control Device Testing	Verification of valve operation (isolation, air release, scour) and control systems under load conditions.	
3.	Automation / SCADA Testing	Functional verification of automation systems, sensors and remote monitoring (where applicable).	
4.	Calibration of Meters & Sensors	Flow meters, pressure gauges, chlorine analyzers and sensors calibrated and Certified.	
5.	Control System Setting	Operational parameters (timing, flow rate, alarms, process variables) set as per design requirements.	
D. Operational Validation			
1.	Trial Run	Entire system operated continuously for not less than 15 days to validate stability and performance.	
2.	Monitoring During Trial Run	Continuous observation of water flow, pressure stability, pump performance and water quality.	
3.	Issue Identification & Rectification	Any operational problems (pressure drops, failures, water quality issues) documented and rectified.	
E. Handover & Community Readiness			
1.	Joint Inspection	Conducted with engineers, contractor, GP/VWSC representatives and community members; joint inspection note signed.	
2.	IEC & Awareness Activities	IEC activities conducted to educate community on water safety, RCL importance and system use.	
F. Documentation & Knowledge Transfer			
1.	Commissioning Report	Comprehensive report prepared including test results, observations and recommendations.	
2.	O&M Manuals & Training	Operation manuals, troubleshooting guides and maintenance schedules handed over; training provided to GP/VWSC staff.	

Annexure-III

Sujalam Shakti Group

Community-based organisations engaged by VWSC including village level Self Help Groups (SHGs, PACs, etc.) for supporting safe, reliable and sustainable rural water services shall be referred to as Sujalam Shakti Groups. This will help align these groups with the objectives of the Jal Jeevan Mission and recognise them as field functionaries for the Mission. States may issue operational guidelines for these groups. These will preferably be women led community support system for long-term functionality and sustainability of rural water supply schemes. Under this convergence framework, identified groups may be oriented and trained to undertake structured roles in coordination with Gram Panchayats and VWSCs, including:

i. Apply Basic Concepts of Rural Water Supply;

ii. Take up relevant WQM&S activities at village/ GP level, Conduct Drinking Water Quality Testing using Field Test Kit (FTK) and Perform Basic Water Disinfection using Chlorine, Bleaching Powder, etc. including:

a. Water quality testing and reporting: Conduct periodic testing of drinking water sources and household supply using FTKs and report results on designated digital platforms for timely action;

b. Remedial action support: Coordinate with VWSCs, Gram Panchayats and technical functionaries for prompt remedial measures in case of contamination or service disruptions;

c. Community surveillance: Support village level vigilance on protection of drinking water sources and infrastructure from contamination, misuse or damage and promote safe water handling practices.

iii. Support utility tariff collection and community engagement:

a. Awareness and facilitation for service delivery charge: Undertake awareness and mobilisation for timely payment of service delivery charge and responsible water use for sustainable O&M, IEC & BCC activities etc;

b. Collection of service delivery charge: Where authorised by the Gram Panchayat/VWSC, SHGs may support collection of service delivery charge and maintenance contributions in a transparent and accountable manner through digital or receipt-based systems.

iv. Use Digital, Financial, Communication Skills and Customer Service including grievance redressal at the Workplace;

v. Implement Source Contamination Prevention and Follow Regional Water Safety Guidelines:

a. Source sustainability: Facilitate community mobilisation for water conservation, recharge and source protection through convergence with VB-G RAM G and other local programmes.

Skill Development and Local O&M Support

The component may include structured skill development of Sujalam Shakti groups to support minor operation and maintenance activities at the village level. Through appropriate training and resource support systems, selected SHG members may be trained in:

i. Basic upkeep and cleaning of water supply infrastructure such as overhead tanks, pump houses and surroundings;

ii. Support for minor repair and preventive maintenance activities under supervision of VWSC/DTU;

iii. Record-keeping and reporting of functionality issues;

iv. Support services for village level water system management.

Cluster-based training platforms and resource augmentation mechanisms etc. may be utilised for periodic capacity building, refresher training and creation of a pool of trained community resource persons. This will not only strengthen sustainability of water supply systems but also create local livelihood opportunities for SHG members.

Financial Incentive Framework

To ensure sustained engagement and accountability, States/ UTs may design a suitable financial incentive mechanism. This may include service-based incentives for water quality testing, facilitation of service delivery charge collection, support for minor O&M activities and performance linked incentives based on functionality, water quality compliance and community participation indicators.

This convergence will enable creation of a trained, community based strengthened women led local institutions and ensure sustained delivery of safe and reliable drinking water services in rural areas.

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Annexure-IV

Indicative list of schemes under which activity/ fund convergence can be prioritised to improve the sustainability of villages

S. No.	Scheme / Grants	Central/ State Government	Components that can be converged
1.	Swachh Bharat Mission-Grameen (SBM-G)	Department of Drinking Water and Sanitation, M/o Jal Shakti	Grey water management soak pits (individual/ community), individual kitchen garden, waste stabilization ponds, etc.
2.	Finance Commission Grants (15th & 16th FC Grants)	Panchayati Raj	Grey water management, drainage systems etc.
3.	Viksit Bharat– Guarantee for Rozgar and Ajeevika Mission (Gramin) Act, 2025 (VB-G RAM G)	Ministry of Rural Development	Construction of recharge structures, check dams, percolation tanks, desilting of water bodies, water conservation works in identified blocks.
4.	Integrated watershed management programme (IWMP)	Department of Land Resources	Watershed management/ RWH/ artificial recharge, creation/ augmentation of water bodies, etc.
5.	Atal Bhujal Yojana (In select States)	Department of Water Resources, River Development & Ganga Rejuvenation	Ground water conservation work from Incentive Component- Focus on sustainable groundwater management, particularly in select north-western states, through community participation.
6.	Repair, Renovation and Restoration of water bodies	Department of Water Resources, River Development & Ganga Rejuvenation	Restoration of larger water bodies.
7.	Rashtriya Krishi Vikas Yojana (RKVY)	Ministry of Agriculture, Cooperation and Farmers Welfare	Watershed related works.
8.	Jal Shakti Abhiyan: Catch the Rain and Amrit Sarovar Mission	Department of Water Resources, River Development & Ganga Rejuvenation	Target interventions in 256 water stressed districts to promote rainwater harvesting and water conservation. Rejuvenate water bodies in each district to enhance water availability and ecosystem health.
9.	WDC-PMKSY 2.0, Department of Land Resources	Ministry of Rural Development	As per recommendations of NITI Aayog 2018, the rejuvenation of spring shed has been incorporated as a new activity.

Annexure-V

Competency and capacity building areas- Community Level

Functional Role	Key Stakeholders	Competency Required	Capacity Building Areas	Expected Outcome
Infrastructure O&M & Asset Management	• GP/VWSC • GP Functionaries • Operators • SHG/ PACS • GP Secretary • Sarpanch	• Understanding of scheme components, routine O&M, preventive maintenance, asset tracking and infrastructure protection.	• Preventive maintenance protocols. • Operational Energy efficiency practices. • Minor repairs Maintaining O&M registers. • Asset register maintenance • Digital asset mapping • Infrastructure protection measures.	• Improved reliability, efficiency and protection of rural water supply infrastructure.
Water Quality, Safety & Health	• GP/VWSC • SHG • AWW • ASHA	• Understanding of water quality standards, testing procedures and public health linkages.	• FTK usage and reporting. • Sanitary inspection. • Awareness on disease water linkages. • Safe water storage practices. • Handwashing promotion. • Diarrhoea prevention awareness. • School and Anganwadi water safety.	• Safe drinking water, improved hygiene behaviour and timely corrective action.
Financial Governance & Accountability	• GP/VWSC • GP • Sarpanch • GP Members • SHG	• Ability to manage service delivery charge systems, financial records and ensure institutional accountability.	• Service delivery charge setting • Collection mechanisms • O&M cost estimation • Bookkeeping and financial records Integration with GP financial systems/ Finance Commission funds Gram Sabha reporting. • Jal Seva Aankalan/ functionality assessment. • Service level benchmarks. • Fund utilization monitoring.	• Financially sustainable schemes with transparent governance and accountability.

Functional Role	Key Stakeholders	Competency Required	Capacity Building Areas	Expected Outcome
Community Leadership, Engagement & Grievance Redressal	• Sarpanch • SHG • GP/VWSC • GP	• Ability to mobilise communities, promote participation and address citizen feedback.	• Leadership and facilitation skills. • Gram Sabha engagement. • Participatory tools (PRA/ Jan Bhagidari). • Gender inclusive governance. • Demand management. • Conflict resolution. • Feedback mechanisms. • Complaint tracking. • Escalation protocols. • Citizen interface systems.	• Greater community ownership and responsive service delivery.
Digital Tools & Source Sustainability	• GP Secretary • GP/ VWSC • Sarpanch • SHG Community Leaders	• Ability to use digital monitoring systems and plan sustainable water resource management.	• Use of Sujalam Bharat platform. • Jal Seva Aankalan . • IMIS and dashboards. • Data interpretation for decision making. • Yield and demand assessment. • Seasonal water risk planning. • Recharge measures • Greywater management.	• Data-driven governance and long-term water security.

Annexure-VI

Competency and capacity building areas- District Level

Functional Role	Key Stakeholders	Competency Required	Capacity Building Areas	Expected Outcome
O&M Systems Strengthening	• Executive Engineers, • DTU • DWSM	• Ability to design, monitor and optimize district-wide O&M systems.	• Preparation of Service Improvement. • Plan from Jal Seva Aankalan data. • Preventive O&M protocols. • Asset lifecycle management. • Energy efficiency.	• Efficient and reliable operation of rural water supply systems.
Water Quality Monitoring & Surveillance	• Lab Personnel • DTU • Executive Engineers	• Ability to implement risk-based water quality management systems.	• WQM&S systems. • Laboratory quality assurance & accreditation. • Water Safety Plan (WSP).implementation. • Fluoride/ arsenic hotspot management. • Water Quality Surveillance protocol.	• Compliance with water quality standards.
Financial & Contract Management	• DWSM Officials • Executive Engineers	• Ability to manage O&M finances and professional contracts.	• Tendering and AMC contract design. • Performance-based contracting. • Third-party monitoring mechanisms. • Financial risk assessment.	• Improved financial discipline and streamlined contract management.
Data & Performance Monitoring	• DWSM • DTU • MIS Personnel	• Ability to use data analytics for scheme monitoring and improvement.	• KPI-based scheme monitoring. • Digital reporting through IMIS. • Energy & water audits. • Performance benchmarking across schemes. • Data validation protocols.	• Enhanced data accuracy and informed performance-driven decision making.

Functional Role	Key Stakeholders	Competency Required	Capacity Building Areas	Expected Outcome
Climate & Infrastructure Resilience	• DWSM • DTU	• Ability to integrate climate resilience into infrastructure planning.	• Source sustainability and recharge measures. • Greywater systems integration. • Drought and flood resilience planning. • Vulnerability mapping.	• Better climate resilient planning and infrastructure preparedness.
Community Interface & Grievance Redressal	• DWSM • DTU	• Ability to ensure district level service responsiveness and oversight.	• Service level communication frameworks • Citizen grievance escalation systems • Participatory oversight mechanisms.	• Greater service responsiveness and improved citizen engagement.
Structural Reforms & Utility-Based Service Delivery	• DWSM • DTU	• Ability to operationalize structural reforms for sustainable and accountable rural water service delivery.	• Implementation of Sujalam Bharat digital architecture and unique scheme IDs. • Jal Seva Aankalan for service assessment and improvement planning. • Utility based service delivery framework for Gram Panchayats/ VWSCs. • Institutional strengthening of VWSCs/ GPs as micro utilities. • Digital asset registry and service benchmarking.	• Institutionalized and accountable rural water service delivery systems with improved transparency, performance monitoring and sustainabilit.

Annexure-VII

Auditor's Observations

State Water and Sanitation Mission _____

Name of the organization receiving grants: _____

S. No.	Issues	Observation of the Auditor
1.	Opening Balance & Closing Balance of the Receipts and Payments account tallies with that of Cash Book.	
2.	Opening Balance adopted tallies with Closing Balance of the last year.	
3.	Whether grantee or other implementing agencies have diverted/ inter-transferred funds from one scheme to another Central Scheme or State funded Scheme during the period in contravention to the existing guidelines? If so, details thereof.	
4.	Are there any mis-utilisation/ unrelated expenditure and misappropriation of funds by the grantee or other implementing agencies during the year? If so, details thereof.	
5.	There is only prescribed number of bank accounts for the scheme.	
6.	There does not exist any minus balance at any stage during the year.	
7.	Where the Sanction Order of the Ministry specifies certain conditions at the time of release of funds, whether the same has been fulfilled.	
8.	Scheme funds are being kept only in savings account.	
9.	Interest earned has been added to the scheme fund.	
10.	Whether interest money is being utilized strictly for the programme purposes as laid down in the existing guidelines.	
11.	State share, as per programme guidelines, for the year has been received during the year.	
12.	All receipts/ refunds have been correctly accounted for and remitted in to the Bank account of the scheme.	
13.	Scheme funds are not being kept in the State Treasury.	
14.	Bank Reconciliation is being done regularly.	
15.	Name and address of the previous Auditor.	

Signature of Competent Authority (Signature of Chartered Accountant with Seal)

Name in full_____
Office seal _____
Membership No._____
Contact No. _____
CAG Empanelment No. & Year _____

Annexure-VIII

Key Areas for Convergence

Ministry/ Department/ Organization	Key area of convergence	Activities under convergence
Water Resources, River Development and Ganga Rejuvenation	Source sustainability	• Mapping, protection and rejuvenation of springs and groundwater sources. • Use of GIS and Decision Support Systems (DSS) for source planning and prioritization. • Technical support for groundwater assessment, monitoring, aquifer mapping and stress mitigation through CGWB. • Support for sustainable development and efficient management of surface water resources through CWC.
Rural Development	Capacity building, employment generation, asset creation and conservation works.	• Convergence of VB-G RAM G funds for drinking water security, source sustainability, groundwater recharge, water conservation and rural water supply works; includes greywater infrastructure and O&M employment. • Monitoring and evaluation through DISHA committee meetings. • Training and skilling for O&M under Nal Jal Mitra Programme via Jal Vitran Sanchalak certification using RSETI infrastructure • Engagement of SHGs and community cadres (via SRLMs) for O&M support, water testing, service delivery charge collection, minor repairs and awareness. • Use of IWMP for augmentation of local water sources through recharge, conservation and watershed works. • Utilization of Finance Commission tied grants and DMF resources for watershed and source sustainability interventions, especially in mining-affected areas. • Capacity building through State Institutes of Rural Development (SIRDs).
Rural Development Skill Development & Entrepreneurship	Skilling and local employment.	• Skilling of personnel for in village O&M, out village O&M and water quality testing under PMKVY.

Ministry/ Department/ Organization	Key area of convergence	Activities under convergence
Panchayati Raj	Local governance & O&M sustainability	• Institutionalization of drinking water O&M, source protection and water security in GPDPs. • Empowerment of GP/ VWSC through appropriate legal and institutional provisions. • Use of Meri Panchayat app and e-GramSwaraj portal for transparency, feedback and GP strengthening. • Ensuring availability of certified operators for water supply schemes. • Support for training and certification of operators. • Utilization of Finance Commission grants for O&M of drinking water supply.
Environment Forests and Climate change	Environmental safeguards & climate resilience	• Facilitation of timely forest, wildlife and environmental clearances for drinking water projects. • Protection and rejuvenation of catchments around drinking water sources through afforestation and conservation measures.
Cooperation	Collective action	• Engagement of Primary Agricultural Cooperative Societies (PACS), LAMPS (Large area multipurpose cooperative societies) cooperatives for Operation & Maintenance (O&M).
Education	Awareness & behaviour change	• Inclusion of WASH in school curriculum; engagement of students as ambassadors. • Introduction of elective courses on water supply systems for engineering students. • Data harmonization.
Health & Family Welfare	Public health surveillance	• Integration of IDSP data at block level for surveillance of water-borne diseases. • Feedback mechanism linking disease surveillance with water quality interventions under JJM. • Data sharing on health status and WaSH services in health institutions for international programmes.
Women & Child Development	Gender participation & behaviour change	• Convergence with nutrition and health programmes to promote behaviour change and improve outcomes for women and children. • Provision of safe drinking water in Anganwadi Centres (AWCs). • Integration of AWC data in JJM IMIS and Sujalam database for harmonization.
Tribal Affairs	Sustainability in tribal areas	• Priority provision of water supply in areas under PM-JANMAN and DA-JGUA, including uncovered populations.

Ministry/ Department/ Organization	Key area of convergence	Activities under convergence
Agriculture	Source sustainability	• Use of AWS and ARG data via API for improved planning of water source sustainability and promotion of "JSJB".
Information	Behavior change communication	• IEC (Information, Education & Communication) related activities for awareness generation and outreach under JJM.
Housing & Urban Affairs	Urban–rural interface	• Water management in peri-urban areas.
Statistics & Planning Implementation	Planning and monitoring	• Leveraging MPLAD funds to support planning, implementation of rural drinking water schemes. • Use of constituency level statistics to prioritize interventions and track progress. • Integration of MPLAD reporting with JJM IMIS.
Coal	Resource utilization and CSR	• Support for source sustainability and water quality interventions in coal-bearing regions.
Mines	Resource convergence	• Utilization of DMF funds for drinking water supply, source strengthening and O&M in mining affected districts • Water quality mitigation and infrastructure augmentation in DMF-notified areas.
Earth Sciences (IMD)	Climate and Disaster Resilience/ Rainfall data	• Provision and validation of rainfall data (via AWS) for water source planning and recharge strategies. • Integration of IMD data with JJM planning and monitoring platforms. • Alignment with Jal Shakti Abhiyan – Catch the Rain for water security in water-stressed districts. • Prioritization of flood-resilient JJM infrastructure by State Disaster Management Authorities.
M/o Road, Transport & Highway	Statutory clearances	• Facilitation of timely clearances for laying of water pipelines and related infrastructure along highways.
Railways	Statutory clearances	• Facilitation of timely clearances for drinking water pipelines crossing railway land and tracks to avoid delay in project execution.
Swachh Bharat Mission - Grameen (SBM-G)	Sanitation and Grey Water Management	• Convergence with SBM-G Phase II for greywater and fecal sludge management to ensure safe wastewater treatment. • Integration of ODF Plus interventions for safe collection, treatment and disposal of liquid waste, preventing contamination and complementing tap water coverage.

Annexure-IX

Monitoring Framework

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
SECTION 1: Institutional Governance & Policy Reforms					
1.1	O&M Policy Notification & Service Standards	• No. of States notified comprehensive O&M Policy. • O&M Policy vis-a-vis 19 integral points of model O&M Policy. • Service standards are in line with the JJM standards.	• Gazette notification of O&M Policy issued. • Review of Policy elements such as grievance redressal resolution time, service standards, O&M arrangements, quality standards, NRW limits, energy KPIs, etc.	• Adherence of district operations with notified O&M Policy. • Periodic compliance review of GP/VWSC with service standards. • Handholding of GPs and VWSCs for following O&M Policy.	• Awareness of O&M Policy among VWSC members. • Adherence to minimum service standards (quantity, quality, regularity). • Display of service standards on GP notice board.
1.2	Drinking Water Allocation for Surface Water-Based Schemes	• No. of States having policy for drinking water allocation for surface water-based schemes. • Tracking of allocation orders on JJM portal.	• State government order for drinking water allocation under State Policy.	• Implementation of water allocation orders at district surface-based schemes. • Monitoring of allocation vs. actual usage.	• Awareness of water allocation norms at village level.
1.3	Inclusion of Drinking Water Services in Public Service Guarantee Act / Citizen Charter	• No. of States with amended Public Service Guarantee Act (PSGA)/ Citizen Charter covering Drinking Water services. • Review of Gazette notifications uploaded on JJM Dashboard	• State government Gazette notification for inclusion of drinking water services in PSGA/ Citizen Charter. • Amended provisions ensuring citizen entitlement for	• Dissemination of PSGA/ Citizen Charter provisions to all GPs in the district. • Monitoring of timelines for service restoration. • Grievance escalation linked to PSGA provisions.	• Display of Citizen Charter/ PSGA provisions at GP office. • Citizens to be aware of entitlements and grievance redressal mechanism. • Complaints under Act to be tracked and

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
1.3			drinking water service standards. • Document to be uploaded on JJM portal/State portal.		resolved within stipulated timeline.
1.4	District Technical Units (DTUs) Establishment	• Districts with operational DTUs reported on JJM portal. • Review of DTU effectiveness across States.	• DTU establishment orders for all districts with staff roles and responsibilities. • Staffing status: designated nodal officer, engineers. • DTU linked to DWSM dashboard.	• DTU operational with full staff as per their roles/ responsibilities defined. • DTU providing technical support to GPs/ VWSCs. • Asset registers maintained and updated by DTU.	• Access to DTU technical support for O&M issues. • Technical grievances escalated to DTU through DWSM. • DTU inspection reports acted upon.
1.5	VWSC as Statutory Standing Committee of GP	• No. of States with notifying recognizing VWSC as statutory standing committee of GPs. • Tracking of Gazette notifications on JJM portal. • No. of VWSCs constituted in the villages.	• State Government Gazette notification and amended PR Act provisions. • Ensure composition of VWSC as per JJM guidelines. • No. of VWSCs constituted in the villages.	• % GPs with constituted VWSC/ GPWSC in the district. • VWSC formation resolutions uploaded on JJM portal. • Composition verified (women ≥50%, AWW, ANM/ ASHA, SHG). • No. of VWSCs constituted in the villages.	• VWSC constituted and functional with prescribed composition. • VWSC resolution passed and uploaded. • VWSC conducting regular meetings and uploading of the minutes on JJM dashboard.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
SECTION 2: Service Utility Framework (Micro & Macro Utilities)					
2.1	Formation of State/Regional Macro Utility (Parastatal Body)	• No. of States with notified State/Regional Macro Utilities. • Review of macro utility O&M performance data from SCSC.	• State Gazette notification forming parastatal macro utility for out-village infrastructure. • Applicable to States with BWS and higher MVS planned. • Macro utility formation order uploaded on JJM portal. • Whether adequate governance structure, funding and capacity support provided. • Clear delineation of functions and asset transfer. • Cost recovery, tariff policy, subsidies defined.	• Coordination with macro utility for district-level O&M support. • Tracking and assessment of the continuity of bulk water supply to villages.	• Systematic monitoring and reporting of bulk water supply breakdowns to the DWSM/ DCSC.
2.2	Performance Benchmarking of Macro Utility (Parastatal Body)	• No. of States notified benchmark guidelines for macro utility performance. • National benchmarks for macro utility O&M efficiency communicated to States.	• State notification of performance benchmark guidelines for macro utility • Guidelines uploaded on State/SWSM dashboard. • KPIs defined: NRW, energy efficiency,	• District monitoring of macro utility performance against notified benchmarks. • Escalation of benchmark non-compliance to SWSM. • Coordination with macro utility for out-	• Established channel of reporting of bulk supply service-level deviations (flow, pressure, hours) and feedback to DWSM on macro utility responsiveness

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			service continuity, financial efficiency, bulk supply reliability, grievance handling for bulk supply, if any. • Compliance report submitted to NJJM periodically. • SOP for rectification of non-compliance • Strategic improvement plan for defaulter Utilities.	village scheme performance review. • Recorded advance communication to micro-utilities for planned outage and/ or disruption of schedule. • Recorded communication to micro-utilities for emergency shutdown/ stoppage of service.	• Recording intra-village distribution performance data.
2.3	Institutionalisation of Micro Utilities (VWSC) for In-Village O&M	• % HGJ villages with notified micro utilities. • Review of State/District notifications and GP resolutions.	• State/District notification of VWSC/GPWSC as micro utility. • Framework for in-village DW supply services adopted.	• District notification for micro utility; % GPs with micro utility status. • Support to GPs for resolution passing and micro utility setup.	• GP resolution passed adopting micro utility framework. • VWSC designated as micro utility with defined role.
2.4	Performance Benchmarking of Micro Utilities and Tripartite Agreement for O&M Contracts	• No. of DTUs notified benchmark guidelines for micro utility performance. • % Micro Utilities benchmarked in State/ UT. • Adoption of benchmarking framework by States/ UTs.	• State notification/guidance to DTUs for benchmark guidelines for micro utilities. • State-level oversight of tripartite agreement compliance where O&M is outsourced.	• DTU notification of performance benchmark guidelines for VWSCs. • If O&M contracted out for in-village, then, Tripartite agreement signed: district + agency/parastatal + GP.	• VWSC aware of performance benchmarks. • Tripartite agreement signed and available at GP if O&M outsourced • Regular reporting of in-village O&M performance data on Sujal.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
				• Monitoring of in-village O&M performance against benchmarks. • % GPs with signed tripartite agreements where O&M is outsourced.	Goan utility application. • Grievance redressal efficiency- % of grievances redressed and % of adherence to committed timelines for the same.
SECTION 3: Scheme Implementation, Technical Compliance & Quality					
3.1	Commissioning & Handing Over Protocol (Jal Arpan)	• No. of schemes in States commissioned with certified 15-day trial run. • No. of schemes in States formally handed over to Gps/ VWSCs (Jal Arpan). • Review of State commissioning compliance reports.	• State/ UT Gazette notification for commissioning protocol adopted. • Commissioning reports (4-phase completion) uploaded on JJM portal. • Service Improvement Plan (SIP) prepared where gaps identified.	• % schemes in district commissioned with trial run compliance. • Joint inspection by DC/ DM and technical officials conducted. • Handover documents verified and uploaded.	• GP/ VWSC participation in 15-day trial run certified. • Formal handing-over acceptance during Jal Arpan • As-built drawings uploaded to e-Gram Swaraj. • Jal Bandhan/Jal Sankalp pledges conducted.
3.2	Quarterly TPIA Effectiveness Reviews by DWSMs	• National review of TPIA deployment effectiveness across States. • No. of States with revised TPIA TOR implemented.	• State-level review of TPIA effectiveness; systemic issues addressed. • ATRs against TPIA observations tracked. • Revised TPIA TOR (mandatory technical tests + DC/ DM oversight).	• Quarterly TPIA effectiveness review reports submitted. • ATR resolution rate for technical deviations. • TPIA observations uploaded with supporting photos and documentation.	• Facilitation of TPIA visits to village infrastructure. • Compliance with TPIA observations at village level. • Reporting to DWSM after scheme inspection.

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3.3	Corrective Action on NJJM Advisory: NWE Observations	• No. of NWE reports submitted on JJM portal per month. • No. of ATRs received from districts against NWE observations. • Advisory issued by NJJM from NWE field observations tracked.	• Advisory issued by State/SWSM based on NJJM advisory from NWE reports. • Advisory and ATRs uploaded on DWSM dashboard. • Systemic issues from NWE reports addressed at State level.	• Advisory received from NJJM/SWSM acted upon within timeline. • ATRs uploaded on DWSM dashboard for all NWE observations. • Monthly: No. of ATRs resolved against NWE observations.	• Field-level compliance with NWE observations and ATR actions. • Facilitation of NWE visits and access to infrastructure. • Corrective measures at village level as directed by DWSM.
3.4	Corrective Action on NJJM Advisory: CPO (NITI Aayog) Observations	• No. of CPO visit reports submitted on JJM portal (quarterly). • No. of ATRs received from districts against CPO observations. • CPO monitoring shifting to impact-assessment forms tracked.	• Advisory issued by State/SWSM based on NJJM advisory from CPO reports. • Advisory and ATRs uploaded on DWSM dashboard. • State-level resolution of systemic issues in Aspirational Districts/Blocks.	• CPO visits facilitated per schedule; ATRs submitted. • Resolution of all CPO observations; % ATRs resolved. • JJM portal module for CPO visits maintained and updated.	• Facilitation of CPO visits to village infrastructure. • Field-level compliance with CPO observations. • Corrective measures as directed by DWSM/SWSM.
3.5	Corrective Action on NJJM Advisory: Jal Seva Adhyayan	• No. of Jal Seva Adhyayan reports by faculty-led teams on JJM portal (bi-annual). • No. of ATRs received from districts against Jal Seva Adhyayan observations.	• Advisory issued by State/SWSM based on NJJM advisory from Jal Seva Adhyayan. • ATRs against observations uploaded on DWSM dashboard. • Systemic engineering assessment findings addressed at State level.	• Advisory from NJJM/SWSM acted upon; ATRs uploaded (bi-annual). • Coordination with AICTE-empaneled institutions for scheme assessments. • % ATRs resolved against Jal Seva Adhyayan observations.	• Facilitation of student team visits for scheme assessment. • Village-level corrective measures as directed by DWSM. • Participation in Jal Seva Adhyayan data collection.

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SECTION 4: Water Quality Governance & Surveillance					
4.1	Community-Based WQ Testing & Digital Reporting	• No. of States/GPs with digital WQ reporting through Sujal Gaon App Coverage of WQ monitoring across rural area • Trends in contamination cases and waterborne disease.	• State WQM&S SOPs notified as per Concise Handbook. • % of village adopting WQ testing protocol. • Integration of WQ data with State dashboards and citizen corner.	• Completeness and timeliness of WQ data via Sujal Gaon App. • Compliance with prescribed testing frequency (per Concise Handbook). • Time taken for technical intervention after contamination detection. • Technical advisories issued to GPs by DTU/ DWSM.	• No. of samples tested at-least for turbidity and Free Residual Chlorine (FRC) per month per village, Schools and Anganwadi centres (compliance should be as per Concise Handbook). • No. of WQ alerts reported and escalated timely. • Sanitary inspection conducted; corrective actions taken.
4.2	Training of Sujalam Shakti Groups (SHGs)-NSQF L1 Certification for FTK Use	• No. of States with Sujalam Shakti groups trained in HGJ GPs/ villages. • Review of NSQF L1 training rollout and certification records.	• All Sujalam Shakti groups (via SHGs) trained in HGJ GPs/ villages. • Training completion records on e-Gram Swaraj/ GP dashboard. • 1–2 women per village certified at NSQF L1 level as part of Sujalam Shakti.	• % HGJ GPs/ villages in district with trained Sujalam Shakti groups Verification of certification records on e-Gram Swaraj. • Refresher training planned as per State/ district schedule	• Sujalam Shakti group formed and trained at village level. • 1–2 women with NSQF L1 certification using FTKs for regular testing. • FTK use demonstrated in Gram Sabha on regular basis.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
4.3	Water Quality Monitoring and Accreditation/ Recognition of all Laboratory as per ISO/IEC 17025	• % WQ labs accredited/ recognized (ISO/IEC 17025) across States. • Monitor State performance with respect to compliance with testing at various supply components as per the Concise Handbook, laboratory performance, remediation & response time in disease outbreak.	• Monitoring of testing infrastructure readiness and performance shall be undertaken, with corrective actions initiated to ensure compliance with testing norms specified in the Concise Handbook. • Ensure all WQ labs accredited/ recognized; upload accreditation certificates on WQMIS • Trained laboratory personnel per accreditation norms. • TAT compliance: T+1 chemical, T+2 bacteriological. • State level oversight of overall water quality compliance, laboratory performance and implementation of timely remedial actions.	• Ensure timely submission of samples to enable completion of all prescribed testing within stipulated timeliness. • Lab Turnaround time (TATs) monitored. • Oversight of all laboratories under district jurisdiction and initiation of corrective actions in cases of noncompliance. • Monitoring WQ alerts, disease outbreaks, monitor remedial actions taken and escalating to State.	• Ensure adherence to water quality testing requirements at sources, WTPs, storage structures and delivery points, as stipulated in the Concise Handbook. • Village level test results of supplied water at service delivery points to be publicly disclosed via GP boards, Sujalam Bharat PM Gatishakti App and SMS. • Immediate relief measures taken post-contamination detection.

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SECTION 5: Source Sustainability & Water Security					
5.1	Source Sustenance Committees (SSCs) at All Levels	• No. of SSCs constituted at State, District, GP levels nationally through JJM IMIS. • Review of State SSC TORs and meeting schedules.	• State-level SSC constituted under SWSM with defined TOR. • Notification of SSC constitution at all three levels. • Review of District SSC TORs and meeting schedules.	• District-level DSSC constituted under DWSM/DTU with defined TOR. • Notification of GP-level VSSCs constituted in district. • Review of VSSCs TOR and meeting schedule.	• GP/ VWSC-level VSSC constituted with TOR. • Source protection activities undertaken. • Local recharge structures constructed/ maintained.
5.2	Source Sustainability Action Plans (SSAPs)	• No. of States with SSAP approved by SWSM through JJM IMIS.	• State-level review of DSSAPs across districts. • Review of DSSAP and DSS data usage. • Submission of SSAP approved by SWSM. • Guidance to districts on DSS usage (CGWB, NRC, IMD NWIC data).	• Review of GP level SSAPs and integration with GPDP. • Submission of DSSAP document approved by DWSM. • Identification of vulnerable sources and seasonal risks. • DSSAP uploaded on DSS Platform. • Convergence with VB-G RAM-G, watershed, FC grants for source protection.	• Submission of GP/VWSC level SSAP and integration with GPDP. • Implementation of local source protection measures. • Reporting of source stress/seasonal variations to DWSM.

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5.3	DSS Training	• State- level training plan for DSS adoption through JJM IMIS. • No. of SWSMs Officials trained on DSS usage. • Coordination with CGWB, NRSC, IMD, NWIC and Ministry of Agriculture.	• Submission of State-level training plan for DSS adoption. • Coordination with regional CGWB, NRSC, IMD and SWIC. • No. of DWSM/DTU officials trained on DSS usage.	• Submission of District level training plan for DSS adoptions. • All HGJ GPs in district trained to use DSS dashboard. • No. of DTUs and HGJ GPs actively accessing DSS for water security planning. • DSS early - warning alerts reviewed and acted upon. • No. of GPs/VWSCs members trained on DSS usage.	• Submission of GP/VWSC levels training programme on DSS usage. • DSS data used for village water budgeting and source planning. • Village Water Security Plans prepared using DSS output.
5.4	Village Water Budgeting: GP-Level Planning Using DSS Data	• State level water budgeting and water security plan. • Review of water budgeting and GPDP Integration.	• State level water budgeting and water security plan. • State-level oversight of district level water budgeting exercise (using DSS data) and water security plan. • DSS integration with State dashboards.	• District level water budgeting and water security plan. • District level oversight of GP level water budgeting exercise (using DSS data) and water security plan . • Verification of GPDP integration for water security components. • No. of GPs with VWSPs uploaded on GP dashboard.	• Preparation of Village water budgeting using DSS. • Integration of VWSP with GPDP • Water budgeting documents uploaded on GP dashboard. • Adoption of grey water management measures.

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SECTION 6: Digital Governance - Sujalam Bharat					
6.1	Sujal Gaon IDs & Source-to-Tap Digital Mapping	• No. of Sujal Gaon IDs created for all GPs on Sujalam Bharat Platform. • % service areas with complete end-to-end digital mapping.	• % schemes with Sujal Gaon IDs assigned and assets geo-tagged.	• Scheme ID linkage verified. • District dashboard active with scheme lifecycle monitoring.	• Sujal Gaon ID assigned and VWSCs sensitized. • Local data entry of O&M activities linked to Sujal Gaon ID.
6.2	GIS-Based Asset Mapping: All assets mapped on Sujalam Bharat platform to be Geotagged on the Sujalam Bharat PM Gati Shakti mobile app	• % assets geo-tagged and approved on Sujalam Bharat Platform. • Integration of JJM-IMIS with Sujalam Bharat PM GatiShakti app for Pipeline and asset mapping reviewed.	• % of Assets mapped on Sujalam Bharat which have been geotagged on Sujalam Bharat PM GatiShakti App in the State.	• % of Assets mapped on Sujalam Bharat which have been geotagged on Sujalam Bharat PM GatiShakti App in the district.	• All in-village assets geo-tagged on Sujalam Bharat PM GatiShakti Mobile App. • Asset records linked to Sujal Gaon ID.
6.3	Sujal Gaon Utility App: Standard Transactional Platform for O&M	• % HGJ GPs with Sujal Gaon App adopted and active. • Daily O&M transaction data reviewed at National level.	• State notification designating Sujal Gaon App as standard O&M platform. • SOP for daily data entries; API integration with JJM-JJM portal confirmed. • % macro and micro utilities with active daily O&M entries.	• % GPs in district with daily O&M entries on Sujal Gaon App. • Support for onboarding and training of VWSC members on app. • Review of transaction completeness and regularity.	• Daily O&M data entered on Sujal Gaon App (supply hours, complaints, WQ tests). • App-based grievance logging and tracking. • Service delivery charge collection data recorded on app.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
6.4	Project Monitoring Module: Physical Progress on Sujalam Bharat	• No. of schemes with regular progress updates on Sujalam Bharat PMM.	• Scheme-wise progress data validated and uploaded on JJM portal. • Commissioning and handover milestones tracked State-wide.	• Component-wise infrastructure creation progress monitored. • Milestone interdependency tracked (completion → commissioning → handover).	• Participation in scheme commissioning verification. • Milestone certification provided before Jal Arpan.
6.5	District Command & Support Centre (DCSC) Under DWSM	• No. of DCSCs established and operationally active. • Response time for technical support requests benchmarked.	• SCSC consolidates DCSC data; API integration with SWSM dashboard. • Escalated complaints and service failures resolved through SCSC.	• DCSC establishment order; operational status confirmed. • Near real-time monitoring of scheme functionality via Sujalam Bharat/Sujal Gaon App/IoT. • Grievance response mechanism for citizen complaints operational. • Spare parts inventory maintained; periodic reports submitted to SCSC.	• Complaints/ grievances logged through Meri Panchayat App/ e-Gram Swaraj. • Breakdown and service restoration events reported to DCSC.

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6.6	State Command & Support Centre (SCSC) Under SWSM: API to NJJM	• API-based data transfer from SCSC to NJJM confirmed.	• SCSC establishment order; API integration with SWSM dashboard. • Grievance response mechanism integrated at SCSC. • State-level performance reported to Ministry via Sujalam Bharat.	• Periodic operational reports submitted to SCSC. • District-level data feeds to SCSC validated.	• Grievance escalation flow (GP→DCSC→ SCSC) understood and used. • Feedback on response time from SCSC.
6.7	Citizen Grievance Interface: Meri Panchayat App & e-Gram Swaraj linked to Sujal Gaon IDs	• National framework for citizen grievance monitoring reviewed. • Citizen satisfaction trends via JSA data.	• Integration of Meri Panchayat App with Sujal Gaon IDs confirmed. • State-level grievance escalation mechanism operational.	• % GPs with active citizen feedback interface linked to Sujal Gaon IDs. • No. of grievances received and resolved monthly on DCSC. • Jan Sunwai/Jal Chaupals organised for citizen redressal	• Complaint register maintained at GP. • Grievances filed via Meri Panchayat App linked to Sujal Gaon ID. • % complaints resolved within prescribed timelines.
SECTION 7: Participatory Governance, Community Engagement & Evaluation					
7.1	Public Disclosure and Community Awareness	• No. of States built mechanism/system for Public Disclosure system (Quarterly) on functionality, water quality status, O&M fund collection & utilization,	• No. of Districts implemented mechanism/system for Public Disclosure system (Quarterly) on key service areas.	• No. of GPs implemented mechanism/ system for displaying (Disclosure system) key service parameters (Quarterly).	• No. of GP built mechanism for Public Disclosure mechanism built on following services at quarterly level: • WQ Testing Status. •Functionality

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		utilization, Source Sustainability actions and grievance mechanisms.			• Functionality Status. • Grievance Redressal Data. • O&M Fund Collection & Utilisation. • Source Sustainability Actions. • No. of Community Orientation organized on Public Disclosure and its importance.
7.2	No. of States implementing Regular Community Engagement through Jal Utsav • Jal Mahotsav • Rajya Jal Utsav/Nadi Utsav • Lok Jal Utsav	• Whether Calendar and Themes of Jal Mahotsav prepared/planned and organized as per Protocol (SoP). • No. of States prepared Rajya Jal Utsav/Nadi Utsav Calendar and organized as per Calendar and protocol (SoP). • No. of Districts prepared Calendar of Lok Jal Utsav and organized Lok Jal Utsav as per Protocol (SoP). • No. of States institutionalized periodical O&M during Jal Utsav. • No. of National level Event organized on Jal Mahotsav.	• State level Calendar and Themes of Rajya Jal Utsav/Nadi Utsav prepared/planned and organized as per Protocol (SoP). • No. of Districts prepared Calendar of Lok Jal Utsav and organized Lok Jal Utsav as per Protocol (SoP). • No. of Districts institutionalized periodical O&M during Jal Utsav. • No. of State level Event organized in convergence with other Departments.	No. of GPs/Villages prepared Calendar of Lok Jal Utsav and organized Lok Jal Utsav as per Protocol (SoP). • No. of GPs declared HGJ during Jal Utsav. • No. of GPs conducting review of O&M during Jal Utsav. • Sharing of Jal Aankalan data and Service Improvement Plan through District level Event. • No. of State level Events organized in convergence with other Departments.	• No. of Jal Arpan planned and organized during Jal Utsav. • Orientation of VWSC done on roles and responsibilities conducted. • Ongoing Scheme Assessment/Review done by. GP/VWSC through Transect walk and Gram Sabha/Jal Chaupal. • Beautification & Painting of All Assets with JJM approved Logo & IEC Messaging. • Activities on Jal Bandhan, Jal Pledge activities done.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
					• No. of Awareness Generation (including digital) Sessions conducted for judicious use of water/water conservation. • No. of people felicitated for exemplary work. • Lok Jal Utsav Calendar prepared.
7.3	Jal Arpan (Community-led Handover)	• No. of states prepared Jal Arpan Annual Calendar & organized as per protocols against planned.	• No. of District level Calendar/ Plan prepared for eligible Village/ GPs and organized against eligible for handing over with protocols. • No. of District level Event organized on Jal Arpan.	• No. of Jal Arpan Calendar/ Plan prepared for eligible Village/ GPs and organized against eligible for handing over with protocols.	• Orientation of VWSC done on roles and responsibilities. • 15-30 Days Trial Run Completed. • VWSC/ PRI members had a transect walk to review/ check/ assess their PWS (Health Audit). • As built Diagram and other handing over document received by GP. • Sanitary inspection done for sources and Delivery point. • Validation through Gram Sabha/Jal Chaupal for hand-over. • Activities on Jal Bandhan, Jal Pledge activities done.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
					• Public Disclosure of All Assets and Scheme related details through IEC. • Beautification & cleaning of All Assets with JJM approved Logo & IEC Messaging.
7.4	Inclusion of WASH Curriculum in Schools	• No. of States with SCERT circular issued for WASH curriculum. • Review of WASH curriculum integration progress nationally.	• SCERT circular issued for implementation of WASH curriculum in schools. • State-level coordination between SWSM and School Education Department. • Annual calendar of school WASH activities linked to Jal Utsav events.	• Dissemination of SCERT circular to all schools in district. • No. of schools with WASH curriculum integrated. • School Jal Sabhas on water testing mandated in all blocks.	• School Management Committee aware of WASH curriculum implementation. • Water quality awareness activities in schools (FTK demonstrations) • Students participate in 'Paani aur Swasthya' pe Charcha activities.
7.5	Jal Seva Aankalan (JSA): Functionality Assessment with SIPs	• National-level benchmarks for JSA assessment parameters reviewed and updated. • Annual Functionality Assessment Report published based on aggregated JSA data: KPI for fund release under JJM 2.0. • JSA periodicity announced by	• JSA institutionalised in all HGJ GPs; progressive expansion to all GPs planned with State-level calendar. • State notification designating JSA as mandatory annual/periodic exercise. • % districts with JSA completed and State	• % GPs in district with JSA completed against target. • DTU technical diagnosis providing specific intervention recommendations for each GP gap. • District Improvement Plan (DIP) prepared based on JSA findings	• JSA submitted digitally through JJM-IMIS Panchayat Dashboard by Panchayat Secretary. • JSA findings discussed and validated in Gram Sabha; resolution passed. • Participation of women, AWW, ASHA in assessment

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
		NJJM; progressive expansion from HGJ GPs to all GPs tracked. • Third-party evaluation findings (NITI Aayog DMEO) cross-referenced with JSA outcomes. • % States with State Improvement Plans (SIP) submitted based on JSA findings. • JSA-based Functionality Assessment Report used as KPI for calibrated fund release.	Improvement Plan (SIP) submitted. • Annual Functionality Assessment Report compiled from aggregated JSA data at State level • Review of JSA coverage, data quality and compliance trends across districts. • State Improvement Plan (SIP) prepared based on aggregated district JSA findings and proposed corrective actions.	with defined responsibilities, timelines and resource allocation. • % corrective actions completed against DIP targets; pending items escalated to SWSM.	process ensured. • Citizen feedback through Meri Panchayat App integrated with JSA improvement planning.
7.6	Two-way Communication through Samvad Platforms	• No. of States covered for DC/ DM Pey Jal Samvad. • No. of States covered for Sujal Gram Samvad.	• No. of Districts covered for DC/ DM Pey Jal Samvad. • No. of Districts covered for Sujal Gram Samvad.	• No. of GP/Villages covered for DC/ Dms Pey Jal Samvad. • No. of GPs/Villages covered for Sujal Gram Samvad.	• No. of GP level good case studies presented during DC/DM Pey Jal Samvad. • No. of GP level good case studies presented during Sujal Gram Pey Jal Samvad. • Citizen feedback through Meri Panchayat App integrated with JSA improvement planning.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
7.7	State Impact Assessment (IA) Report	• No. of State conducted Impact Assessment in convergence with MoHFW, MoWCD, MoE for inter-ministerial impact data. • No. of National event organized on the issues. • Tools for impact assessment developed in convergence with other relevant ministries.	• No. of districts conducted Impact Assessment in convergence with MoHFW, MoWCD, MoE for inter-ministerial impact data. • No. of State event organized with convergence.	• Inclusion of WASH Curriculum in Schools.	• Public Disclosure of All Assets and Scheme related details through IEC. • Beautification & cleaning of All Assets with JJM approved Logo & IEC Messaging.
SECTION 8: Capacity Building Framework					
8.1	Training of HGJ Village Field Functionaries	• No. of States which have identified and finalized Nodal Training Institute at State level for different Stakeholders. • No. of States prepared and submitted approved AAP with Calendar. • Standardised Content prepared for different stakeholders. • No. of States completed the Training Programmes as per AAP and uploaded on	• Identification and finalization of Nodal training Institute(s). • No. of Districts identified and finalized Training Institutes for different Stakeholders. • Collaboration with different partners/ line Ministries/Deptt. • AAP and Calendar prepared based on District Action Plan. • Customization & creation of Training contents as per need considering local context and language.	• No. of Institutes identified at district level for different stakeholders. • Preparation of District Action plan for different stakeholders including DTU/PRI/ Sujalam Shakti Groups etc. • Number of Trainings completed as per Calendar and details of targeted participants covered. • No. of HGJ GPs with all field functionary.	• No. of participants nominated by Gram Panchayat for different categories of Training for District Action Plan. • No of GP level field functionaries (category wise) completed the training programme. • Gaps identified through Jal Seva Aankalan and other appropriate assessment (if done). • No. of participants.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
		Training dashboard/IMIS/ e-GramSwaraj. • No. of States conducted Gap Assessment (post Training). • No. of States conducted Refresher Training. • Monitor HGJ GPs with all field functionary training completed.	considering local context and language. • No. of ToT completed as per AAP/Calendar and uploaded on Training dashboard/IMIS/ e-GramSwaraj. • No. of trainings completed as per AAP/Calendar and uploaded on Training dashboard/IMIS. • Gap Assessment conducted (post training) by all Districts. • No. of Districts completed Refresher Training through Master Trainers. • No. of State level Officials/Functio naries trained. • Monitor field functionary training in all Har Ghar Jal villages with at least one trained and certified Nal Jal Mitra per Gram Panchayat.	training completed as per CIRP. • Gap Assessment of trained stakeholders conducted. • No. of Refresher training conducted for the identified stakeholders based on Gap Assessment. • No. of District level Officials/Functio naries trained. • No. and details of GP/Village level field functionaries (category wise) trained. • Monitor HGJ GPs in district with all field functionary training completed. • Monitor training covers O&M, water quality, financial governance and digital tools.	identified based on the gap assessment, nominated and completed the Refresher Training. • All VWSC members and field functionaries trained. • Record of all Training to be maintained at GP level and upload on e-GramSwaraj/GP Dashboard.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
SECTION 9: Skilling And Human Resource Development					
9.1	District-wise O&M Human Resource (HR) Gap Assessment & Training Plans	• Monitor state-wise plan, progress, enrolment, training and certification of skilled human resource for O&M of PWS.	• Undertake assessment of human resource requirements at all levels for O&M of schemes and compute district-wise training gaps. • District-wise progress with respect to NJM, technician (out-village), operator, Sujalam Shakti Groups' enrolment, training and certification for both in-village and out-village O&M. • Monitor quality of training delivery through assessment pass rates, dropout rates and feedback from certified personnel and Gram Panchayats.	• District-wise O&M HR Gap assessment completed and submitted in Action plan for NJM, technician, operator, Sujalam Shakti Group for in-village and out-village both.	• Monitor GP-level requirement of HR for in-village O&M needs to be identified for training and reported in action plan.
9.2	Adoption and roll-out of training Courses	• Coordinate for development of new skill courses (Sujalam Shakti etc.) in alignment with NSQF.	• Monitor adoption and roll out the training courses across identified training institutions in district.	• District to coordinate with SSDMs, ITIs, Polytechnics, RSETIs and other implementing	• Mobilise and recommend candidates for skill training.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
		• Review adoption of NSQF Level-4 Jal Vitaran Sanchaalak course and other NSQF aligned courses. • No. of States conducted Gap Assessment (post Training). • No. of States conducted Refresher Training. • Monitor HGJ GPs with all field functionary training completed.	• Monitor adoption of NSQF Level-4 Jal Vitaran Sanchaalak course and other NSQF aligned courses. • Monitor appointment of 'Programme Implementing Agencies (PIA)' for planning, coordination and delivery of all training. • Ensure adequacy of training infrastructure including accredited institutions, operational batches and geographic coverage across blocks and districts.	agencies for preparation of proposal and training delivery If required appoint 'Programme Implementing Agencies (PIA)' for delivery of all training.	
9.3	HR Policy notification and deployment	• State-wise status of HR policy notification to be monitored. • Monitor deployment status of certified skilled HR O&M requirements.	• Formulate and notify appropriate policy and guidelines for engagement of human resources at rural water service utility levels. • Monitor deployment status of	• DWSM dashboard tracking of skilled HR deployment. • Monitor deployment status of certified Nal Jal Mitras and Multi-skill Technicians – mapped against scheme-wise	• Certified NJM deployed for in-village O&M of water supply schemes.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
			certified Nal Jal Mitras and Multi-skill Technicians- mapped against scheme-wise O&M requirements for each district.	requirements.	
SECTION 10: Operational & Financial Sustainability					
10.1	Financial Monitoring Tool for O&M	• Review adoption status of structured financial monitoring tools linked with Sujal Gaon ID.	• Develop or customise a State-level financial planning and monitoring tool linked to Sujal Gaon ID and SUJALAM Bharat platform. • Monitor key financial indicators (O&M cost recovery, collection efficiency, NRW, utilisation). • States to adopt structural reforms alongside providing funds, function and functionaries for sustainable O&M of rural piped water supply infrastructure. • Monitor service delivery charge collection and service delivery charges for bulk water transfer.	• O&M Budget Utilisation monitored (Actual ÷ Approved budget) for every panchayat. • Timeliness of financial reporting to State; fund convergence tracked. • Monitor key financial indicators (O&M cost recovery, collection efficiency, NRW, budget utilisation, energy efficiency). • Monitor service delivery charge collection and service delivery charges for bulk water transfer.	• Maintain financial data of micro utility (billing, collection, expenditure) accurately and timely. • Maintain transparent records of receipts and payments. • Support digital billing, collection and expenditure tracking. • O&M accounts maintained by VWSC. • Report O&M Cost Recovery Ratio (service delivery charges ÷ O&M expenditure). • Report collection efficiency (amount collected ÷ amount billed). • Report electricity cost

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
10.2	Annual O&M Financial Provisioning	• Review of State-level O&M financing policy. • Review of State-level O&M budget provisioning.	• Assess fund availability (service delivery charges, FC grants, State funds, CSR, DMF). • Ensure funding gaps are addressed through State financial resources. • Provision of annual O&M funds in separate State budget head.	• Compile village O&M budgets and bulk system O&M needs. • Validate estimates and submit consolidated district demand to State. • Track release and utilisation of allocated O&M funds. • Monitoring NRW at the district level; corrective action for high-NRW schemes.	• Prepare annual scheme-wise O&M budget estimates using SWSM formats. • Identify gap between service delivery charge collection and required O&M cost. • Place O&M budget in GP/VWSC meetings for approval and monitoring. • Non-Revenue Water (NRW) data to be submitted to DWSM.
10.3	Preventive Maintenance (PM) Schedule for All Schemes	• Review of predictive maintenance adoption across States.	• Monitor planning and adoption of preventive maintenance schedule. • Number of districts with a preventive maintenance schedule for all schemes.	• Prepare scheme-wise preventive maintenance schedule with the help of DTU. • Ensure schedules are shared with operators and VWSCs. • Asset register updated with condition data, repair history, lifecycle planning. • Inventory management for spares ensured. • Engage the community in regular surveillance and vigilance to identify disruptions early.	• Follow the preventive maintenance schedule on ground.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
10.4	Operational efficiency	Monitor formation and adoption of SOPs for operational efficiency.	• Develop SOPs for pump operation, chlorination, valve control, water quality testing and customer grievance handling. Real-time monitoring tools such as flow meters, pressure sensors, online water quality devices and SCADA help detect issues early and improve decision-making.	• Monitor energy use, water production, service hours and breakdowns as per SOPs. • Daily data entry of operational activities on Sujal Gaon App.	• Routine system checks; repair of leakages within defined timelines. • Report energy use, water production, service hours and breakdowns etc. as per SOPs.
SECTION 11: Research, Innovation & Knowledge					
11.1	State Technical Committee (STC): Notified with TORs	• NTC evaluated innovative technologies to SWSM/STC. • STC constitution and alignment with NTC TORs.	• STC constituted and notified with TOR for technology appraisal, pilot approvals, scale-up. • STC reporting to SWSM • TOR aligned to NTC framework. • SWSM/ STC to discuss/adopt NTC recommended technologies for district/regional technological needs. • Pilot implementation of the technologies in the district	• Assessment and submission of district specific technology proposals through the STC. • Pilot implementation sites identified in district.	• Awareness of new technologies for O&M and WQ. • Participation in pilot technology implementation at village level.

Sec. No.	Monitoring Parameter	NJJM (National Level)	SWSM (State Level)	DWSM (District Level)	GP / VWSC (Village Level)
11.2	Outcome-Based Financing (OBF) of State & UT for R&D Programme	• STC/ SWSM proposals for R&D in line with Annual Action plan reviewed/approved by NTC/ NJJM. • OBF Framework by States/ UTs in line with NJJM.	• State-led R&D and OBF framework notified. • Call for proposals shortlisting/ approval. • Monitoring of R&D projects.	• Pilot sites in district geo-tagged and monitored. • Progress reports pilot submitted	• Participation in field validation of pilot technologies. • Feedback provided to DTU/ STC on technology performance.

Annexure X

District Level Impact Evaluation Matrix (Indicative)

District impact indicators are used to understand broad changes and trends linked to improved drinking water services by combining JJM data, community assessments and data from other sectors. These indicators show overall contribution and progress and are not meant to prove direct cause-and-effect outcomes at the district level.

IMPACT AREA 1: Health Impact

Level	Indicators	Data Source
Output	• Scheme details with Har Ghar Jal Status • Functionality status of schemes. • Citizen Grievance Linked Incidences and nature of issues highlight. • Percentage Har Ghar Jal receiving regular drinking water supply. • Percentage of HHs with coverage under each habitation. • Types of treatment/disinfection technology used in the piped water supplied. • Digital service logs validating continuity and chlorination status at habitation level.	• Primary questionnaire on citizen perception about the impact and ground-level information collection. • Scheme data DWSM dashboard and JJM IMIS. • Functionality from Jal Seva Aankalan. • Citizen grievance through Meri Panchayat App and GP dashboard.
Outcome	• Reduction in diarrhoeal/water-borne incidence among under 5 children due to increase in piped water supply coverage. • Establishing whether piped water supply provision has an impact on diarrhoeal/ water-borne incidences. • Matching health outcomes with overall functionality of Sujal Gaon.	• Primary questionnaire on citizen perception about the impact and ground-level information collection. • On-ground primary questionnaires data, Data records from AWCs, ASHA and ANM. • Digital records of Poshan Tracker Health data, NHP, NFHS, IDSP, SRS (Sample Registration System) Office of the Registrar General & Census Commissioner of India, other relevant health data.
Impact	• Savings in Medical out of Pocket expenditure • Reduced under 5 mortality linked to water borne diseases due to increase in piped water supply coverage. • Impact on overall Life Expectancy of the rural population linked to DALYs (Disability Adjusted Life Years). • Possible magnitude of impact of reliable drinking water service records on health outcomes.	• Time series analysis of all the above-mentioned data can be carried out to assess the impact.

IMPACT AREA 2: Nutrition and Child Development

Level	Indicators	Data Source
Output	• Scheme details with Har Ghar Status. • Functionality status of schemes. • Citizen Grievance Linked Incidences and nature of issues highlight. • Percentage Har Ghar Jal receiving regular drinking water supply. • Percentage of HHs with coverage under each habitation. • Types of treatment/disinfection technology used in the piped water supplied. • Digital service logs validating continuity and chlorination status at habitation level.	• Primary questionnaire on citizen perception about the impact and ground-level information collection. • Scheme data DWSM dashboard and JJM IMIS. • Functionality from Jal Seva Aankalan. • Citizen grievance through Meri Panchayat App and GP dashboard.
Outcome	• Reduction in malnutrition levels (stunting, wasting, underweight). • Enables district wise comparison between Har Ghar Jal status villages and non-Har Ghar Jal status villages.	• Primary questionnaire on citizen perception about the impact and ground-level information collection. • Digital records from NFHS, Poshan Tracker, etc.
Impact	• Overall improved nutrition levels. • Establishing possible impact of drinking water supply delivery coverage on nutrition levels. • Time series service continuity data for long term improvement and impact assessment.	• Time series analysis of all the above-mentioned data can be carried out to assess the impact.

IMPACT AREA 3: Education and School Attendance

Level	Indicators	Data Source
Output	• Percentage of schools with functional water supply. • Percentage of AWCs with functional water supply.	• Primary questionnaire on citizen perception about the impact and ground-level information collection. • Scheme data DWSM dashboard and JJM IMIS. • Functionality from Jal Seva Aankalan. • Citizen grievance through Meri Panchayat App and GP dashboard.
Outcome	• Increased enrolments of children in schools. • Increased attendance and retention of girls.	• Primary questionnaire on citizen perception about the impact and ground-level information collection. • On-ground primary questionnaires data, Data records from AWCs, School Attendance, Annual Status of Education Reports (Rural). • Digital records from UDISE+ data, Poshan Tracker, NFHS.
Impact	• Increase in overall education levels of students attributable to piped water supply provision. • Increase in years of schooling and potential lifetime earnings especially for girls. • Time series data records linked to piped water supply service area delivery.	• Time series analysis of all the above-mentioned data can be carried out to assess the impact.

IMPACT AREA 4: Reduction in Off-premises Water Collection linked Drudgery

Level	Indicators	Data Source
Output	• Scheme details with Har Ghar Status • Regularity and Functionality. • Citizen Grievance Linked Incidences and nature of issues highlight. • Percentage Har Ghar Jal receiving regular drinking water supply. • Percentage of HHs with coverage under each habitation.	• Primary questionnaire on citizen perception about the impact and ground-level information collection. • Scheme data DWSM dashboard and JJM IMIS • Functionality from Jal Seva Aankalan. • Citizen grievance through Meri Panchayat App and GP dashboard.
Outcome	• Reduction in time spent for fetching water for Men, Women and Children.	• Primary questionnaire on citizen perception about the impact and ground-level information collection. • On-ground primary questionnaires data, Data records from AWCs, School Attendance, ASHA and ANM, SRLM data on engagement of women in livelihood activities and other data from livelihood programs running in the districts. • Annual data on Micro-credit mobilization of the District (District Lead Bank). Digital records from UDISE+ data, Poshan Tracker, NFHS.
Impact	• Increased participation of women in education, livelihoods and other economic activities. • Impact on Labour Force Participation Rates.	• Time series analysis of all the above-mentioned data can be carried out to assess the impact.

IMPACT AREA 5: Economic Development & Livelihoods

Level	Indicators	Data Source
Output	• Local jobs created due to O&M of schemes. • Number of locally trained workers (plumbers, pump operators, FTK testers) deployed. • SHGs engaged in water source sustainability works (soak pits, recharge, greywater reuse) • SHG members trained in O&M / water quality testing. • Scheme details with Har Ghar Status. • Citizen Grievance Linked Incidences and nature of issues highlight. • Percentage Har Ghar Jal receiving regular drinking water supply. • Residential/Commercial/ Institutional/ Industrial developments enabled to due provision of piped water supply schemes in rural and peri-urban areas.	• Primary questionnaire on citizen perception about the impact and ground-level information collection. • Scheme data DWSM dashboard and JJM IMIS. • Functionality from Jal Seva Aankalan. • Citizen grievance through Meri Panchayat App and GP dashboard.
Outcome & Impact	• Indicator on increase in local rural entrepreneurship due to rural water supply infrastructure. • Increase in livelihood created due to growth of local O&M and market ecosystem. • Increase in women's participation in income generating activities due to time savings. • Increase in land value of rural and peri-urban areas due to provision of piped water supply. • Impact on overall Ease of Living. • Increased contribution in agricultural and allied sector activities.	• Primary questionnaire on citizen perception about the impact and ground level information collection. • JJM data on number of people skilled and employed in O&M activities. • District Town Planning Records for Peri-Urban Areas (Baseline), PRI feedback, District Registrar Office for assigning the enhancement in Land Value due to water infrastructure. • NRLM and SRLM data, District employment data on rural employment of women, Data on increased participation of women in agriculture and allied sector – Agriculture data. • PLFS (district trends), qualitative district assessment.

IMPACT AREA 6: Rural Tourism Enablement

Level	Indicators	Data Source
Output	• Percentage of HHs with coverage under each village. • Percentage of rural population involved tourism activities in the village. • Number of Villages in Districts with Tourism as a product under the 'One District One Product' (ODOP). • Number of villages updated and classified under Best Tourism Villages. • Availability of water in public facilities (schools, AWCs, homestays, community spaces).	• Primary questionnaire on citizen perception about the impact and ground-level information collection. • JJM IMIS data. • State Department of Tourism reports, District Verification reports, District Tourism reports.
Outcome	• Increase in rural livelihood activities in villages (homestays and local services). • Increase in tourism linked activities in the villages. • Linking overall functionality of Sujala Gaon with tourism schemes convergence.	• Primary questionnaire on citizen perception about the impact and ground-level information collection. • District Tourism Dept., ODOP data, Ministry of Tourism.
Impact	• Percentage of rural population involved tourism activities. • Long-term value capture in tourism due to provision of piped water supply to villages. • Per capita increase in income due to increase in tourism activities linked to piped water supply coverage increment. • Improvement in hygiene and visitor friendly conditions in villages.	• Time series analysis of all the above-mentioned data can be carried out to assess the impact, with focus on Har Ghar Jal status.

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Note:

Har Ghar Jal
Jal Jeevan Mission

Building Partnership
Changing Lives

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